

FEDERAL HIGHWAY ADMINISTRATION

TURNER-FAIRBANK HIGHWAY RESEARCH CENTER REPORT

***I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation:
Mechanical Behavior of Powers Fasteners Power-Fast+ Adhesive Anchors***

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CHAPTER 1. INTRODUCTION

Late in the evening of July 10, 2006, ten two-ton concrete panels suddenly fell from above the roadway of the eastbound I-90 Seaport Access Tunnel in Boston, Massachusetts. The National Transportation Safety Board (NTSB) is leading the investigation to determine the cause of the failure. In support of this NTSB effort, the Federal Highway Administration (FHWA) deployed staff and contractors to Boston on July 15, 2006. This initial work by FHWA is summarized in a report titled *“I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Load and Load-Displacement Testing Results.”*¹

In the portal of the I-90 Seaport Access Tunnel, the concrete panels are hung from the roof with hanger brackets. The hanger brackets are attached to the ceiling with threaded studs which are embedded into the concrete and held in place with a two-part epoxy. This threaded stud and epoxy system is more commonly known as an adhesive anchor. Initial observations at the accident site indicated that the suspended panels fell when one or many of these adhesive anchors pulled out the holes in which they were embedded.

The adhesive anchors used in the tunnel, installed between July and December 1999, were in service about seven years at the time of the accident. The adhesive anchors used are an eight inch long threaded stud embedded approximately five inches into the concrete ceiling. The stud is 5/8 inch in diameter with 11 threads per inch, and made from AISI 316 stainless steel. The stud was adhered to the concrete hole with a two-part epoxy supplied by Powers Fasteners of Brewster, NY. The epoxy Powers used was formulated by the Sika Corporation (Lyndhurst, NJ) and they reported no significant changes in the epoxy formulation between 1999 and 2006. Sika also sells the Power-Fast epoxy anchor system directly as Sikadur AnchorFix-3 and Sikadur AnchorFix-4. Construction invoices researched by the State of Massachusetts show that the epoxy used was a fast-set formula of the Powers Foil-Fast epoxy. This product is still available from Powers Fasteners by the current brand name Power-Fast+ Fast-Set Formula.

In collaboration with the NTSB, FHWA staff at the Turner-Fairbank Highway Research Center (TFHRC) performed laboratory testing on the Power-Fast+ adhesive anchor system. The purpose of this testing was to evaluate the effect of certain installation parameters (hole type, preparation, etc.) on the adhesive anchor’s mechanical behavior. Static tension load-displacement tests were used as the means to evaluate mechanical behavior. This document summarizes the procedures and results from this laboratory parametric testing.

CHAPTER 2. EXPERIMENTAL PROGRAM

The performance of the adhesive anchors was evaluated through static tension load-displacement tests that simultaneously monitored the load and displacement behavior of the anchor. This chapter summarizes how the experiments were conducted in the TFHRC Structures Laboratory.

TEST SLAB

A rectangular concrete test slab was designed to mimic the tunnel roof. The ASTM E488-96² standard was used for guidance to size the slab. The standard dictates the test slab should be no thinner than 1.5 times the embedment depth. In addition, adjacent anchors or support reactions should be spaced a minimum of one embedment depth from any anchor. The embedment holes in the Seaport Tunnel were cored to 5.625 inches deep and the stainless steel threaded rod was embedded to a depth of 5 inches (verified by NTSB³). In order to reproduce these characteristics in the lab, the specifications require a minimum 7.5 inch thick slab with a 5-inch minimum clearance between anchors.

Anchors in the tunnel were installed in groups of either two or four with 10 inch center-to-center spacing. Anchors in the test slab were spaced every 10 inches in the transverse direction and every 6 inches in the longitudinal direction which is slightly above the minimum requirement. The overall dimensions of the slab were 76.5 inches by 104 inches. The slab was designed to be 20 inches thick so that anchors could be installed and tested on both plan faces. Each face was used for a separate phase of testing. Phase 1 used the bottom side of the slab (i.e. the side cast against the plywood form) and Phase 2 used the top side of the slab (i.e. the side that was trowel-finished during casting).

The test slab reinforcement was designed and detailed to prevent flexure and shear cracking during testing. This required an upper and lower reinforcing steel (rebar) mat of #5 bars spaced at 6 inches longitudinally and at 10 inches transversely. A #11 rebar was placed in the center of upper and lower mats in the longitudinal direction because the planned testing parameters included holes cored directly through this bar. The #5 bars in the transverse direction are the primary reinforcement for bending. To fully develop this reinforcement the bars were bent into a closed hoop. Straight bars were used in the longitudinal direction. A schematic of the rebar cage is shown in Figure 1.

Also, shown in Figure 1 are four large diameter holes created by casting 3-inch poly vinyl chloride (PVC) pipe into the slab. These holes facilitate the maneuvering and manipulation of the approximately 13,500 pound slab within the laboratory.

Casting

The test slab was cast on September 25, 2006 using a 4000 psi, #57 mix supplied by a local ready-mix producer. Concrete arrived at the lab at 7:53am, 1.18 hours after being batched. A

representative sample was taken from the middle of the pour to perform slump tests and make cylinders. The slump was measured to be 2.75 inches. Twenty 4x8 cylinders were made, compacted by vibration in 2 layers according to ASTM C192-05⁴, and covered with plastic caps. Finish troweling was complete by 8:11am and the slab was covered by plastic. The slab remained in the formwork to cure, being wetted daily and kept covered by plastic. The test slab forms were stripped on October 2, 2006, and the cylinders were removed from their molds on October 3, 2006.

Concrete Properties

The twenty cylinders cast during the slab pour were tested over a five month period to characterize the concrete properties of the test slab. Primarily the cylinders were tested for compressive strength, however a small series of split tensile strength tests were also conducted. All cylinders had their ends ground smooth for even bearing and to ensure the ends were parallel. The compression tests were done in accordance with ASTM C39⁵ and the split tensile tests with ASTM C496⁶. Testing was done in a 1000 kip Forney cylinder testing machine with a calibration traceable to NIST. Compression tests were conducted two weeks after casting to ensure the necessary concrete strength had been attained before hole coring began on the slab. Additional cylinders were compression tested after 28 days in accordance with ASTM C33. Finally, compression and split tensile tests were conducted on the first day of adhesive anchor testing. The results of all this testing are summarized in Table 1 and 2.

Coring

A Field Engineers Daily Report (FEDR)⁷ created during construction of the Seaport Access Tunnel indicated, and interviews conducted by NTSB confirmed that a Hilti DD-100 core drilling rig with a water collection kit was used to core the holes in the tunnel ceiling. Figure 2 shows this rig fitted with a 3/4-inch Hilti BI-P4 diamond-tipped core bit. In an effort to best reproduce the installation practices used during tunnel construction, the test slab was positioned such that holes were cored with the aforementioned core drilling rig in the overhead position, with the anchors installed overhead also.

The core drill had a depth stop gauge that was set to 5.625 inches. In general, 30 seconds were needed to core holes to this depth. However, the five holes that were purposely drilled through a #11 rebar took approximately 90 minutes each to core. The majority of that time was spent coring through the steel.

Phase 1

Coring for the installation of anchors for Phase 1 of testing began on October 11, 2006. Holes for 10 anchors were cored October 11th and 61 additional holes were cored on October 17th, of the holes cored October 17th, two unintentionally hit a #5 rebar and one which was intended to go through the center of a #11 rebar only grazed it. The remaining holes were drilled on October 18, 2006. A pachometer was used to more precisely locate the #11 rebar so that the remaining

four holes intended to pass through the #11 rebar would be cored through its center. Also on October 18th, ten holes were hammer drilled with carbide-tipped bit using a Milwaukee 1 ½-inch Rotary Hammer drill.

Phase 2

Most of the holes for Phase 2 were cored on January 16, 2007 except for those reserved for the alternative adhesive and damp installation testing parameters. Eight holes were cored on January 25, 2007 and the last five holes were drilled on January 26, 2007.

LOAD-DISPLACEMENT TESTING APPARATUS

The anchors were not pull-tested with the slab overhead because of the difficulty supporting a jacking rig overhead. Rather, the slab was flipped over using the lab's overhead cranes so the jacking rig could rest upon the slab and pull the anchors out in an upward direction.

The adhesive anchors were incremental load-displacement tested using the unconfined method in full compliance with the ASTM E488 Specification. This was accomplished using the 3-post jack stand detailed in Figure 3. The jack stand supported an Enerpac JDH-1508 double acting, hollow-core hydraulic ram (herein referred to as a jack) which was used to apply load and a 100 kip Strainert model FL100U(C)-2DGKT (S/N 08905-16) flat load cell which was used to monitor load. A picture of the entire testing apparatus is shown in Figure 4. The jack was run in load control by adapting a servo valve into the system via a custom distribution manifold. Hydraulic pressure of 3000 psi was provided by MTS pumps and producing a jack capacity of 90 kips. Closed-loop control was maintained by a MTS 458.20 controller monitoring the load cell output. The load cell was calibrated with the MTS 458.11DC conditioner using a load cell traceable to a NIST load standard. The MTS controller also utilized a 458.91 MicroProfiler to permit a precise loading history regime to be applied to the adhesive anchor.

Three loading regimes were programmed into the MicroProfiler. The regimes generally produced a load/time history similar to the one detailed in Figure 5. Each test started by ramping the load on the anchor up to a proof load and holding this load for 2 minutes. The majority of the anchors were proofed with 3250 pounds of load. However, a few tests were conducted with proof loads of 6300 pounds. After the proof load was removed, the anchor was left unloaded for 5 minutes to allow elastic recovery, if any, to occur. Then load was applied in steps with 2 minute hold periods. The ASTM Specifications call for load steps of not more than 15% of the anticipated ultimate load resisted by the anchor. As this target load was unknown prior to testing each of the three load regimes programmed into the MicroProfiler used different load step increments; 1000, 1500, and 2000 pounds. In general, the regime with 1500 pound load steps was used which resulted in test durations of approximately 30 minutes.

Data were collected at a rate of 2 Hz using a Data Translation 16-bit USB data acquisition unit and HP Vee software installed on a Dell laptop. The displacements of the adhesive anchors were monitored with brackets holding either two or three linear variable differential transformers

(LVDT) (model RDP DCTH400AG). Initially a bracket holding three LVDTs was used however failure of the anchors often included a shallow concrete cone which was affecting the displacement measurements recorded. Therefore, altering the apparatus to include a bracket that held two LVDTs positioned further from the anchor was necessary to avoid this interaction. The aluminum two LVDT bracket originally fabricated for Phase 1 proved too delicate and a more robust steel bracket was used during Phase 2. A schematic of all three LVDT brackets is shown in Figure 6.

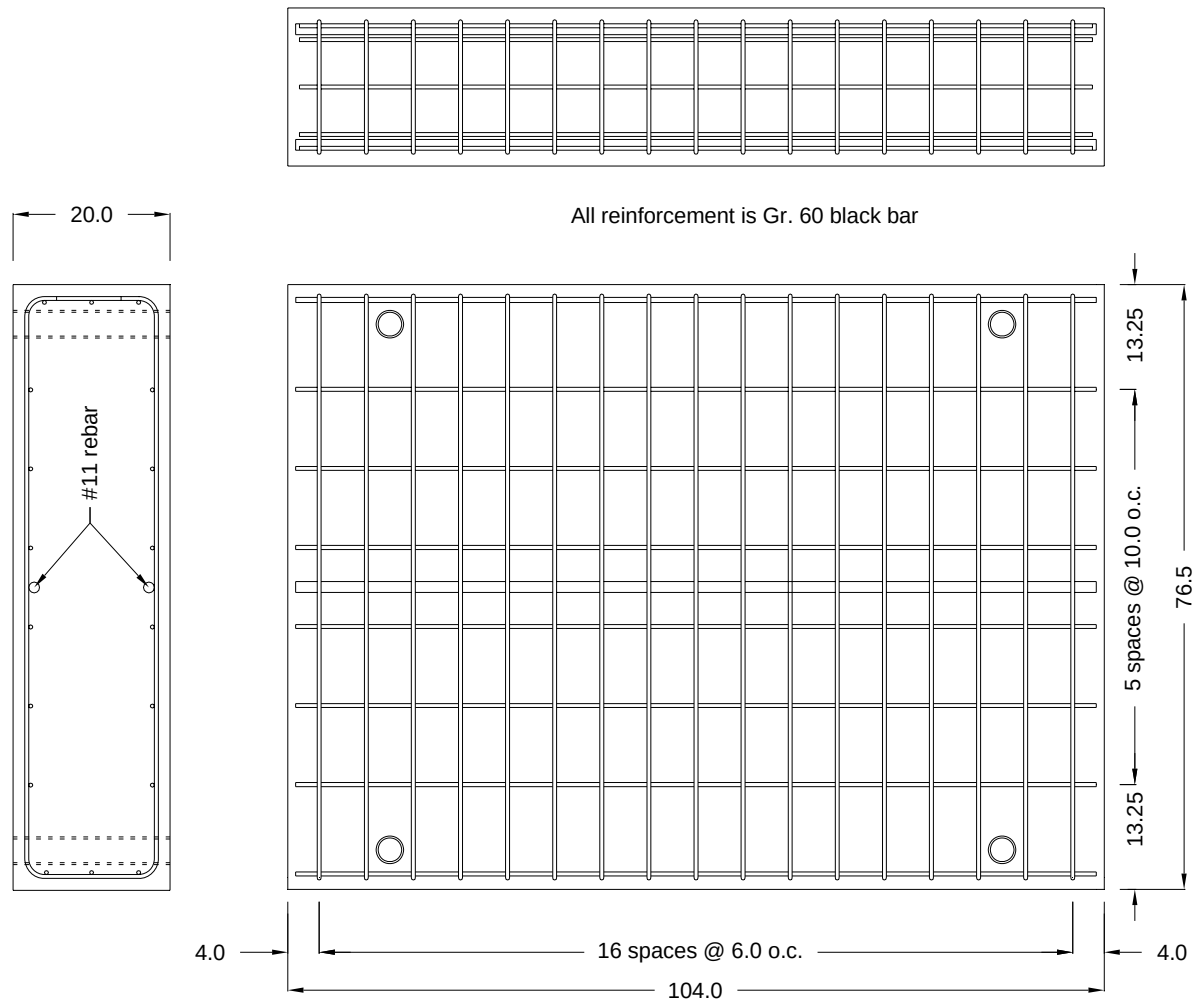


Figure 1. Reinforcement detailing.

Table 1 Concrete Cylinder Compressive Strength

Specimen	Test Date	Length (inch)	Diameter (inch)	Peak Load (pounds)	Peak Stress (psi)
CA/T #2	10 Oct 2006	7.891	4.000	58940	4690
CA/T #13	10 Oct 2006	7.817	4.026	61280	4810
CA/T #14	10 Oct 2006	7.837	4.024	59680	4690
CA/T #16	10 Oct 2006	7.789	4.027	61740	4850
CA/T #18	10 Oct 2006	7.801	4.028	59960	4700
CA/T #19	10 Oct 2006	7.925	4.028	59160	4640
Average	-	-	-	-	4730
CA/T #9	23 Oct 2006	7.980	3.997	63540	5060
CA/T #10	23 Oct 2006	7.977	3.999	62520	4980
CA/T #17	23 Oct 2006	7.760	4.026	66840	5250
CA/T #20	23 Oct 2006	7.902	4.027	63320	4970
Average	-	-	-	-	5065
CA/T #8	17 Nov 2006	8.013	4.000	60120	4780
CA/T #1	17 Nov 2006	7.883	4.025	63220	4970
CA/T #6	17 Nov 2006	7.945	4.000	62900	5010
CA/T #3	17 Nov 2006	7.832	4.024	65060	5120
Average	-	-	-	-	4970
CA/T #5	5 Apr 2007	7.979	4.000	58640	4667
CA/T #7	5 Apr 2007	7.892	3.998	60720	4836 ^a
Total Average	-	-	-	-	4891
^a – specimen out-of-spec because of oval shaped cross-section and not considered in average.					

Table 2 Concrete Cylinder Split-Tensile Strength

Specimen	Test Date	Length (inch)	Diameter (inch)	Peak Load (pounds)	Peak Stress (psi)
CA/T #4	17 Nov 2006	7.935	3.999	28600	574
CA/T #11	17 Nov 2006	7.790	4.023	29140	592
CA/T #12	17 Nov 2006	7.835	4.028	33680	679
CA/T #15	17 Nov 2006	7.752	4.024	30500	622
Average	-	-	-	-	596^a
^a – Specimen CA/T #12 was omitted from the average because it was misshapen and therefore out-of-spec					



Figure 2. Hilti DD-100 drill with water collection kit.

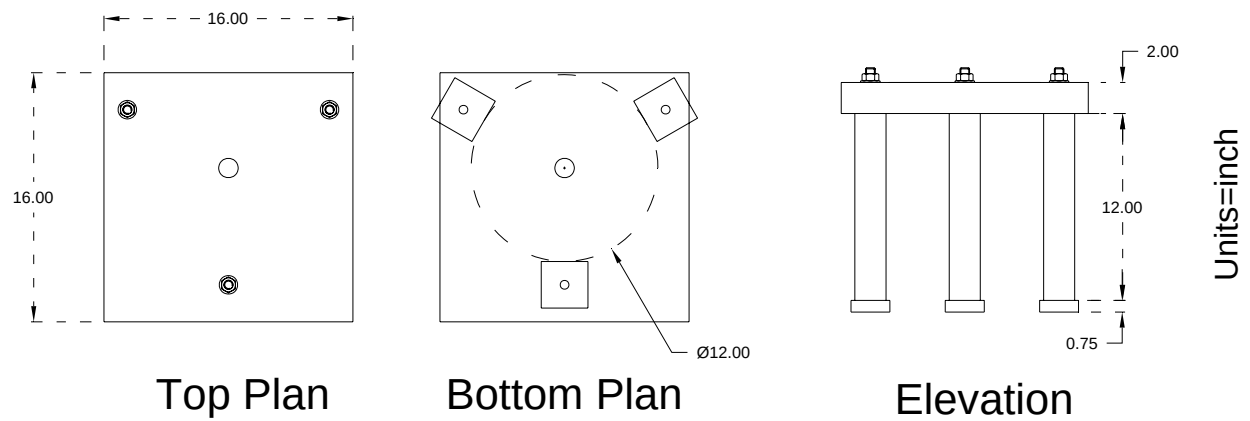


Figure 3. Schematic of jack plate.



Figure 4. Picture of jacking test frame attached to an epoxy anchor.

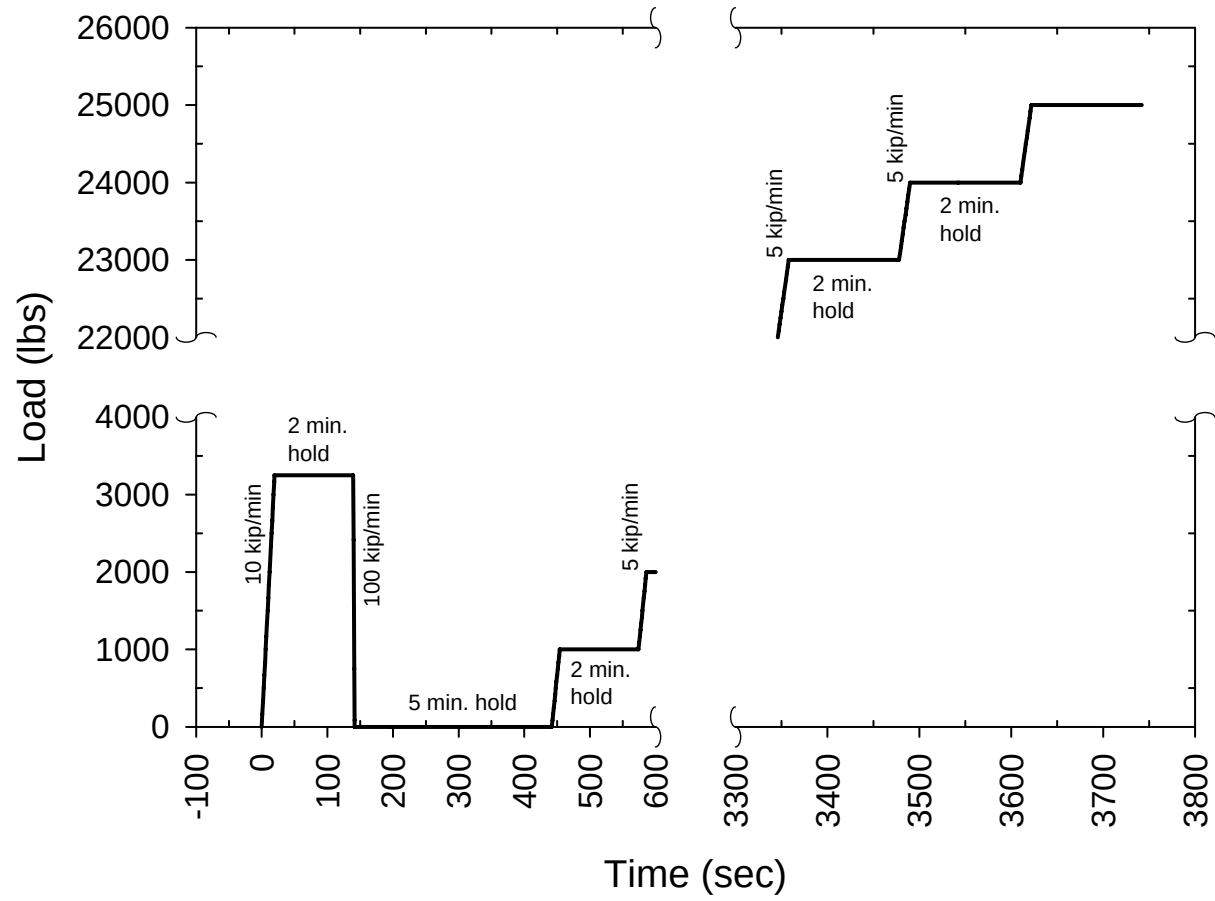
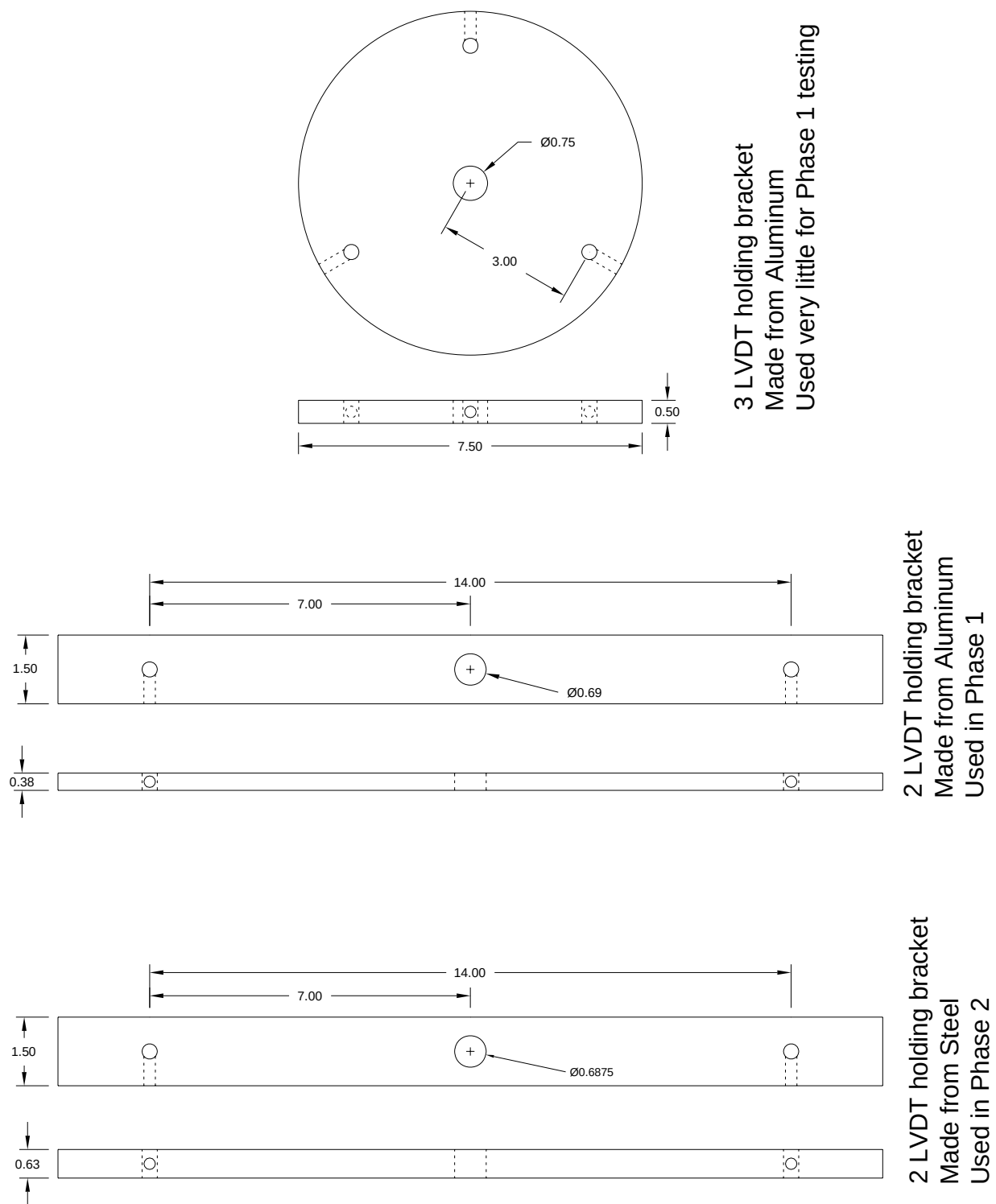


Figure 5. Loading protocol used for static pull tests.



CHAPTER 3. INSTALLATION PARAMETERS

Load-displacement testing was conducted to characterize the influence that variations in installation practices had on adhesive anchor performance. This chapter details the specimen matrices constructed and the protocols used for loading and data collection.

PHASE 1 TEST MATRIX

Collaboratively with NTSB, FHWA identified 11 installation parameters that potentially could have influenced adhesive anchor capacity. The Phase 1 test matrix was constructed to investigate each of these parameters individually. Each of the 11 installation parameters studied in Phase 1 are described in the following sections. Unless otherwise noted, all anchors were installed in a hole that was cored with a diamond tipped core bit, did not pass through rebar, and had been cleaned out using a blast of compressed air followed by a brushing and finally a second blast of compressed air. Epoxy (Power-Fast+ Fast Set Formula) was injected in the holes overhead starting at the top of the hole and slowly working down as the hole filled. Each threaded rod was simultaneously inserted and twisted approximately 180 degrees into the epoxy filled hole to achieve a 5 inch embedment length.

Parameter Descriptions

ICC Procedure, Downward Install — 7 specimens

Powers publishes design aids outlining the strength of their epoxy anchors for installation in different media and loadings. The experimental data is certified by the International Code Council (ICC) and the certification is based on specific installation procedures. Therefore, the installation process and procedures used by ICC were expected to produce ideal anchor performance. In order to replicate ICC procedures, these anchors were installed in holes created with hammer-drill using a carbide tipped bit. Also, both the drilling and the installation of the anchors were done down-hand, i.e. not overhead.

ICC Procedure, Overhead Install — 10 specimens

These anchors were also installed using the ICC procedure. However, the holes which were again created with a hammer-drill and carbide-tipped bit, and the anchor installation were done overhead.

Cored Hole - Best Practices — 10 specimens

The holes created for the installation of the suspended ceiling within the Seaport Access Tunnel were cored, despite that the ICC procedure only qualified the anchors using holes created with a hammer drill. These specimens represent the best possible installation that could have been performed in the Seaport Access Tunnel.

Screen Tube Installations— 5 specimens

The installation procedures originally provided by the finish contractor outlined that screen tubes should be used in anchor installation. Typically, screen tubes are used for anchor installation into masonry block where the anchor could span through large voids. The screen tube then limits the amount of epoxy that can travel into these large cavities. Though no screen tubes were observed in any of the anchors pulled from the Seaport Access Tunnels they were added to the experiment matrix. For these specimens, the epoxy was first injected into a screen tube, then the screen tube was inserted into the hole, followed by the anchor.

Dirty Hole — 5 specimens

The epoxy was injected into these holes without cleaning the hole with compressed air or a brush.

Brush-Only Hole — 5 specimens

The epoxy was injected into the hole after the hole was only brush-cleaned, no compressed air was used.

Air-Only Hole — 5 specimens

The epoxy was injected into the hole after it had only been cleaned with compressed air, no brushing was performed.

Through #11 Rebar — 5 specimens

The holes for these specimens were purposely cored through a #11 rebar to see if bonding between the epoxy and steel significantly altered anchor performance. Figure 7 shows a picture of the 5 steel cores created from coring through the #11 rebar.

Unturned Installation — 5 specimens

The anchors for these specimens were installed without twisting the anchor 180 degrees as it was inserted into the epoxy.

46.6% white / 53.3% gray Volume Proportions — 5 specimens

53.3% white / 46.6% gray Volume Proportions — 5 specimens

The Power-Fast+ epoxy is a two-part system, one component is white and one component is gray. Proper use requires these components to be mixed in equal volumes. However, it is possible that air bubbles within the cartridges or other features could cause altered mixing ratios. To study the effect of misproportioning, these installations required hand-mixing of the two epoxy parts with each specimen set using the proportion of components indicated. The resulting misproportioned epoxy was then injected into the anchor hole with a large-bore syringe.

40.0% white / 60.0% gray Volume Proportions — 5 specimens

60.0% white / 40.0% gray Volume Proportions — 5 specimens

These two sets of specimens were hand-mixed with either a 60/40 volumetric ratio of white and gray components or gray and white components as indicated.

Void and Unturned Installation — 10 specimens

During the field testing in the Eastbound and Westbound Seaport Access Tunnels, it was observed that a large portion of the anchors removed from the ceiling had one long, longitudinally oriented void in the epoxy along the anchor. These voids were measured and it was found that they reduced the bond surface area between the rod and concrete by as much as 40%⁸. Therefore it became important to replicate this type of voiding in order to characterize its effects on anchor performance.

To determine how these and other voids are created, mock-up holes were fabricated using clear acrylic pipe in which clear silicone caulk representing the epoxy could be injected⁹. The purpose of this mock-up was to allow visual inspection of the flow of the silicone and the creation of voids as a threaded rod was pushed into the tube that represented the cored hole. The 0.75 inch inside diameter acrylic tubes were cut to a length of 5.625 inches and capped to replicate the cored holes in the concrete tunnel ceiling.

A variety of installation variations were investigated in an attempt to create voids. The best reproduction of the voids observed in the tunnel was created by leaving an approximate 0.5-inch air gap at the top of hole and then completely filling the hole with epoxy below that. Shown in Figure 8 are before and after pictures that represent this condition. Clearly, as outlined with the black dashed lines, the single long air void is created with these circumstances.

These specimens were installed by purposely inducing a 0.5-inch air gap or bubble at the top of the hole. Each threaded rod was then inserted straight into the epoxy, with the intent of creating the longitudinally oriented air void seen on anchors tested in the tunnel. To create the bubble, the end of the epoxy mixing nozzle was held about 0.5-inch from the top of the hole, while epoxy injection began. The nozzle location was briefly maintained near the top, and then slowly drawn out of the hole at a rate that left the hole properly filled with epoxy over its remaining length. A verification of the presence of the intended void could not take place until testing was complete, 10 specimens were installed to insure an adequate statistical base for the investigation of this parameter.

Void with Twist Installation — 10 specimens

Additional mock-up installations with the clear acrylic tubes identified that the size of the longitudinally oriented void was exacerbated if the rod was twisted after void creation. A twist of the anchor tended to smear the void over a larger area. As with the previous group, these

specimens purposely induced a large longitudinally oriented void along the anchor that is broadened by twisting the rod 180 degrees.

Void with Twist Install and 6300 lb Proof Load — 5 specimens

In 1999, the proof load used to verify proper installation of the adhesive anchors in the tunnel was raised from 3250 pounds to 6350 pounds. These specimens were produced using the same procedure for Cored Hole with Induced Void, Twist Installed specimens but were subjected to a 6300 pound proof loading prior to pull testing to failure. This parameter was entered into the matrix to investigate whether increasing the proof load would have been an effective identifier of potentially flawed installations or if the higher proof load would have caused damage.

Standard Set Epoxy — 5 specimens

These specimens were installed using the Power-Fast+ Standard Set formula, herein referred to as Standard Set. However, as these specimens were a late addition to the matrix, they were installed in the downward position in five unused holes from Phase 1 which were cored and cleaned overhead.

Torque Relaxation — 4 specimens

During Phase 1, six locations were identified on the slab where laboratory technicians could practice coring holes and injecting epoxy. Holes at two of these locations were not cored deep enough to accommodate the full embedment depth of the anchor and were abandoned. The four remaining holes had anchors installed within them using best practice procedures. These anchors were used to quantify tension transferred to the epoxy by wrenching a nut onto the anchor at a specified torque which would react off the concrete. A load cell was used to monitor the change in load present in these anchors for approximately 16 hours after nuts had been torqued to 125 ft-lbs.

General Notes

All epoxy for this phase of testing was injected into holes by Tim Tuggle (lead laboratory technician), and all threaded rods were inserted by Justin Ocel (project engineer). The majority of the anchors were installed on October 31, 2006. In general, the cored holes did not retain much dust in them, as in the coring operation so much water is being flushed through the hole that it is essentially being cleaned during the coring operation. Holes made with the hammer-drill were left in a dramatically different condition as the process packs the side of the hole with a very fine dust that is difficult to remove. Therefore the cleaning process of brushing and blowing the hammer-drilled holes continued until dust was adequately removed from the hole.

Figures 9, 10 and 11 outline the placement of anchors on the test slab by parameter and notes of interest from installations conducted for Phase I. Figure 9 shows the placement of all anchors installed in the overhead position. Figures 10 and 11 show the placement of anchors installed in

the downward position. Also indicated on these figures are the first hole that used a new cartridge of epoxy, the last hole to use a cartridge of epoxy and holes that were not filled adequately to cause squeeze out of the epoxy during installation of the threaded rod.

All the Power-Fast+ Fast Set epoxy used in Phase 1 came from Lot #062570I1LC and the Power-Fast+ Standard Set came from Lot #062700H1LC.

PHASE 2 SPECIMENS

Once the results from Phase 1 were analyzed, a matrix of specimens was compiled for Phase 2 testing. Phase 2 investigated parameters additional to those studied in Phase 1 and coupled some of the individual Phase 1 parameters.

Parameter Descriptions

Simpson SET22 Epoxy — 5 specimens

An alternative adhesive was included in this phase of testing to provide a behavioral comparison for the Power-Fast+ products. The SET adhesive manufactured by Simpson Strong-Tie Anchor Systems was chosen for this comparison. The SET adhesive is a two-part epoxy system like the Power-Fast system. The system was purchased in 22oz cartridge system such that the same injecting gun could be used for all the Phase 2 installations. These anchors were installed overhead in overhead cored holes using full manufacturer recommendations.

Damp Dirty Hole — 5 specimens

This parameter was meant to investigate the mechanical behavior of an anchor installed immediately after the coring operation without cleaning the hole. Installing the anchor immediately after coring meant the concrete within the hole was wet. These anchors were overhead installed in an overhead cored hole (not cleaned) with a twist install and no induced voids.

Torqued to 50 ft-lbs — 5 specimens

Torqued to 90 ft-lbs — 5 specimens

Torqued to 125 ft-lbs — 5 specimens

These sets of specimens studied the effect of pretensioning the epoxy by applying a specified torque to a nut that reacted the anchor rod off the concrete test slab. The anchors were installed overhead in an overhead cored hole using full manufacturer recommendations.

In the tunnel, two or four anchors were used to hold structural tee brackets to the tunnel ceiling. The structural tee sections had a flange thickness of approximately 1-inch with 13/16-inch diameter holes for the anchors to pass through. To duplicate the compliance this steel bracket had on the pretensioning process, 1-inch thick steel plates were fabricated to replicate the tee

brackets during torquing. These steel plates were 5-inches by 6-inches in plan and had a 13/16-inch diameter hole located to maintain the same edge distance as that on the structural tee brackets from the tunnel. A schematic of the pretensioning setup is shown in Figure 12. A sealing washer, which was a thin stainless steel washer bonded to neoprene, and a stainless steel lock washer were provided between the steel plate and the stainless steel nut. The nuts were torqued using a Proto 6016 torque wrench with a 30-150 ft-lb range.

The anchors were allowed to cure for at least one week before pretensioning. The 15 pretensioned specimens are shown in Figure 13 prior to torquing which occurred on the afternoon of February 2, 2007. Once tensioned, the anchors were left untouched for a period of approximately 65 hours after which any remaining tension was relieved by removal of the pretensioning nut and setup on the morning of February 5, 2007.

3 white : 2 gray Ratio Epoxy — 5 specimens

2 white : 3 gray Ratio Epoxy — 5 specimens

Due to the broad range of results for this parameter produced during the Phase 1 testing, these specimens were again included in Phase 2. However, in order to have a more control over epoxy proportioning during this phase of testing, special injection equipment that mixed the two components at a precise 3:2 ratio (e.g. 60%/40% by vol.) was used. A picture of a 3:2 ratio injection tube is shown in Figure 14 along with its 20-element static mixing tube. Each of these off-ratio mixing cartridges only held enough epoxy to fill one hole. These anchors were installed overhead in overhead cored holes using full manufacturer recommendations except for the proportioning of the epoxy parts.

Baked Epoxy — 5 specimens

This parameter was meant to investigate the effect sustained elevated temperature had on the epoxy. Prior to installation, a cartridge of epoxy was placed standing upright in a 140°F oven for 56 hours to simulate seven 8-hour periods (one week) of elevated temperature in storage. These anchors were installed overhead in an overhead cored hole using full manufacturer recommendations except for the tempered epoxy.

Incompletely Mixed Epoxy — 5 specimens

A standard mixing nozzle contains 18 mixing elements (e.g. paddles) to properly mix the two parts of the epoxy at the time of use. Each paddle is rotated 90 degrees from its neighbor to cause the two parts of the epoxy to fold upon each other as it is pushed through nozzle. The intent of these specimens was to study the effect on anchor performance of reduced or incompletely mixed epoxy as a result of a laborer cutting the mixing tube to a reduced length to facilitate epoxy injection.

A used mixing tube was cross-sectioned at TFHRC and a needle penetrometer was used to characterize the epoxy at each cross section. A picture of the cross-sectioned mixing tube is shown in Figure 15 along side a whole mixing tube. Shown in Figure 16 is a graph of the 1-minute penetration results for a 750 gram needle as a function of location along the mixing tube. The data indicates that the epoxy enters a transition behavior region at about 2-inches or four mixing paddles from the nozzle tip. Therefore, the mixing condition at this location was selected for study.

For these installs, a mixing nozzle was modified by removing four of the internal paddles. A picture of this modified mixing tube can be seen in Figure 17. These anchors were installed overhead in an overhead cored hole using full manufacturer recommendations except for using the modified mixing tube shown in Figure 17.

Worst Possible Install — 10 specimens

Building upon the “Incompletely Mixed Epoxy” parameter above, these specimens studied a coupling of parameters to produce a worst case installation. The mixing tube used for this parameter is shown in Figure 18. This modified tube had four internal mixing paddles removed and was cut to a length of 5 inches from the stiffening fins at the base of the mixing tube. Upon exiting the modified mixing nozzle, the epoxy has the same amount of mixing as the “Incompletely Mixed Epoxy” specimens. However, with the holes cored to 5.625 inches, the 5 inch long epoxy tube created an air bubble that caused a longitudinally oriented void when the anchor was installed. In addition, these anchors were installed in dirty holes. Therefore, these tests coupled the effects of voids, incompletely mixed epoxy, and dirty hole installation. All these anchors were installed overhead in an overhead cored hole.

General Notes

All Phase 2 anchors were installed on January 26, 2007. Lab technicians Tim Tuggle and Rob Montgomery performed all the installations. All the epoxy was injected into the holes using a hand-powered injection gun, unlike the pneumatic assisted gun used in Phase 1. The Power-Fast+ epoxy used in Phase 2 came from manufacturer Lots #062890I1LC and #062570I1LC. The Simpson Strong-Tie SET adhesive was marked “187412II”.

Figure 19 outlines the layout of the Phase 2 anchors on the test slab and indicates notes of interest from the installation process. During Phase 2 installations, multiple new mixing tubes were used on the same epoxy cartridge. It was not realized until after the first three anchors had been installed that the injection tube for the 60:40 specimens did not reach the top of the hole. The subsequent anchors installed in this series and in the 40:60 series used a rubber hose to extend the mixing tube to the top of the hole.



Figure 7. Plugs of steel created after drilling through #11 rebar.

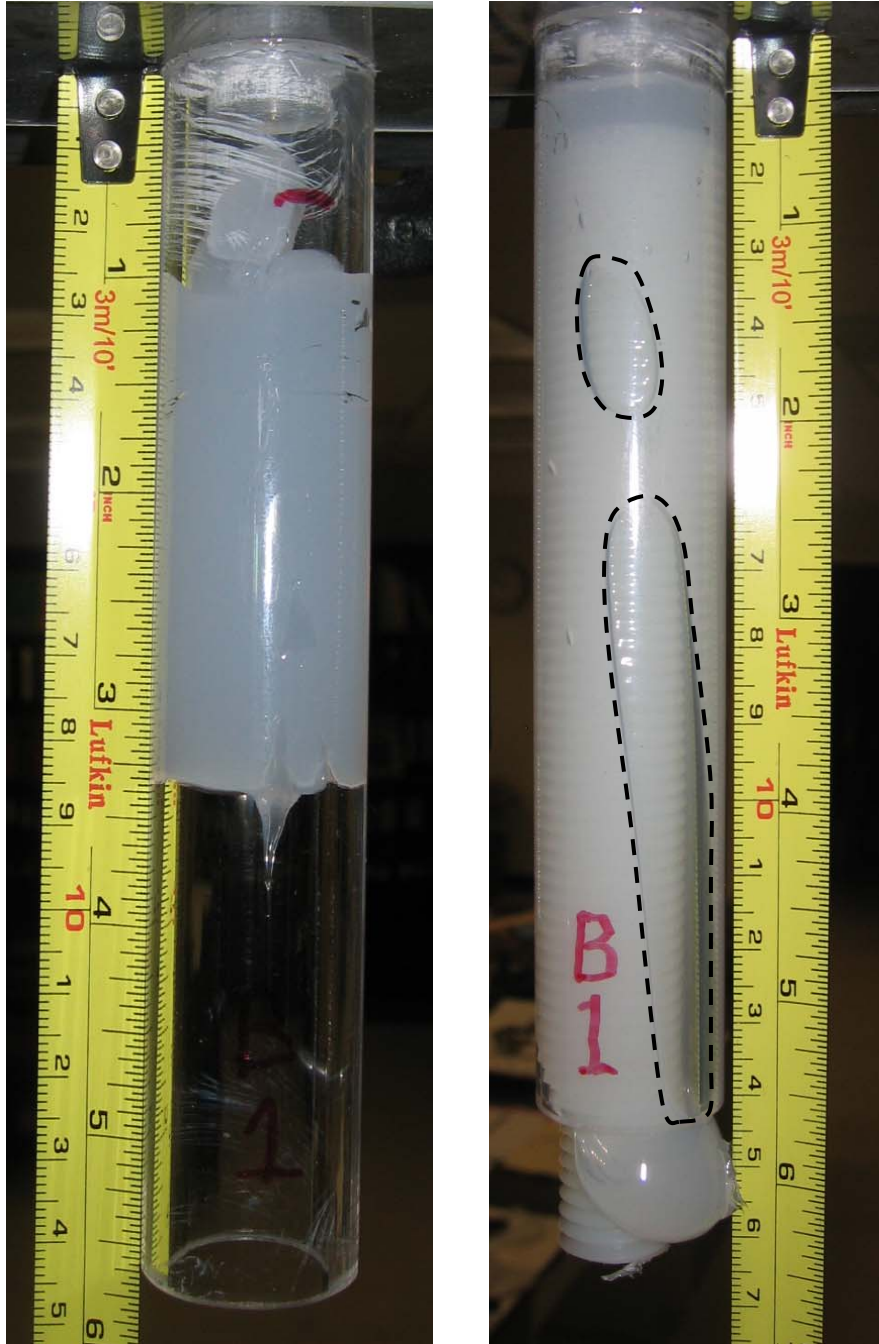
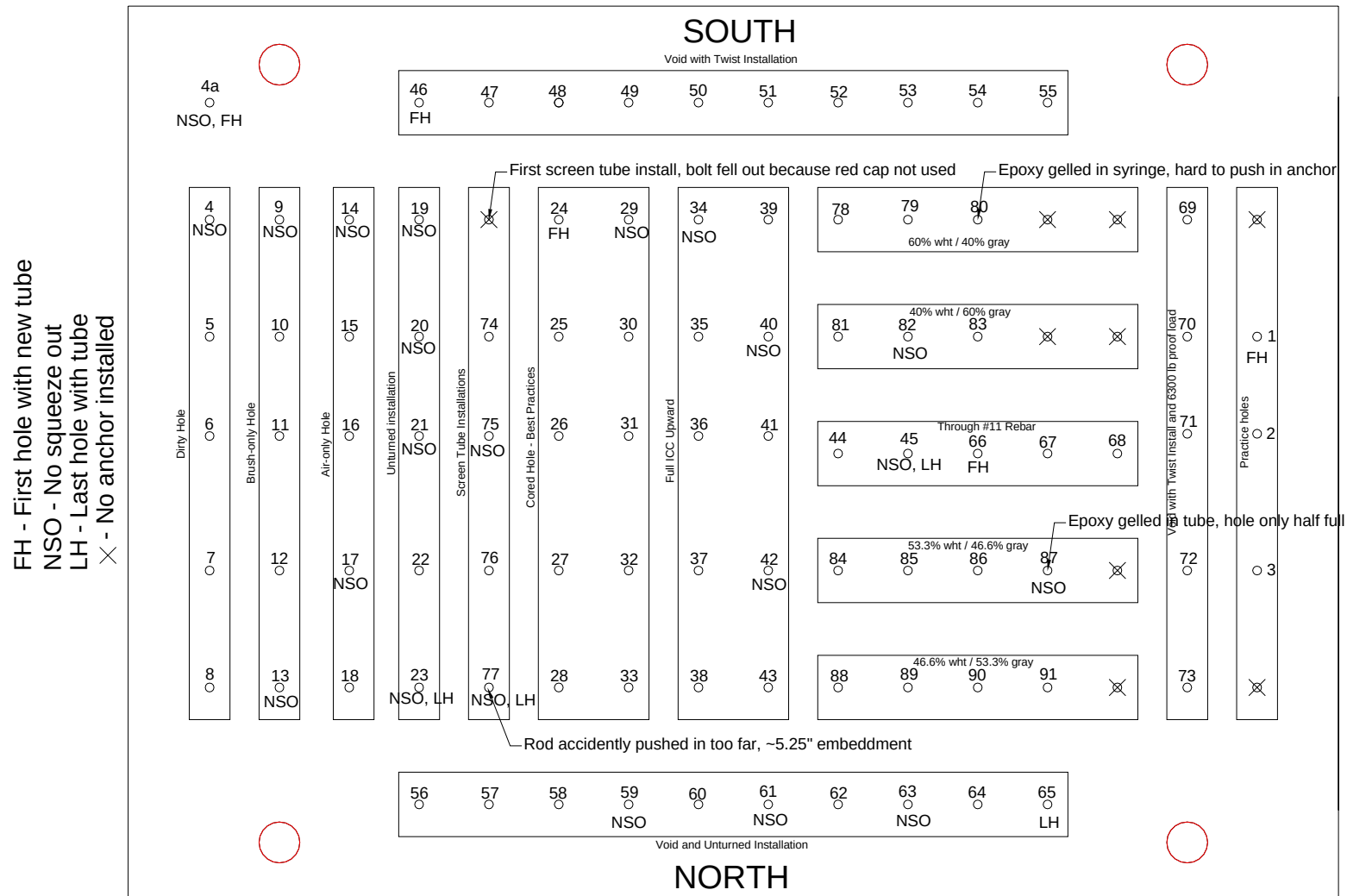


Figure 8. Left: Silicone within clear tube with an air gap at the top. Right: Result of pushing a 5/8" rod into the silicone with an air gap, note the long void created which is outlined.

Anchors installed in overhead position



Anchors 1-77 installed by Tim Tuggle and Justin Ocel on 31 October 2006
Anchors 78-98 installed by Tim Tuggle and Justin Ocel on 2 November 2006

Figure 9. Phase 1 overhead installation sequence and associated notes

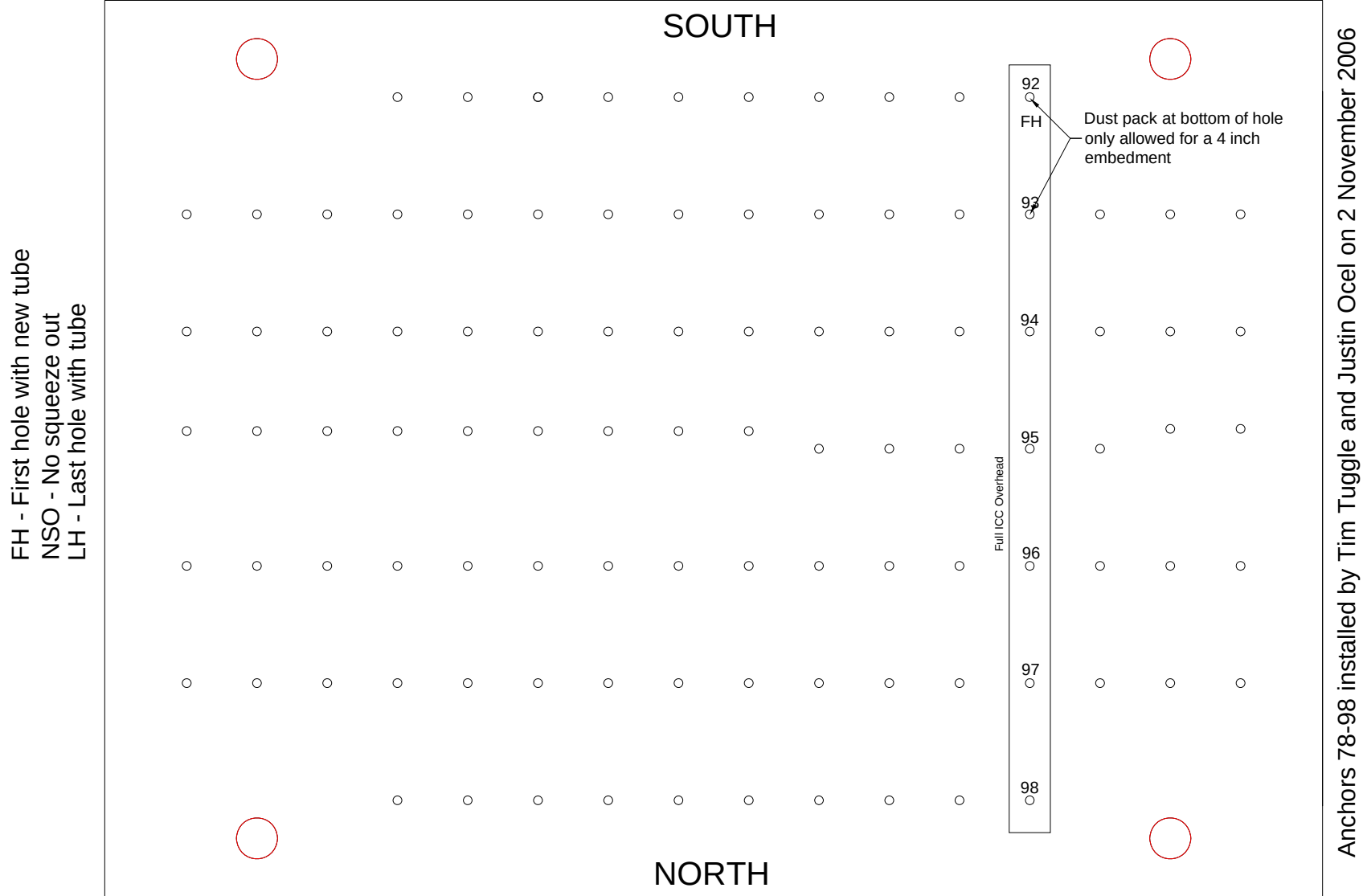
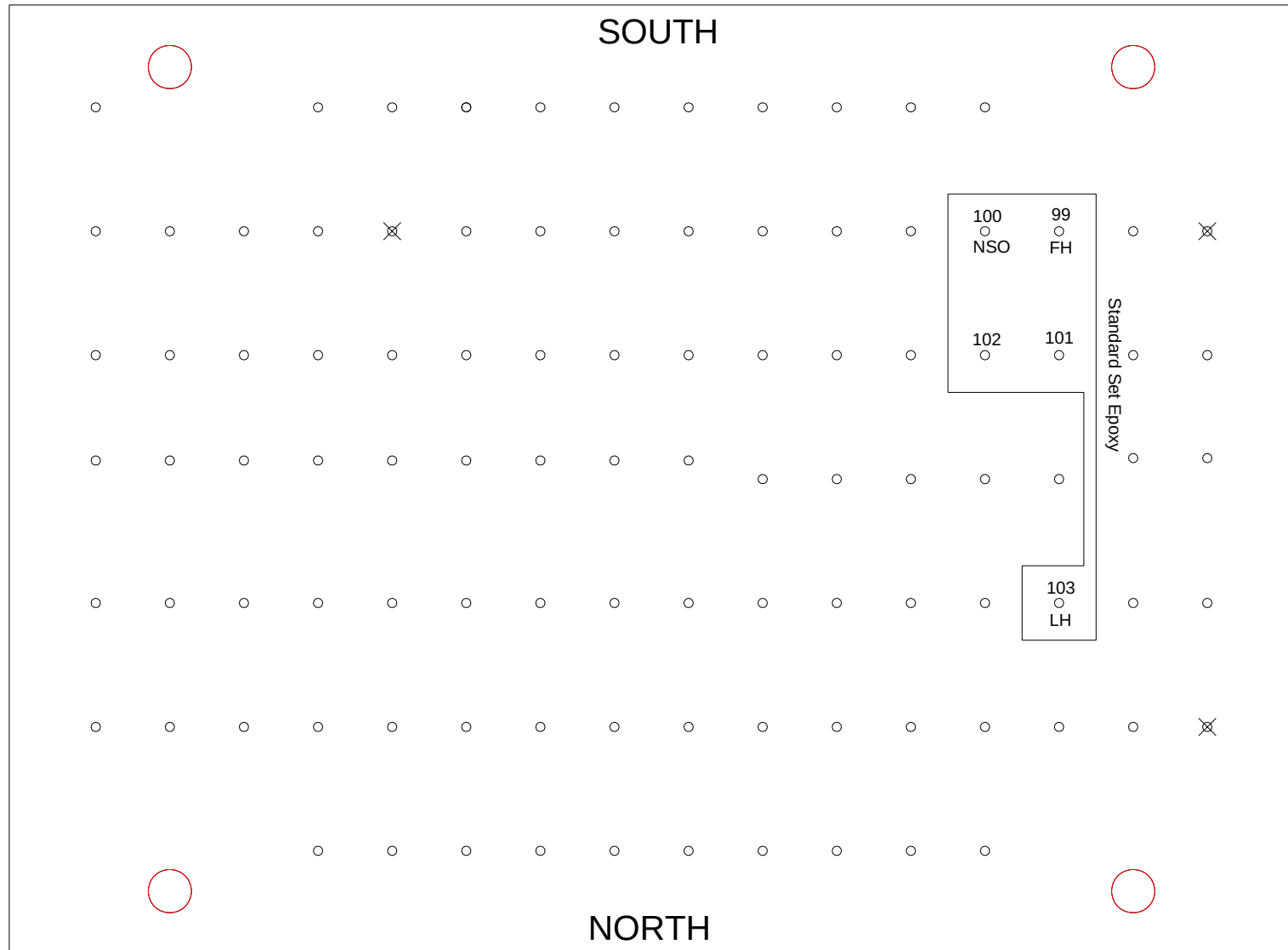


Figure 10. Phase 1 downward installation sequence and associated notes.

FH - First hole with new tube
 NSO - No squeeze out
 LH - Last hole with tube



Anchors 99-103 installed by Tig Tuggle on 7 December 2006

Figure 11. Phase I downward installation of standard set epoxy.

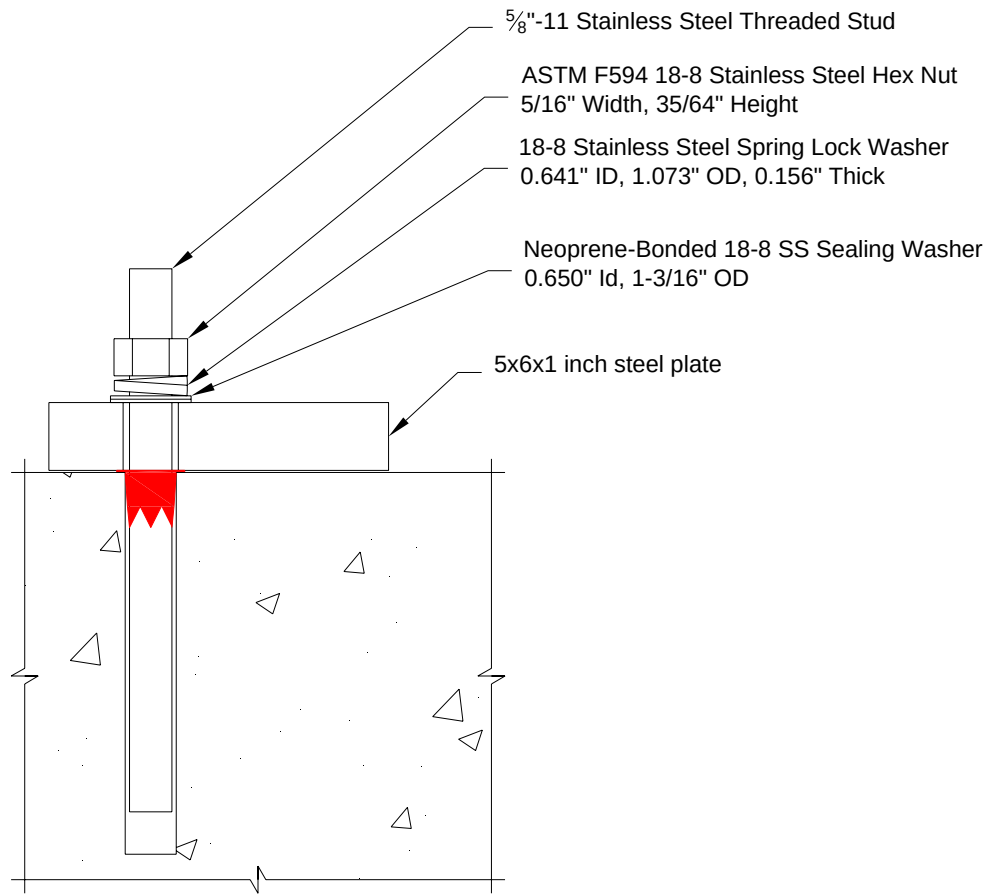


Figure 12. Schematic showing arrangement of nut, washers, and spacer plate for pretensioning.



Figure 13. Picture of 15 anchors prior to pretensioning.

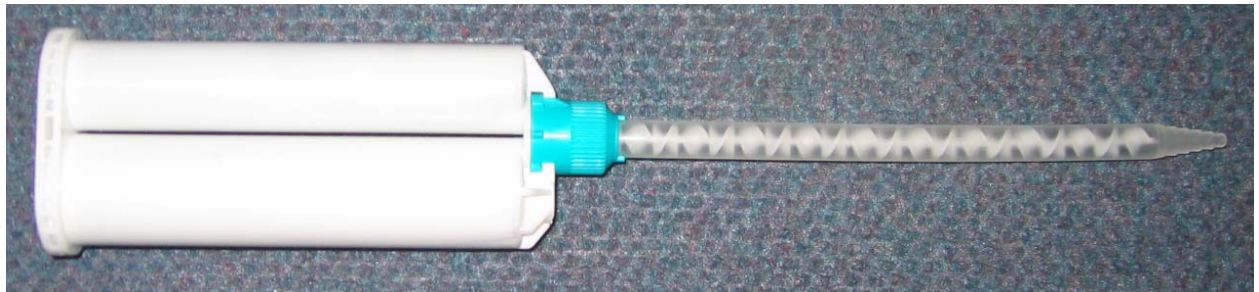


Figure 14. 3:2 volume ratio mixing tube.



Figure 15. Top: Entire mixing tube. Bottom: Mixing tube sliced in ~0.5 inch increments.

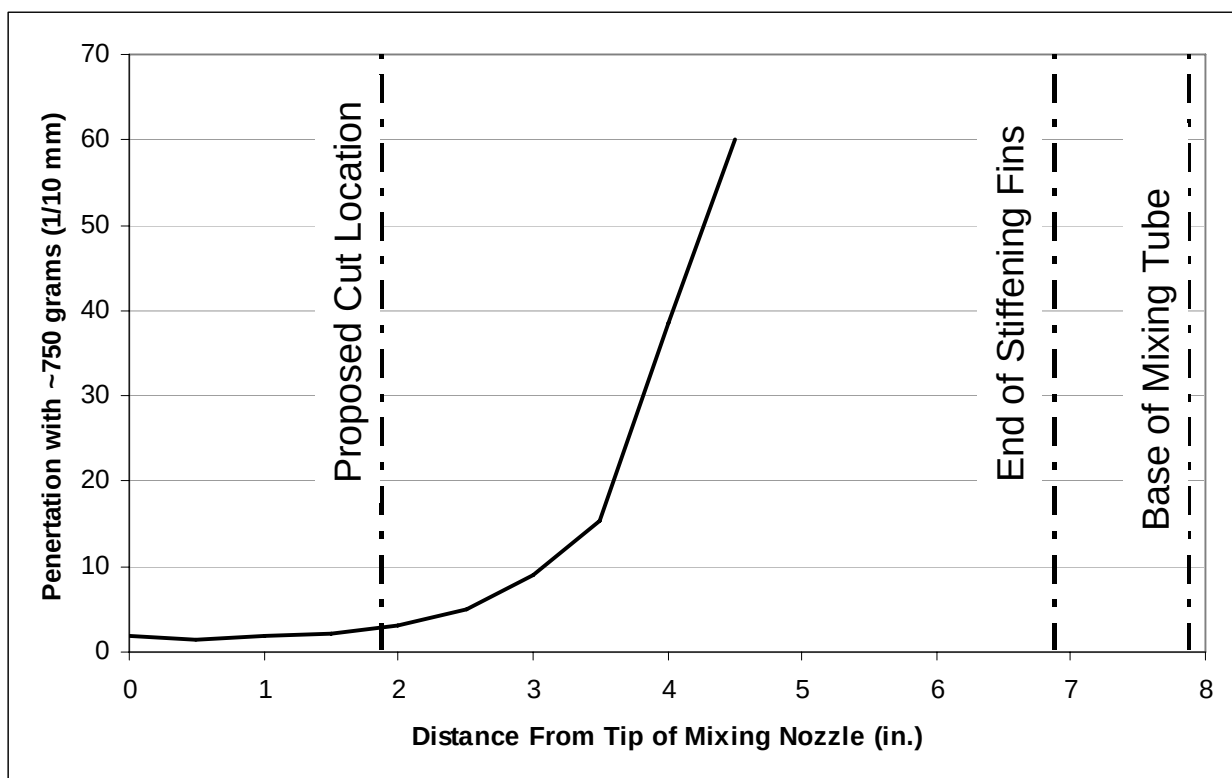


Figure 16. Penetration results of cross-sectioned mixing tube.

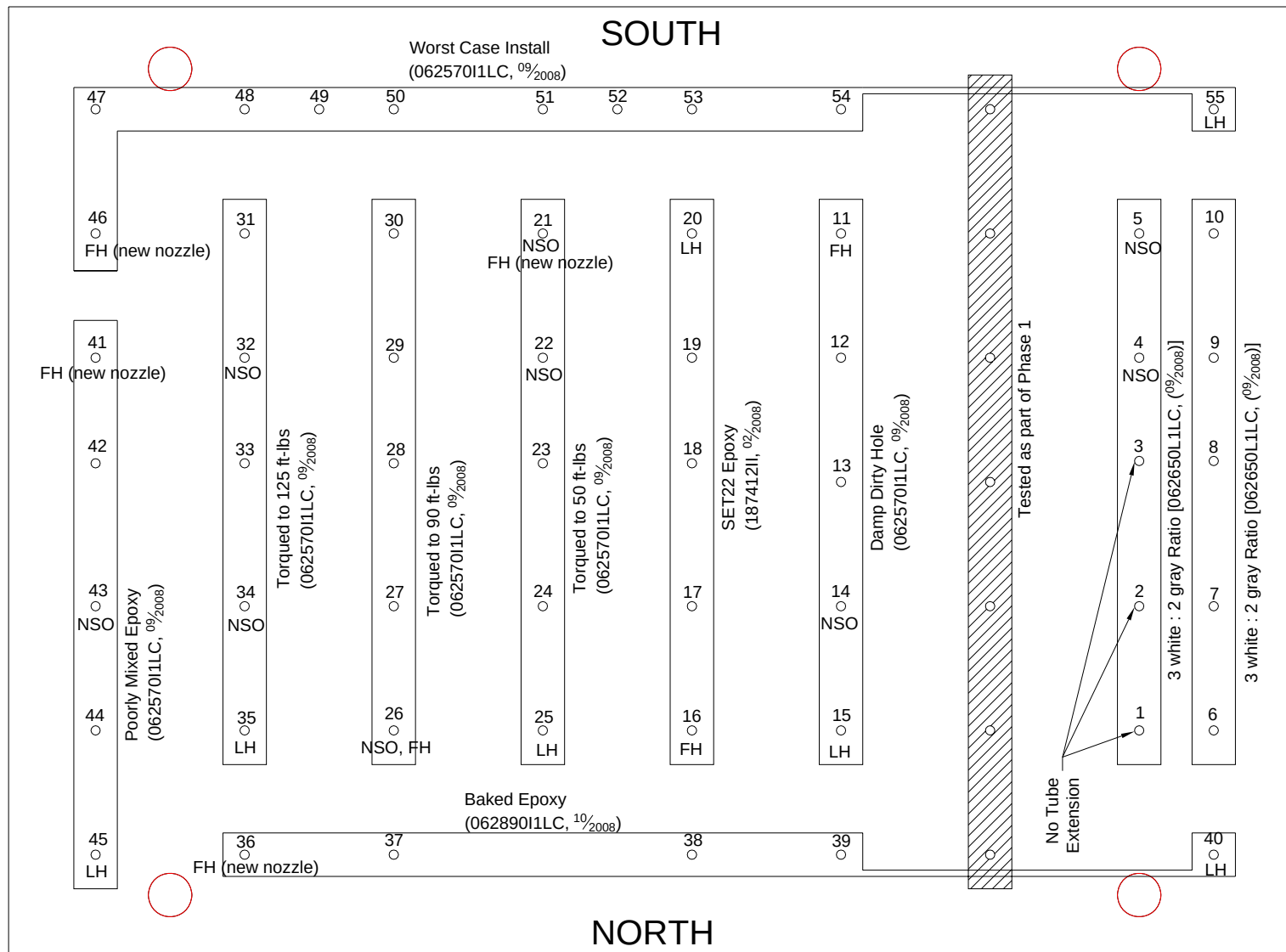


Figure 17. Modified mixing tube for “Incompletely Mixed Epoxy.”



Figure 18. Modified mixing tube for “Worst Case Install.”

FH - First hole with new tube
 NSO - No squeeze out
 LH - Last hole with tube



Anchors 1-55 installed by Tim Tuggle and Rob Montgomery on 26 January 2007

Figure 19. Phase 2 installation notes.

CHAPTER 4. EXPERIMENTAL RESULTS

This chapter will outline the results of the incremental load-displacement tests conducted on the 166 anchors included in Phase 1 and 2 of this study. This chapter only presents results in terms of peak loads and stresses resisted by each anchor. Complete load/displacement and time/load history plots along with pre- and post-test pictures for each anchor are reported in Appendices 1 and 2 for Phases 1 and 2 respectively.

Tables 3 through 29 display the peak load, peak stress, and statistical information (mean and coefficient of variation) for each of the installation parameters tested in Phases 1 and 2. The mean represents the average of data, and the coefficient of variation (COV) is the standard deviation of the data divided by the mean (an indication of the spread of the data about the mean). In Phase 1, the effect of voids on anchor performance was included as an individual parameter in the test matrix. However, many anchors intended to isolate the effects of other installation parameters also had voids despite the best efforts of the installers. Therefore, two sets of peak load statistical information is provided for parameter groups when excessive voiding coupled with the primary parameter investigated influenced the test results. In those cases, the first mean and COV reported represented all the data from anchors tested for that particular parameter. These statistics are meant to show the inherent variability in the performance of the anchor system. The second set of statistical information reported for a parameter group neglects those specimens whose performance was influenced by the presence of voids to better understand the effect of the individual installation parameter. Unfortunately, for many installation parameters, producing statistical information for a group neglecting specimens affected by voids was not possible due to insufficient test data.

The bond stresses reported were calculated for each anchor by dividing the peak load by the measured bond area, assuming a uniform bond stress model. The bond area surface for all anchors was assumed to be at the concrete/epoxy interface. Anchors properly embedded had a calculated bond area of 10.17inch^2 after accounting for the debonded area created by the red adhesive retaining cap. All anchors with voids had their bond areas determined by Jim Wildey of NTSB with a graphical method. The bond area calculation does not account for adhesive or cohesive failure in the epoxy, nor account for cone failures in the concrete. This method of calculating peak bond stresses was utilized in an attempt to normalize the statistics across the different installation parameters because of unintentional voids. The calculated peak bond stress was not intended to represent true bond stress.

The discussions in this chapter will only discuss normalized loads and stresses. All results were normalized to “Cored Hole – Best Practices” parameter which had an average peak load of 22084 lbs without the presence of voids or a peak bond stress of 2210 psi (see Table 5). This parameter was used as the basis because it represented the best possible anchor performance that could be expected from the anchors installed in the Seaport Access Tunnel. Normalized load and stress values were attained by dividing peak loads or stresses from individual parameters by that

from the basis. The normalized value then represents a percent increase or decrease in strength from the “Cored Hole – Best Practice” parameter. Normalized loads greater than 1.0 represent installation parameters that out performed the “Cored Hole - Best Practice” specimens. Likewise normalized loads less than 1.0 represent parameters which resulted in a reduced anchor capacity.

The epoxy manufacturers publish expected ultimate loads for different sized anchors, embedment depths and concrete strengths based on their own testing. The anchors in this study were installed to an embedment depth of 4.32 inches (5.0 inches minus 0.68 inches for the red cap) and the concrete compressive strength was measured to be approximately 5000 psi. Using the tables published by Powers, 5/8” diameter anchors installed using Standard Set epoxy are expected to have an ultimate pull-out strength of 18700 pounds, and this value is reduced 25% for the Fast Set epoxy, or 14025 pounds. Using the same embedment and concrete strength, anchors installed with the SET22 epoxy would be expected to pull out at 25286 pound using the data published by Simpson Strong-Tie Company. The published strengths for the three epoxies would have normalized loads of 0.847, 0.635, and 1.145 for the Power-Fast + Standard Set, Power-Fast+ Fast Set, and SET22 epoxies respectively.

PHASE 1

Figures 20 and 21 show the specimen labeling scheme for the Phase 1 anchors. These figures can be cross-referenced to the installation notes shown on Figures 9, 10 and 11 to identify poor performing anchors that resulted from installation sequence. Despite discarding the first several ejections of epoxy from each cartridge per the manufacturer’s recommendations, the performance of some anchors installed with a new epoxy cartridge were still affected. Specifically, specimen #3 of the “Through #11 Rebar” set was neglected because it was the first anchor installed with a new cartridge and demonstrated abnormally low pull-out resistance. It cannot be determined if the low pull-out resistance was from incompletely mixed, or improper ratios of epoxy first coming out of a new cartridge. Specimen #4 of the “Cored Hole - Best Practice” set was also neglected because it was the second installed anchor from a new cartridge and pulled at an abnormally low load. There is a possibility that the hand notes taken during installation could be incorrect and this really was the first installed anchor but regardless the specimen was considered an outlier via the Modified Thomson Technique¹⁰ and removed from the statistics.

Figure 22 summarizes the Phase 1 results in terms of normalized peak loads and Figure 23 by normalized peak load and normalized bond stress. Two bars are presented for each parameter in these figures; black bars represent the average data using all specimens within a set (in terms of peak load for Figure 22 and peak bond stress for Figure 23) and the gray bars represent the average of peak load data neglecting those specimens with voids. For some parameters a gray bar is not shown because too many specimens in that set had voids for a reliable average to be calculated decoupling the effect of voids. Also shown are error bars which represent a normalized coefficient of variation (calculated as the standard deviation of an individual parameter divided by the average from the “Cored Hole - Best Practice” condition). For clarity

only the upper error bar is shown understanding there is a lower symmetric bar to define the size of the scatter band. All the “Cored Hole - Best Practice” specimens used as the normalization basis exhibited adhesive failure (epoxy/concrete interface failed and both threaded rod and epoxy were pulled from within the hole) and nine of these ten anchor tests also pulled out a shallow cone of concrete. All concrete cones had a depth of no more than the concrete cover depth of approximately 1.5 inches.

As shown in Figure 22, there is little difference between the performance of anchors installed in cored holes and those installed in holes created with a hammer-drill (the “ICC Downward” and “ICC Overhead” parameters). When comparing the normalized average peak loads, the “ICC Overhead” parameter was 9.8% stronger, and the “ICC Downward” parameter only 1% stronger than the “Cored Hole - Best Practice” results. However, after accounting for the scatter bands, there is virtually no difference between anchor performances in a hammer-drilled or cored hole in terms of load capacity. All “ICC Overhead” and “ICC Downward” anchors exhibited adhesive failures coupled with a shallow concrete cone.

The Power-Fast+ Standard Set formula epoxy produced a 12.1% increase in strength. The four Standard Set specimens that did pull out during testing exhibited adhesive failures in the confined region of concrete below the #6 rebar layer, and concrete delamination (cone-like failure) in the concrete cover region above the #6 rebar.

Installing an anchor through the #11 rebar had no effect on performance as the normalized average peak load was 1.01 for this parameter. All the “Through #11 Rebar” specimens had adhesive failures in the concrete cover above the #11 rebar and cohesive failures through the #11 on into the confined concrete below the #11 rebar. However, as can be seen in Figure 7, the hand-drilled core leaves a stepped pattern through the #11 rebar that could be contributing to the failure mechanism at and below that level.

When anchors were installed with screen tubes, the load capacity was reduced approximately 16%. However, all four of the screen tube specimens tested contained a large void which certainly affected performance. The normalized stress data of 0.95 shown on Figure 23, indicates that the screen tubes only slightly diminished performance. The “Screen Tube Installation” specimens also exclusively exhibited adhesive failures.

Hole preparation had a small but significant effect on anchor load capacity. The normalized peak loads for anchors installed in a dirty hole, brush-only cleaned, and air-only cleaned hole were 0.83, 0.94 and 0.85 respectively. The lower performance for the “Dirty Hole” specimens should be considered with less weight because too many of the specimens had voids to calculate an average decoupling out the effect of voids. Therefore, a better measure of hole preparation performance would be looking at the normalized peak stresses in Figure 23. In terms of stress, the “Dirty Hole,” Brush-Only Hole,” and “Air-Only Hole” had normalized stress values of 0.97, 0.88, and 0.88 respectively. Based on the normalized load and stress values, it appears that

improper hole preparation causes about a 5-12% reduction in strength. All these anchors exhibited adhesive failures.

When voids were intentionally induced into the epoxy during installation, those anchors installed without a twist had a normalized peak load of 0.68 and those installed with a twist of 0.60. Therefore, twisting the anchor after the void was created did exacerbate the effect of the void on pull-out performance, as expected from the observations made during the clear tube mock-up testing. It was expected in terms of normalized stress these anchors would produce values near 1.0. Interestingly, the normalized stresses (see Figure 23) are 0.88 when installed without a twist and 0.78 when installed with a twist, which mirror the trend of the normalized load data but are not close to 1.0. Normalized stress values less than 1.0 indicate that the lower pull-out strengths are not associated strictly with a reduced bond area. Since the voids are longitudinally oriented on one side of the anchor they cause non-uniform bond stresses around the circumference of the anchor leading to stress concentrations which prematurely fail the bond resulting in a reduced load capacity. All 20 specimens intentionally installed with voids displayed adhesive failures. Two of the specimens installed with a twist slid out of the hole without causing a shallow concrete cone.

Increasing the proof load to 6300 pounds on “Void with Twist and 6300lb Proof” parameter did not affect anchor load capacity. These specimens produced a normalized peak load of 0.65 which compares very favorably to the 0.60 normalized peak load of identically prepared specimens proofed at 3250 pounds. Therefore, the higher proof load does not appear to be a better screening tool to identify poorly installed anchors (none of the five anchors failed during the proof load), nor does it appear to decrease the strength performance of the anchor.

The greatest difference in performance observed during the Phase 1 testing was produced by specimens installed with mis-proportioned epoxy. An epoxy mix of 53.3% white component and 46.6% gray component by volume resulted in anchor rod fracture during testing. This behavior was dramatically different than that displayed by anchors installed with an epoxy mix of 46.6% white and 53.3% gray which resulted in an average load capacity reduction of 44%. When the mis-proportion is increased to 60% white to 40% gray, and 40% white to 60% gray, the results diverge even further. The 60/40 white to gray epoxy installed also fractured the anchor rod during testing while the strength reduction for the 40/60 white to gray installs increased to 89%. All anchors installed with epoxy mixed predominately with the gray component exhibited adhesive failures and did not produce shallow concrete cones.

Torque Relaxation

Shown in Figure 24 are the load decay curves for the four specimens pretensioned by a 125 ft-lb torque and allowed to relax for approximately 16 hours. For these measurements, the load cell beared on a thin steel washer, and not a thick steel plate as was used in Phase 2. Specimens 1 and 2 used a $\frac{3}{4}$ " washer under the load cell which flattened and tore the flange of the red adhesive retaining cap. Specimens 3 and 4 used a washer with an inside diameter larger than

outside diameter of the red cap flange to avoid interacting with the cap during pretensioning. As the curves in the figure show, a 125 ft-lb torque imparted large loads to the adhesive anchors. The peak loads recorded ranged from 8400 to nearly 15000 pounds. Also, the time step measurements indicate that these loads decay dramatically within the first couple hours, converging towards a long-term residual load in the 3000 to 3600 pound range.

After these anchors were allowed to relax, the load cell was removed and each anchor was load-displacement tested. The test results can be found in Appendix 1.

PHASE 2

Figure 25 shows the specimen labeling scheme for the Phase 2 anchors. This figure can be cross-referenced to the installation notes shown on Figure 19 to identify poorly performing anchors that resulted from installation sequence.

Tables 20 through 29 display the peak load, peak stress, and statistical information for each of the Phase 2 installation parameters. This data is graphically displayed in the bar charts of Figures 26 and 27 following the same coloring scheme used in Figures 22 and 23.

The SET22 adhesive was included in the Phase 2 matrix to compare the behavior of a similar two-part epoxy adhesive that was used for the majority of this testing. As shown in Figure 26 the SET22 adhesive specimens averaged a 17% increase in load capacity over those similarly installed with Powers Power-Fast+ epoxy, but all the specimens had void. In terms of the normalized stress, the SET22 epoxy showed a 40% increase. However, the Powers epoxy anchor tests produced adhesive failures whereas the SET22 epoxy anchor tests exhibited cohesive failures. These cohesive failures were characterized by a pull-out of the threaded stud from the epoxy (i.e. along the surface defined by the tips of the threads).

The effect of damp, unclean hole installations is reflected in the results of the “Damp Dirty Hole” parameter. In Phase 1, the “Dirty Hole” installation tests achieved a normalized average peak stress of 0.97 whereas the “Damp Dirty Hole” installation was 0.81. Bond stress was used to compare these two parameters because too many specimens in the “Dirty Hole” parameter had voids. After accounting for scatter in the data, it appears that installing the anchor immediately after coring (without cleaning) causes a 20% reduction in strength, and a 3% reduction when installed in a dry and dirty hole. The failure mode of the “Damp Dirty” specimens was adhesive.

The 2:3 ratio specimens of Phase 2 were meant to reproduce the results of the 40%/60% volume specimens from Phase 1 with more precision on proportioning. In Phase 1, specimens installed with epoxy containing a predominance of gray component all failed during the proof-load (i.e. normalized peak load of 0.11) portion of the loading regime. Similarly installed specimens in Phase 2 had a normalized peak load of 0.48 indicating that the hand-mixed proportions in Phase 1 were probably slightly higher than 40/60 percentage target. Regardless, as can be seen in Appendix 2, the Phase 2 specimens exhibited undesirable behavior with large inelastic displacements occurring during load step hold periods.

Phase 2 specimens installed with epoxy comprised of predominately the white component failed with an average normalized peak load of 0.99, lower than the Phase 1 result. The failure mode of four of these Phase 2 specimens included a shallow concrete cone while the fifth failed by cleanly sliding out of the hole. The difference in failure mode and strength can primarily be attributed to the voids present in these specimens because a tube extension was not used when injecting the epoxy for three of the specimens, thus producing an air bubble in the epoxy. As in Phase 1, when the epoxy consisted of more gray component the failure was adhesive and when the epoxy consisted of more white component the failure was cohesive.

The “Incomplete Mixed Epoxy” specimens exhibited normalized peak loads of 1.02 which is essentially the same failure load of the “Cored Hole - Best Practice” specimens. However, the failure modes were different. In both cases shallow concrete cones accompanied the anchor failure. Below the concrete cone, the “Cored Hole - Best Practices” specimens exclusively produced adhesive failures while the “Incomplete Mixed Epoxy” specimens typically exhibited adhesive failure in the confined concrete, and cohesive failures in the concrete cover. In addition, the epoxy in the “Incomplete Mixed Epoxy” specimens tended to flake off the threaded rod as if the epoxy had hardened non-uniformly. Despite the differences in failure surface, the similarity of the peak loads indicates that the reduced epoxy mixing resulting from eliminating four paddles from the nozzle was inconsequential on the load capacity of a well installed anchor.

The “Worst Case Install” specimens were expected to have voids on all ten specimens, however only five ended up having voids. The average normalized peak load was 0.80 for specimens without voids and 0.67 for specimens with voids. Recall the “Worst Case Install” parameter was meant to couple together the effects from the “Dirty Hole,” “Incomplete Mixed Epoxy,” and “Void with Twist Install” parameters. Therefore, the normalized load of 0.80 without the effect of voids really represents the coupled effect between dirty holes and incomplete mixed epoxy, but individually these two parameters do not account for this 20% reduction but highlights the scatter in the data. When comparing normalized stress values, it is apparent that the reduced strength of the “Worst Case Install” parameter is from the presence of voids only.

Torqued Specimens

As shown in Figure 26 the specimens subjected to 50, 90, and 125 ft-lb torques exhibited average normalized peak loads similar or greater than the baseline “Cored Hole - Best Practice” results. In terms of bond stress (shown in Figure 27) the statistics were more counter-intuitive with higher pretensioning torques yielding lower peak bond stresses. However, the average bond stresses calculated were still 15-27% higher than the “Cored Hole - Best Practice” condition.

The torqued specimens also produced qualitative results which could be relevant to the failure investigation. Shown in Figure 28 and Figure 29 are pictures of the red adhesive retaining cap and sealing washer after specific torque levels had been applied to the anchor and subsequently released. Applying torque to the nut compresses the spacer plate into the red cap and the lock washer into the sealing washer, both of which incur some damage as part of the process. As the

pictures show, the red cap is relatively undamaged after a 50 ft-lbs torque, but is visibly deformed after experiencing a 125 ft-lbs torque. Likewise, the sealing washer is minimally affected by the lock washer after 50 ft-lbs, but heavily indented after the 125 ft-lbs torque. While the level of damage to the red cap and sealing washer is only described subjectively herein and is a product of the interaction of these elements and the concrete slab, the pictures in the figures could be used to categorically sort anchors by level of torque they had previously experienced on site.

Baked Epoxy

The average results for the “Baked Epoxy” specimens presented in Figures 26 and 27 are misleading, and the relatively high coefficient of variation indicates an abnormality. Averaging results for this parameter does not provide a good indicator of anchor performance. When these anchors were installed, a substantial amount of light brown liquid and white gel first filled the mixing nozzle. The brown liquid came from the chamber with the gray component. However, the liquid did not get purged from the mixing tube and instead mixed quite well with the white component. Despite discarding the first epoxy from the mixing tube as usual, the first anchor installed with this epoxy cartridge failed the proof load portion of the load-displacement test regime and simply slid out of the hole. The epoxy on this first specimen appeared as if it had not fully reacted and was still soft. The second installed anchor with this epoxy cartridge failed at 16476 pounds. The third and fourth anchors installed from this epoxy cartridge had load capacities in excess of 19400 pounds. The fifth anchor reached approximately 18000 pounds at failure, but subsequent investigation determined that the hole was only half filled with epoxy. When considering the peak stress results from these tests, there is a consistent trend in performance that correlates to the sequence of installation, or more precisely, to the location within the cartridge from where the epoxy was ejected. Therefore, the segregation due to heating of the brown liquid from the grey component has a significant effect on anchor load capacity, but this effect would most likely be limited to the first one or two anchors installed with an epoxy cartridge which had experienced a sustained elevated heating.

DATA ANALYSIS

Figure 30 and Figure 31 plot the normalized loads and stress for all the Phase 1 and 2 data.

Table 30 provides a comparison of all statistics collected from Phases 1 and 2. The second column contains the average and COV for all the data produced investigating each parameter. The third column is the average and COV for the parameter data neglecting those specimens with voids. The fourth column contains the average and COV of results assuming a uniform bond stress. The last column shows the mean minus two standard deviation strength that would be expected based on the statistics of the data. This calculation assumes the data is normally distributed and represents the load at which 97.5% of a population would be expected to exceed. The table has been sorted by putting the mean minus two standard deviation strengths in ascending order. Because the scatter of the test data affects the result of this calculation, more parameters are expected to produce lower load capacities relative to the “Cored Hole - Best

Practice” specimens than when solely considering the average peak load results. Also, due to the insufficiency of test results for the “Torqued to 50ft-lbs,” “Unturned Installation,” “Dirty Hole,” “Baked Epoxy,” and “Screen Tube Installation” specimens, the results presented should be considered a lower-bound. Of the remaining results, installation parameters shown to be most significant are those with misproportioned epoxy containing more gray component, those with intentional voids, and those tests described as “Damp Dirty Holes,” “Brush-Only Holes”, and “Worst Possible Install”.

Load-displacements histories for each anchor tested can be found in Appendices 1 and 2. What is evident in all of these plots is that the epoxies tested creep significantly during the 2-minute holding periods of the loading regime. The load level at which these creep displacements became significant is variable. For example, the “Cored Hole - Best Practice” specimens installed with the Power-Fast+ Fast-Set epoxy exhibited noticeable creep once a load level of approximately 5000 pounds was reached while the anchors installed with the Power-Fast+ Standard Set formula and SET22 epoxies did not display noticeable creep until approximately the 15000 pound threshold.

The second column in 0 contains the average displacement for all specimens at the end of the 12000 pound load step. If the specimen did not reach this load step then the data shown is for the concurrent peak load displacement. The third column in this table normalizes these displacements by the results of the “Cored Hole - Best Practice”, the same parameter used to normalize the peak load results. If the parameters with knowingly high variation are neglected (Unturned Installation and Dirty Hole) the parameters that most affect anchor displacement performance at this load level are those with artificially high gray component proportions, those with purposely induced voids, and the “Damp Dirty Hole” installation. In general, the installation parameters that most affected load capacity also produced anchors that crept at lower load thresholds.

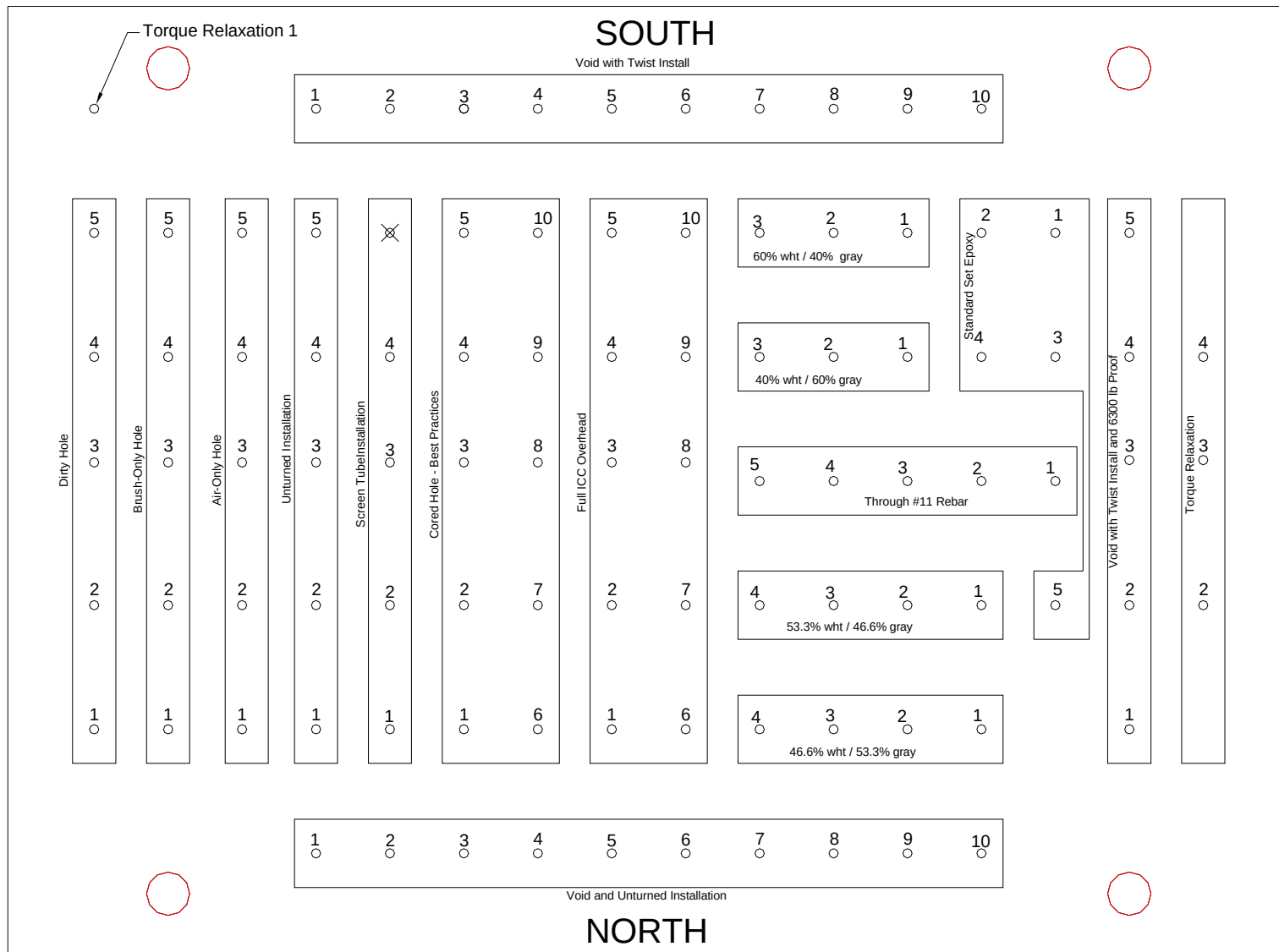


Figure 20. Anchor labeling scheme for Phase 1 anchors.

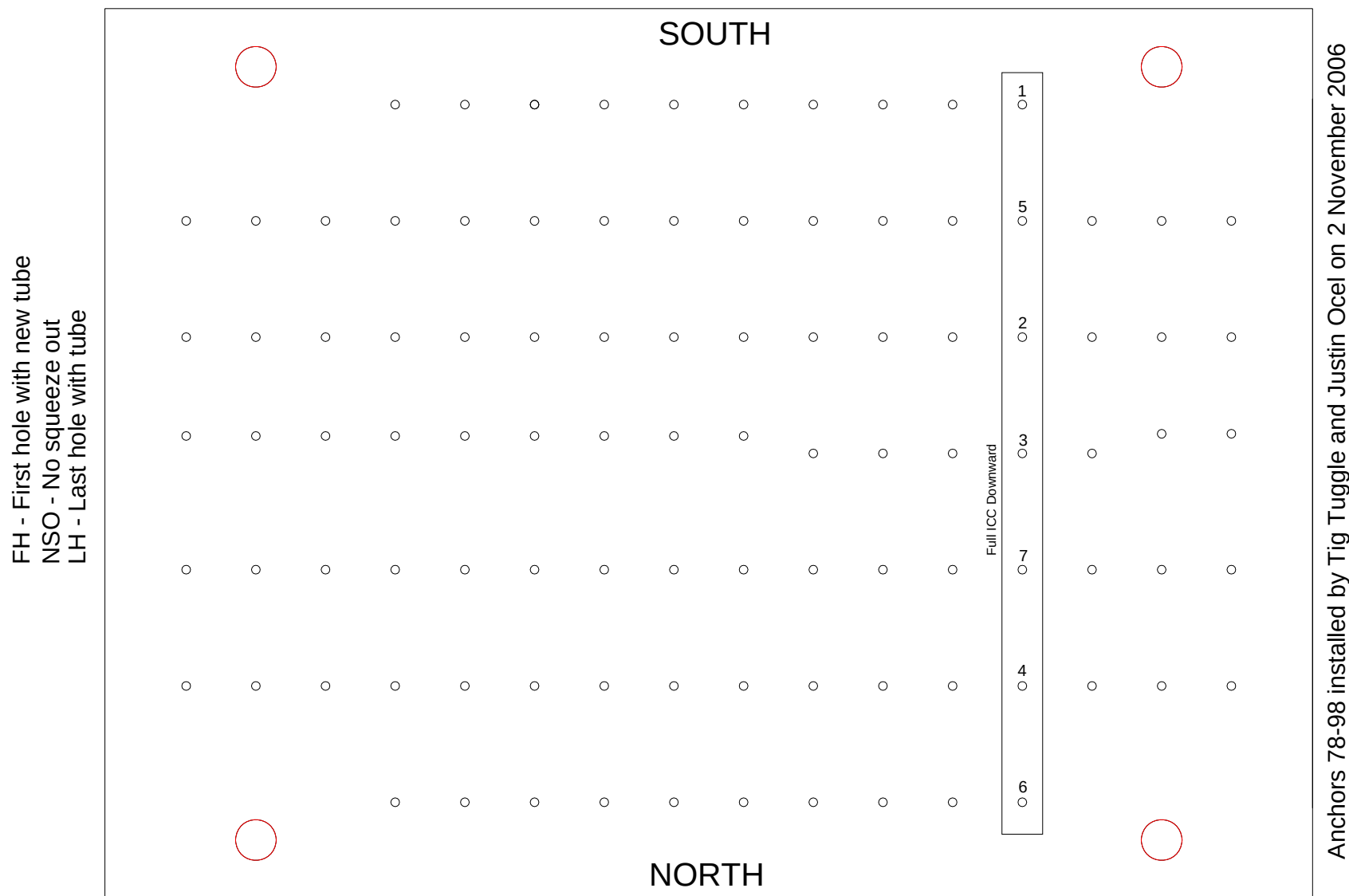


Figure 21. Anchor labeling scheme for Phase 1 anchors (cont.).

Table 3 Peak Load and Bond Stress Data for Full ICC Downward Specimens

Specimen Full ICC Downward	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	4.000	14206	7.82 in ²	1816	13
2 ^D	3.031	23858	10.11 in ²	2361	13
3 ^D	2.938	21779	10.66 in ² [0.09 in ²]	2127	14
4 ^D	2.938	20874	10.33 in ²	2021	15
5 ^D	3.031	23151	9.95 in ² [0.16 in ²]	2326	14
6 ^D	2.906	21832	10.40 in ²	2099	15
7 ^D	4.000	c	c	c	14
μ, COV	not applicable	22299, 0.053 ^d	not applicable	2187, 0.068 ^d	not applicable
^a - Bond area was calculated as $0.75\pi(8.0-\text{protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c - Values not known as power outage caused the jack to rapidly pull the anchor without data collection. ^d – Neglected #1 from statistics because embedment was only 4.0 inches. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 4 Peak Load and Bond Stress Data for Full ICC Upward Specimens

Specimen Full ICC Upward	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.000	23905	10.18 in ²	2349	36
2 ^D	2.875	25861	10.47 in ²	2469	36
3 ^D	3.000	22880	10.18 in ²	2248	36
4 ^D	3.125	23916	9.88 in ²	2420	36
5 ^D	2.969	23941	10.25 in ²	2335	36
6 ^D	3.063	23900	10.03 in ²	2382	37
7 ^D	3.000	24830	10.18 in ²	2439	38
8 ^D	3.063	25930	10.03 in ²	2585	38
9 ^D	3.000	23367	10.18 in ²	2296	38
10 ^D	2.938	23936	10.33 in ²	2318	38
μ, COV	not applicable	24247, 0.041	not applicable	2384, 0.041	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 5 Peak Load and Bond Stress Data for Cored Hole – Best Practice Specimens

Specimen Cored Holes	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.063	20754	10.03 in ²	2069	35
2 ^D	3.031	21958	10.11 in ²	2173	35
3 ^D	3.031	19998	10.11 in ²	1979	35
4 ^D	3.031	14729	10.11 in ²	1458	35
5 ^{D, V}	2.563	23172	10.59 in ² [0.56 in ²]	2188	35
6 ^{D, V}	3.125	19894	8.93 in ² [1.39 in ²]	2332	38
7 ^D	3.000	21945	10.18 in ²	2156	38
8 ^D	3.063	23917	10.03 in ²	2384	38
9 ^D	3.000	23932	10.18 in ²	2351	38
10 ^{SO, PF}	2.875	19970	8.84 in ²	2260	38
μ, COV	not applicable	21027, 0.129 22084, 0.073 ^c	not applicable	2210, 0.061 ^d	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Represents average and standard deviation neglecting #5 and #6 (because of voids), #10 because the hole was only partially filled with epoxy, and #4 which was considered an outlier via the Modified Thompson τ Technique ¹⁰ . ^d - #4 was omitted from average because it was considered an outlier via the Modified Thompson τ Technique. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 6 Peak Load and Bond Stress Data for Screen Tube Installation Specimens

Specimen Screen Tube	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO}	2.688	22306	10.92 in ²	2044	35
2 ^{D, V}	2.969	18009	8.53 in ² [1.71 in ²]	2111	35
3 ^{D, V}	3.000	17887	8.16 in ² [2.01 in ²]	2191	35
4 ^{D, V}	3.063	16027	7.77 in ² [2.30 in ²]	2064	35
μ, COV	not applicable	18557, 0.143	not applicable	2102, 0.031	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 7 Peak Load and Bond Stress Data for Dirty Hole Specimens

Specimen Dirty Hole	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.031	19429	9.52 in ² [0.59 in ²]	2041	18
2 ^{D, V}	3.063	19622	8.97 in ² [1.08 in ²]	2188	21
3 ^{D, V}	2.938	19565	8.42 in ² [1.87 in ²]	2323	21
4 ^D	3.031	20798	10.11 in ²	2058	21
5 ^{D, PF}	3.031	11890	7.07 in ²	1682	21
μ, COV	not applicable	18261, 0.197	not applicable	2153, 0.061 ^c	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c - #5 was omitted from average because it was only partially wetted with epoxy. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 8 Peak Load and Bond Stress Data for Brushed-Only Hole Specimens

Specimen Brush-Only Holes	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO, PF}	3.125	11961	5.89 in ²	2031	25
2 ^D	3.063	21688	10.03 in ²	2162	25
3 ^D	3.031	19932	9.75 in ² [0.35 in ²]	2043	25
4 ^{D, V}	3.000	13202	8.26 in ² [1.91 in ²]	1598	25
5 ^{D, V}	3.063	15969	7.99 in ² [2.07 in ²]	1999	25
μ, COV	not applicable	16550, 0.254 20810, 0.060 ^c	not applicable	1951, 0.126 ^d	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Represents the average between #2 and #3 because #1 was only partially wetted with epoxy and #4 and #5 had large voids. ^d -#1 omitted from average because it was only partially wetted with epoxy. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 9 Peak Load and Bond Stress Data for Air-Only Hole Specimens

Specimen Air-Only Holes	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.031	17915	9.56 in ² [0.55 in ²]	1874	31
2 ^{D, V}	3.031	15919	7.33 in ² [2.79 in ²]	2170	31
3 ^{D, V}	3.063	15898	8.87 in ² [1.18 in ²]	1792	31
4 ^D	3.000	19908	10.18 in ²	1956	31
5 ^{SO, PF}	3.000	18724	9.57 in ²	1956	31
μ, COV	not applicable	17673, 0.100 18849, 0.053 ^c	not applicable	1950, 0.072	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – #2 and #3 were omitted from the average because of the presence of large voids. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 10 Peak Load and Bond Stress Data for Through #11 Rebar Specimens

Specimen Through #11 Rebar	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.063	22316	10.03 in ²	2225	29
2 ^D	2.938	21887	10.33 in ²	2120	30
3 ^D	2.969	17924	10.25 in ²	1748	30
4 ^{SO, PF}	2.969	21904	8.17 in ²	2680	30
5 ^D	2.875	23813	10.47 in ²	2274	30
μ, COV	not applicable	21569, 0.101 22101, 0.014 ^c	not applicable	2172, 0.034 ^c	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c - #3 was omitted from averages because this hole was from the first epoxy from a new tube and the failed anchor appeared as if it was mis-mixed or mis-proportioned. #5 was omitted because a complete core of steel was not removed from the #11 rebar. #4 was omitted because the hole was only partially filled with epoxy. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 11 Peak Load and Bond Stress Data for Unturned Installation Specimens

Specimen Unturned Installation	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO, PF}	3.063	12615	5.60 in ²	2254	31
2 ^D	3.000	21944	10.18 in ²	2156	32
3 ^{SO, PF}	2.969	17914	7.80 in ²	2295	32
4 ^{SO, PF}	3.000	21514	8.54 in ²	2519	32
5 ^{D, V}	3.031	17930	9.05 in ² [1.06 in ²]	1981	32
μ, COV	not applicable	18383, 0.204	not applicable	2241, 0.088	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 12 Peak Load and Bond Stress Data for 46.6% wht / 53.3% gray Specimens

Specimen 46.6% wht / 53.3% gray	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.219	10489	9.66 in ²	1085	22
2 ^D	3.125	11812	9.88 in ²	1195	22
3 ^{SO}	3.000	14849	10.18 in ²	1459	22
4 ^D	3.000	11948	10.18 in ²	1174	22
μ, COV	not applicable	12274, 0.150	not applicable	1228, 0.131	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 13 Peak Load and Bond Stress Data for 53.3% wht / 46.6% gray Specimens

Specimen 53.3% wht / 46.6% gray	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.250	19373	8.65 in ² [1.00 in ²]	2241	26
2 ^{RF}	3.094	24888	9.96 in ²	2499	26
3 ^{RF}	3.188	25139	9.74 in ²	2582	26
4 ^{RF}	3.125	25205	9.88 in ²	2550	26
μ, COV	not applicable	23651, 0.121 25078, 0.007 ^c	not applicable	2544, 0.016 ^c _d	not applicable
^a - Bond area was calculated as $0.75\pi(8.0 - \text{protrusion} - 0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c - #1 was neglected from the averages because the epoxy had begun to gel in the injection syringe and was only half filled before the rod could be inserted, plus there was a void. ^d - Since #2, #3, and #4 fractured the rod, the peak bond stress is not a failure stress. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 14 Peak Load and Bond Stress Data for 40% wht / 60% gray Specimens

Specimen 40% wht / 60% gray	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO}	3.125	3268	9.88 in ²	331	27
2 ^{SO}	3.000	2101	10.18 in ²	206	27
3 ^{SO}	3.000	1938	10.18 in ²	190	27
μ, COV	not applicable	2436, 0.298	not applicable	242, 0.317	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 15 Peak Load and Bond Stress Data for 60% wht / 40% gray Specimens

Specimen 60% wht / 40% gray	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{RF}	3.063	25164	10.03 in ²	2509	27
2 ^{RF}	2.750	28780	10.77 in ²	2673	27
3 ^{RF}	3.438	25017	9.15 in ²	2735	28
μ, COV	not applicable	26320, 0.081	not applicable	2639, 0.044 ^c	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – All specimens failed by rod fracture, therefore peak bond stress was not a failure stress. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 16 Peak Load and Bond Stress Data for Voids and Unturned Installation Specimens

Specimen Voids and Unturned Installation	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.063	13928	7.91 in ² [2.15 in ²]	1761	21
2 ^{D, V}	3.000	16038	7.60 in ² [2.57 in ²]	2109	21
3 ^{D, V}	3.063	14017	7.53 in ² [2.53 in ²]	1861	22
4 ^{SO, PF}	3.063	2141	1.18 in ²	1817	22
5 ^{D, V}	3.000	15552	7.64 in ² [2.53 in ²]	2035	22
6 ^{D, V}	3.000	12020	6.96 in ² [3.22 in ²]	1728	22
7 ^{D, V}	3.125	15963	7.85 in ² [2.09 in ²]	2033	22
8 ^{D, V}	3.094	15800	8.28 in ² [1.72 in ²]	1909	22
9 ^{D, V}	3.000	15788	7.98 in ² [2.20 in ²]	1979	22
10 ^{D, V}	3.031	15953	7.56 in ² [2.56 in ²]	2110	22
μ, COV	not applicable	13688, 0.102 15007, 0.093 ^c	not applicable	1934, 0.072	not applicable

^a- Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB.

^b- Peak stress represents the peak load divided by the bond area.

^c - #4 was omitted from average because virtually no epoxy was injected into the hole.

Failure Type:

RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy

Table 17 Peak Load and Bond Stress Data for Voids with Twist Installed Specimens

Specimen Voids and Twist Install	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.000	14810	7.42 in ² [2.76 in ²]	1997	22
2 ^{D, V}	3.000	11872	7.42 in ² [2.76 in ²]	1600	23
3 ^{D, V}	3.031	13885	8.00 in ² [2.12 in ²]	1736	23
4 ^{D, V}	3.000	13451	7.62 in ² [2.55 in ²]	1764	23
5 ^{SO, V}	2.969	13475	8.73 in ² [1.51 in ²]	1544	23
6 ^{SO, V}	3.281	13479	8.06 in ² [1.56 in ²]	1673	23
7 ^{D, V}	3.281	13409	7.64 in ² [2.00 in ²]	1755	23
8 ^{D, V}	2.875	12033	7.40 in ² [2.98 in ²]	1626	23
9 ^{D, V}	3.063	11941	7.17 in ² [2.91 in ²]	1666	23
10 ^{D, V}	3.000	13519	7.41 in ² [2.77 in ²]	1825	23
μ, COV	not applicable	13188, 0.072	not applicable	1719, 0.075	not applicable
^a - Bond area was calculated as $0.75\pi(8.0-\text{protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 18 Peak Load and Bond Stress Data for Voids with Twist Install and 6300lb Proof Load Specimens

Specimen Voids with Twist Install and 6300 lb Proof Load	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.125	16453	7.75 in ² [2.20 in ²]	2124	32
2 ^{SO}	3.938	10510	7.97 in ²	1319	32
3 ^{D, V}	3.000	16463	7.66 in ² [2.51 in ²]	2148	32
4 ^{D, V}	3.188	11980	7.42 in ² [2.42 in ²]	1615	32
5 ^{D, V}	3.063	12310	7.36 in ² [2.71 in ²]	1673	32
μ, COV	not applicable	14301, 0.174 ^c	not applicable	1890, 0.151 ^c	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – #2 was omitted from averages because prior to testing this anchor was overtorqued and spun freely within its hole. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 19 Peak Load and Bond Stress Data for Standard Set Epoxy Specimens

Specimen Standard Set Epoxy	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.063	23911	10.03 in ²	2384	7
2 ^{RF}	3.094	25135	9.96 in ²	2524	7
3 ^{D, V}	3.156	23962	8.74 in ² [1.11 in ²]	2742	7
4 ^D	3.063	24973	10.03 in ²	2489	7
5 ^{RF}	3.125	25042	9.88 in ²	2534	7
μ, COV	not applicable	24605, 0.025 24765, 0.023 ^c	not applicable	2534, 0.051	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.625)$ minus the area of any voids. The 0.625 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c - #3 omitted from average because of a large void Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

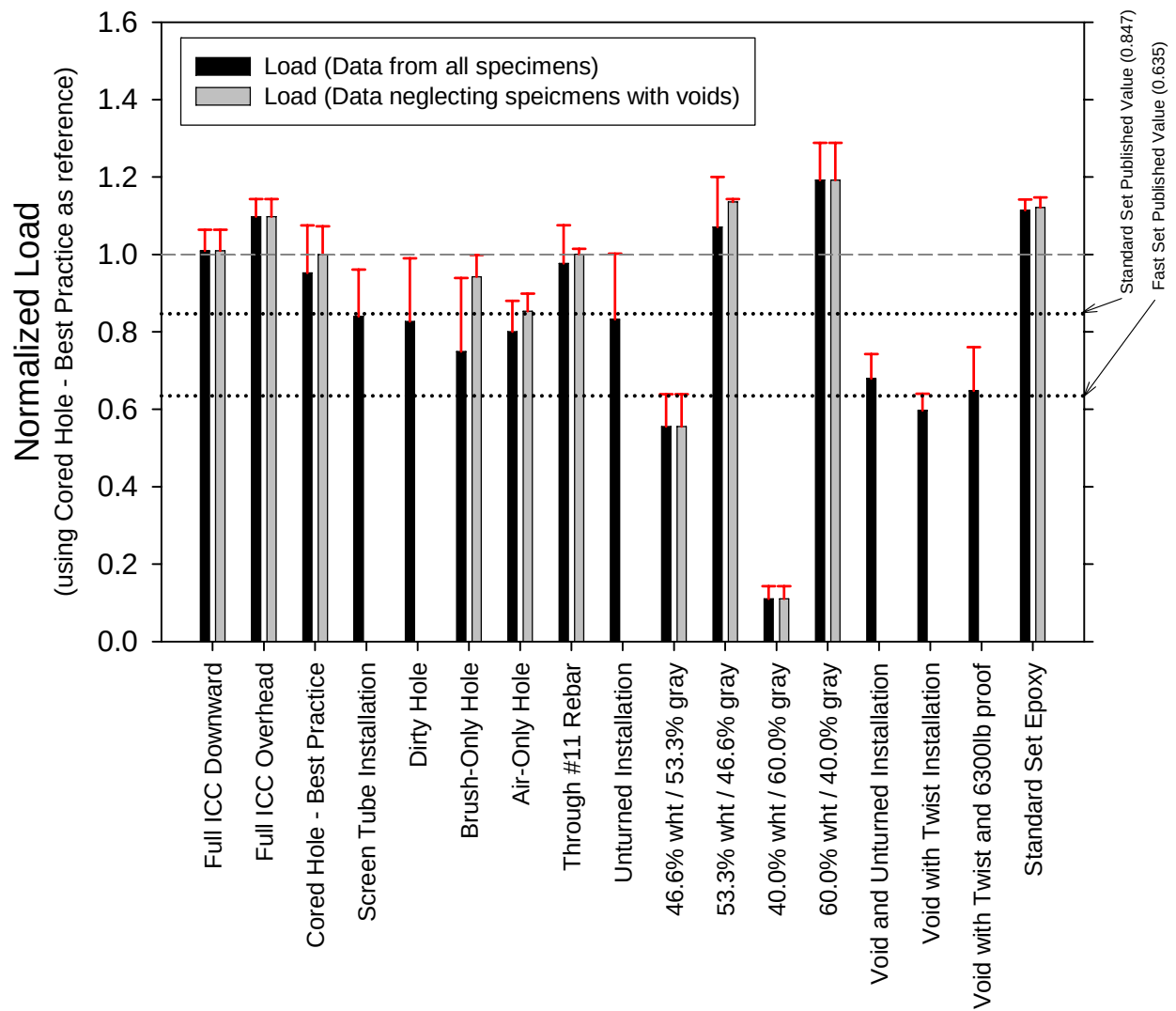


Figure 22. Phase 1 normalized load data with and without voids.

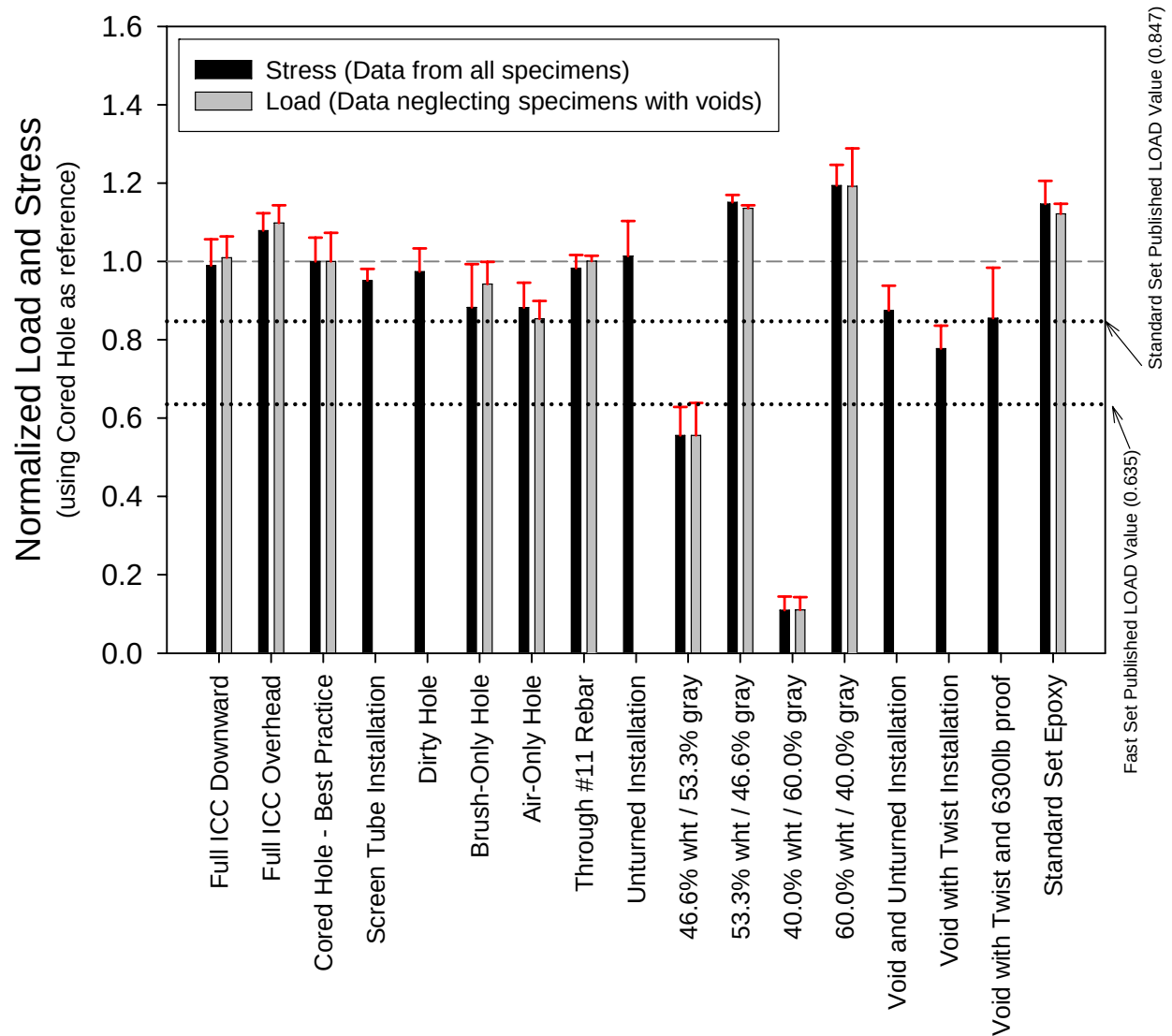


Figure 23. Phase 1 normalized load and stress data.

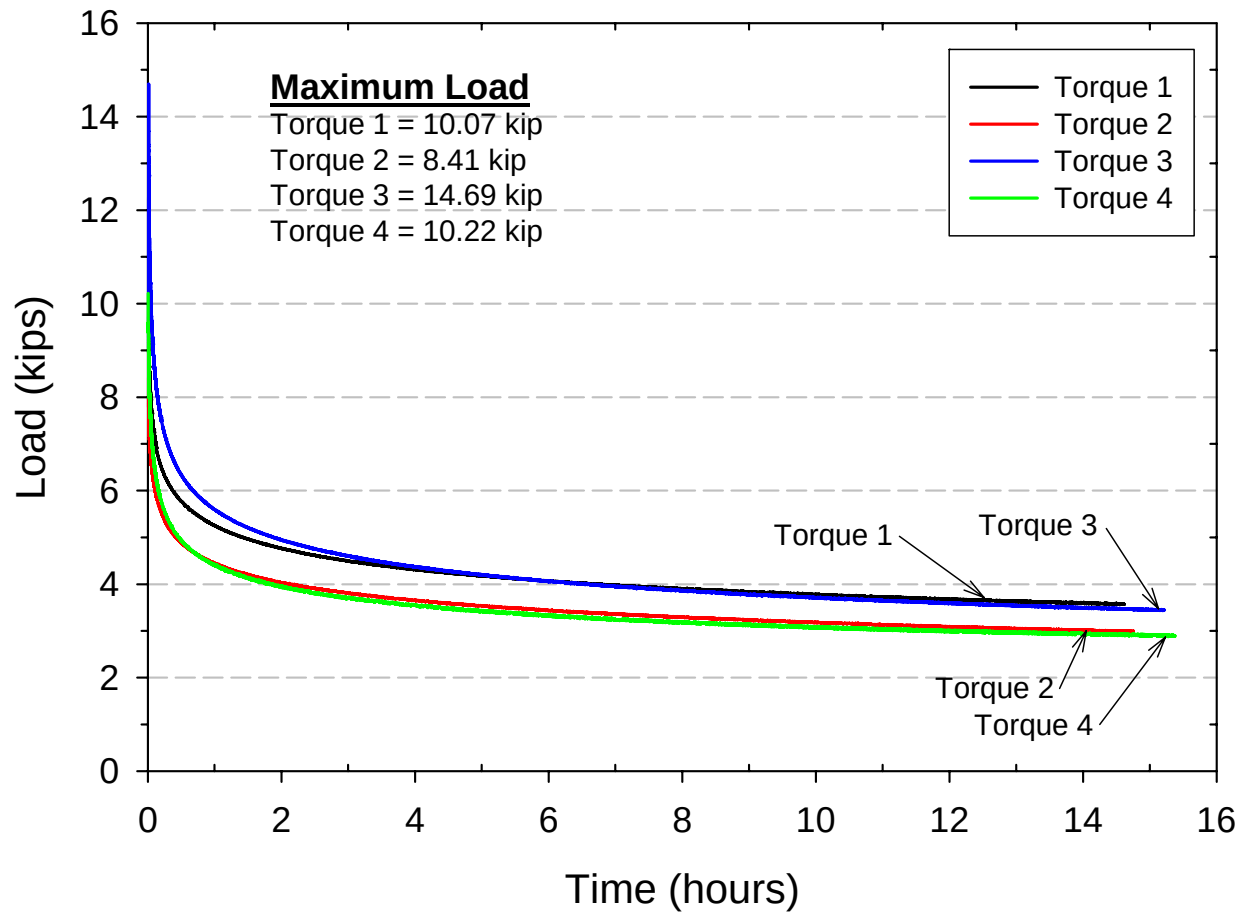
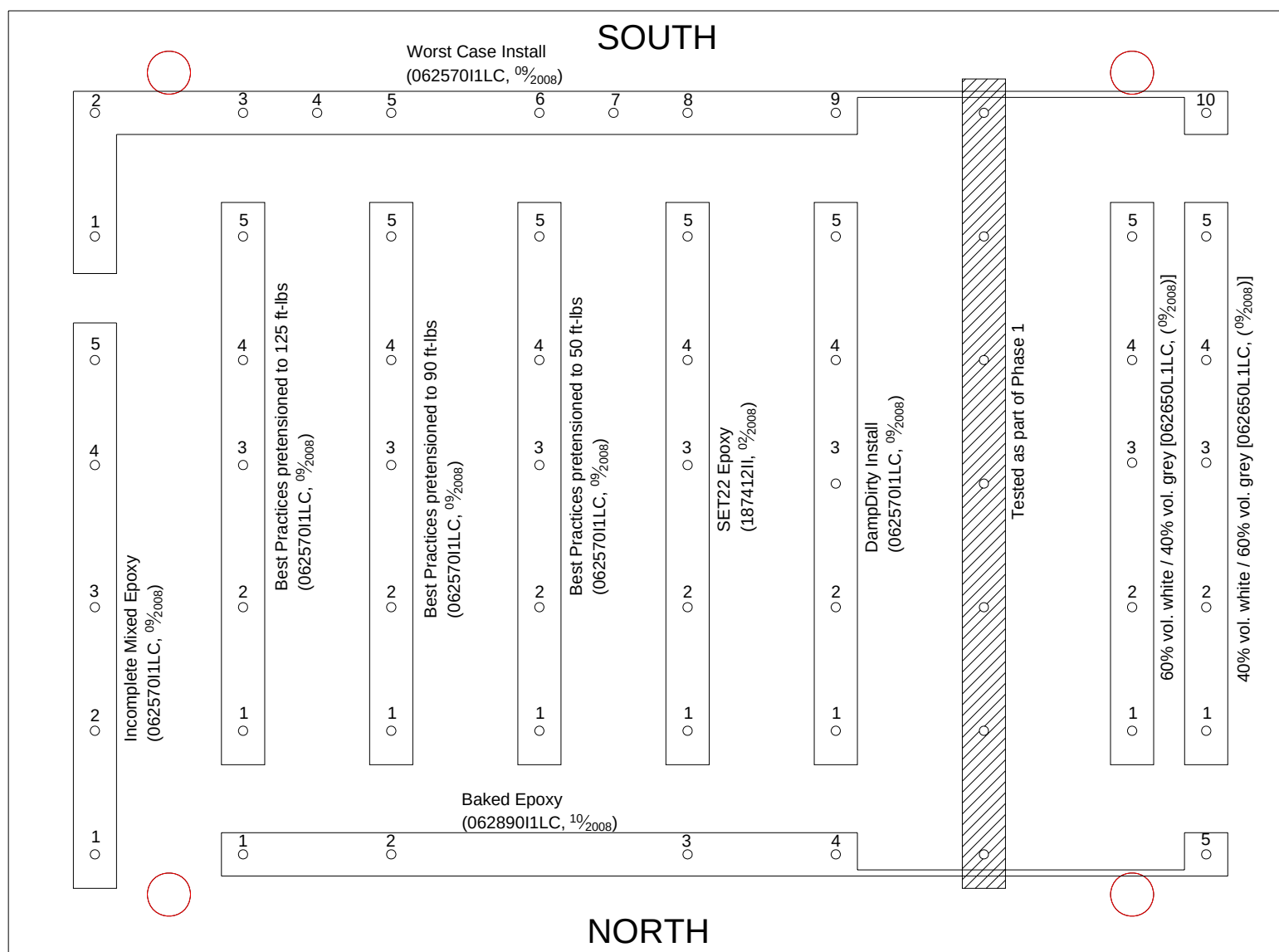


Figure 24. Torque relaxation curves.



Anchors 1-55 installed by Tim Tuggle and Rob Montgomery on 26 January 2007

Figure 25. Anchor labeling scheme for Phase 2 anchors.

Table 20 Peak Load and Bond Stress Data for SET22 Epoxy Specimens

Specimen SET22 Epoxy	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.125	27489	8.17 in ² [1.77 in ²]	3366	12
2 ^{D, V}	3.094	27703	8.46 in ² [1.53 in ²]	3275	13
3 ^{D, V}	3.156	25976	7.43 in ² [2.46 in ²]	3494	13
4 ^{D, V}	3.063	25165	9.57 in ² [0.47 in ²]	2630	13
5 ^{D, V}	3.125	23338	8.61 in ² [1.32 in ²]	2712	14
μ, COV	not applicable	25934, 0.069 ^c	not applicable	3096, 0.128	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Average includes effect of void because all SET22 specimens had voids Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 21 Peak Load and Bond Stress Data for Damp Dirty Hole Specimens

Specimen <i>Damp Dirty Hole</i>	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.031	17980	10.11 in ²	1779	14
2 ^{SO, PF}	3.031	13383	7.80 in ²	1715	14
3 ^{D, V}	3.156	19472	8.89 in ² [0.95 in ²]	2190	14
4 ^D	3.031	19488	10.11 in ²	1928	14
5 ^{D, V}	3.031	11985	8.73 in ² [1.38 in ²]	1373	14
μ, COV	not applicable	16462, 0.215 18734, 0.057 ^c	not applicable	1797, 0.167	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – average is only for #1 and #4 because #3 and #5 had voids and #2 was only partially filled with epoxy Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 22 Peak Load and Bond Stress Data for Specimens Torqued to 50 ft-lbs

Specimen Torqued to 50 ft-lbs	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	3.188	25983	8.93 in ² [0.84 in ²]	2910	11
2 ^{D, V}	3.031	25567	7.53 in ² [2.59 in ²]	3395	11
3 ^{RF}	3.063	25828	10.03 in ²	2575	11
4 ^{SO, PF}	3.125	12004	5.45 in ²	2203	11
5 ^{D, V}	3.094	13349	7.78 in ² [2.23 in ²]	1717	11
μ, COV	not applicable	20546, 0.350 ^c	not applicable	2560, 0.252	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Average includes effect of voids because all specimens had reduced bond area, and #3 fractured the rod Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 23 Peak Load and Bond Stress Data for Specimens Torqued to 90 ft-lbs

Specimen Torqued to 90 ft-lbs	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO, PF}	3.031	25738	8.17 in ²	3149	10
2 ^D	3.031	27761	10.11 in ²	2747	11
3 ^{RF}	3.094	25726	9.96 in ²	2584	11
4 ^D	3.125	25932	9.88 in ²	2624	11
5 ^{D, V}	3.000	23973	8.66 in ² [1.52 in ²]	2770	11
μ, COV	not applicable	25826, 0.052 26473, 0.003 ^c	not applicable	2775, 0.081	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Represents average of specimens without voids Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 24 Peak Load and Bond Stress Data for Specimens Torqued to 125 ft-lbs

Specimen Torqued to 125 ft-lbs	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.094	23971	9.96 in ²	2407	10
2 ^{D, V}	3.000	22927	8.54 in ² [1.64 in ²]	2685	10
3 ^{D, V}	3.094	21988	7.72 in ² [2.29 in ²]	2849	10
4 ^{SO, PF}	2.938	23977	6.04 in ²	3971	10
5 ^{D, V}	3.031	19995	9.35 in ² [0.76 in ²]	2138	10
μ, COV	not applicable	22572, 0.074	not applicable	2810, 0.250	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 25 Peak Load and Bond Stress Data for 3wht : 2 gray Ratio Specimens

Specimen 3 wht : 2 gray Ratio Epoxy	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{D, V}	2.938	17999	8.77 in ² [1.53 in ²]	2052	12
2 ^{D, V}	3.063	25242	8.18 in ² [1.87 in ²]	3085	12
3 ^{D, V}	3.063	21975	8.29 in ² [1.77 in ²]	2651	13
4 ^{SO, PF}	2.969	19976	5.89 in ²	3391	13
5 ^{D, V}	3.094	23978	7.77 in ² [2.24 in ²]	3087	13
μ, COV	not applicable	21834, 0.134	not applicable	2853, 0.182	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 26 Peak Load and Bond Stress Data for 2 wht : 3 gray Ratio Specimens

Specimen 2 wht : 3 gray Ratio Epoxy	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.000	8999	10.18 in ²	884	13
2 ^{SO, V}	3.125	7366	8.79 in ² [1.13 in ²]	838	13
3 ^{SO}	3.000	11972	10.18 in ²	1176	13
4 ^{SO, V}	3.125	7474	7.97 in ² [1.97 in ²]	938	13
5 ^{SO, V}	3.031	8314	9.73 in ² [0.37 in ²]	854	13
μ, COV	not applicable	8825, 0.213 10486, 0.200 ^c	not applicable	938, 0.148	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Represents average of specimens without voids Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 27 Peak Load and Bond Stress Data for Baked Epoxy Specimens

Specimen Baked Epoxy	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete^a [Void Area]	Peak Stress^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO, V}	3.031	3255	8.40 in ² [1.72 in ²]	388	7
2 ^D	3.063	16476	10.03 in ²	1642	7
3 ^D	3.125	19461	9.88 in ²	1969	10
4 ^D	3.063	19359	10.03 in ²	1930	10
5 ^{SO, PF}	3.063	17995	6.70 in ²	2686	10
μ, COV	not applicable	15309, 0.447 18432, 0.092 ^c	not applicable	1723, 0.487	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Represents specimens without voids Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 28 Peak Load and Bond Stress Data for Incompletely Mixed Epoxy Specimens

Specimen <i>Incomplete Mixed Epoxy</i>	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^D	3.125	22469	9.88 in ²	2273	7
2 ^D	3.063	22474	10.03 in ²	2240	7
3 ^{D, PF}	2.750	19478	8.25 in ²	2362	7
4 ^D	3.094	22487	9.96 in ²	2258	7
5 ^{D, V}	3.031	19474	8.89 in ² [1.23 in ²]	2192	7
μ, COV	not applicable	21277, 0.077 22477, 0.000 ^c	not applicable	2265, 0.027	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Represents specimens without voids Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

Table 29 Peak Load and Bond Stress Data for Worst Case Installed Specimens

Specimen Worst Case Install	Initial Protrusion (inch)	Peak Load (pounds)	Bond Area to Concrete ^a [Void Area]	Peak Stress ^b (psi)	Age of Epoxy Joint when Tested (days)
1 ^{SO, V}	3.125	14983	9.32 in ² [0.58 in ²]	1607	12
2 ^{D, V}	3.125	14797	8.52 in ² [1.40 in ²]	1737	12
3 ^{D, V}	3.000	13298	7.48 in ² [2.70 in ²]	1779	12
4 ^{D, V}	3.094	14984	7.55 in ² [2.45 in ²]	1983	12
5 ^D	3.031	14981	10.11 in ²	1482	12
6 ^D	3.031	22467	10.11 in ²	2223	12
7 ^D	3.094	14990	9.96 in ²	1505	12
8 ^{D, V}	3.094	16476	8.99 in ² [0.99 in ²]	1833	12
9 ^D	3.031	17976	10.11 in ²	1779	12
10 ^D	3.125	17978	9.88 in ²	1819	12
μ, COV	not applicable	16293, 0.161 14907, 0.076 ^c 17678, 0.173 ^d	not applicable	1775, 0.124	not applicable
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b - Peak stress represents the peak load divided by the bond area. ^c – Average load statistics of anchors with voids. ^d – Average load statistics of anchors without voids. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy					

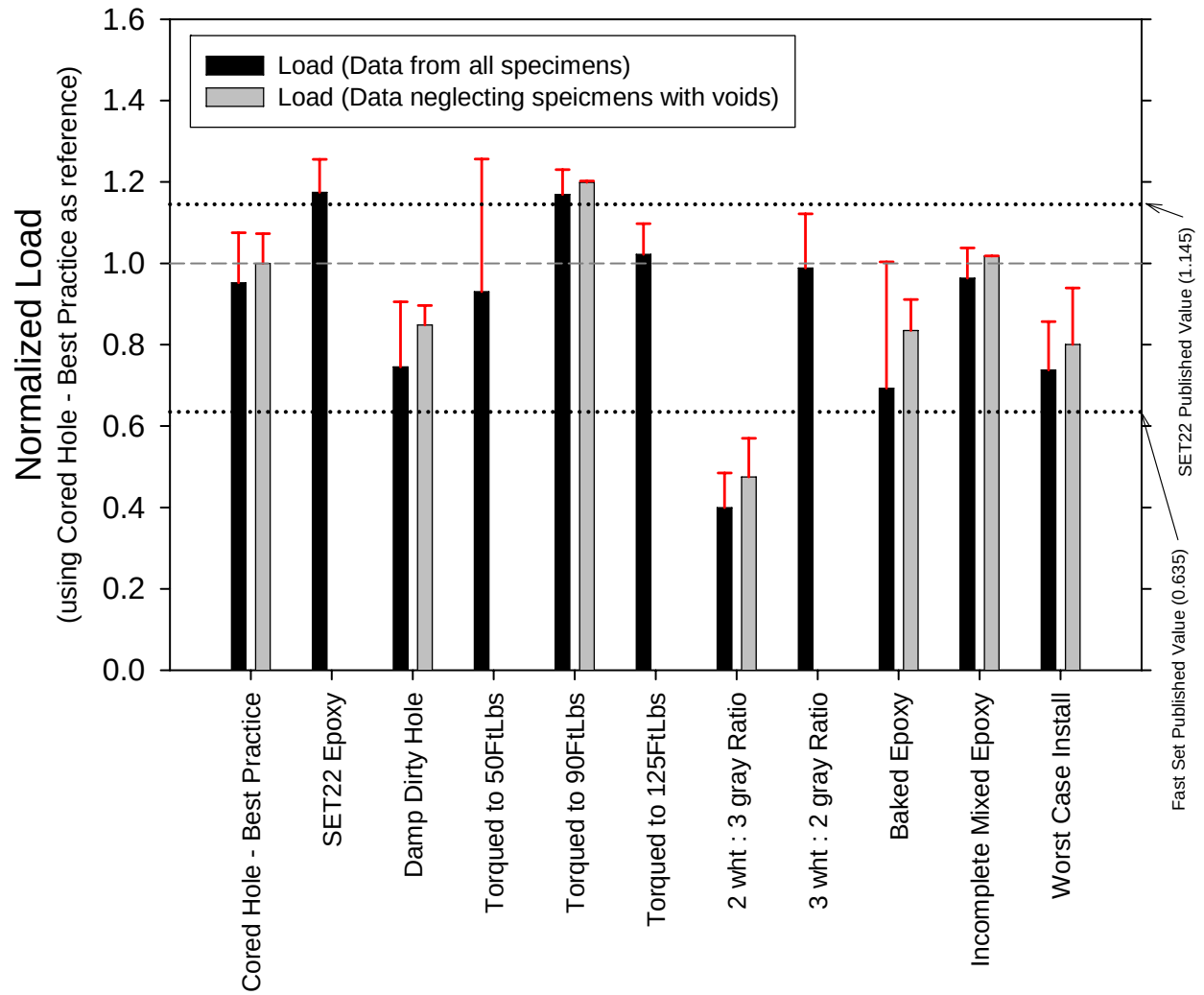


Figure 26. Phase 2 normalized load data with and without voids.

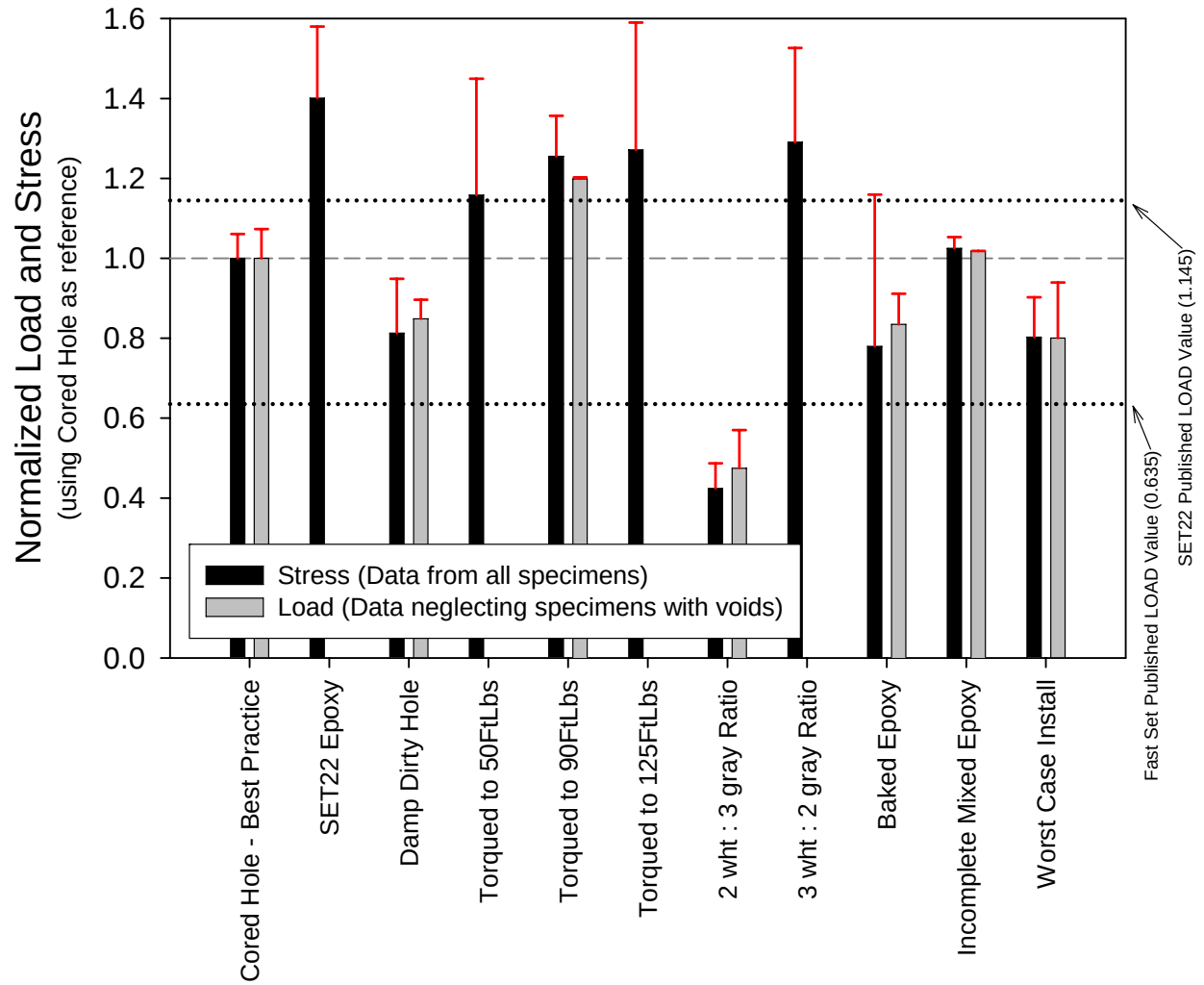


Figure 27. Phase 2 normalized load and stress data.



50 ft-lbs



90 ft-lbs



125 ft-lbs

Figure 28. Contrasting difference between red caps after torque. Top: 50 ft-lbs. Middle: 90 ft-lbs. Bottom: 125 ft-lbs.



50 ft-lbs



90 ft-lbs



125 ft-lbs

Figure 29. Contrasting difference between sealing washers after torque. Top: 50 ft-lbs. Middle: 90 ft-lbs. Bottom: 125 ft-lbs.

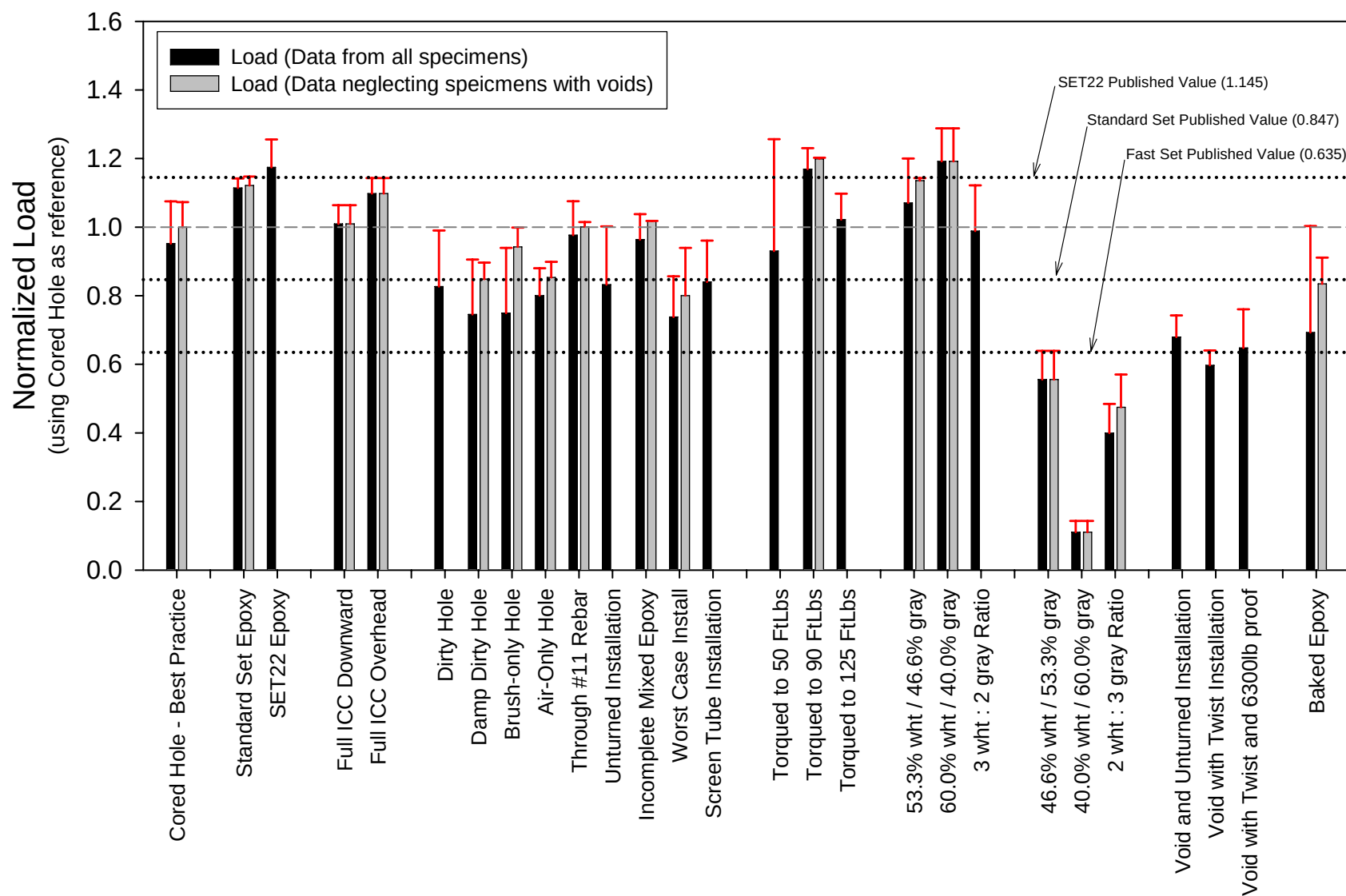


Figure 30. Normalized loads for Phases 1 and 2.

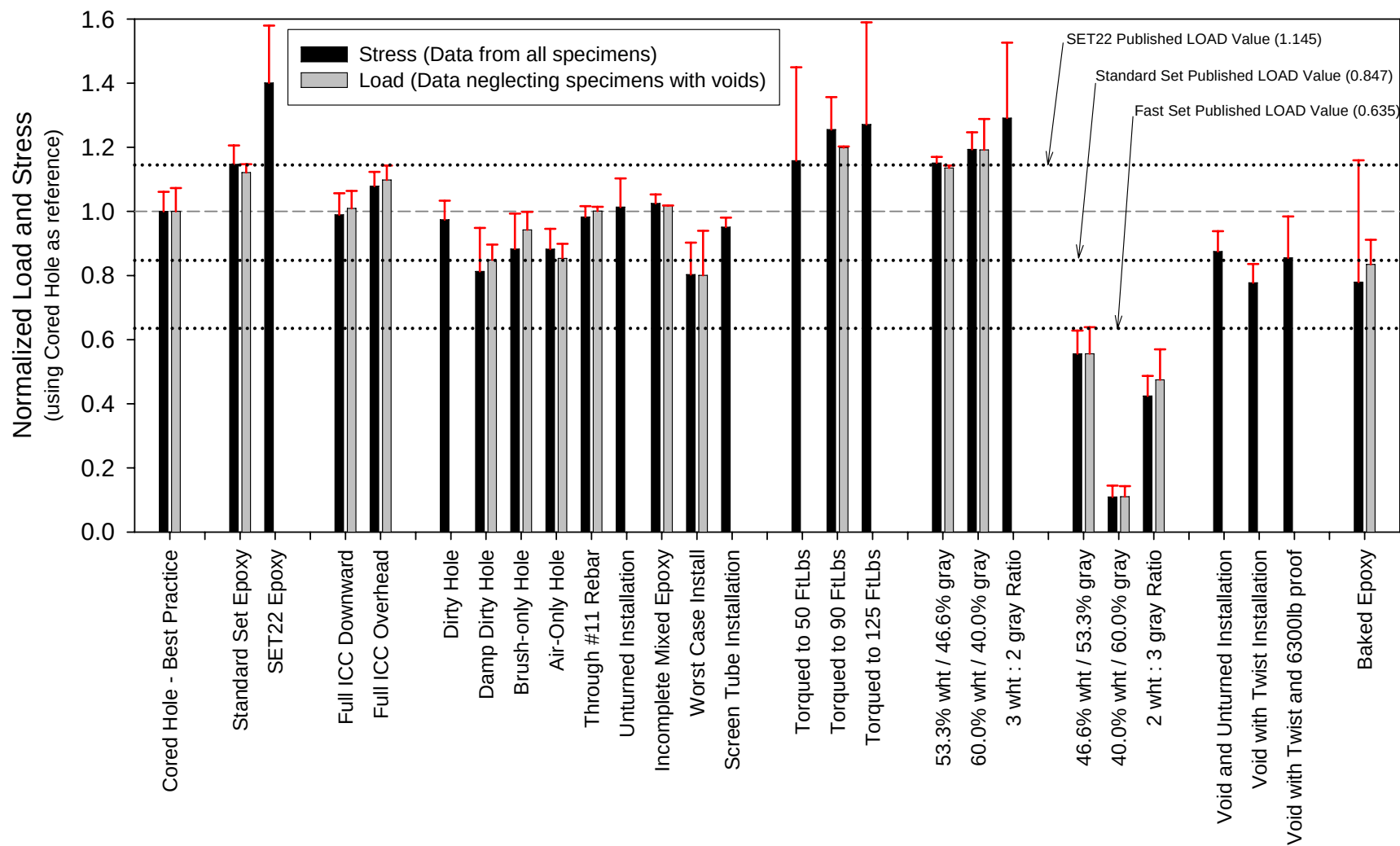


Figure 31. Normalized load and stress for Phase 1 and 2 specimens.

Table 30 Comparison of Peak Load and Bond Stress Statistics

Specimen Series	Peak Load from All Specimens μ (lbs), COV	Peak Load Neglecting Specimens with Voids μ (lbs), COV	Peak Stress μ (psi), COV	97.5% Survival Load Neglecting Voids ^b (lbs)
40.0% wht / 60.0% gray	2436, 0.298	2436, 0.298 ^a	242, 0.317	984
Torqued to 50 ft-lbs	20546, 0.350	a	2560, 0.252	6164
2 wht : 3 gray Ratio	8825, 0.213	10486, 0.200	938, 0.148	6292
46.6% wht / 53.3% gray	12274, 0.150	a	1228, 0.131	8592
Void w/Twist and 6300lb Proof	14301, 0.174	a	1890, 0.151	9324
Unturned Installation	18383, 0.204	a	2241, 0.088	10883
Dirty Hole	18261, 0.197	a	2153, 0.061	11066
Void w/Twist Install	13188, 0.072	a	1719, 0.075	11289
Worst Possible Install	16293, 0.161	17678, 0.173	1775, 0.124	11561
Void and Unturned Installation	15007, 0.093	a	1934, 0.072	12216
Screen Tube Installations	18557, 0.143	a	2102, 0.031	13250
Baked Epoxy	15309, 0.447	18432, 0.092	1723, 0.487	15041
3 wht : 2 gray Ratio	21834, 0.134	a	2853, 0.182	15982
Damp Dirty Hole	16462, 0.215	18734, 0.057	1797, 0.167	16598
Air-Only Hole	17673, 0.100	18849, 0.053	1950, 0.072	16851
Brush-Only Hole	16550, 0.254	20810, 0.060	1951, 0.126	18313
Cored Hole - Best Practice	21027, 0.129	22084, 0.073	2210, 0.061	18860
Torqued to 125 ft-lbs	22572, 0.074	a	2810, 0.250	19231
Full ICC Downward	22299, 0.053	a	2187, 0.068	19935
Through #11 Rebar	21569, 0.101	22101, 0.014	2172, 0.034	21482
60.0% wht / 40.0% gray	26320, 0.081	a	2639, 0.044	22056
Full ICC Overhead	24247, 0.041	a	2384, 0.041	22259
SET22 Epoxy	25934, 0.069	a	3096, 0.128	22355
Incomplete Mixed Epoxy	21277, 0.077	22477, 0.000	2265, 0.027	22477
Standard Set Epoxy	24605, 0.025	24765, 0.023	2534, 0.051	23626
53.3% wht / 46.6% gray	23651, 0.121	25078, 0.007	2544, 0.016	24727
Torqued to 90 ft-lbs	25826, 0.052	26473, 0.003	2775, 0.081	26314
^a – Either no specimens or too many specimens in the set had voids to calculate individual statistics considering the effect with and without voids. Therefore the statistics reported represents all specimens in a set.				
^b – If statistics neglecting specimens with voids was nonexistent, the statistics using all specimen data was used.				

Table 31 Comparison of Average Displacement Data

Specimen Series	Average Displacement after 12 kip Load Step (inches)	Normalized Displacement (using “Cored Hole – Best Practice” as basis)
2 wht : 3 gray Ratio	0.364	20.541
40.0% wht / 60.0% gray	0.252	14.206
46.6% wht / 53.3% gray	0.149	8.384
Void w/Twist and 6300lb Proof	0.113	6.387
Brush-Only Hole	0.031	1.748
Void with Twist Install	0.027	1.508
No Twist Installation	0.024	1.367
Damp Dirty Hole	0.021	1.182
Void Unturned Install	0.019	1.092
Dirty Hole	0.018	1.009
Cored Hole - Best Practice	0.018	1.000
Baked Epoxy	0.017	0.939
Torqued to 50 ft-lbs	0.016	0.914
Screen Tube Installation	0.015	0.864
Air-Only Hole	0.014	0.791
Full ICC Overhead	0.013	0.757
Worst Possible Install	0.013	0.717
Full ICC Downward	0.013	0.707
Through #11 Rebar	0.011	0.623
53.3% wht / 46.6% gray	0.010	0.587
3 wht : 2 gray Ratio	0.010	0.562
Torqued to 125 ft-lbs	0.009	0.530
SET22	0.009	0.524
Standard Set Epoxy	0.009	0.489
Torqued to 90 ft-lbs	0.009	0.485
Incomplete Mixed Epoxy	0.008	0.462
60.0% wht / 40.0% gray	0.006	0.326

CHAPTER 5. SUMMARY

This investigation examined the effect of certain installation parameters on the load-displacement performance of the adhesive anchors used in the construction of the ceiling system of the I-90 Seaport Access Tunnel in Boston, Massachusetts. The research was conducted with every intention to create specimens that only varied one installation parameter at time, except for those parameters in Phase 2 that exclusively looked at cross-coupling. However, the introduction of voids appears to be inevitable when installing anchors in the overhead position. A total of 166 anchors were tested during both phases of this study. Of those anchors, 30 were attempted to be installed with voids, but only 24 were actually successful. Of the 136 anchors that should not have had voids, 35 (25.7% of 136) ended up having voids, and 15 (11.0% of 136) were unintentionally only partially filled with epoxy. Looking at the values presented in Table 30, the COV for the 27 installation parameters range from 0.025-0.447 with an average of 0.150 when all data within a set is considered. When specimens with voids are neglected, the range of COV decreases to 0.000-0.350 with an average of 0.107. This shows the inherent variability involved with overhead installations mainly from the formation of voids which even in a laboratory environment proved inevitable. Of the 12 anchors installed in the downward position only one had a large void. Although it is difficult to draw conclusions from so few specimens, observation of all the overhead and downward installations that were a part of this program suggests that downward installations are less apt to produce voids. Recall that voids are formed by trapping an air bubble in the hole while injecting epoxy. This often occurs when the injection nozzle tip moves around during injection. In a downward installation the tip of the mixing tube can rest at the bottom of the hole and the epoxy can fill in around it. In an overhead installation the operator must bear the weight of the injection gun which makes holding the tip of the mixing nozzle in place more difficult. This is exacerbated with manually operated injection guns because the tip tends to move even more as a result of the hand pumping.

When the anchors were installed in Phase 1, the red adhesive retaining cap was installed first, and then the mixing tube was pushed through the red cap to fill the hole with epoxy. In this order, the red cap prevented visual inspection to ensure the hole was properly filled with epoxy and is a reasonable explanation for the many partially filled holes from Phase 1. For Phase 2 the epoxy was injected without the red caps in place so a visual inspection could determine if the hole was properly filled which reduced the number of partially filled holes for Phase 2. Also, there was no appreciable difference between the pneumatic (Phase 1) and manual squeeze (Phase 2) injection guns with regard to producing voids.

In general, the group of parameters that most affected performance had misproportioned epoxy with more gray component than white (60% reduction in load capacity), followed by the group of parameters that intentionally created voids in the epoxy (30-40% reduction in load capacity). Perhaps the most surprising finding of this research is these types of epoxies can exhibit creep behavior during a short duration (30 to 60 minute) test. The creep behavior was noted with all

three epoxies (Power-Fast+ Standard Set, fast set, and Simpson SET22), though the magnitude of displacement was most evident for anchors using the Power-Fast+ fast set formula.

CHAPTER 6. REFERENCES

- 1 Hartmann, J., Ocel, J., Shipp, J., Wright, W., and Fuchs, P. “*I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Load and Load-Displacement Testing Results*,” Federal Highway Administration, Turner-Fairbank Highway Research Center, McLean, VA, October 2006.
- 2 ASTM E488-96(2003), “Standard Test Methods for Strength of Anchors in Concrete and Masonry Elements,” American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 3 NTSB, “Materials Laboratory Factual Report, Report No. 06-073,” National Transportation Safety Board, Washington D.C., December 13, 2007.
- 4 ASTM C192/C192M-05, “Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory,” American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 5 ASTM C39/C39M-05, “Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens,” American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 6 ASTM C496/C496M-04e1, “Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens,” American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 7 Central Artery/Tunnel Project Contract C09B2, Field Engineers Daily Report (FEDR), Date Range: 01-Jan-1999 to 01-Jan-2004.
- 8 NTSB, “Materials Laboratory Factual Report (Draft), Report No. 06-091,” National Transportation Safety Board, Washington D.C., February 14, 2007.
- 9 NTSB, “Materials Laboratory Report No. 06-092,” National Transportation Safety Board, Washington D.C., January 10, 2007.
- 10 Wheeler, A. J., and Ganji, A. R.; “*Introduction to Engineering Experimentation, Second Edition*,” Prentice Hall, Upper Saddle River, NJ, pp. 149-150, 2004.

APPENDIX 1

In the time between testing and report writing, the names of the parameters either changed or were streamlined. In this appendix, pre- and post-pull pictures were taken of each anchor but the labels in the pictures do not match what is used with the test of this report. The following table serves as a cross-reference between the picture label and how it was referred to in the report text.

Installation Parameter	Picture Label in Appendix 1 Tables
Full ICC Downward	Full_ICC_Downward
Full ICC Overhead	Full_ICC_Upward
Cored Hole – Best Practice	BestPractices
Standard Set	RegularSet
Screen Tube Installations	ScreenTubes
Dirty Hole	Uncleaned
Air-Only Hole	AirOnly
Brush-Only Hole	BrushOnly
Unturned Installation	NoTwist
46.6% wht / 53.3% gray	47wht_53grey
53.3% wht / 46.6% gray	53wht_47grey
40.0% wht / 60.0% gray	40wht_60grey
60.0% wht / 40.0% gray	60wht_40grey
Void Unturned Installation	Bubble_woTwist
Void with Twist Installation	Bubble_wTwist
Void with Twist and 6300lb Proof	Bubble_wTwist_6300
Through #11 Rebar	ThroughRebar
Torque Pull	TorquePull

Table 32 Full ICC Downward 1

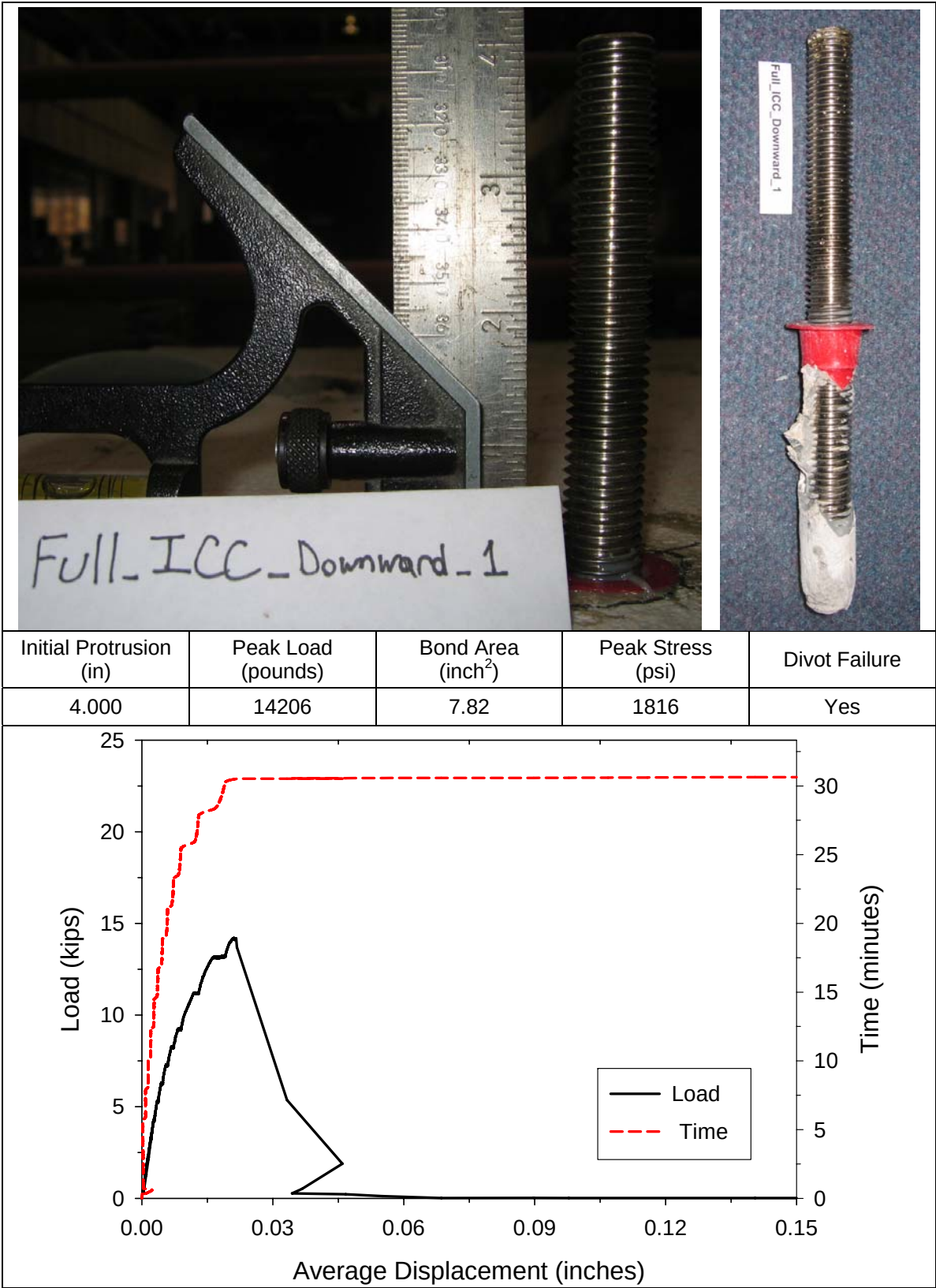


Table 33 Full ICC Downward 2

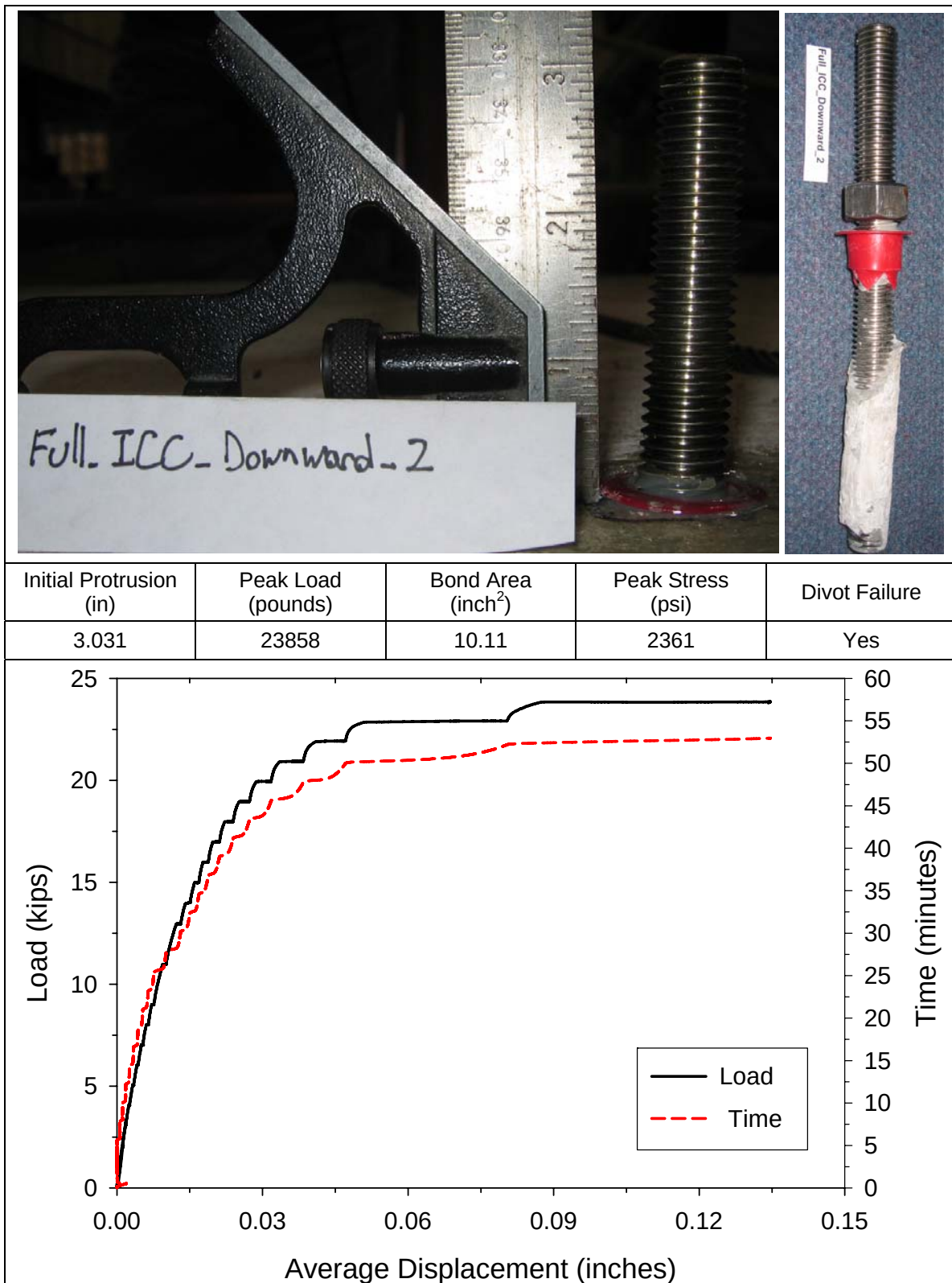


Table 34 Full ICC Downward 3

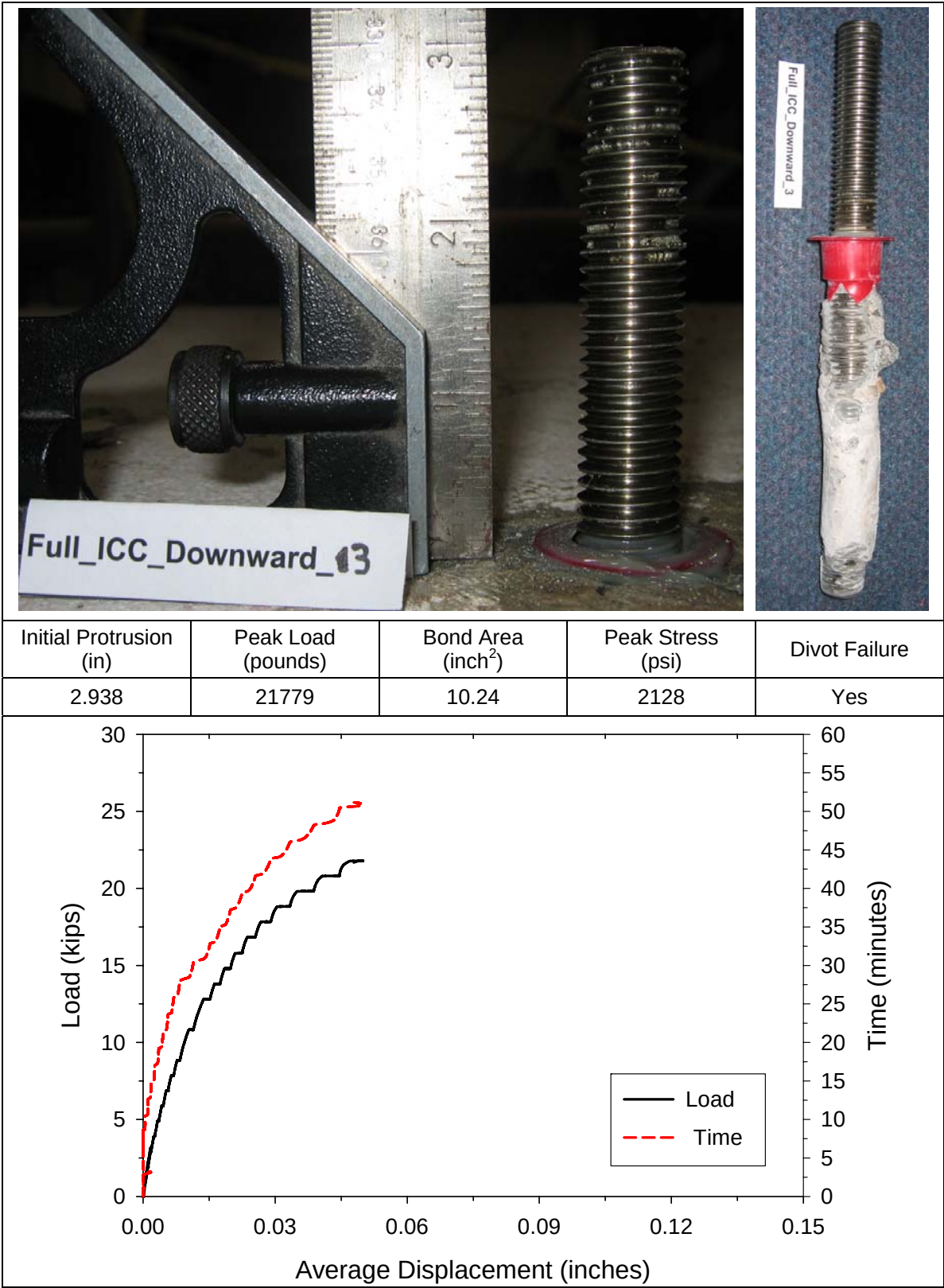


Table 35 Full ICC Downward 4

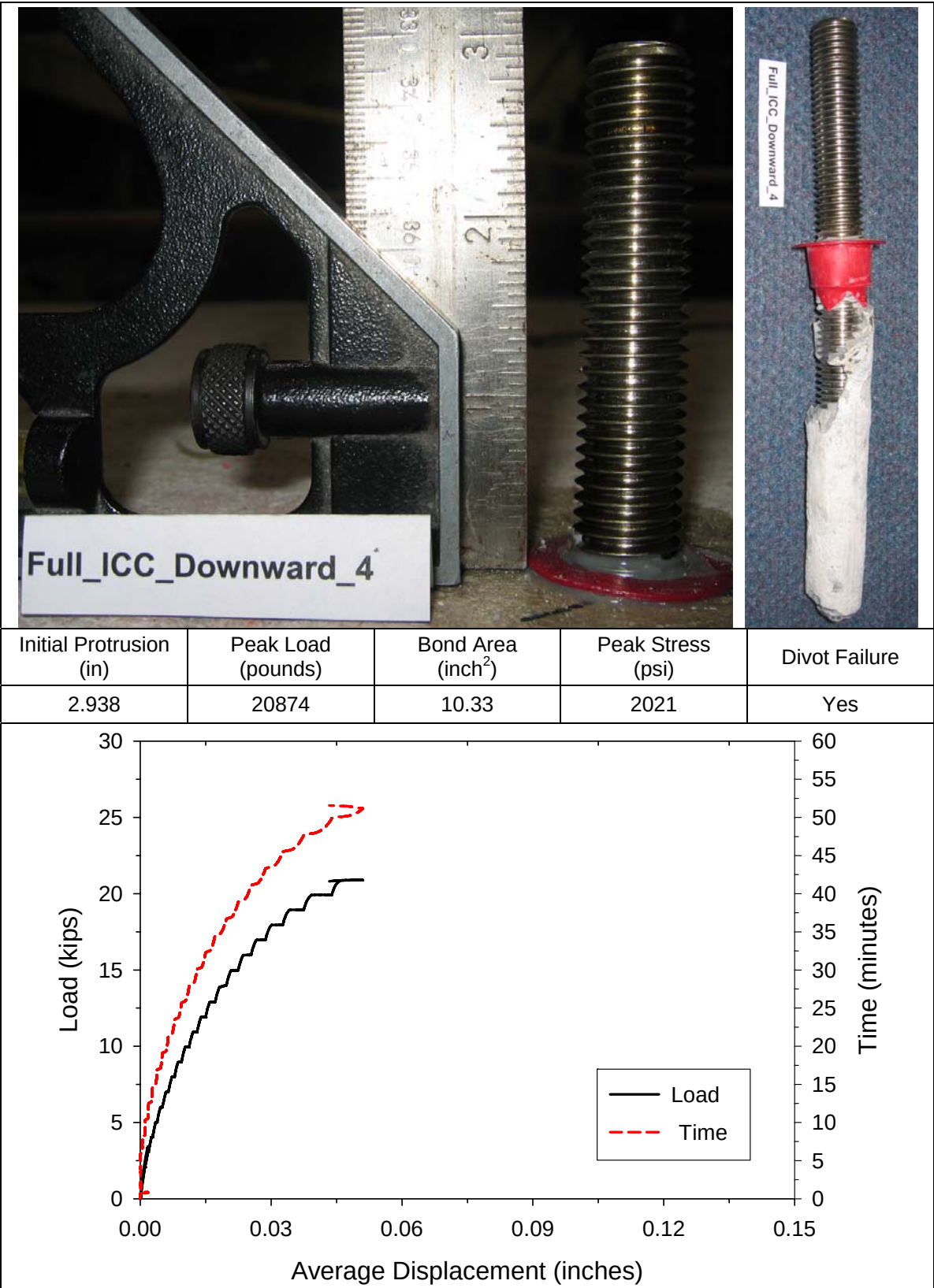


Table 36 Full ICC Downward 5

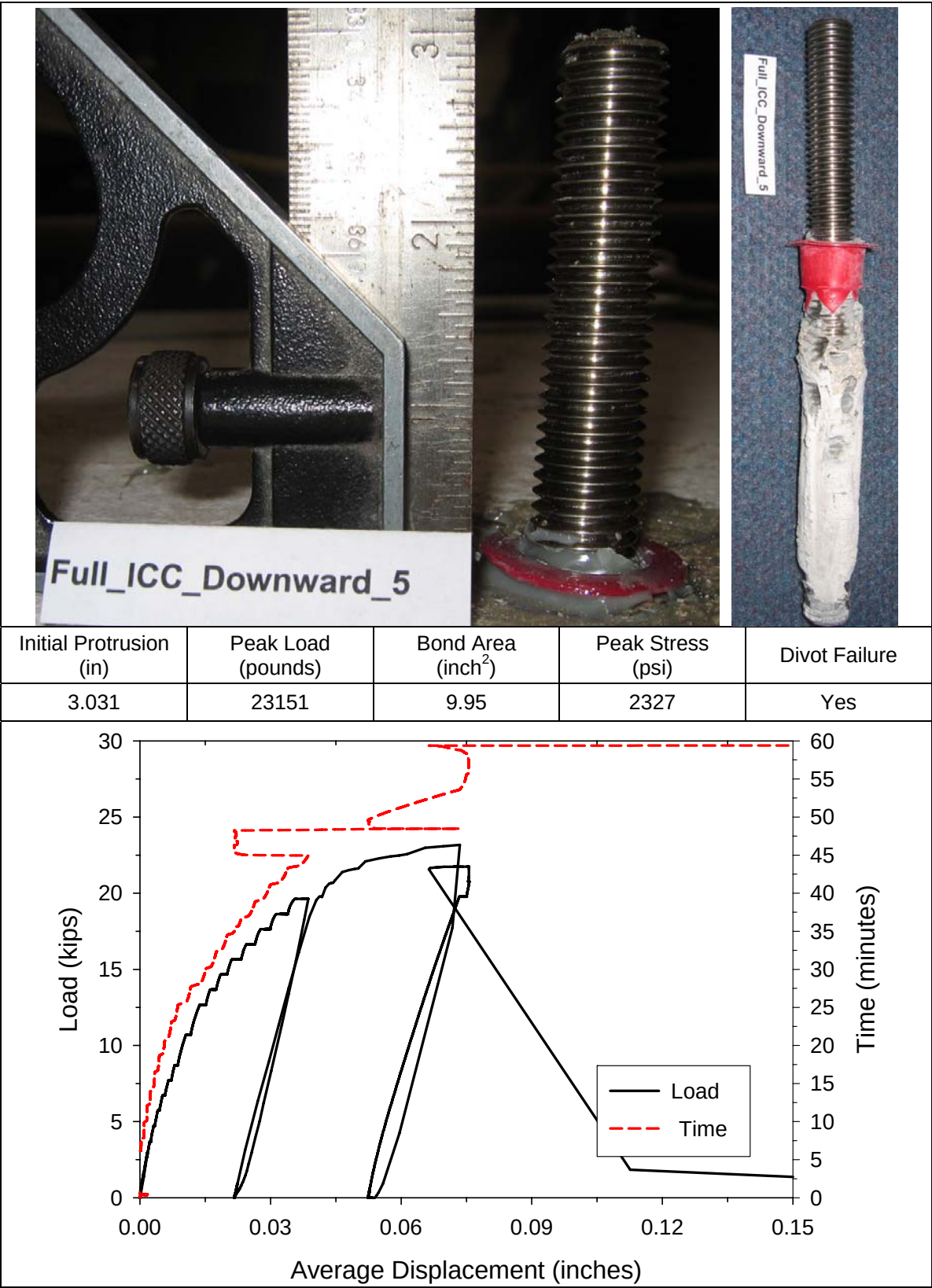


Table 37 Full ICC Downward 6

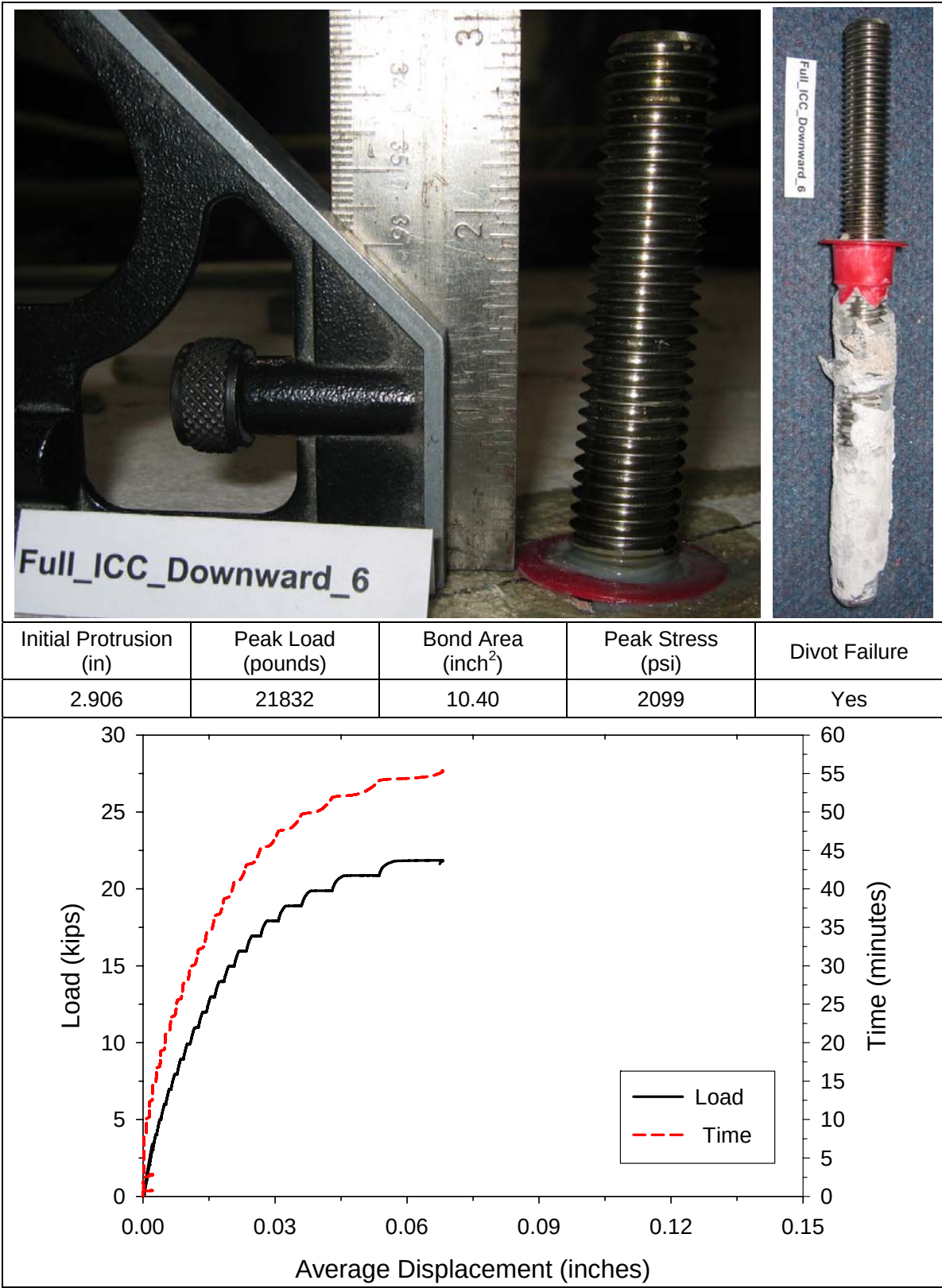


Table 38 Full ICC Downward 7

				
Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
4.0	Not Available	Not Available	Not Available	Yes
Load/displacement data not collect because power outage cause a hydraulic spike that suddenly pulled the anchor to failure without time to start data collection.				

Table 39 Full ICC Overhead 1

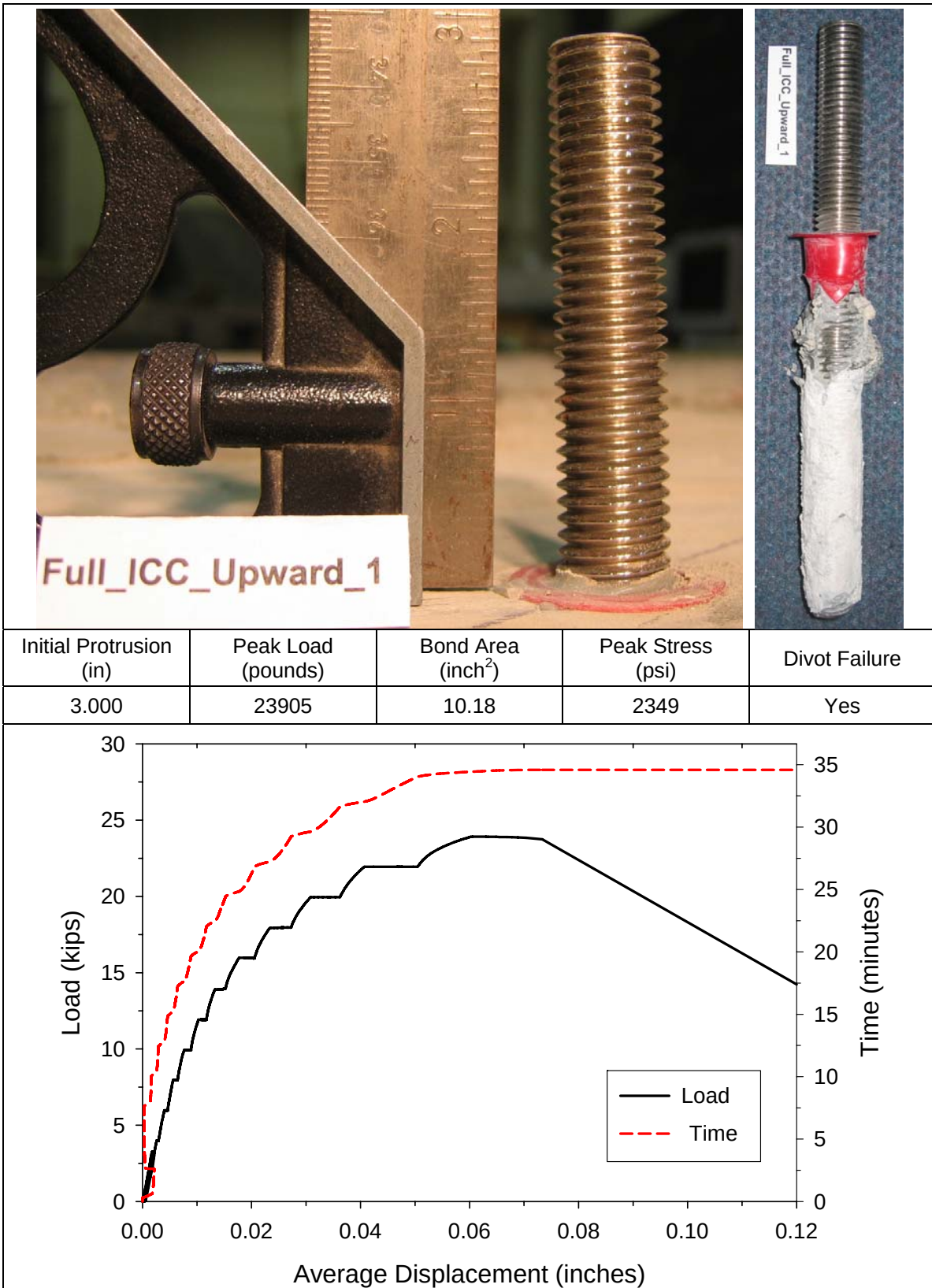


Table 40 Full ICC Overhead 2

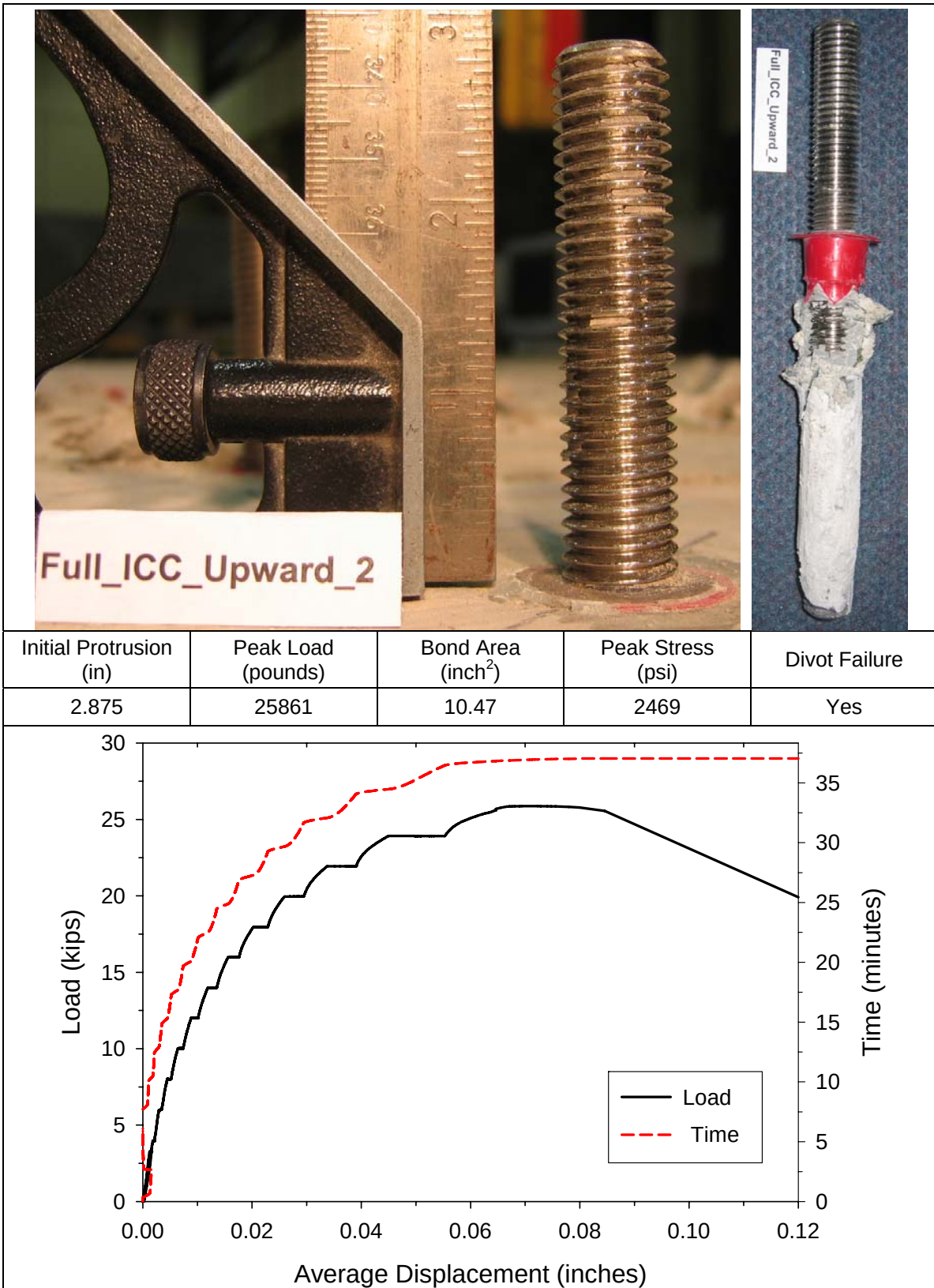


Table 41 Full ICC Overhead 3

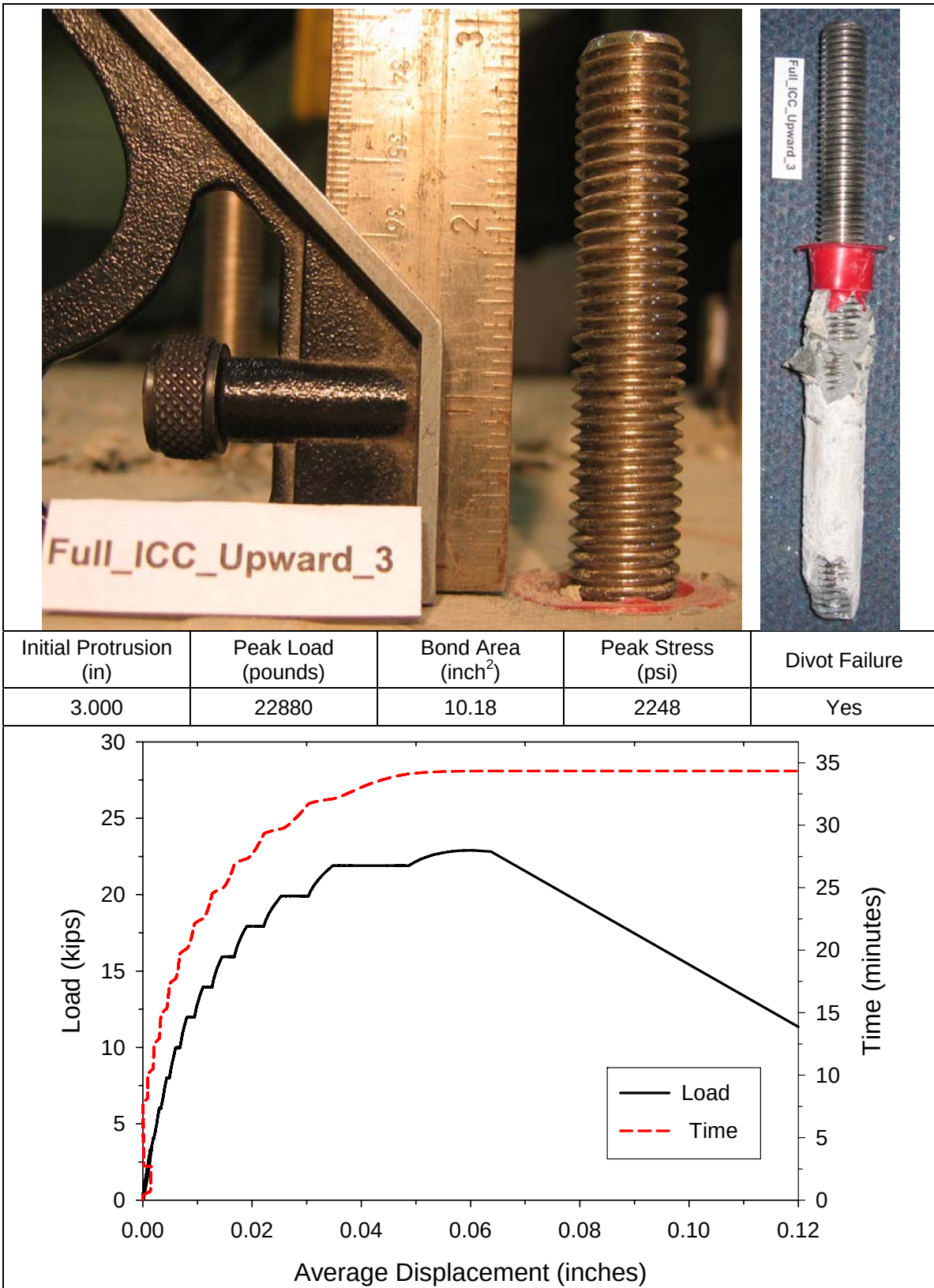


Table 42 Full ICC Overhead 4

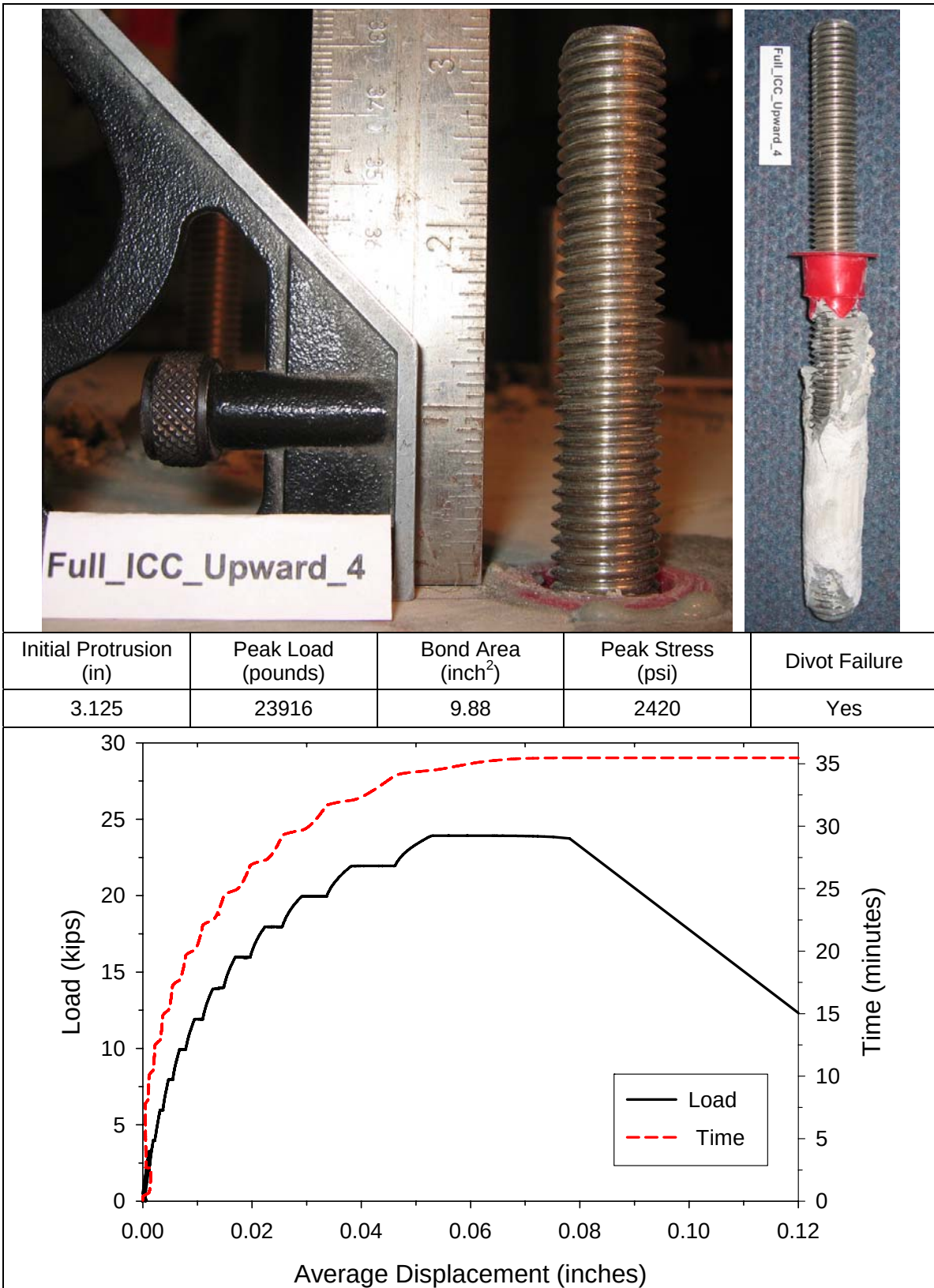


Table 43 Full ICC Overhead 5

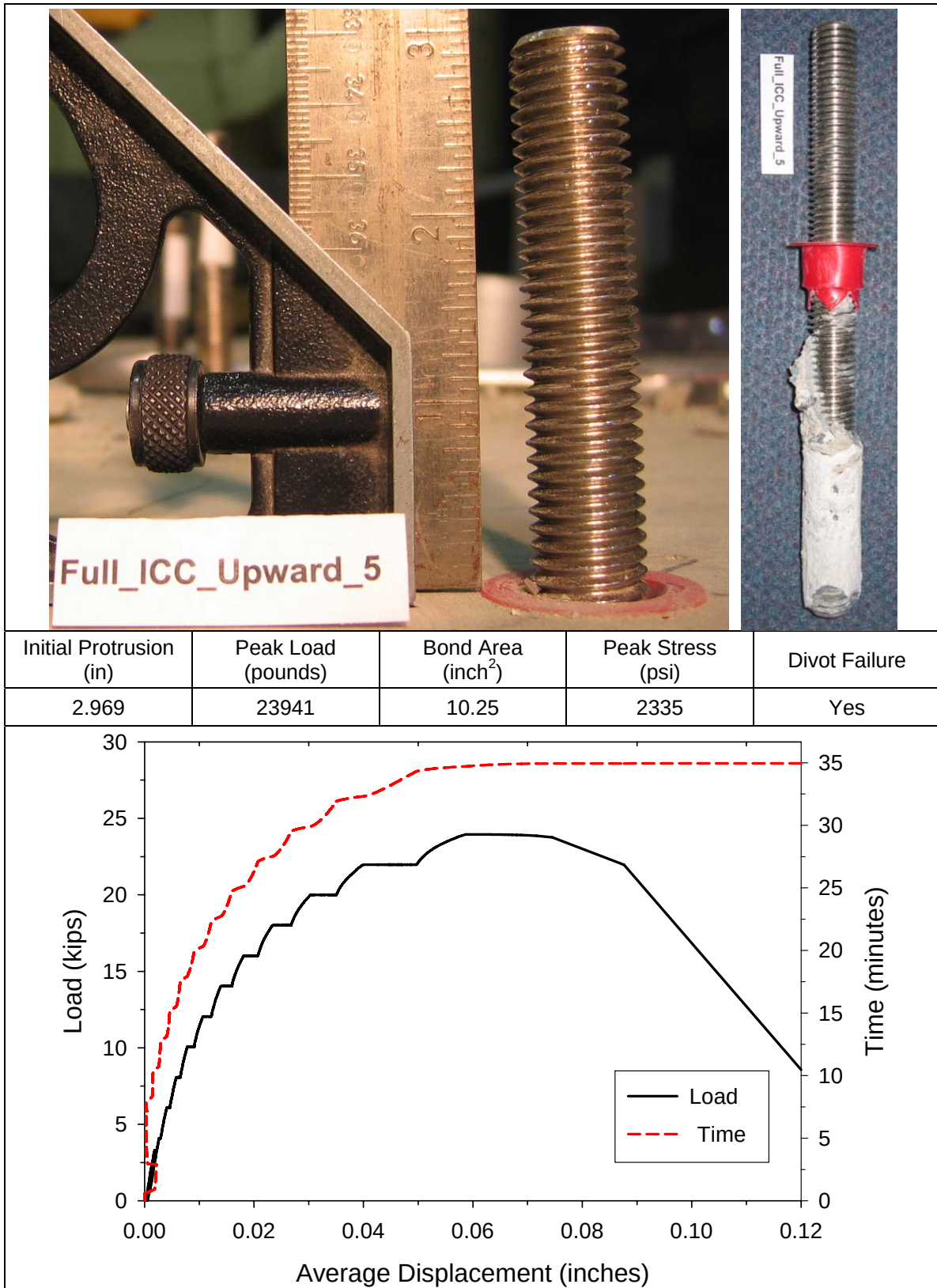


Table 44 Full ICC Overhead 6

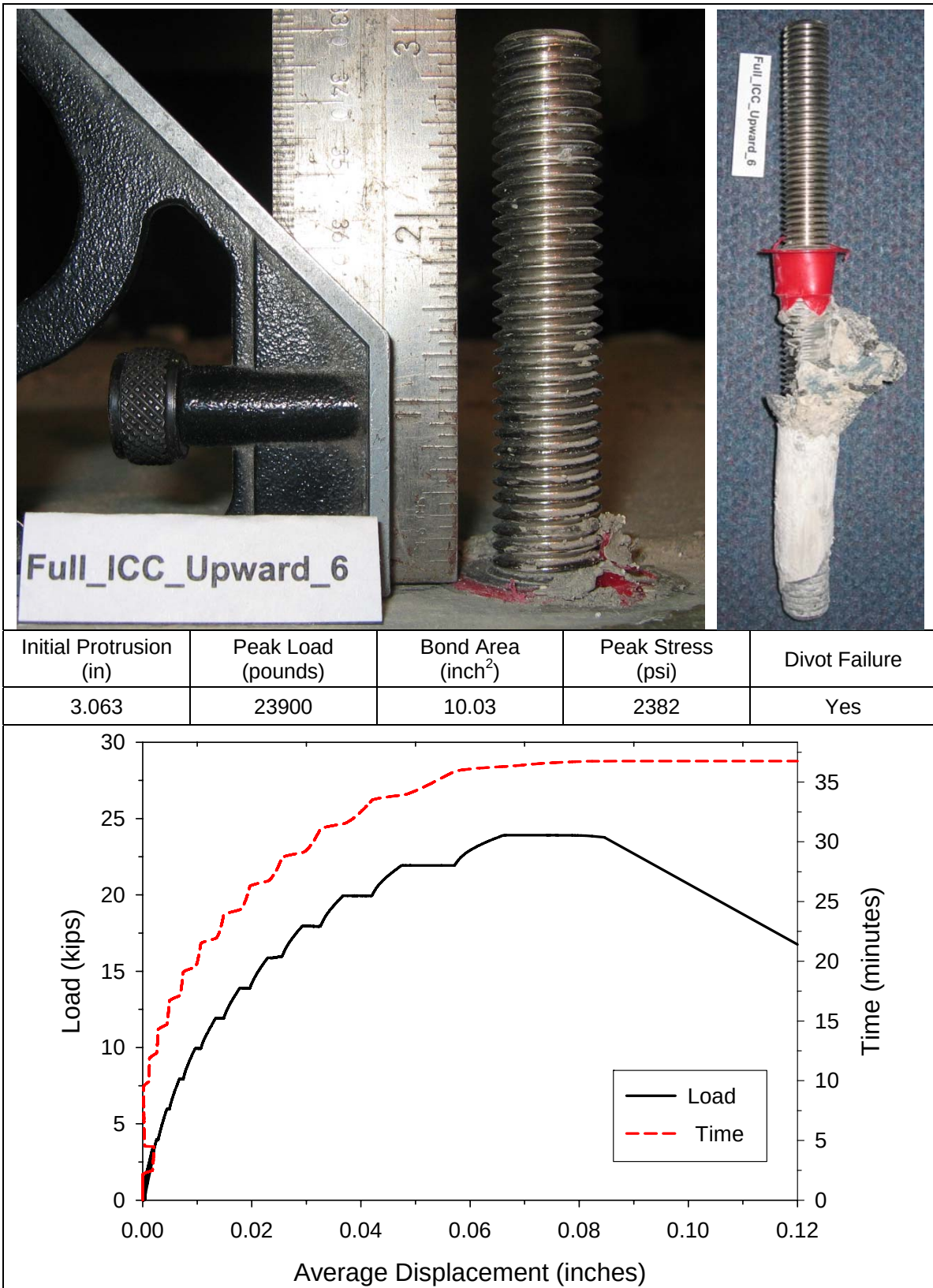
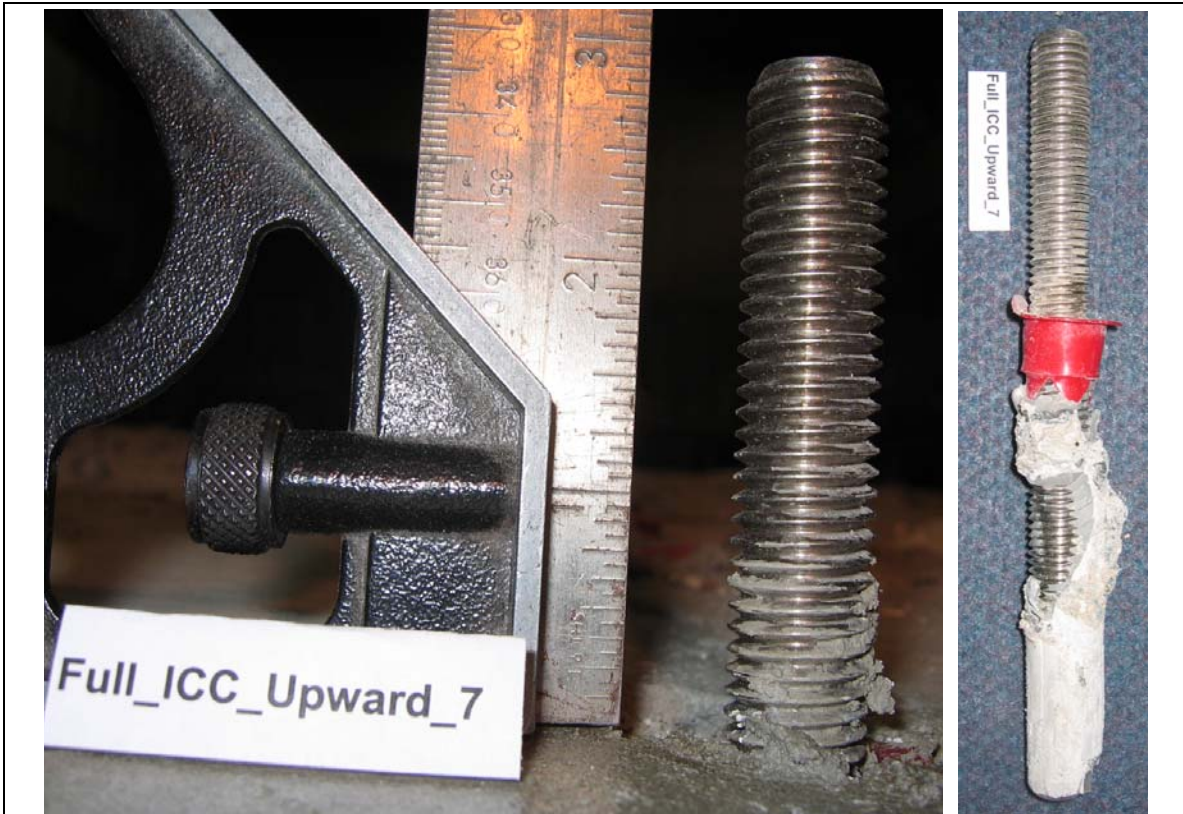


Table 45 Full ICC Overhead 7



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.000	24830	10.18	2439	Yes

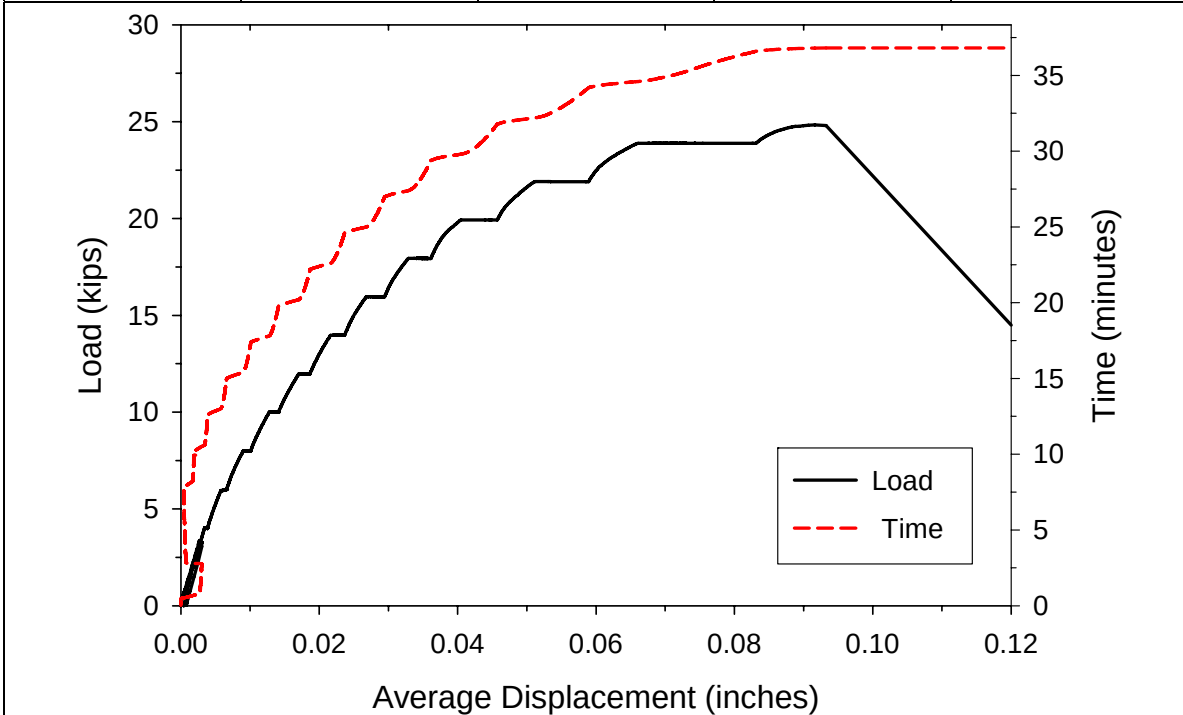


Table 46 Full ICC Overhead 8

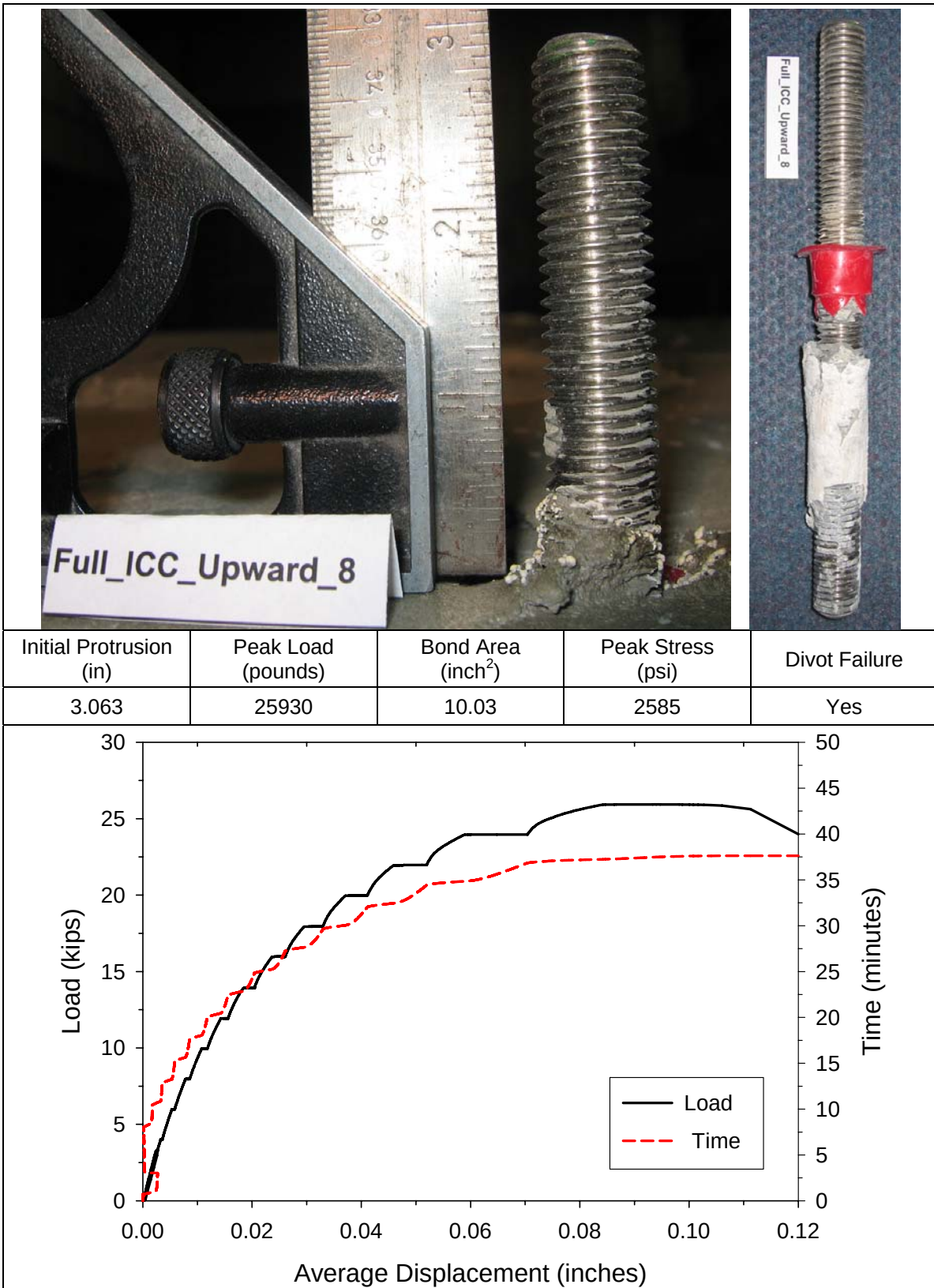


Table 47 Full ICC Overhead 9

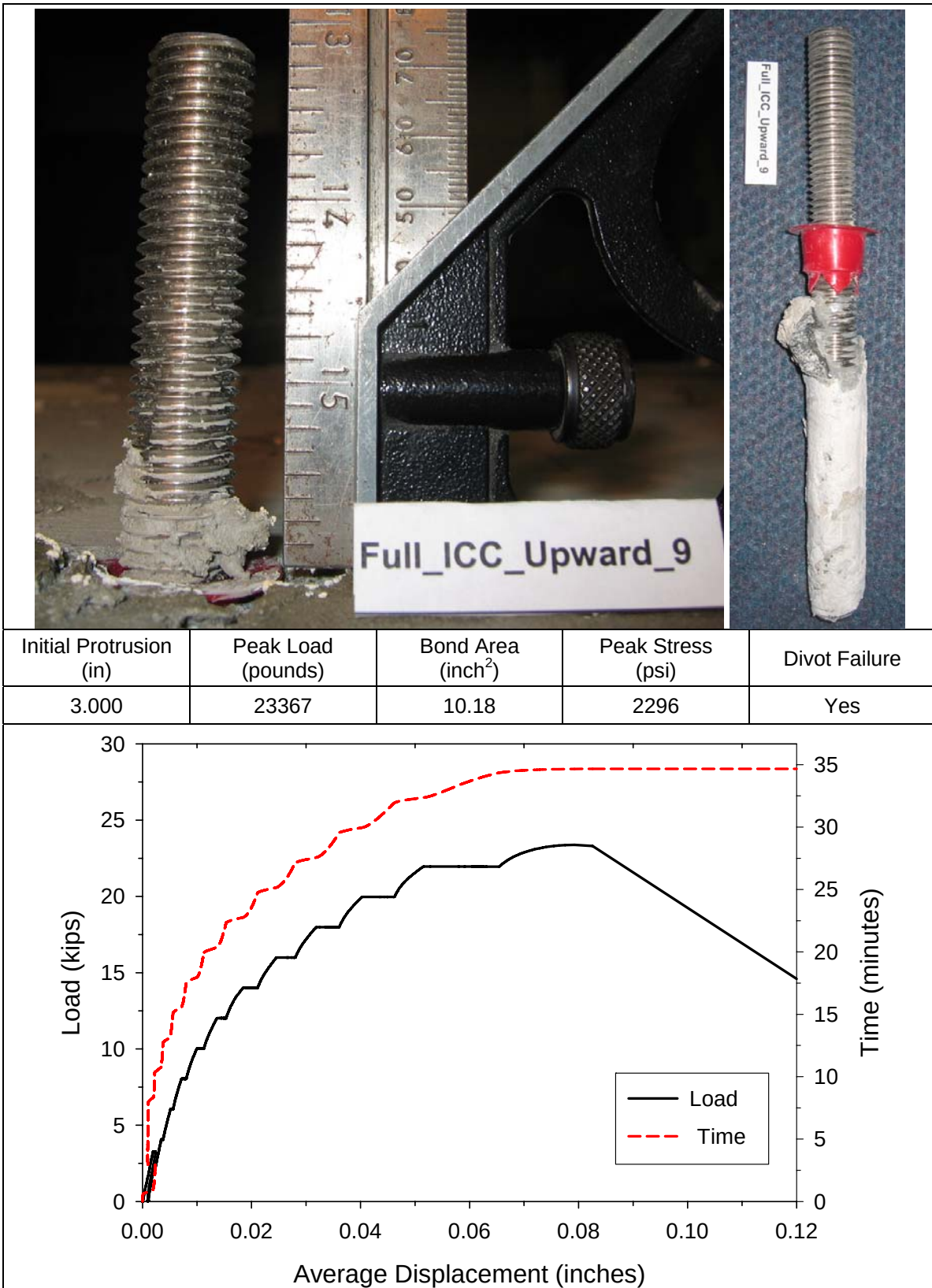


Table 48 Full ICC Overhead 10

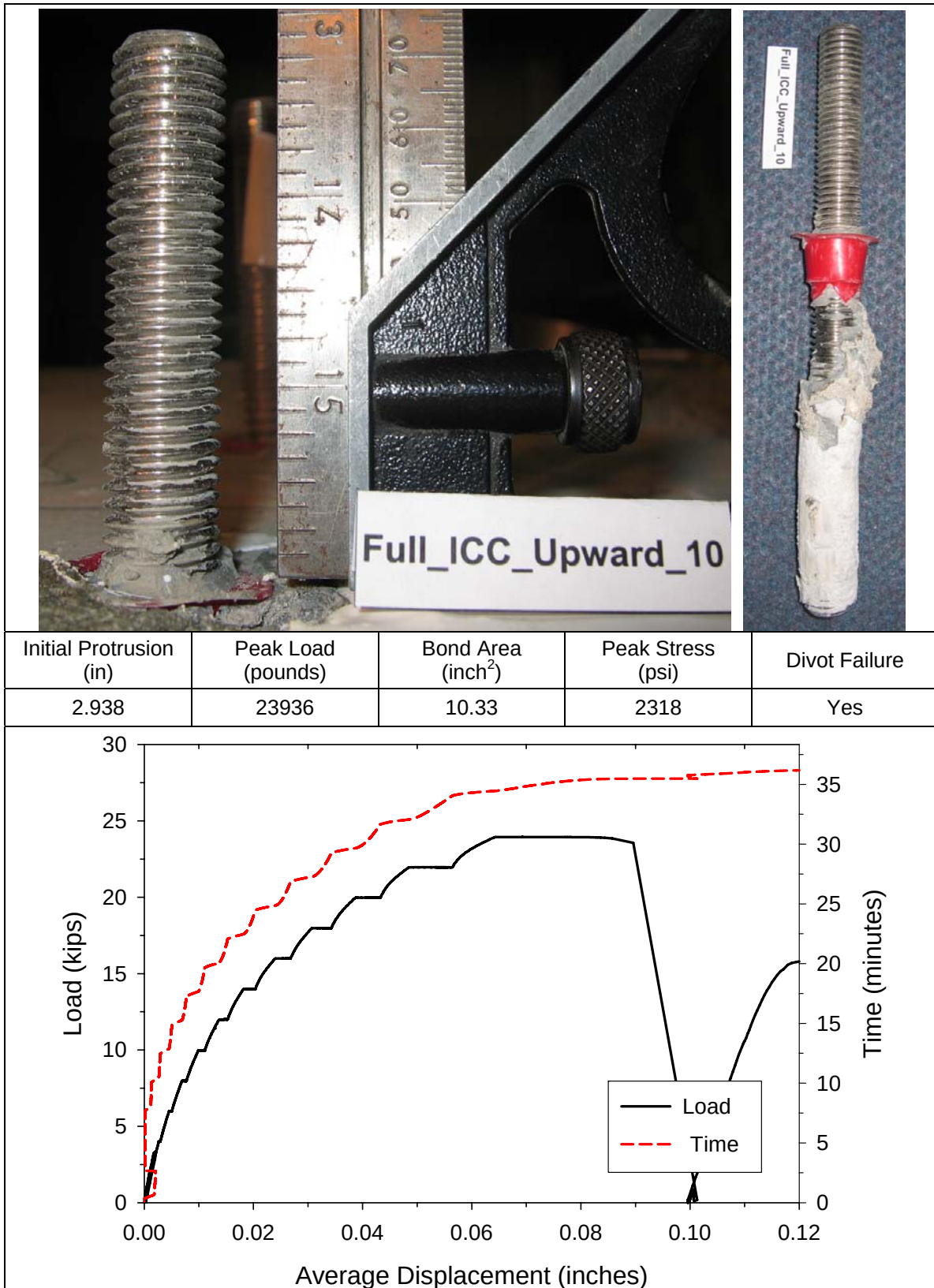


Table 49 Cored Hole – Best Practices 1

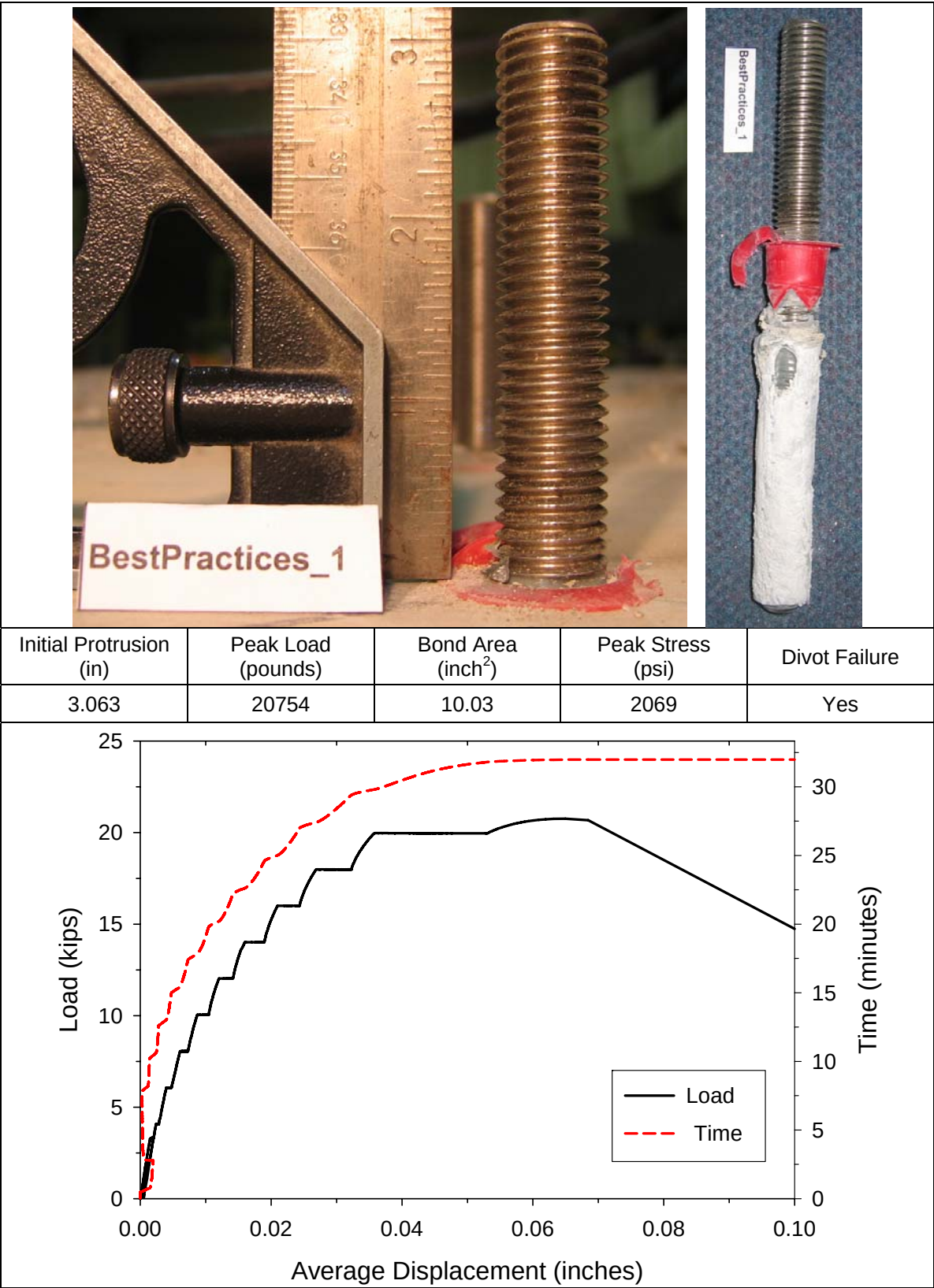


Table 50 Cored Hole – Best Practices 2

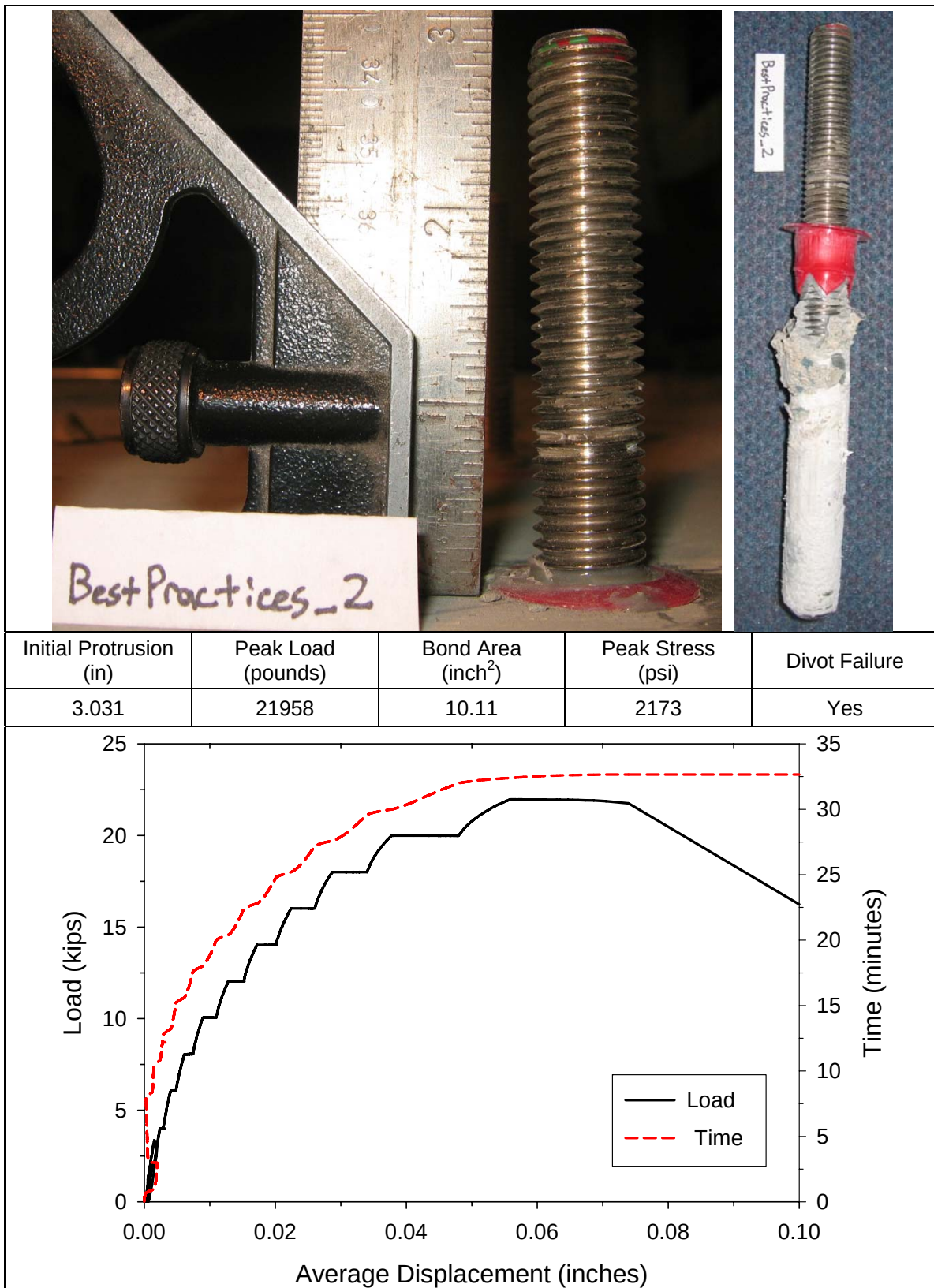


Table 51 Cored Hole – Best Practices 3

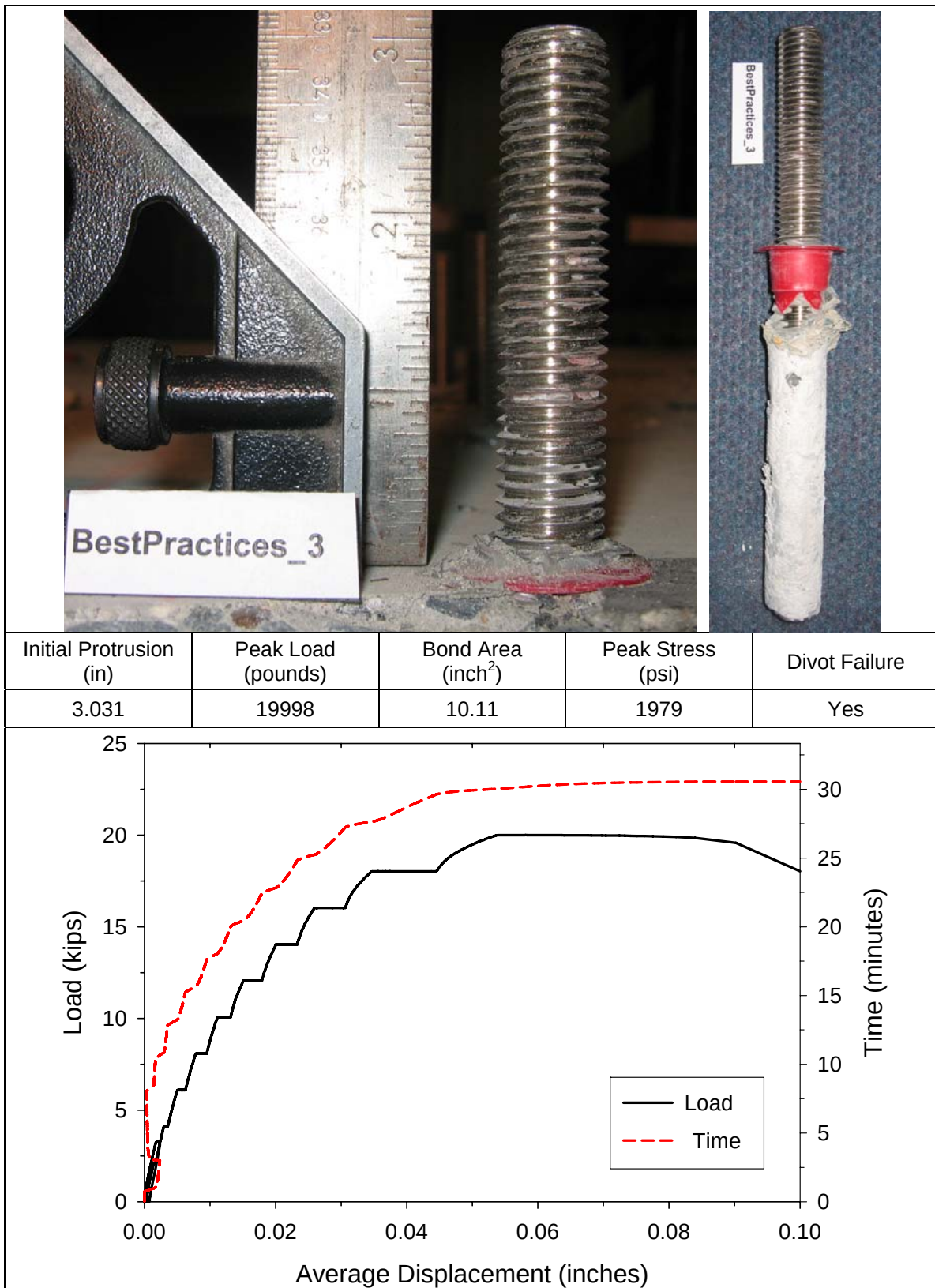


Table 52 Cored Hole – Best Practices 4

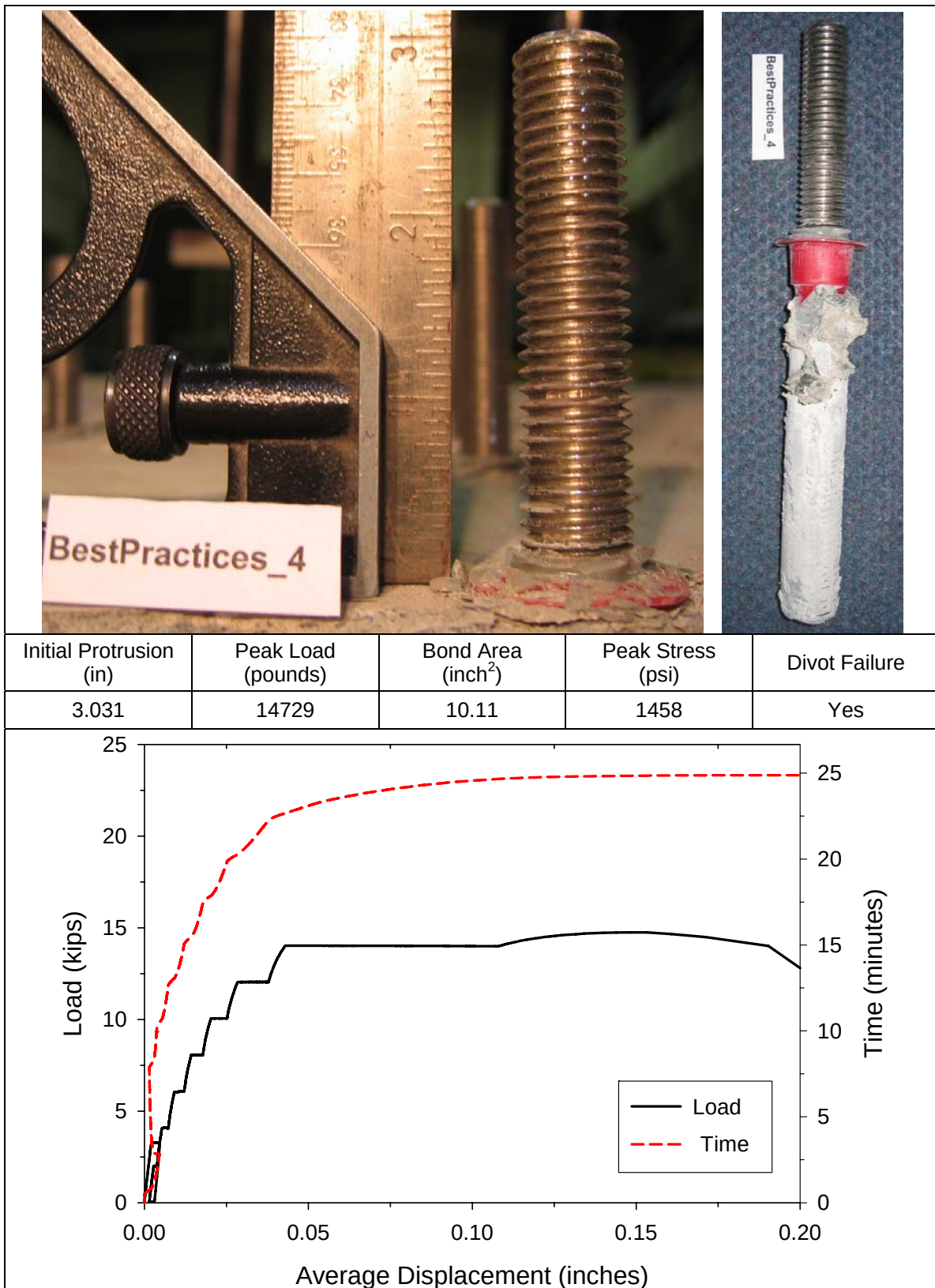


Table 53 Cored Hole – Best Practices 5

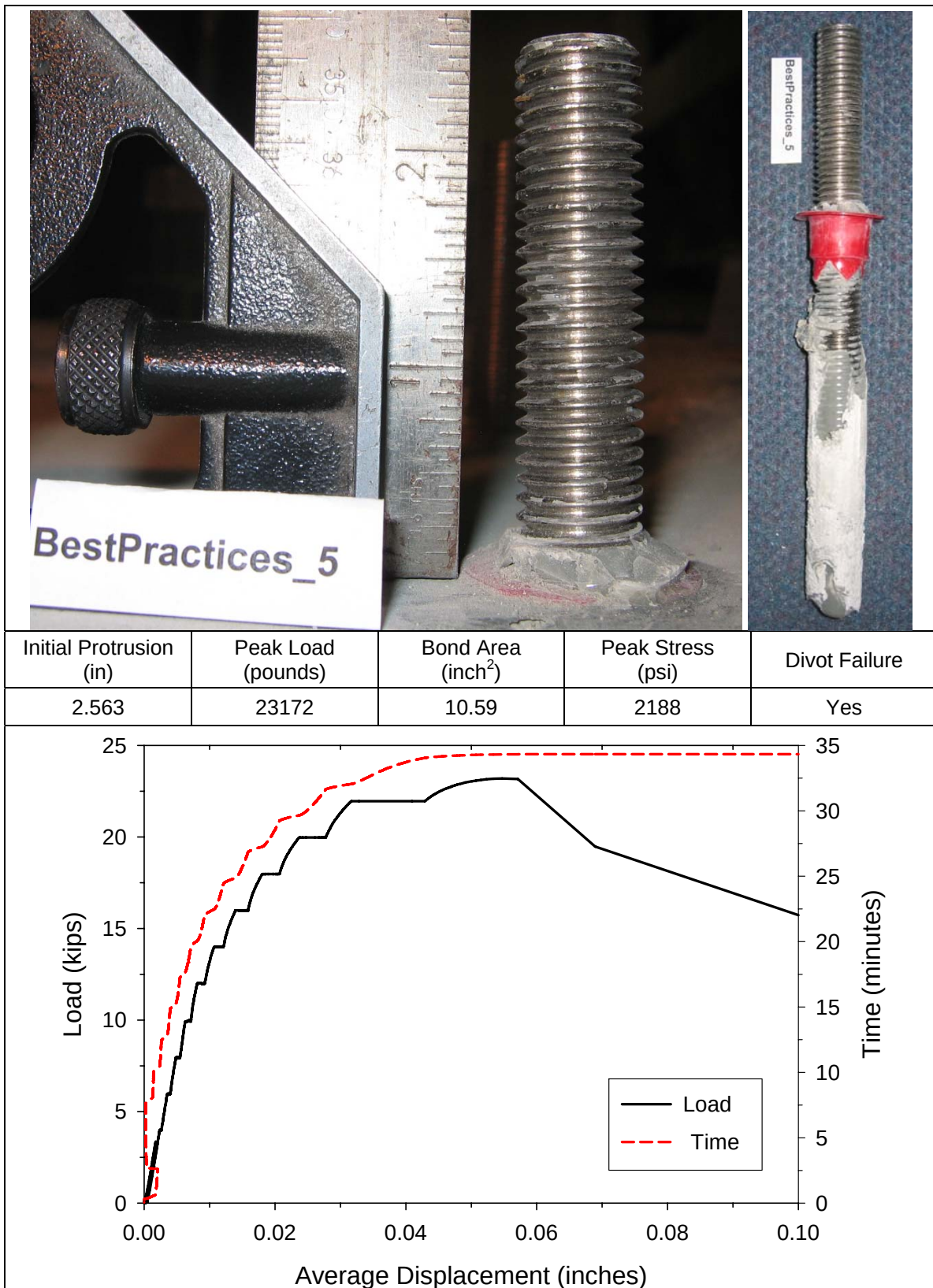


Table 54 Cored Hole – Best Practices 6

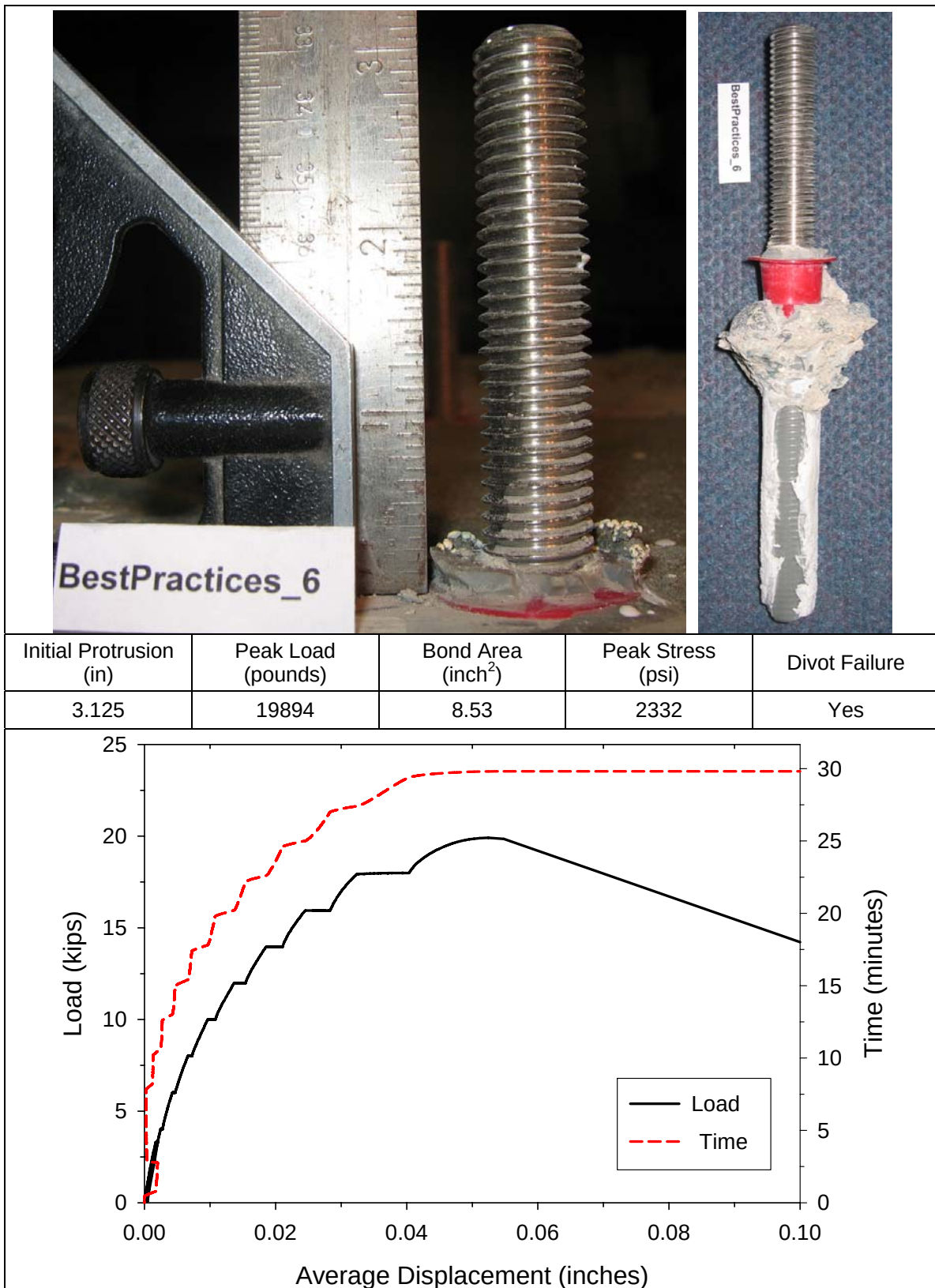


Table 55 Cored Hole – Best Practices 7

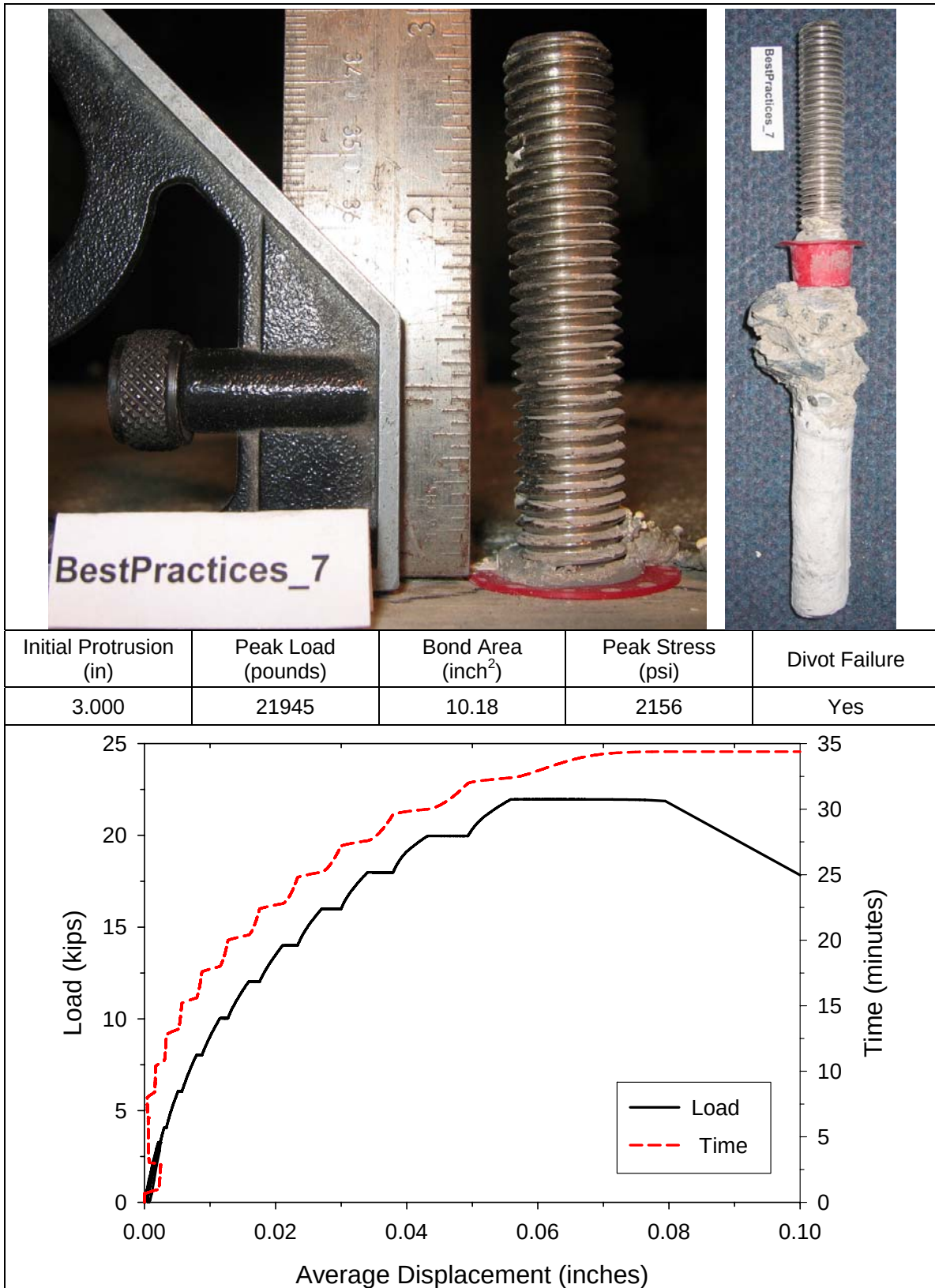


Table 56 Cored Hole – Best Practices 8

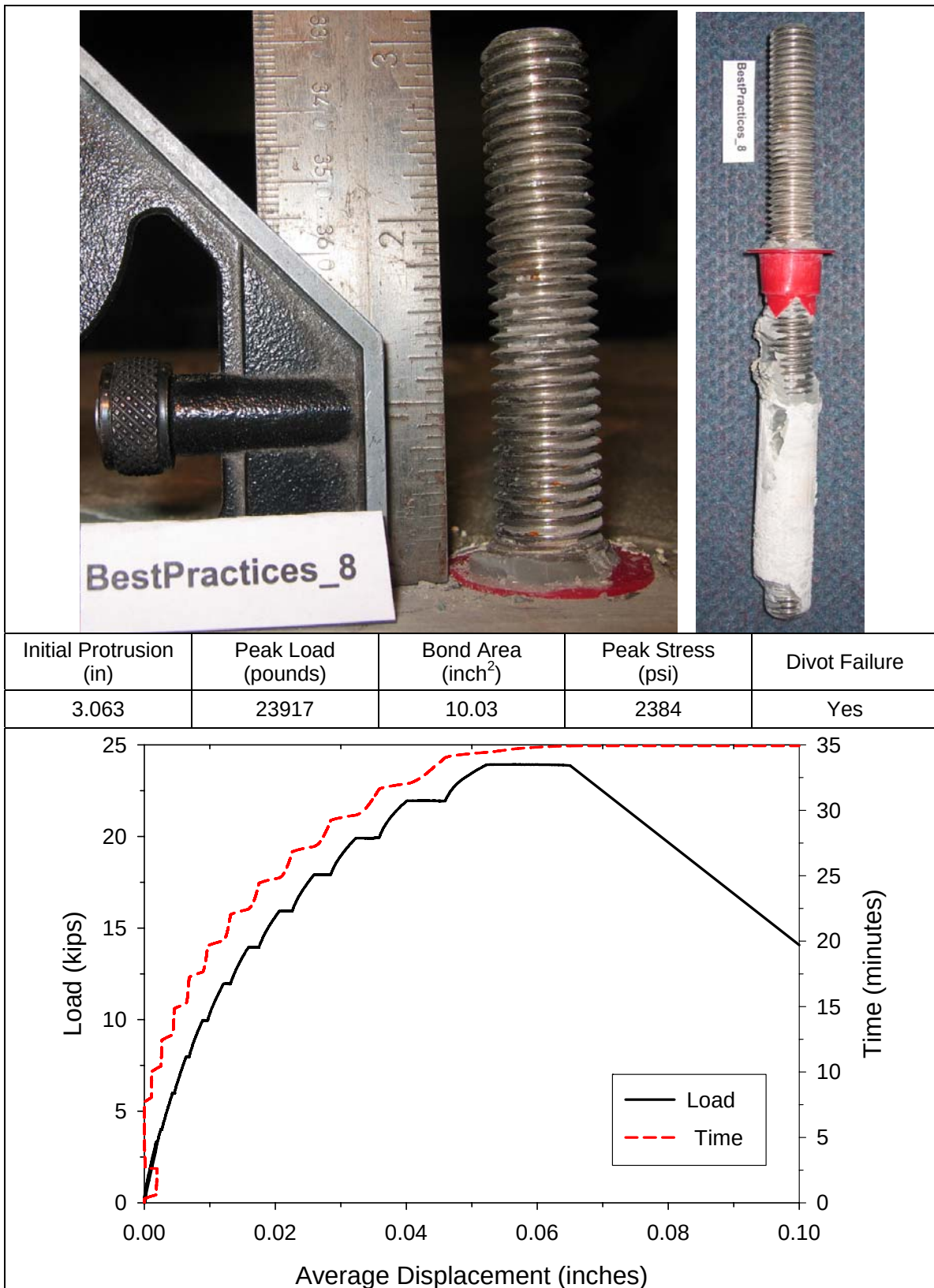


Table 57 Cored Hole – Best Practices 9

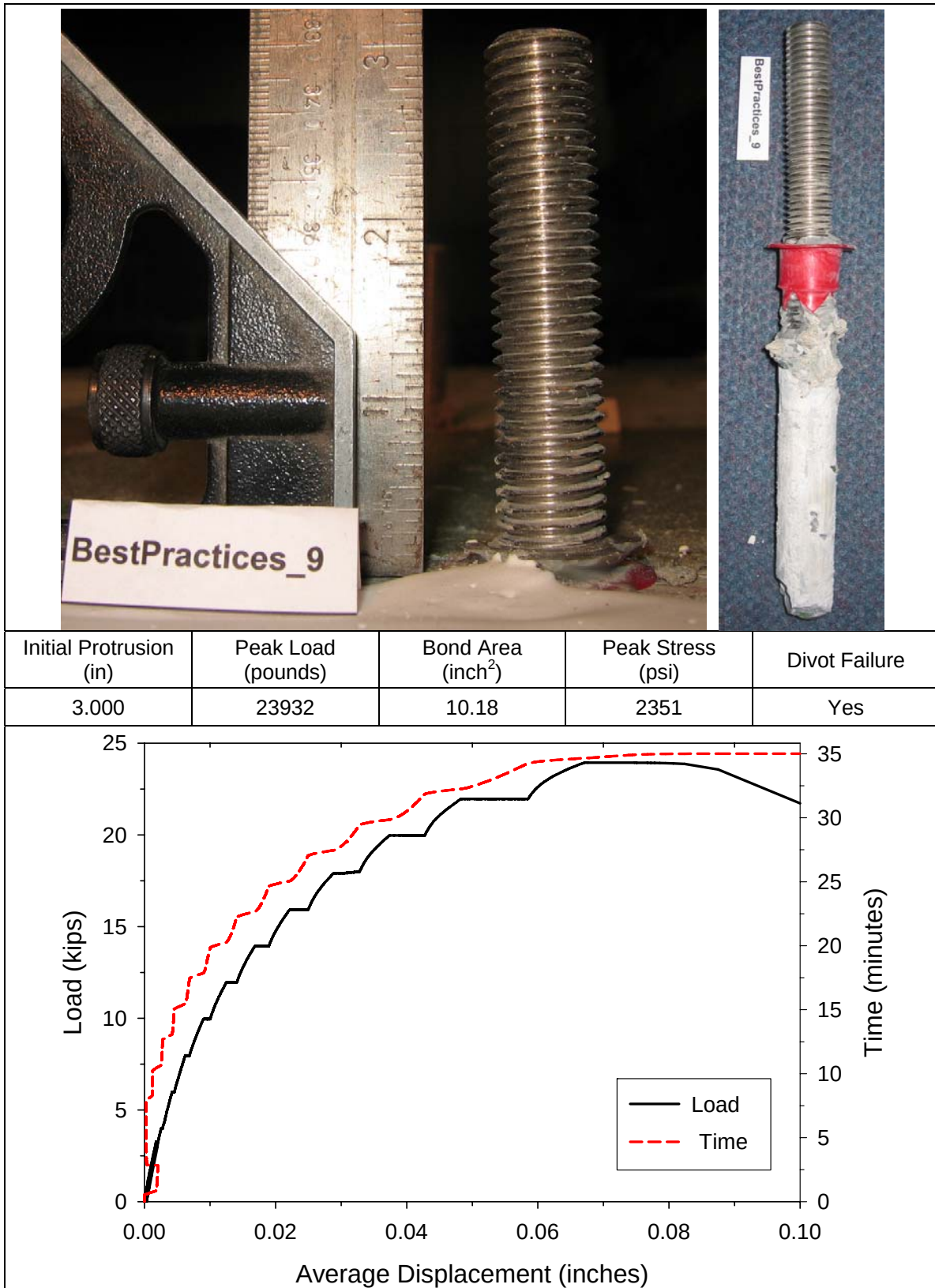


Table 58 Cored Hole – Best Practices 10

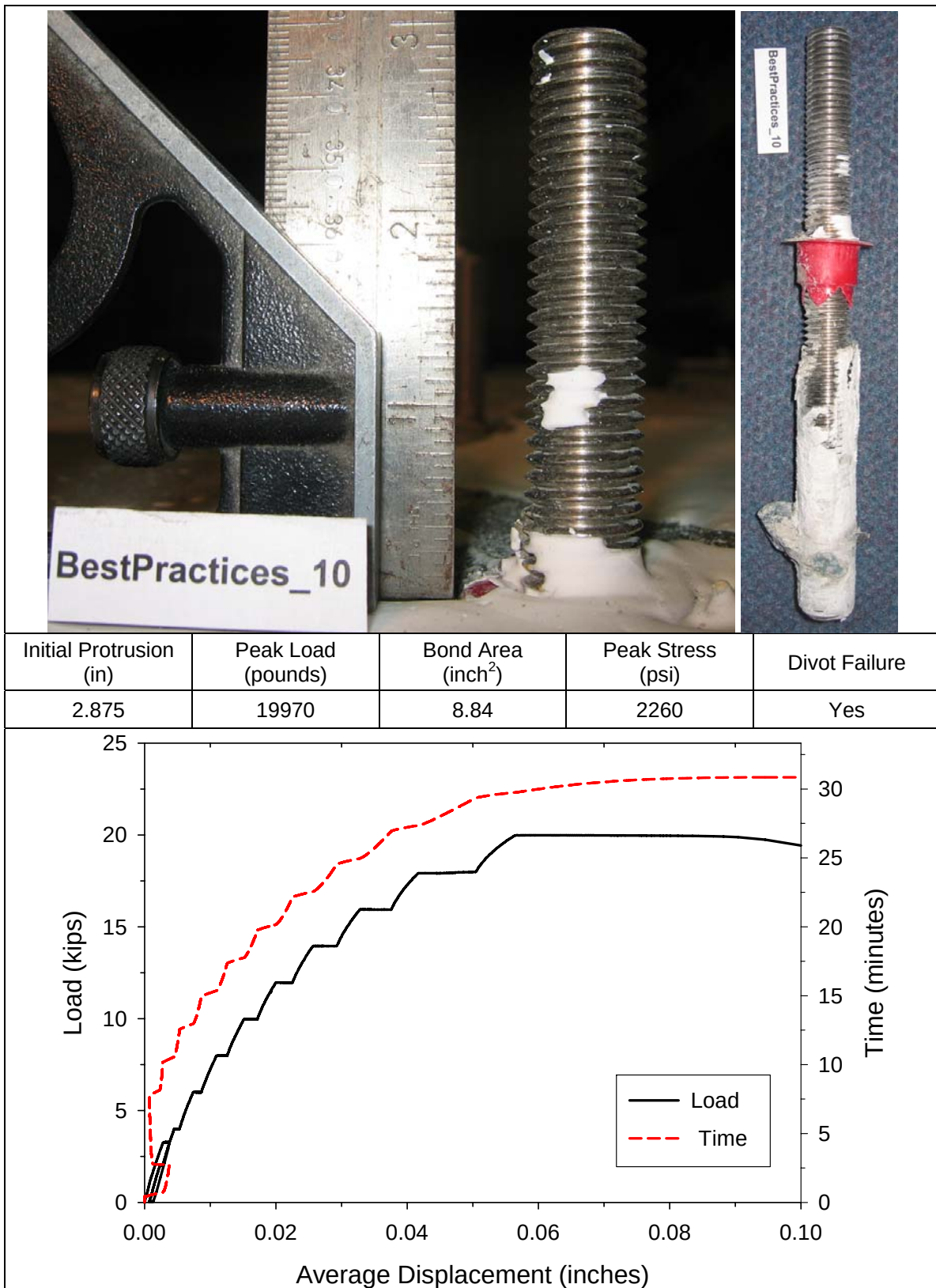


Table 59 Standard Set 1

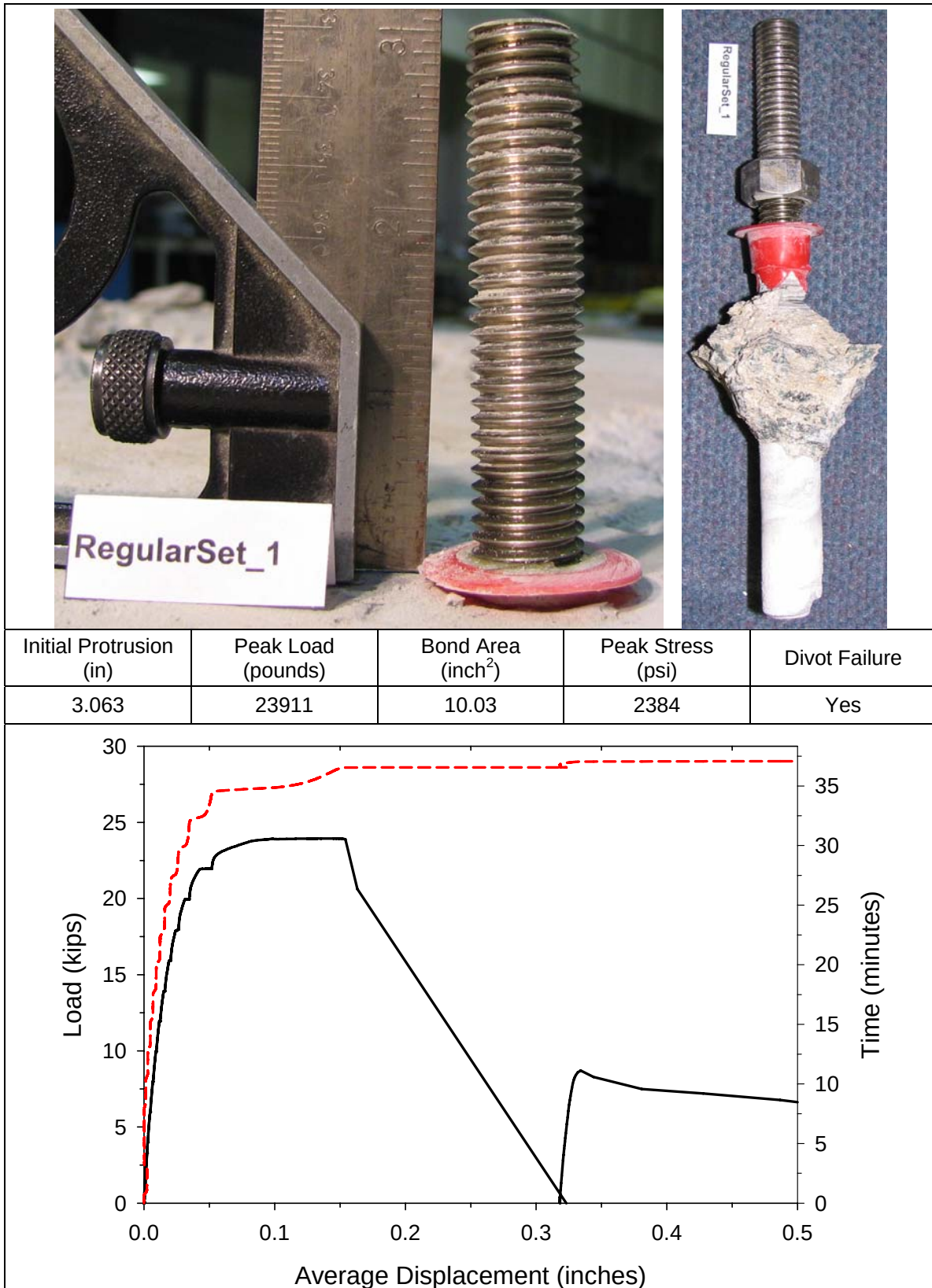


Table 60 Standard Set 2

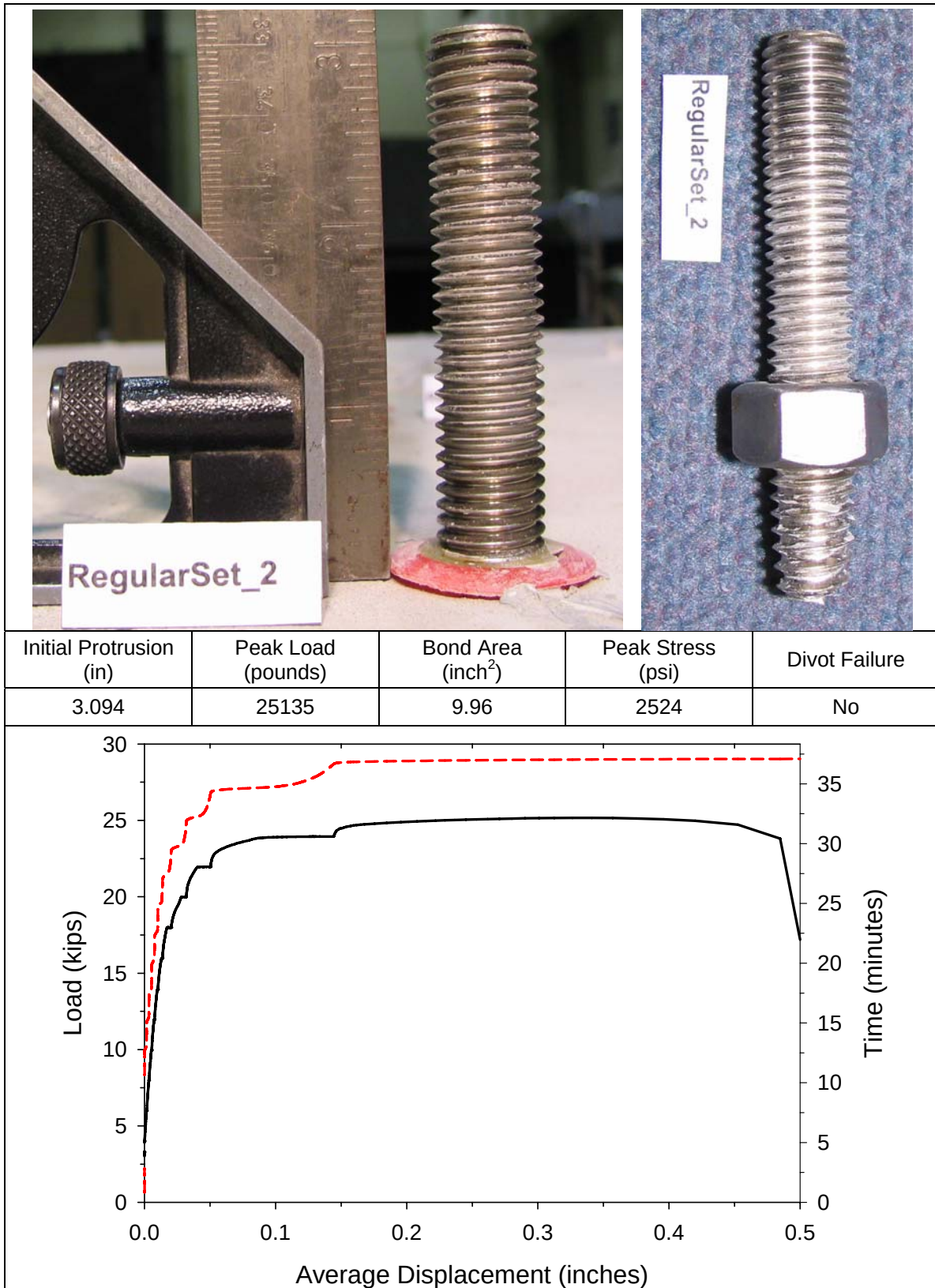


Table 61 Standard Set 3

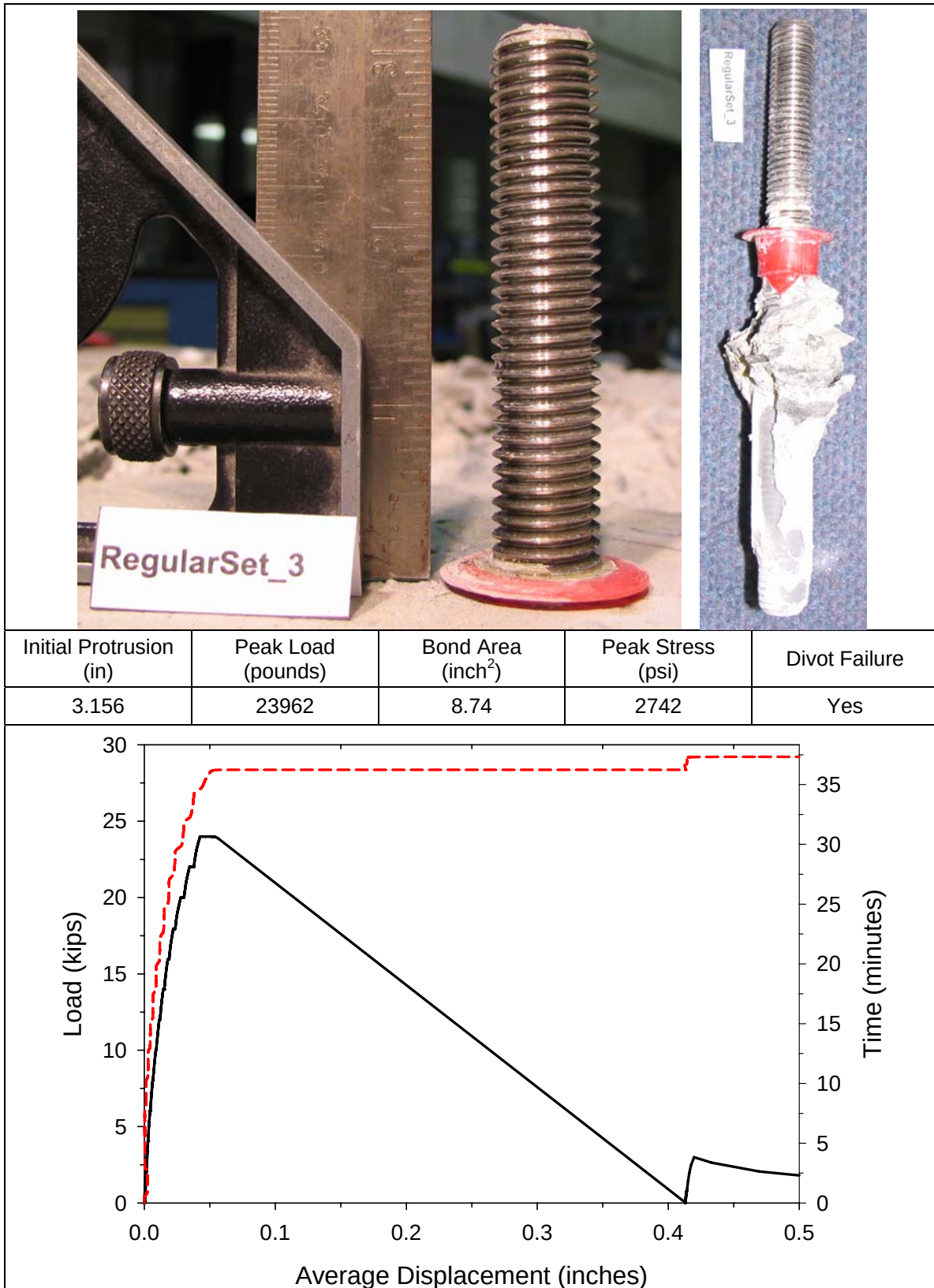


Table 62 Standard Set 4

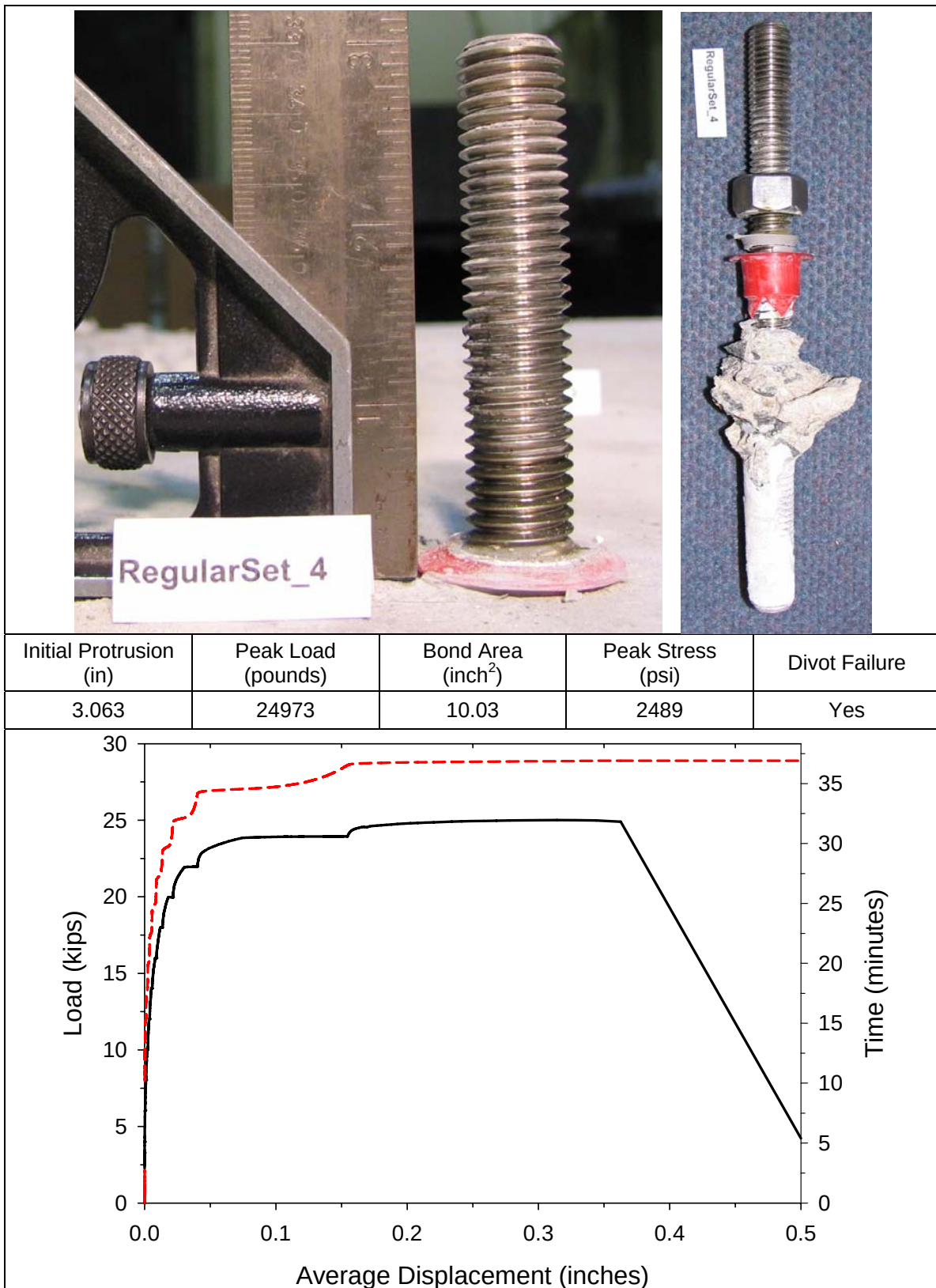


Table 63 Standard Set 5

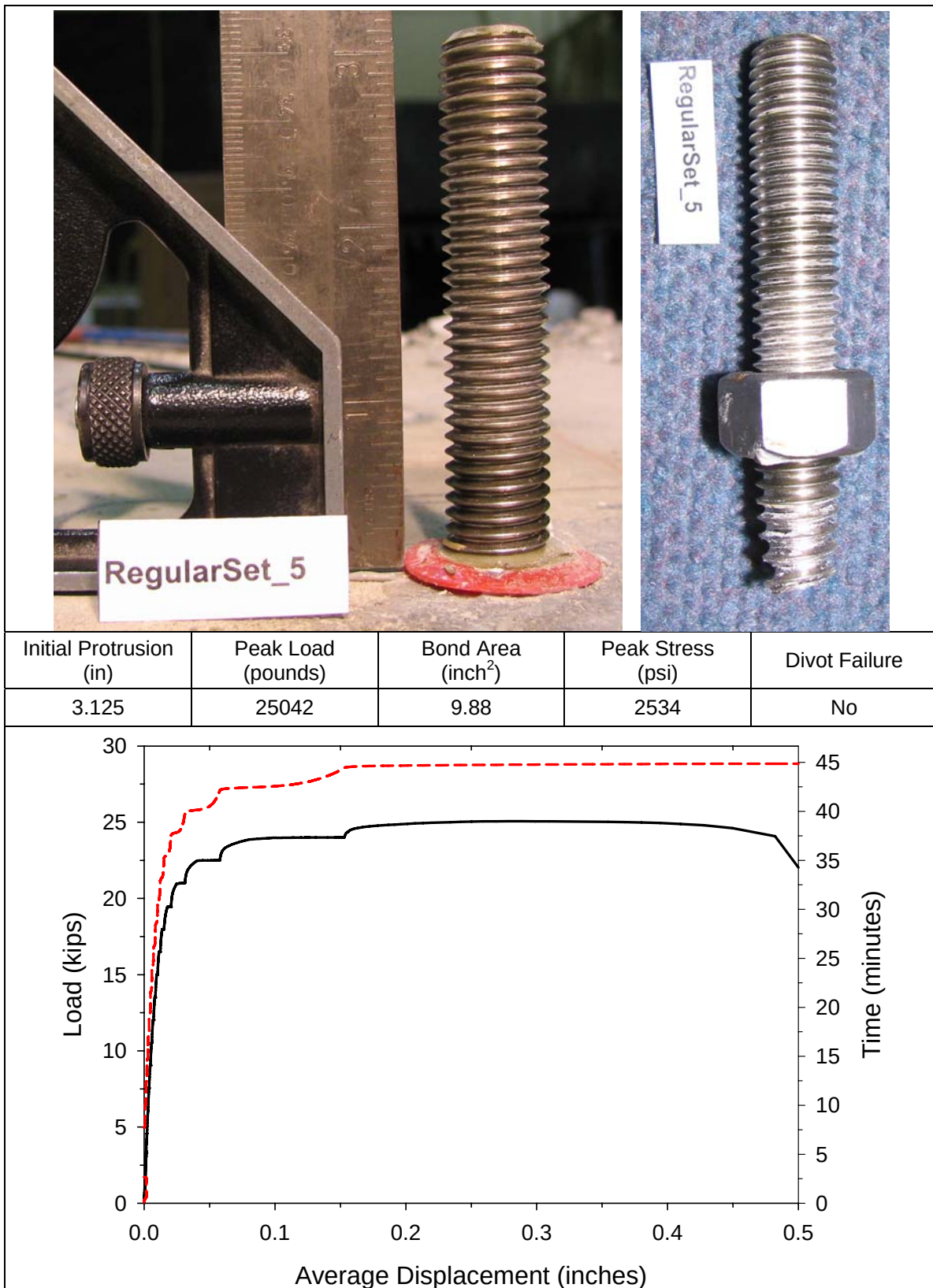


Table 64 Screen Tube Installations 1

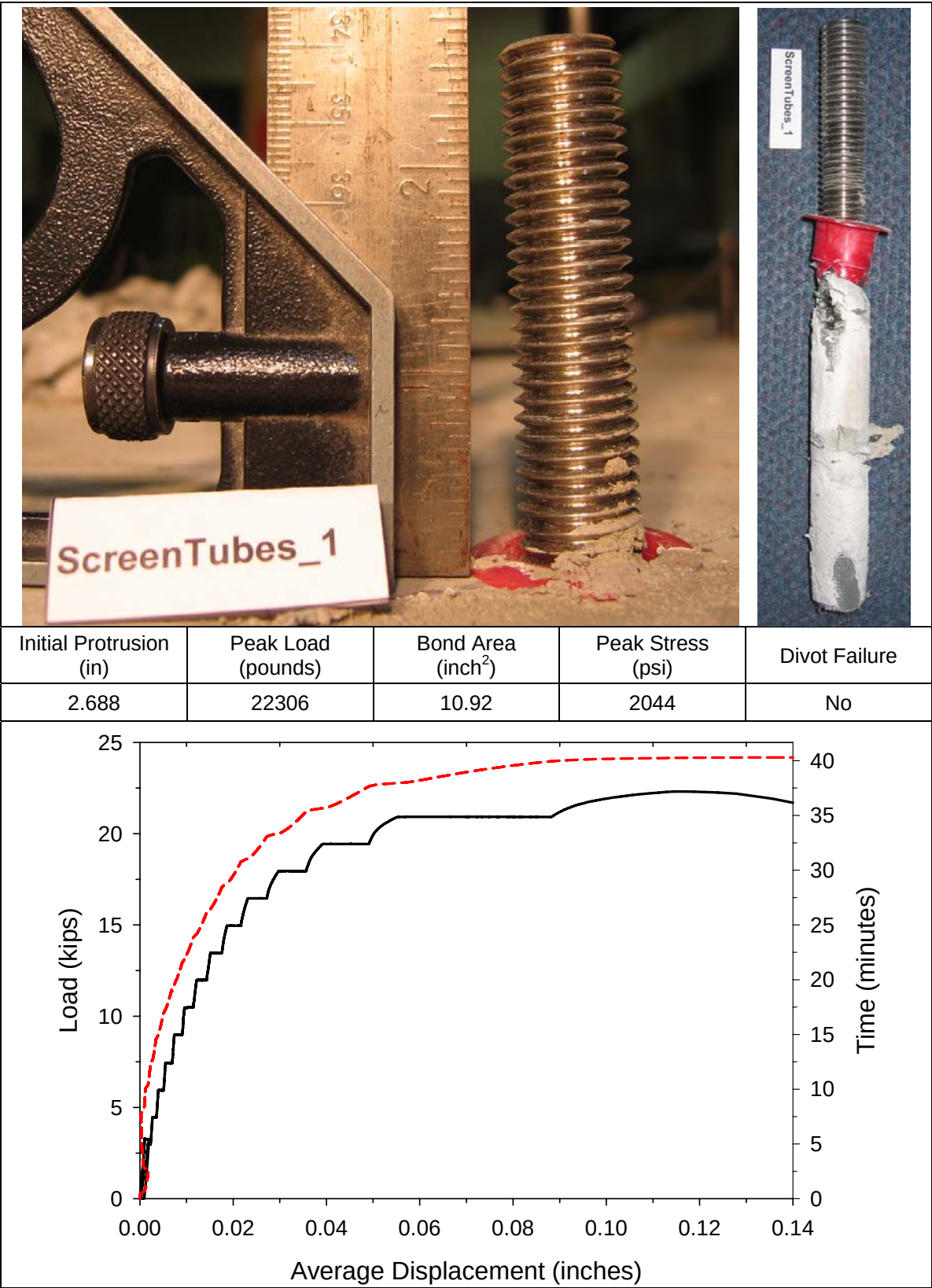


Table 65 Screen Tube Installations 2

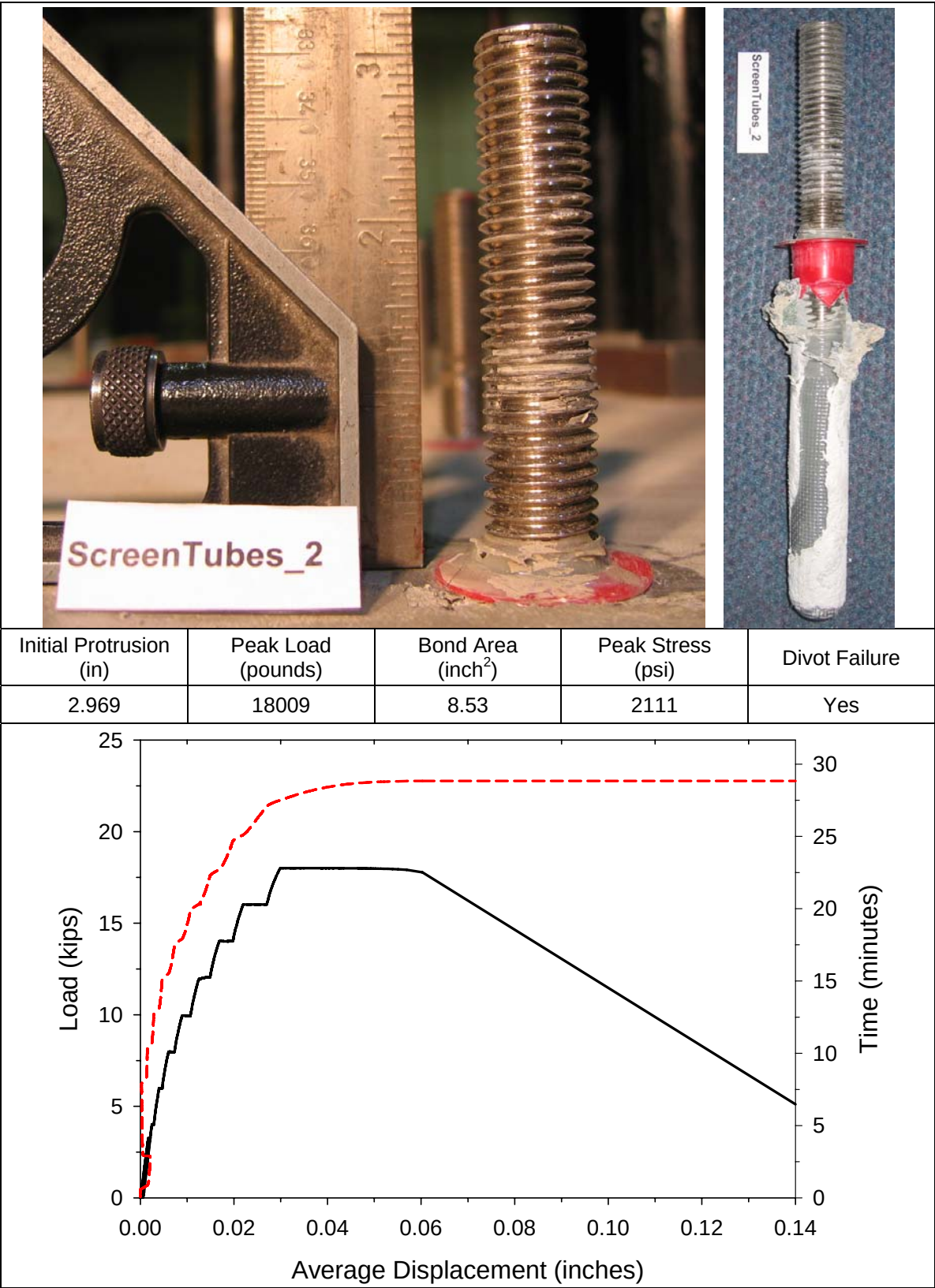


Table 66 Screen Tube Installations 3

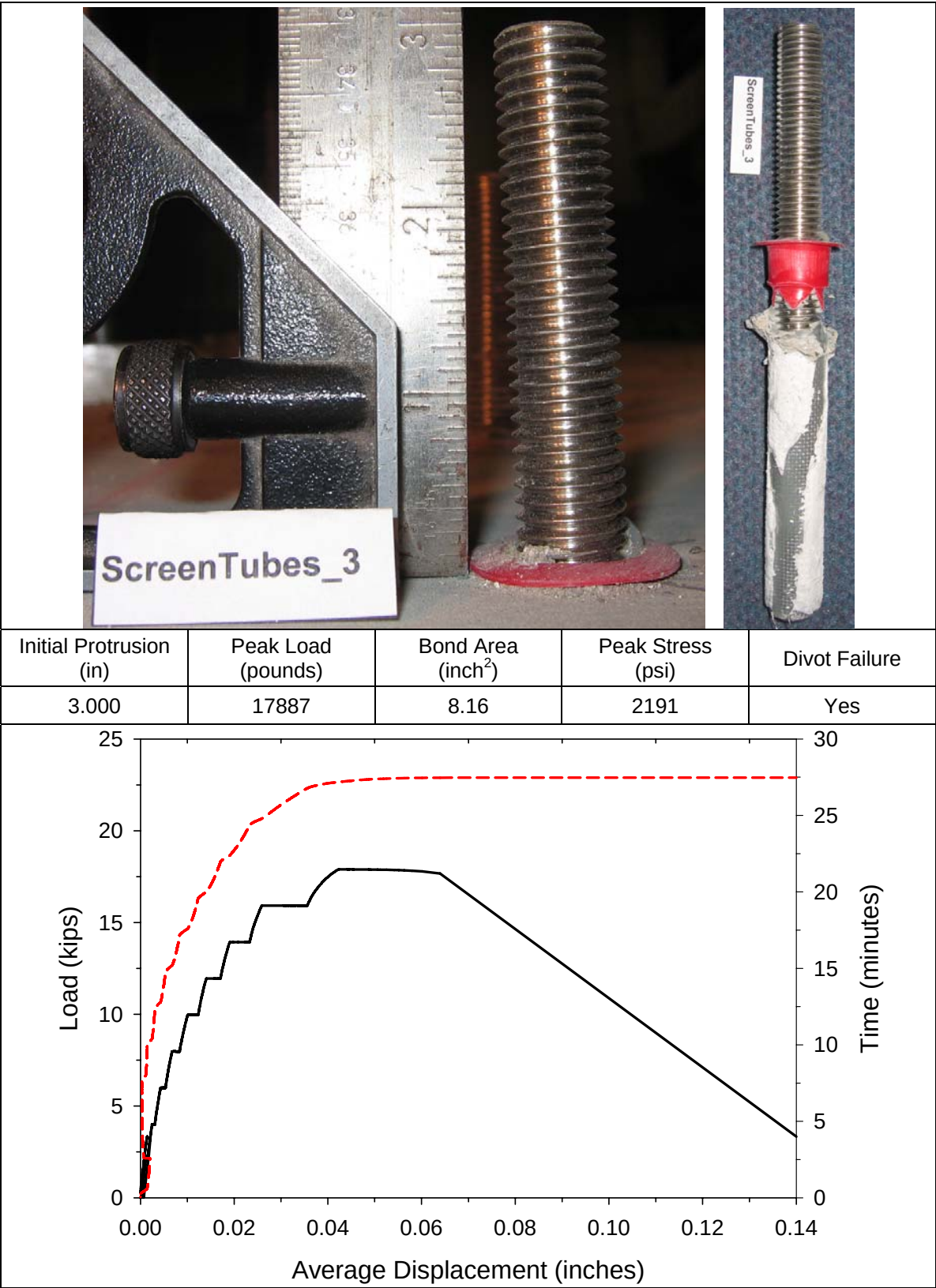


Table 67 Screen Tube Installations 4

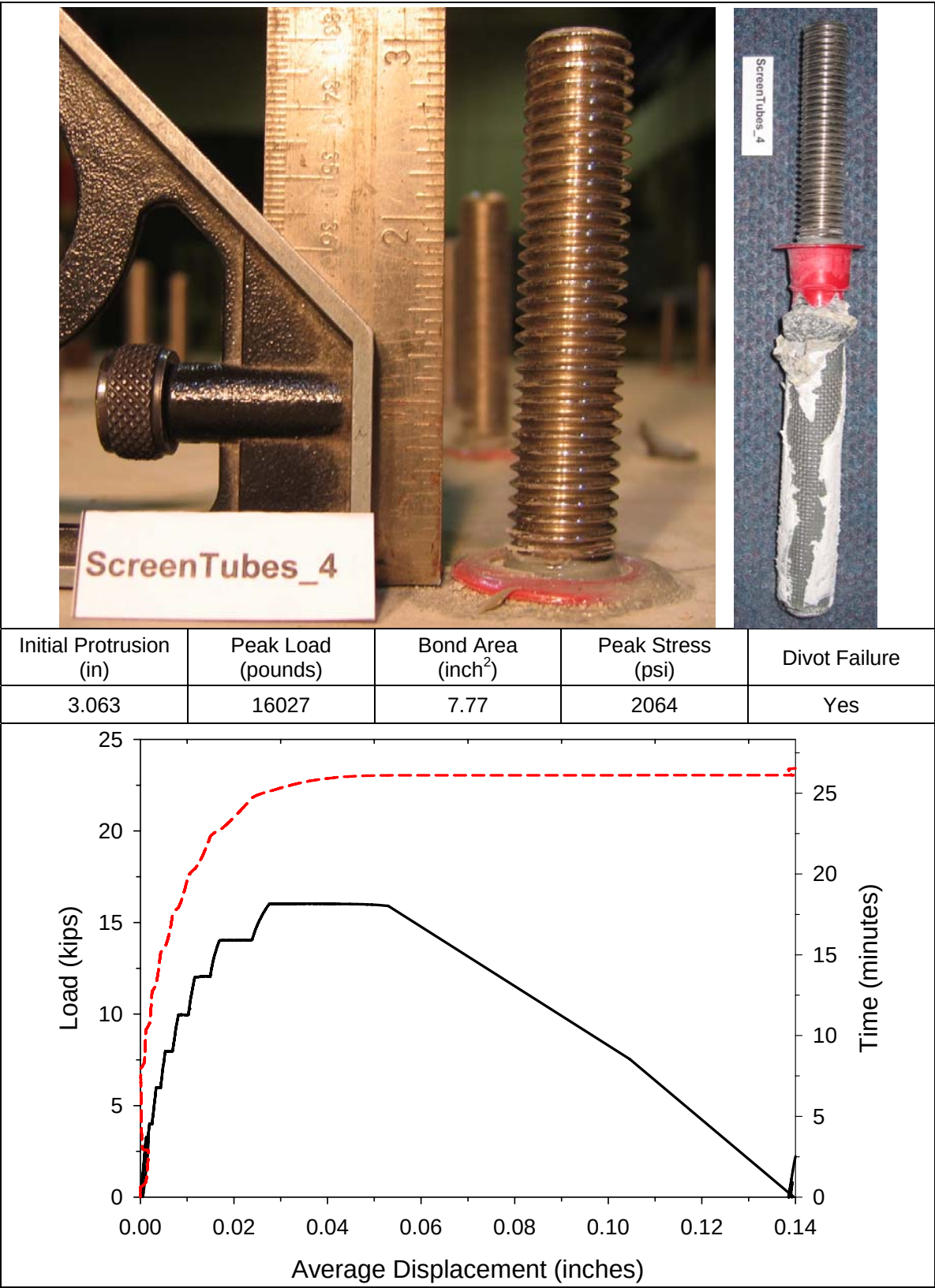


Table 68 Dirty Hole 1

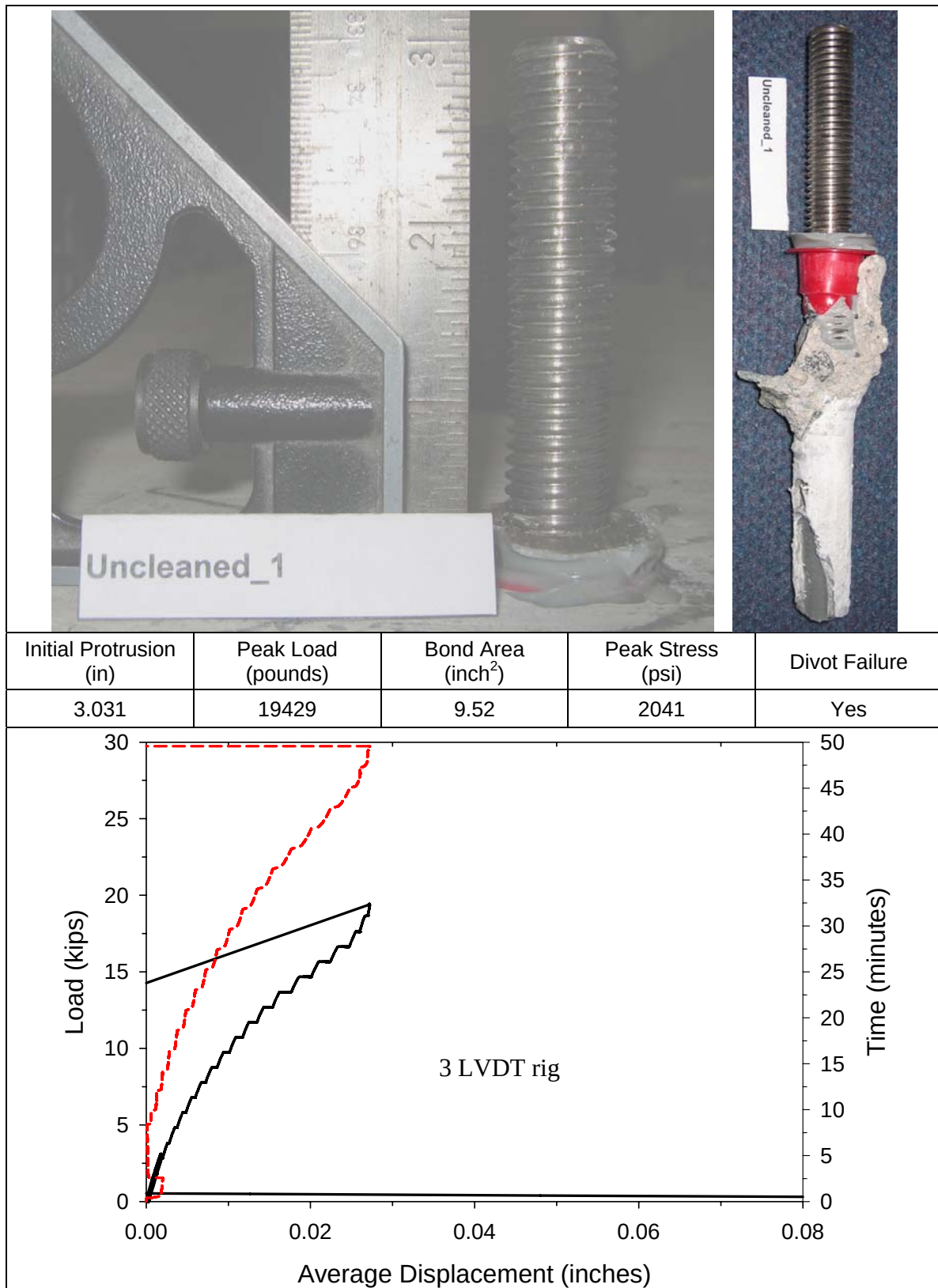


Table 69 Dirty Hole 2

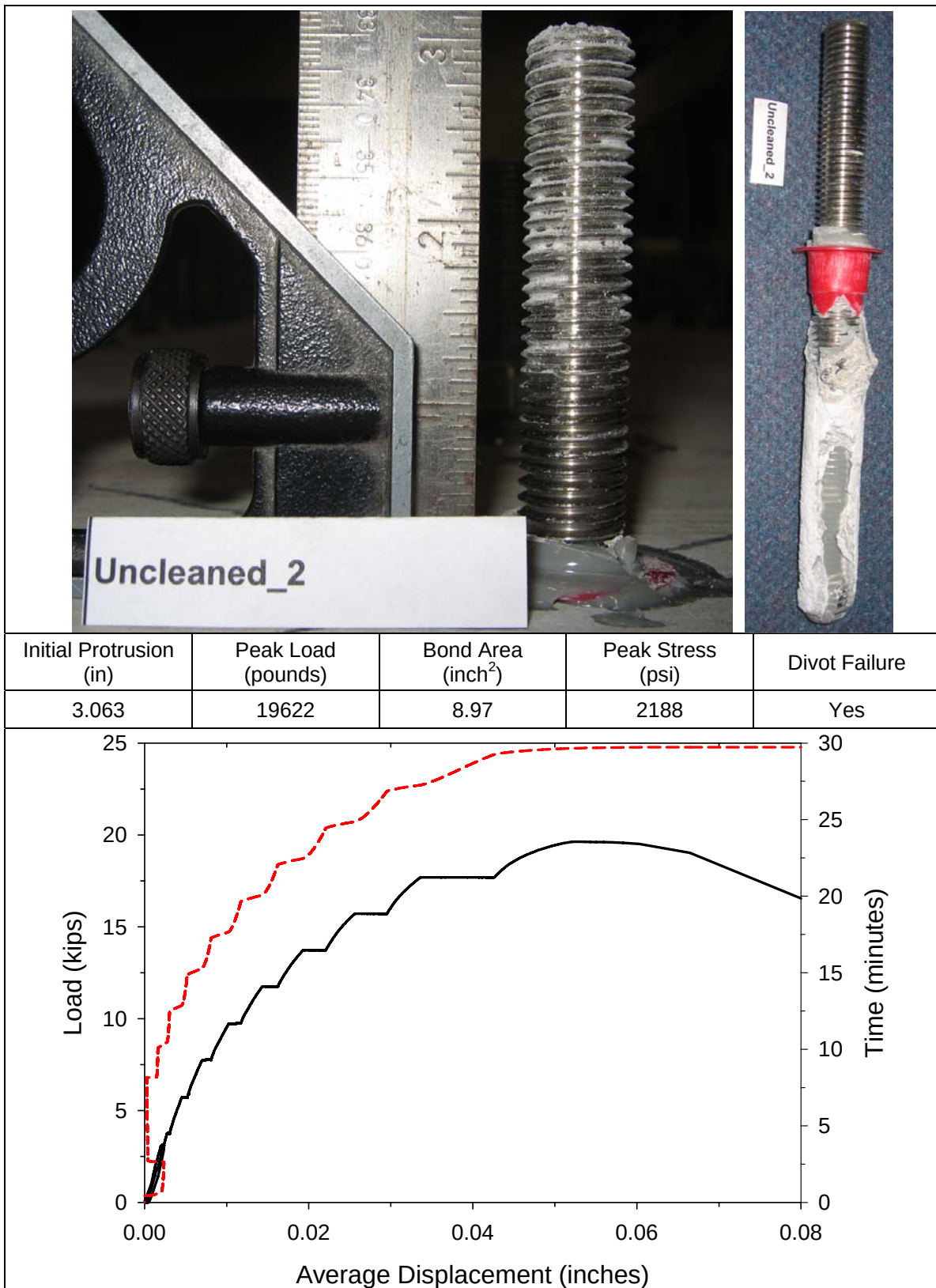


Table 70 Dirty Hole 3

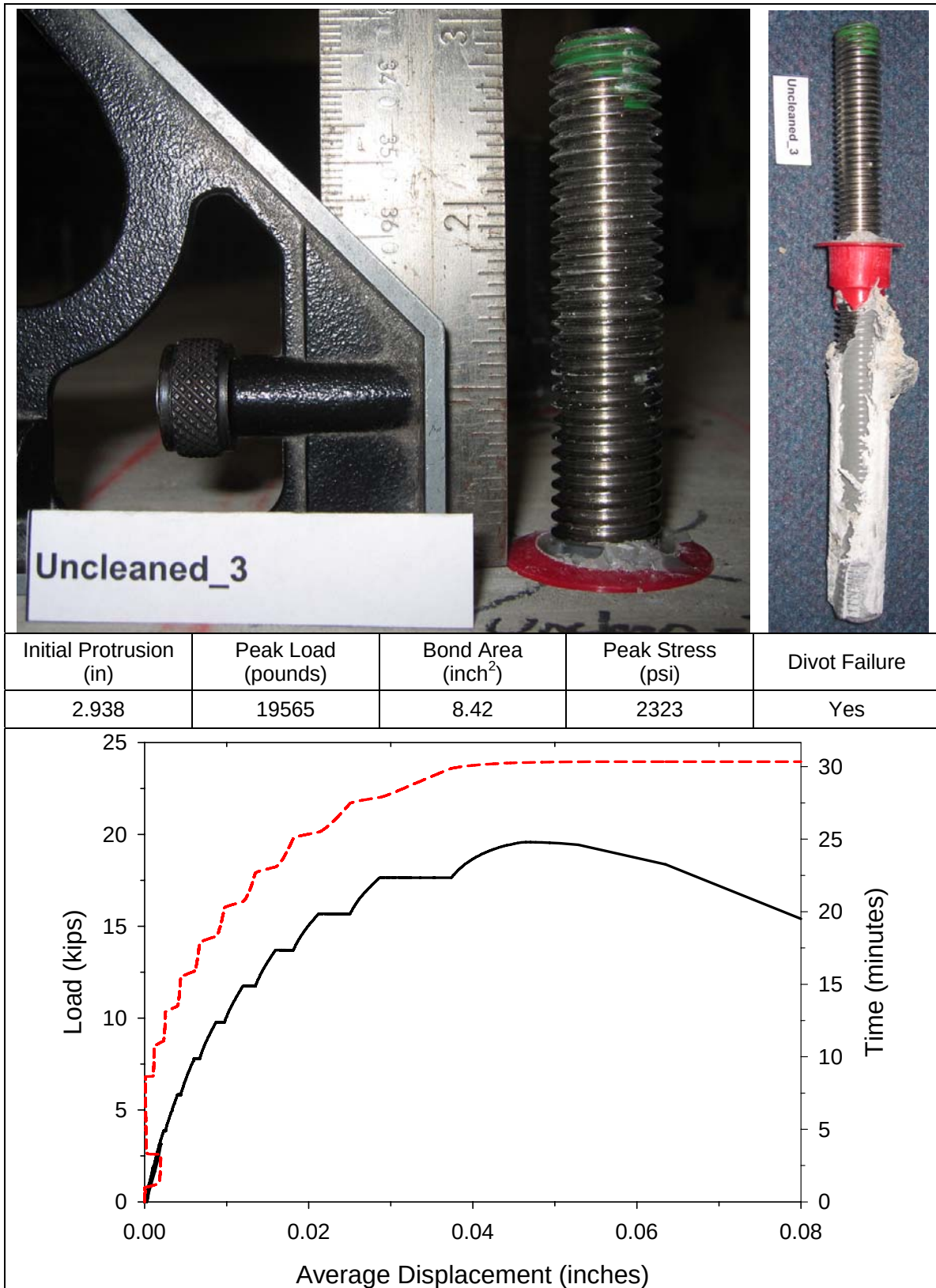


Table 71 Dirty Hole 4

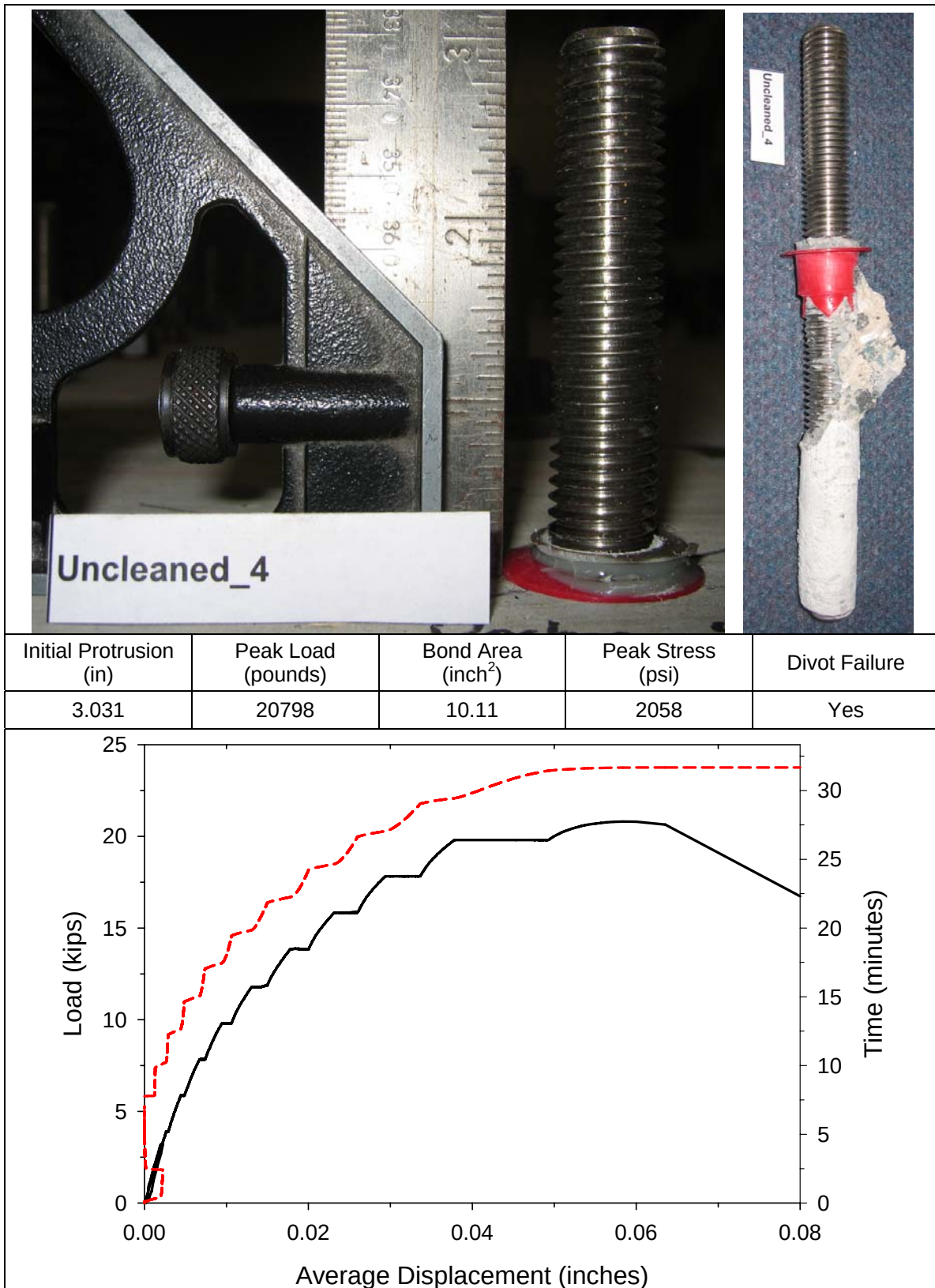


Table 72 Dirty Hole 5

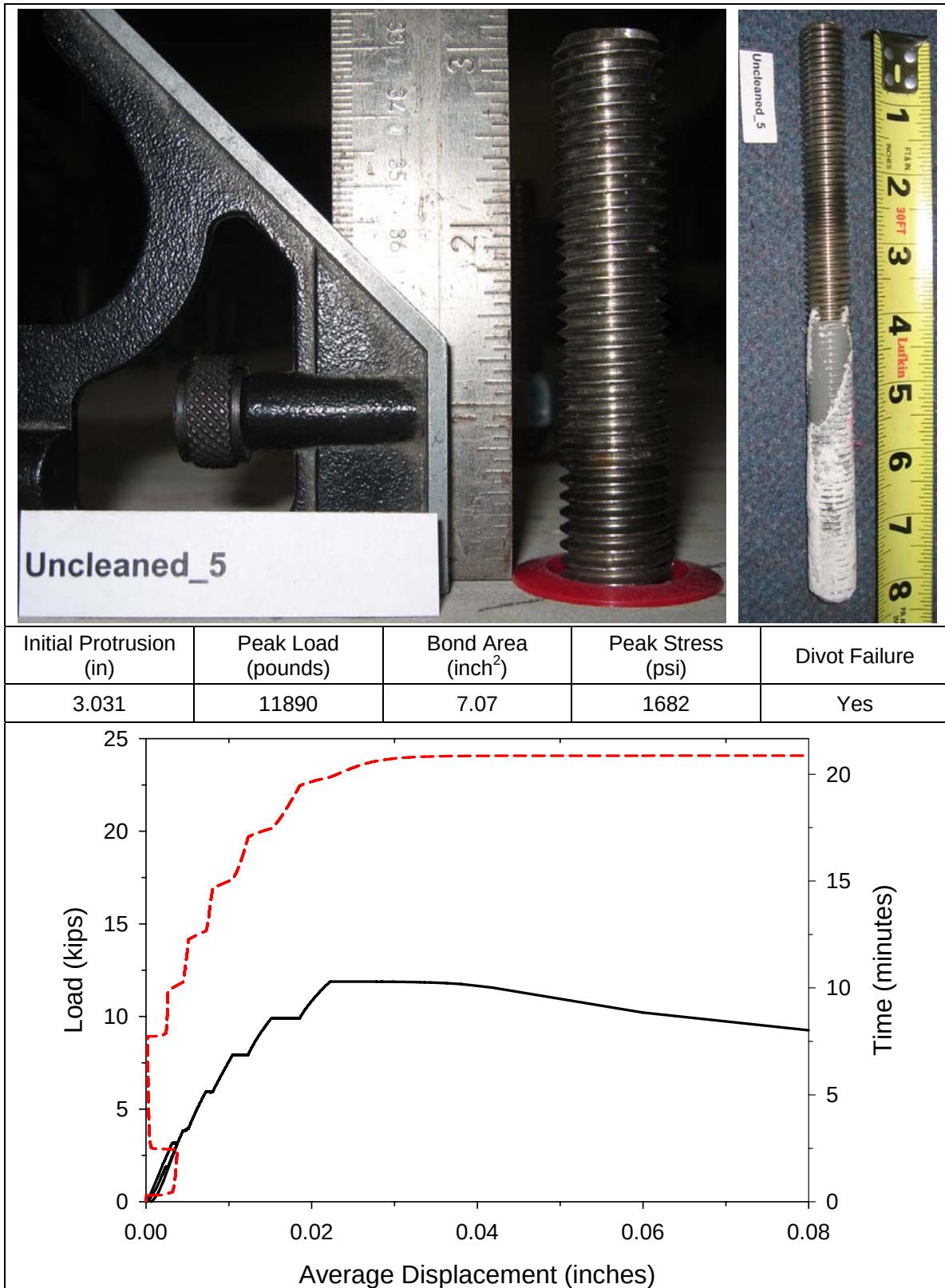


Table 73 Air-Only Hole 1

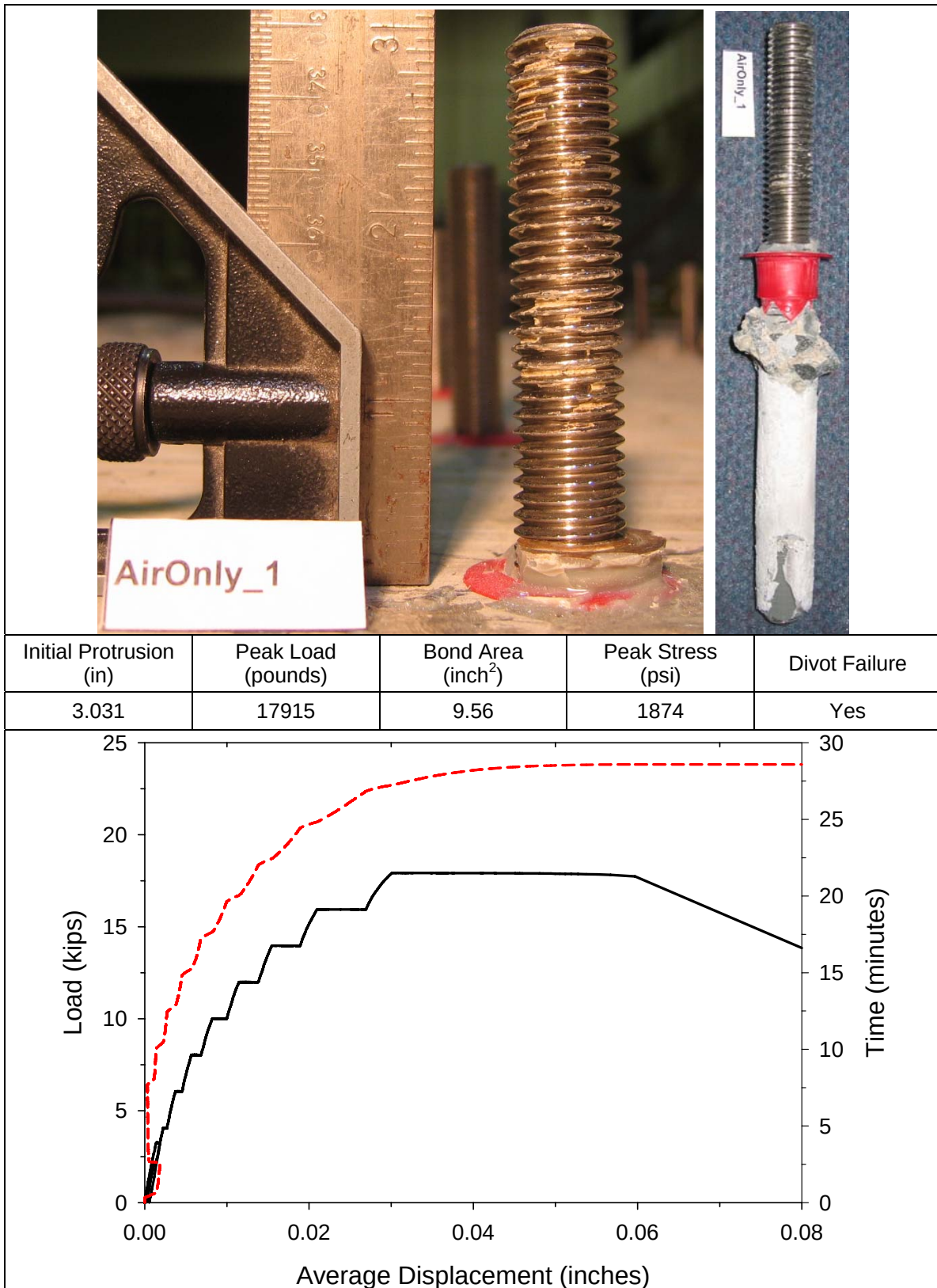


Table 74 Air-Only Hole 2

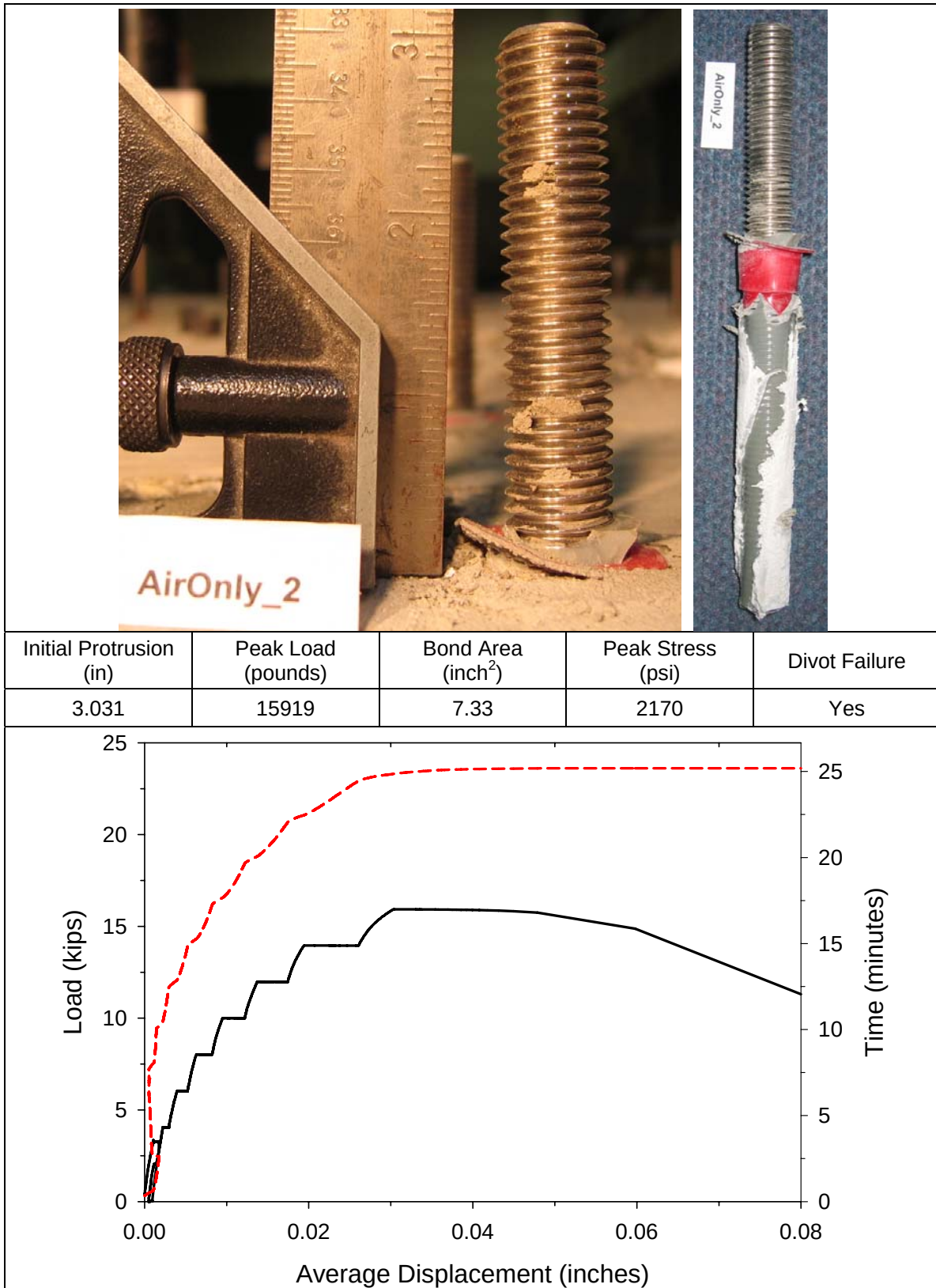


Table 75 Air-Only Hole 3

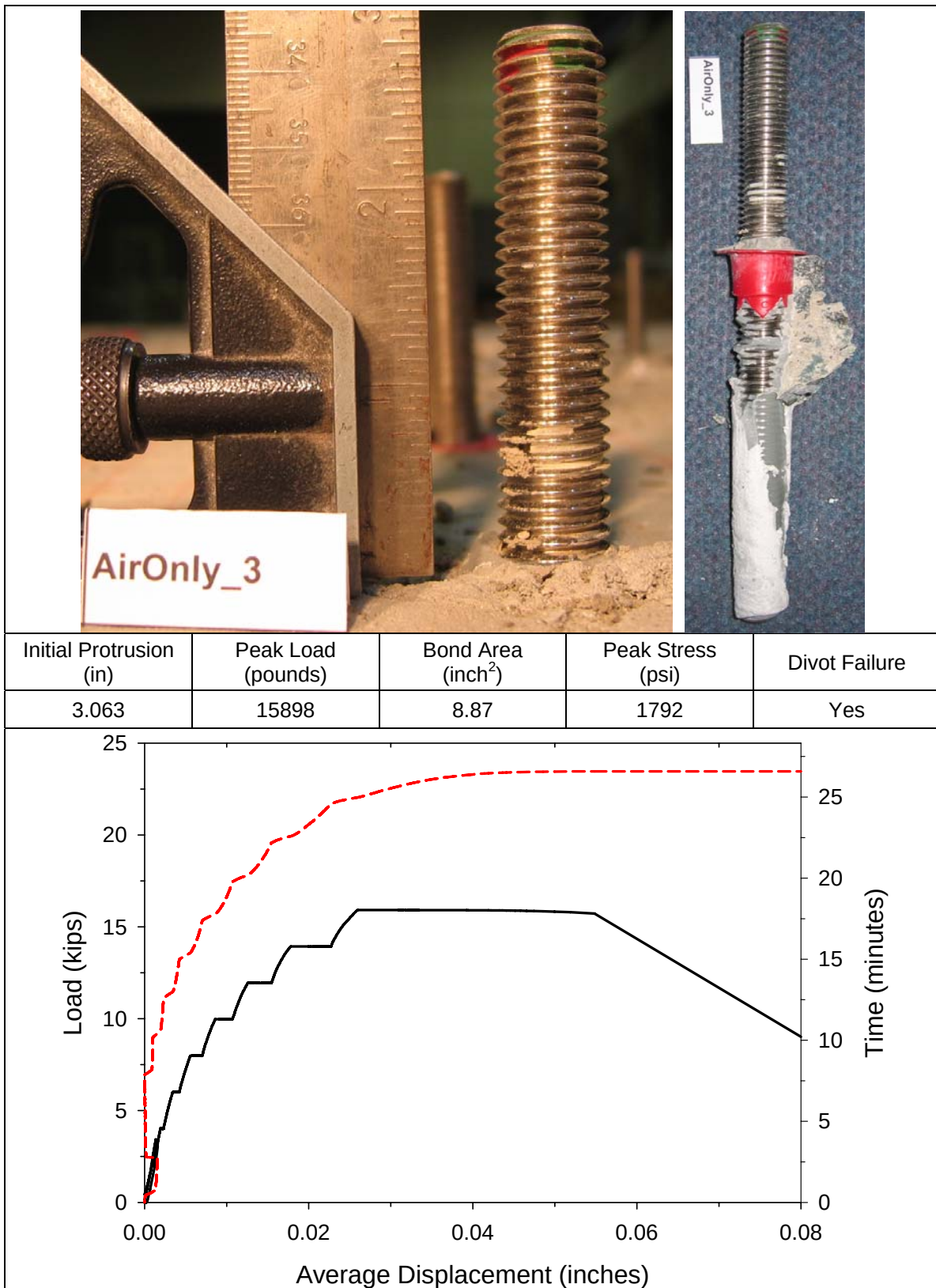


Table 76 Air-Only Hole 4

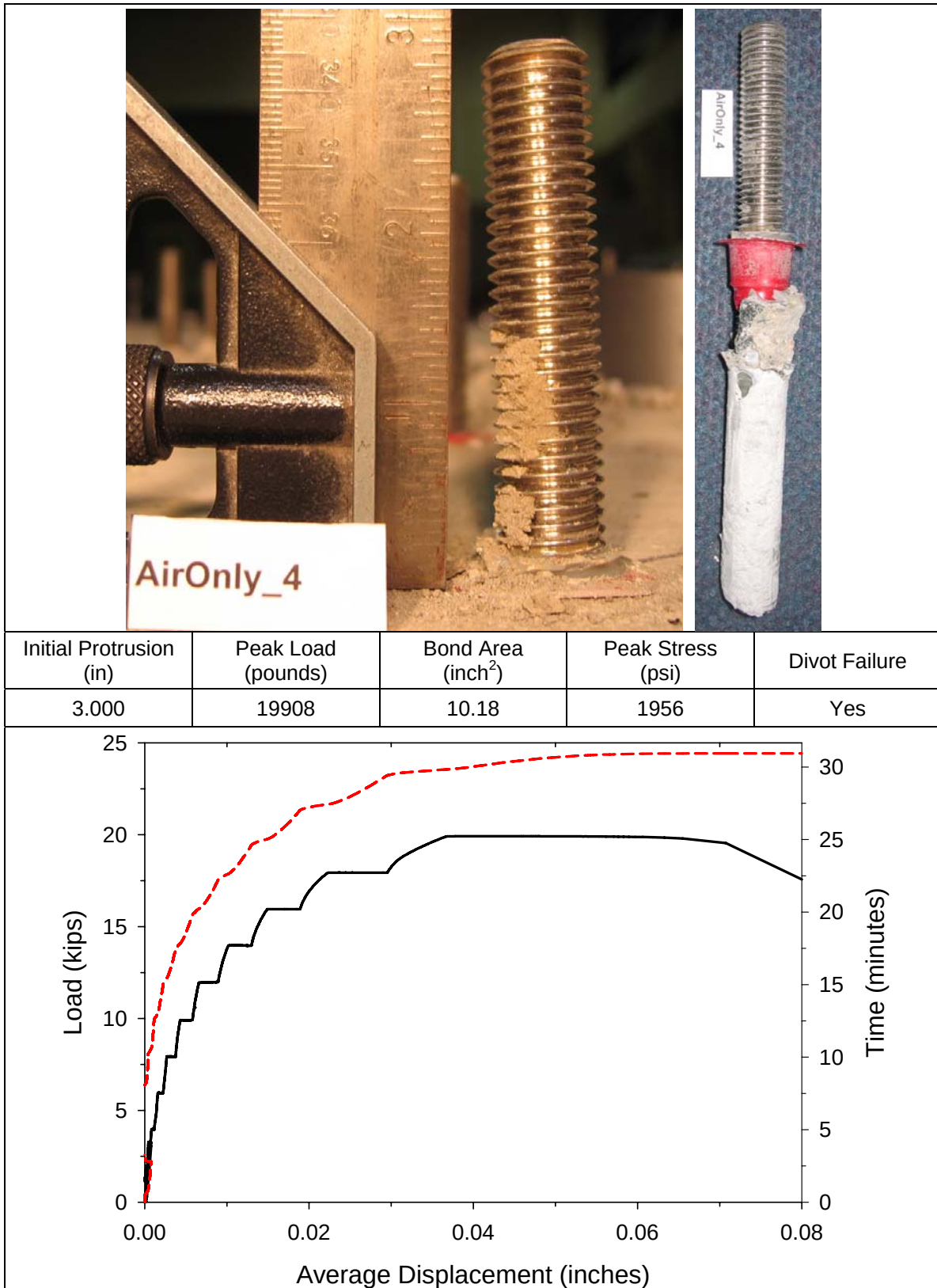


Table 77 Air-Only Hole 5

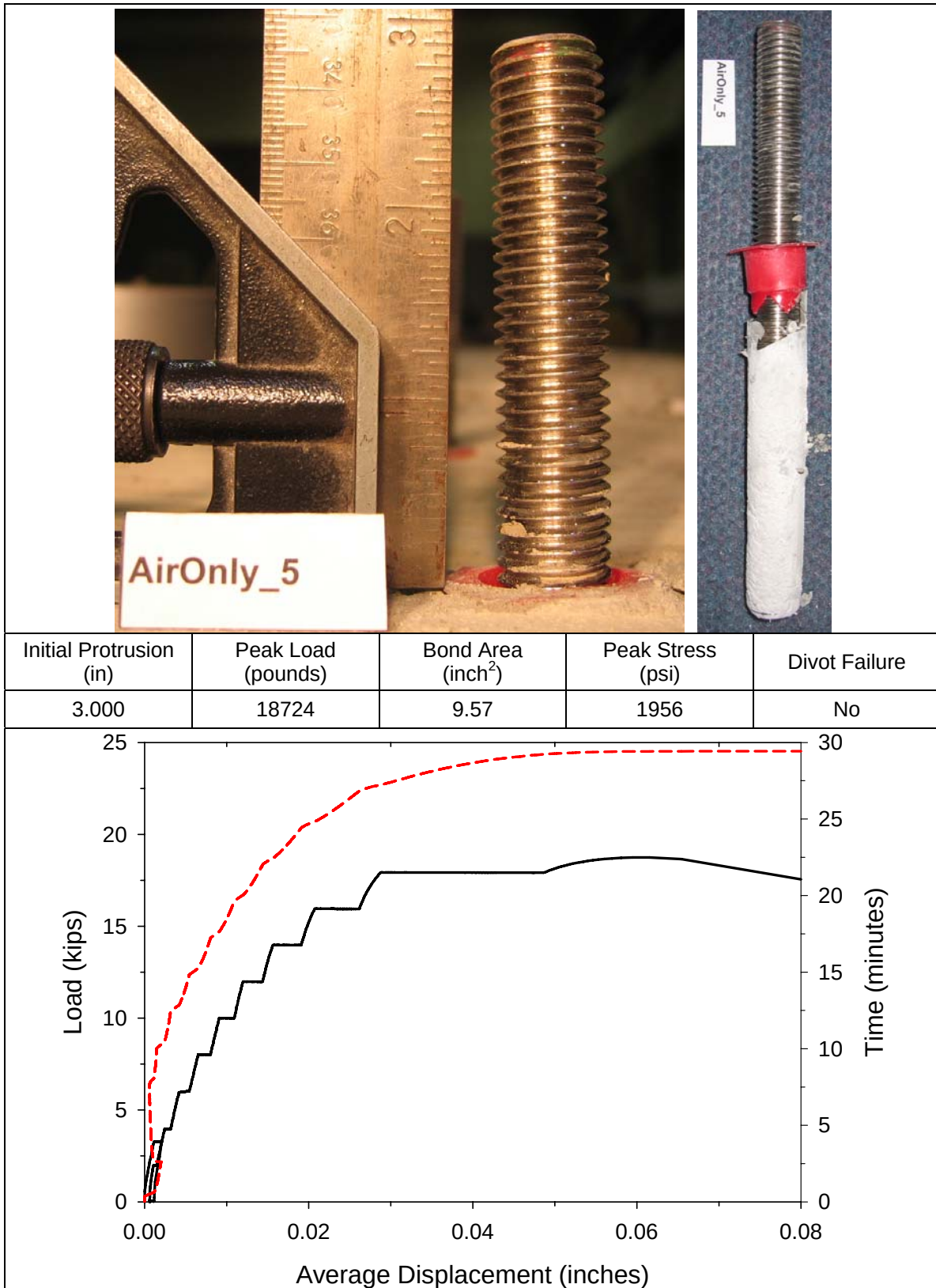


Table 78 Brush-Only Hole 1

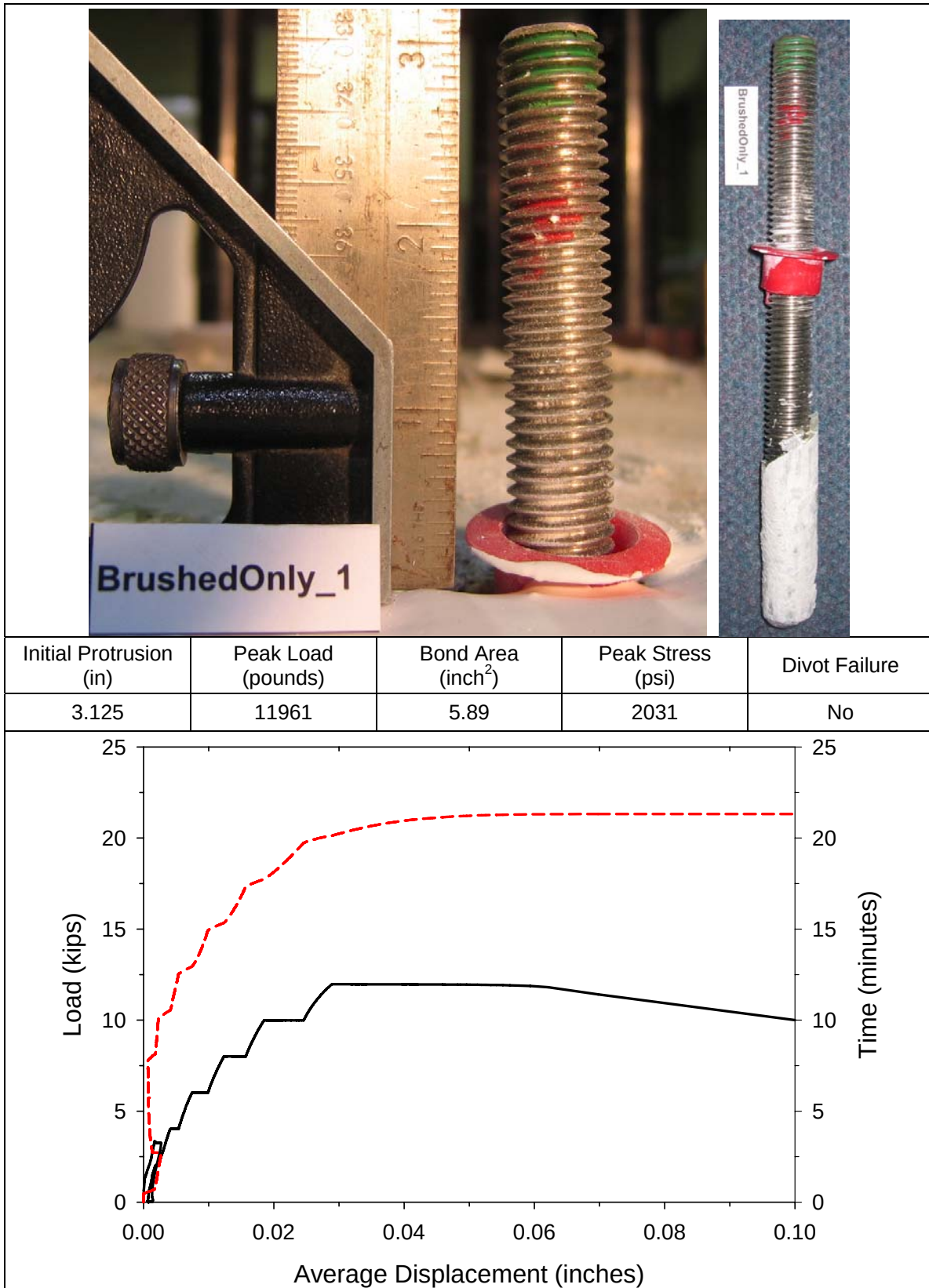


Table 79 Brush-Only Hole 2

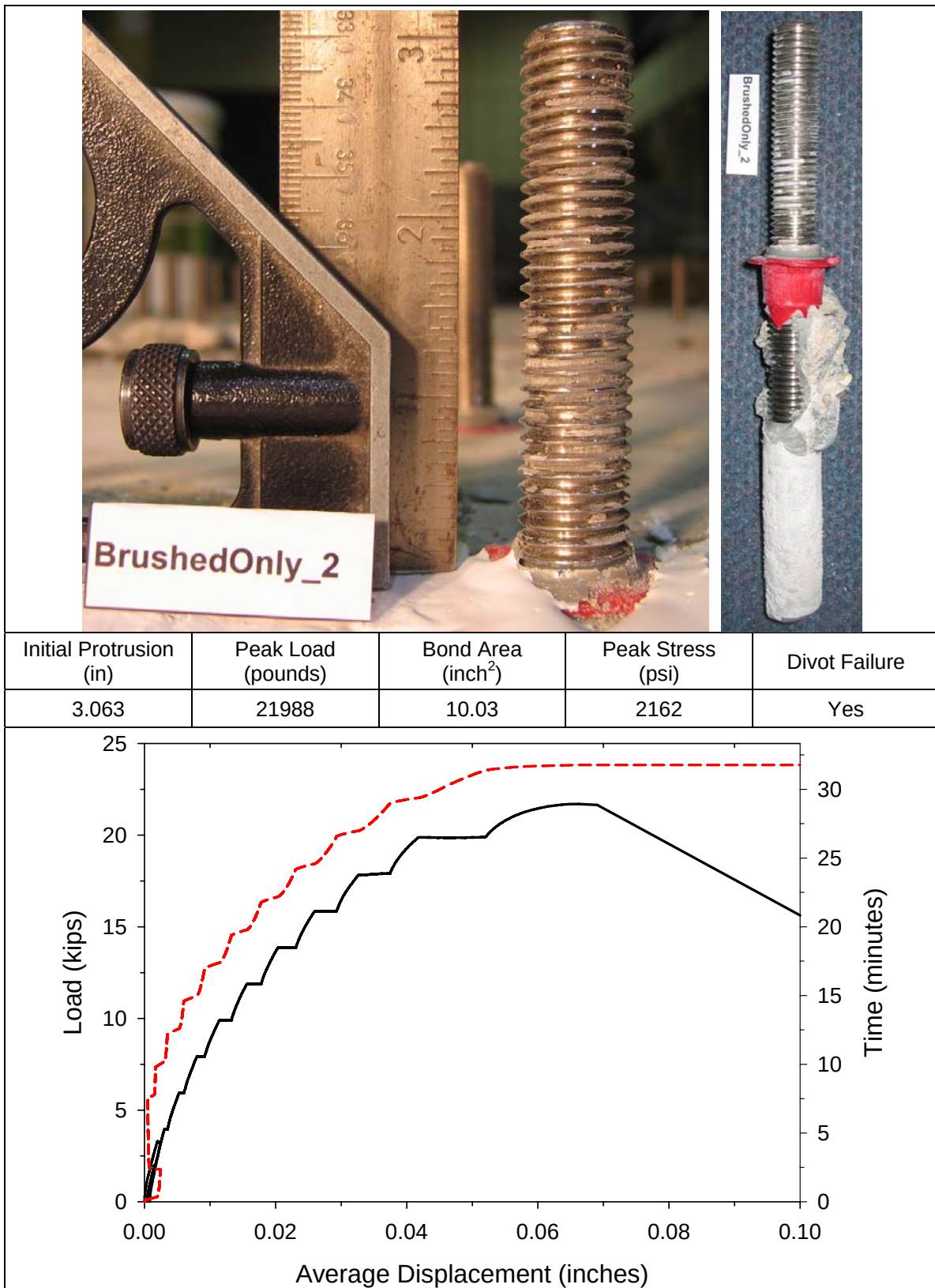


Table 80 Brush-Only Hole 3

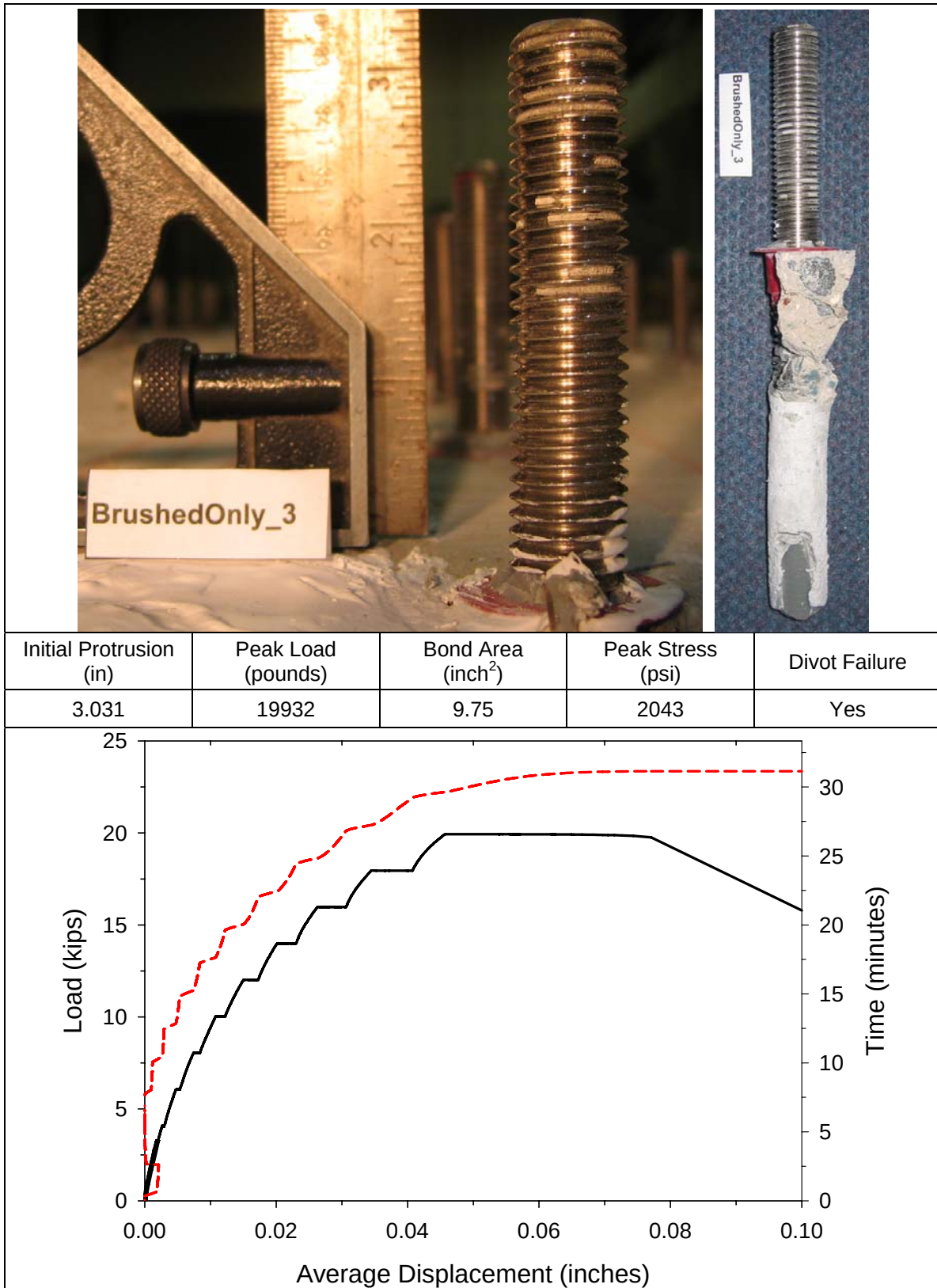


Table 81 Brush-Only Hole 4

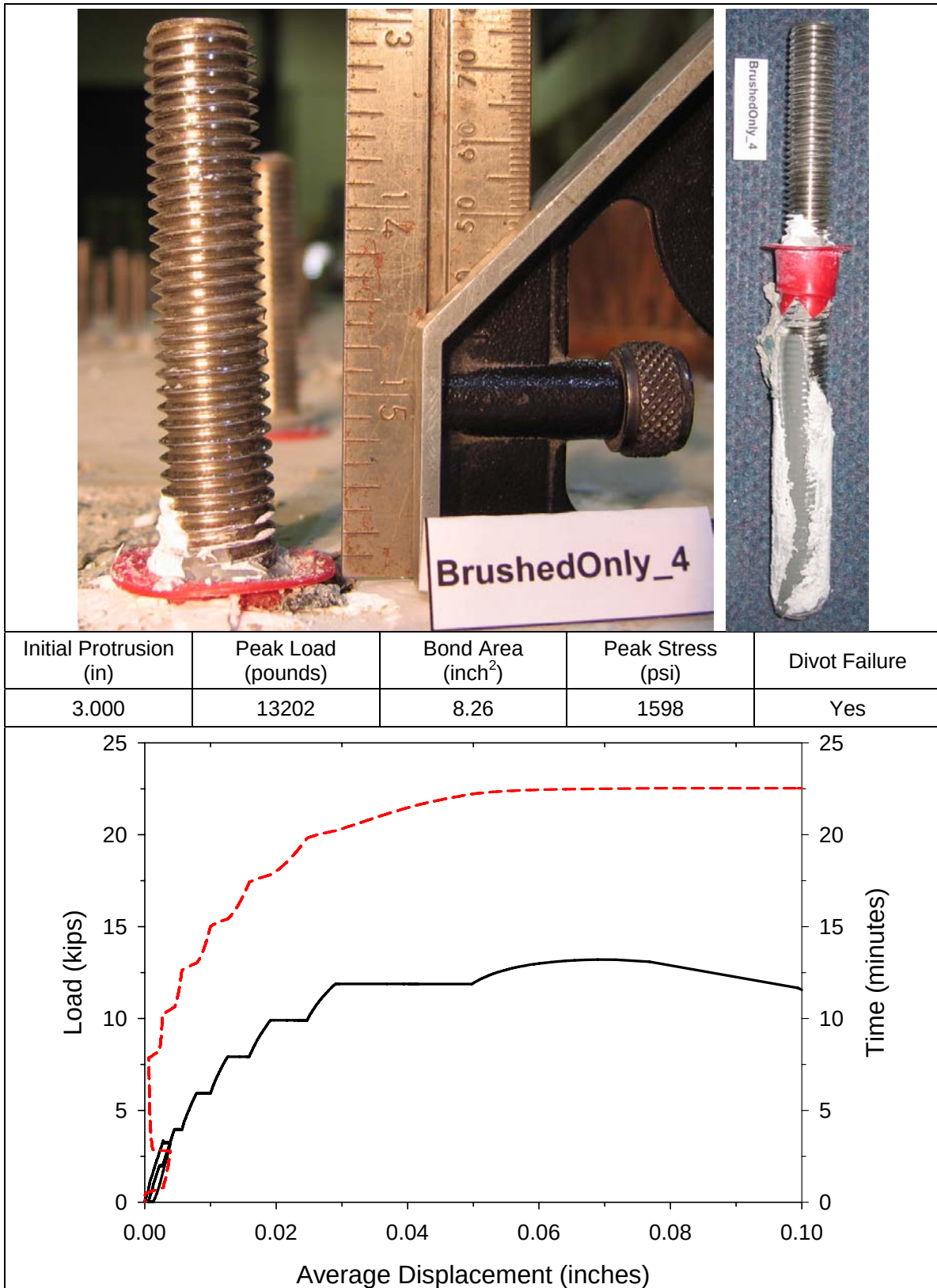


Table 82 Brush-Only Hole 5

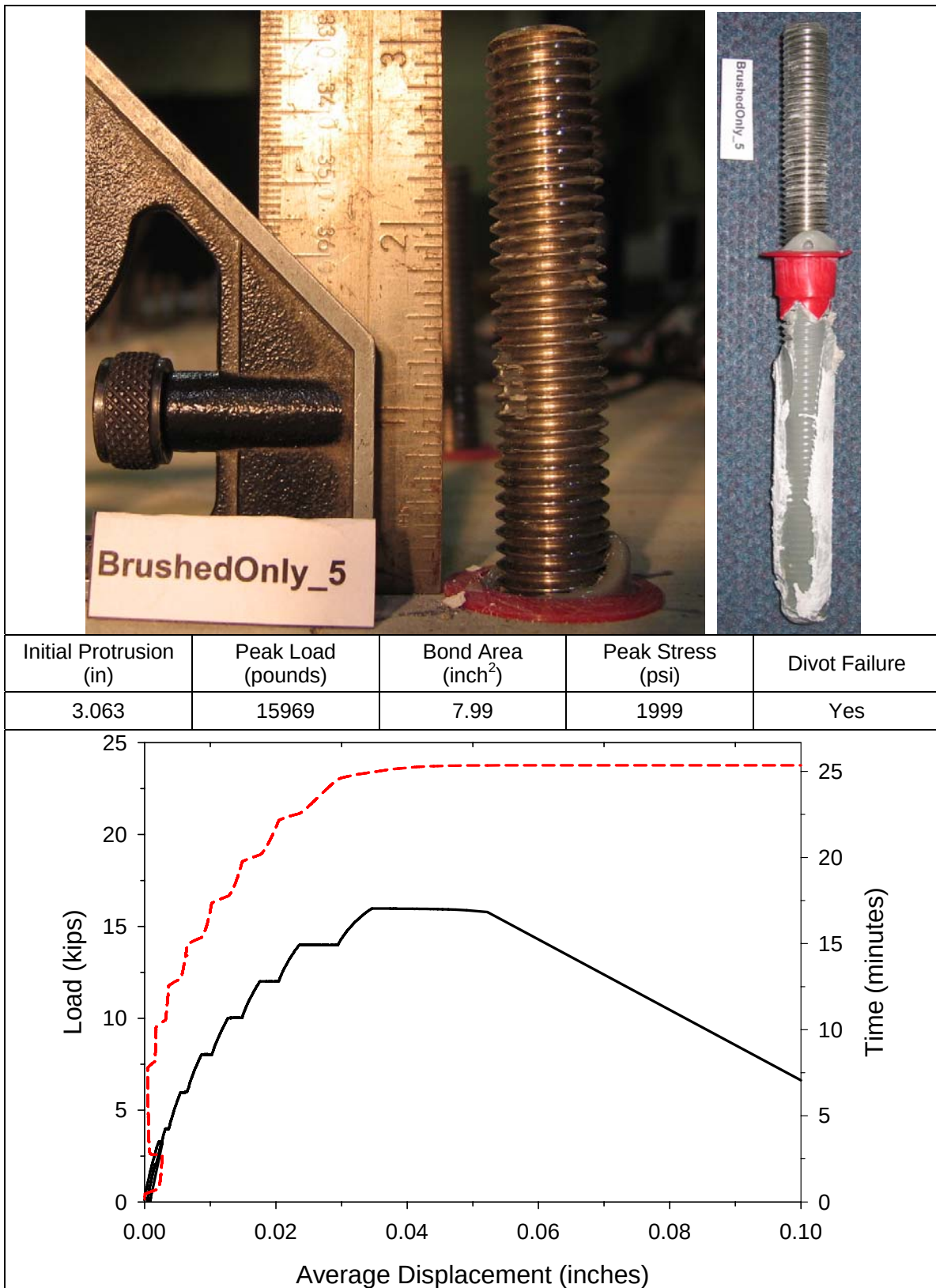


Table 83 Unturned Installation 1

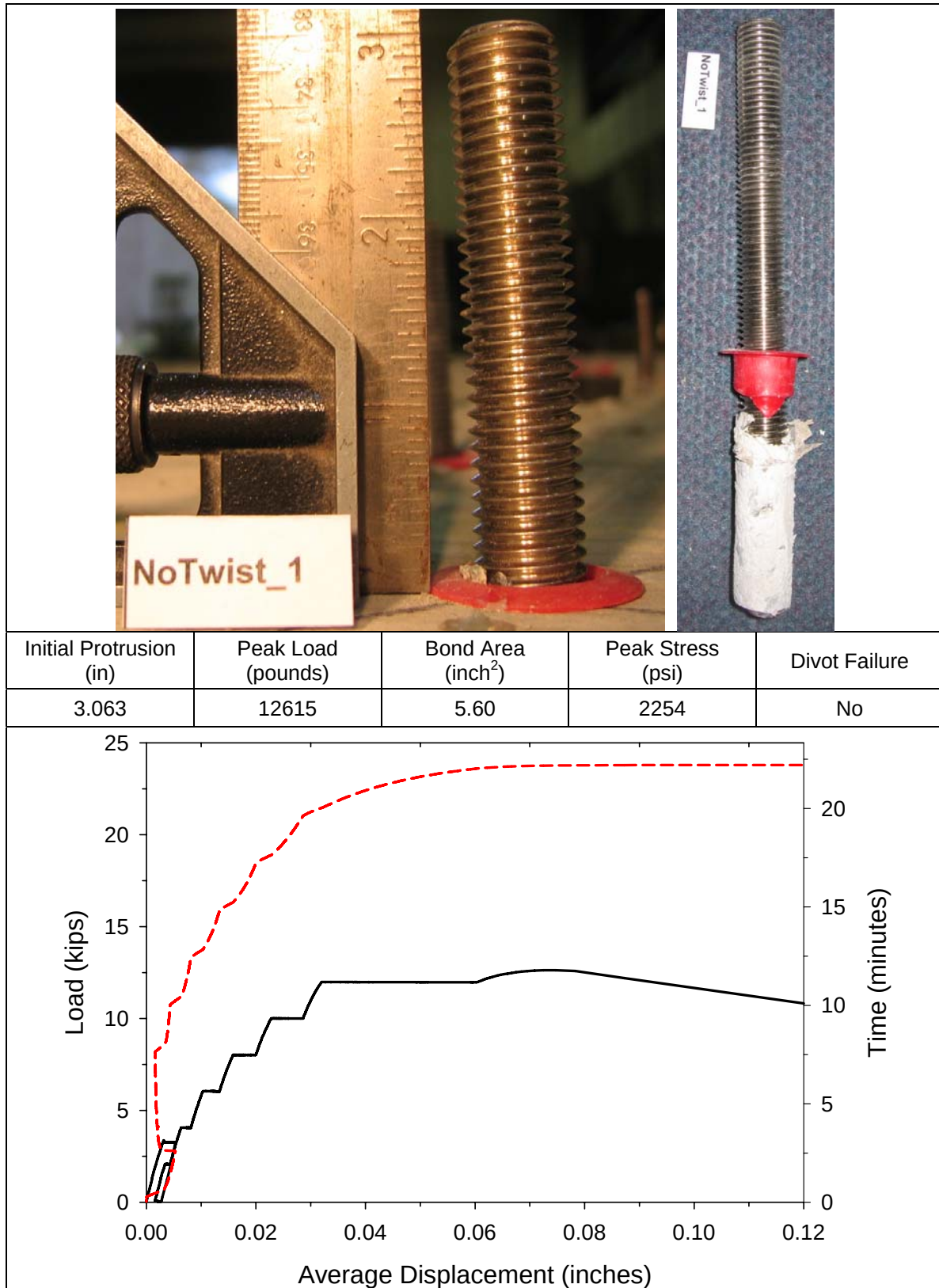


Table 84 Unturned Installation 2

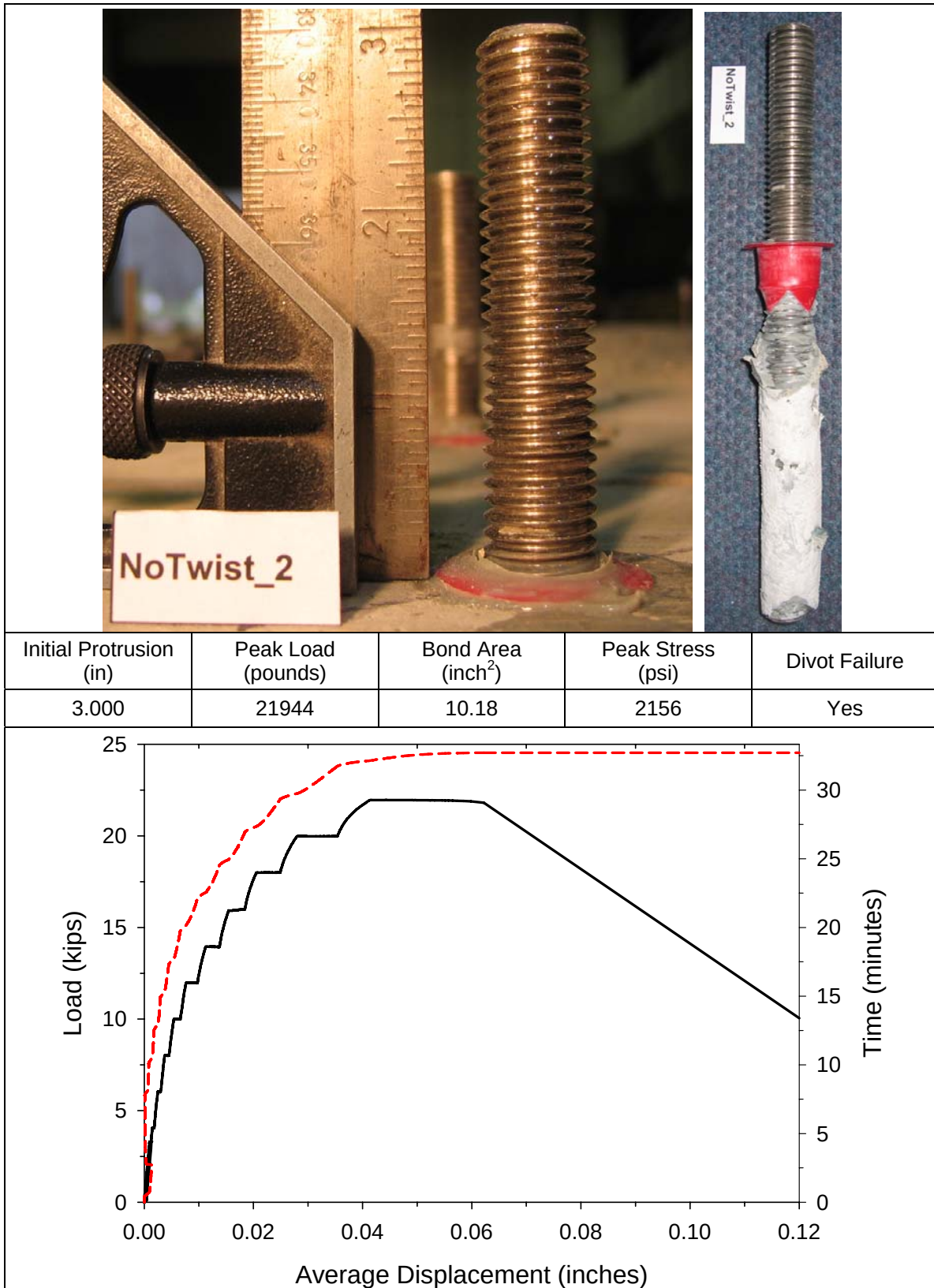


Table 85 Unturned Installation 3

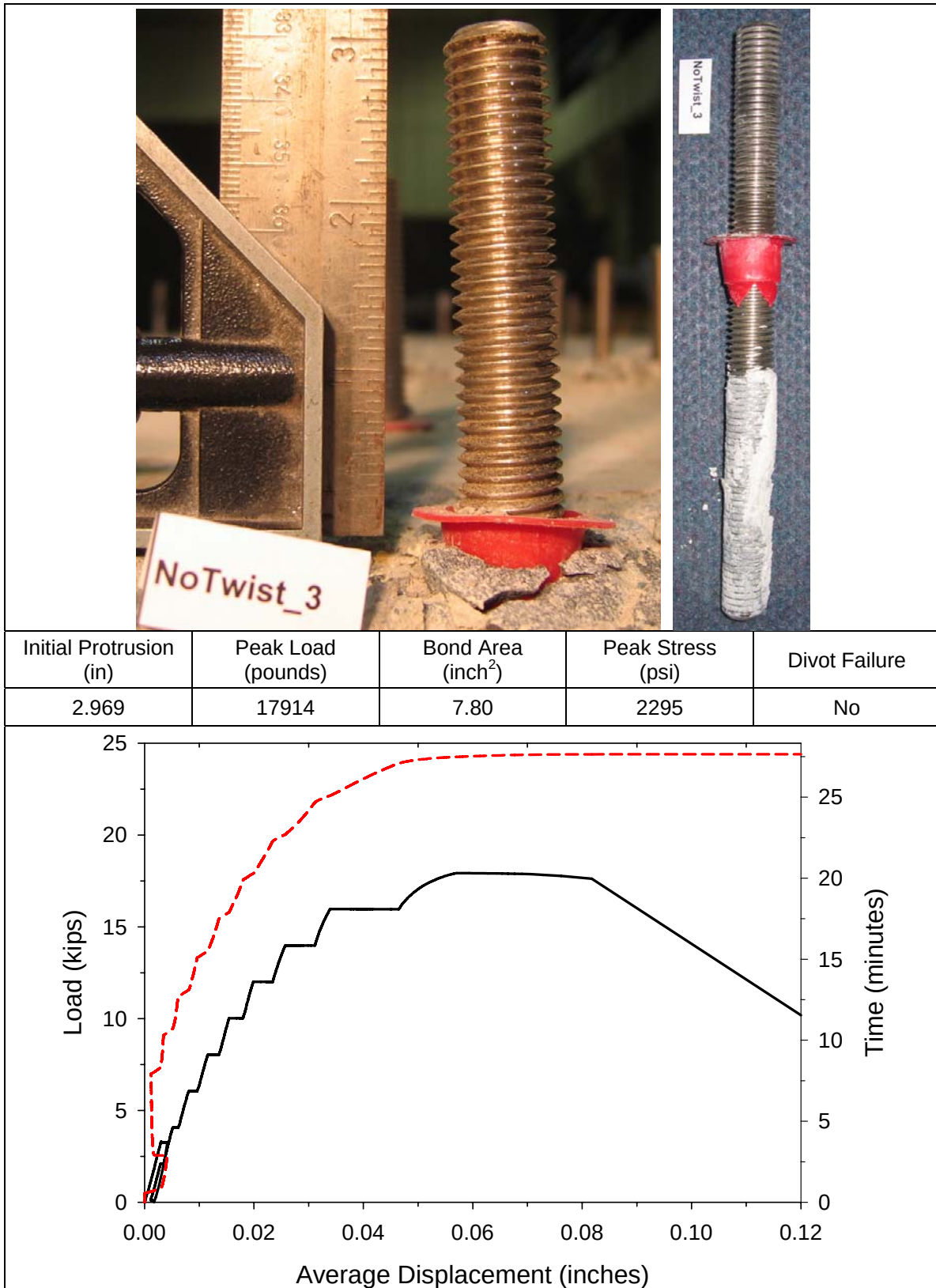


Table 86 Unturned Installation 4

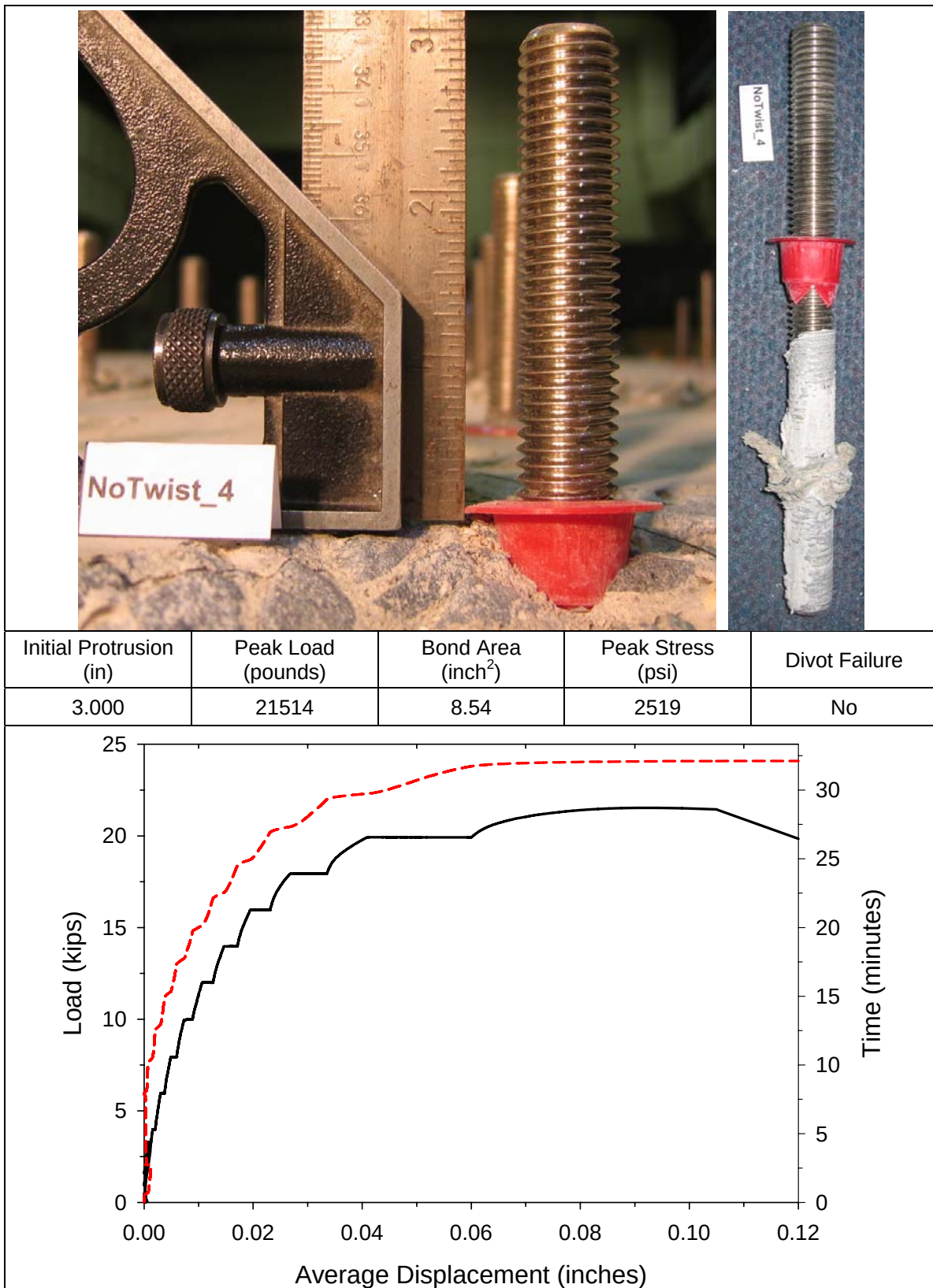


Table 87 Unturned Installation 5

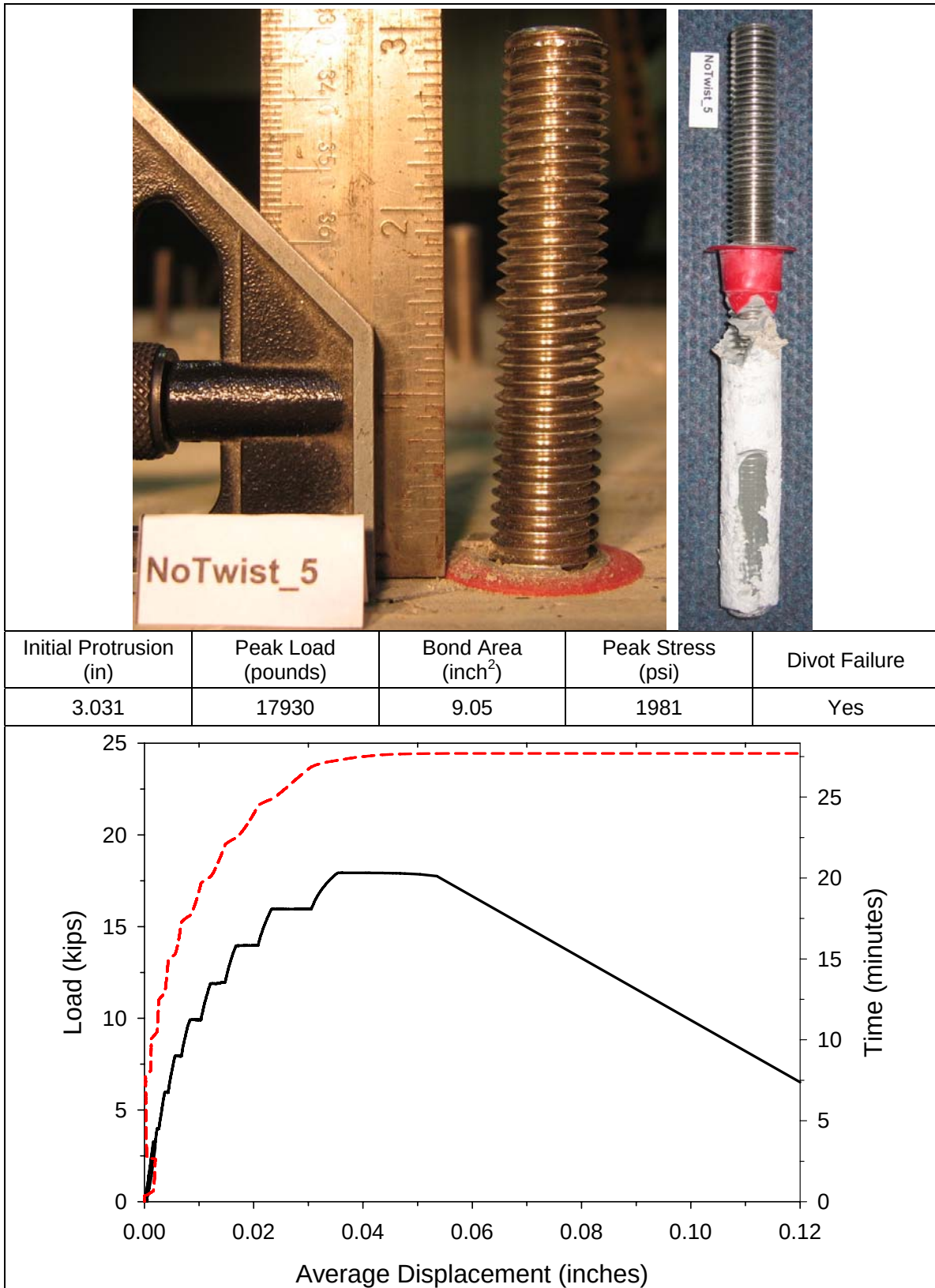


Table 88 46.6% wht / 53.3% gray 1

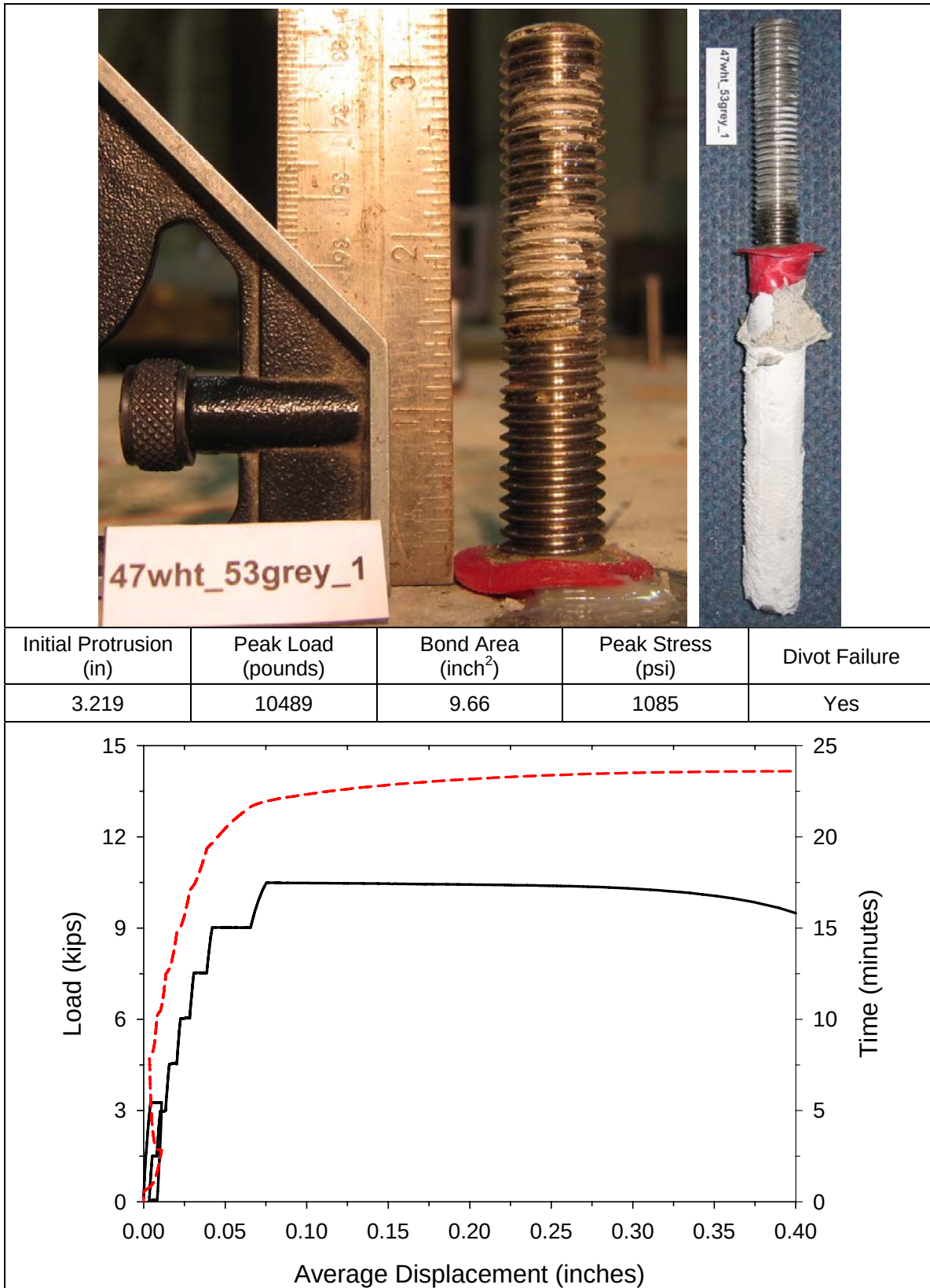


Table 89 46.6% wht / 53.3% gray 2

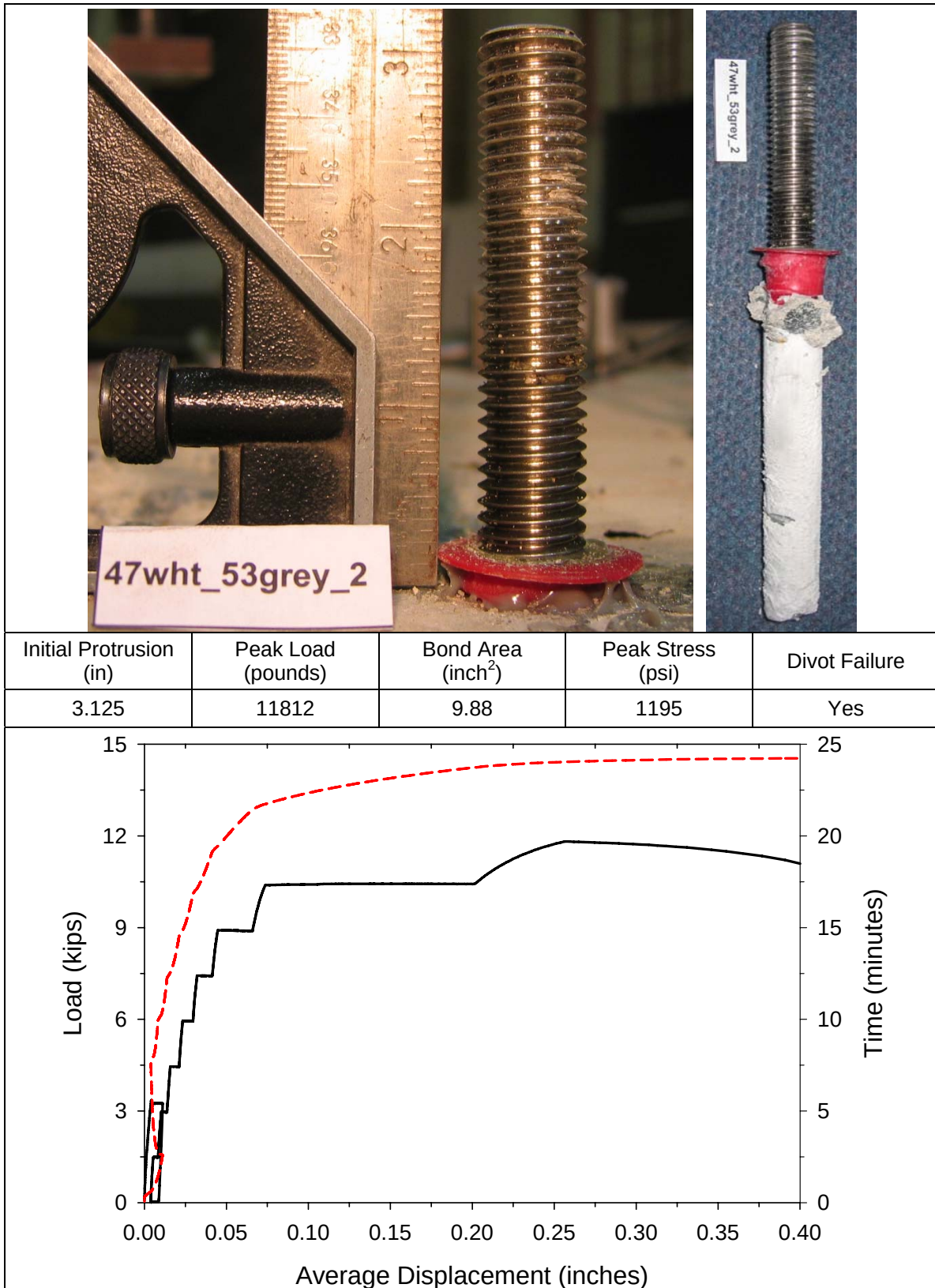


Table 90 46.6% wht / 53.3% gray 3

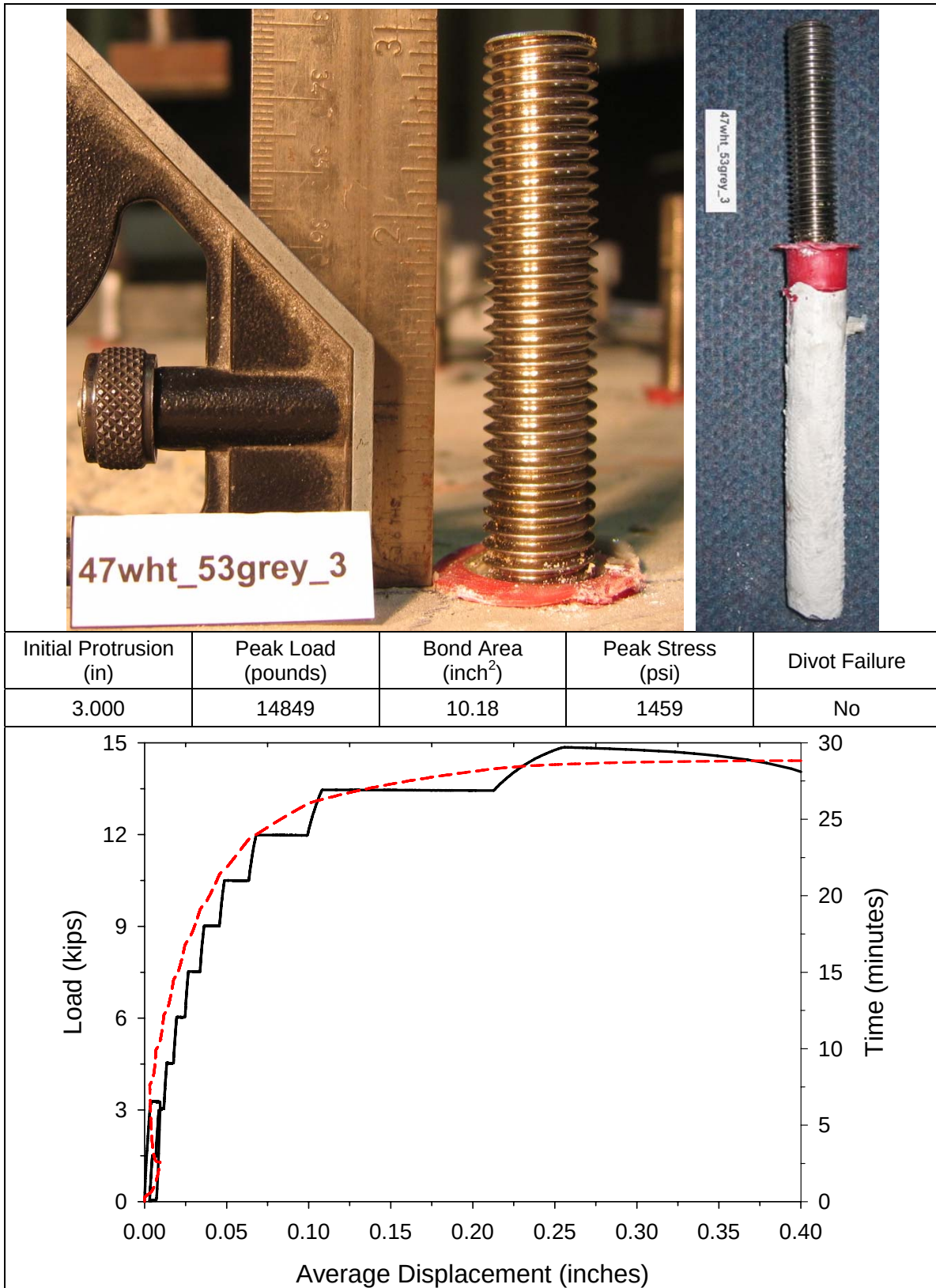


Table 91 46.6% wht / 53.3% gray 4

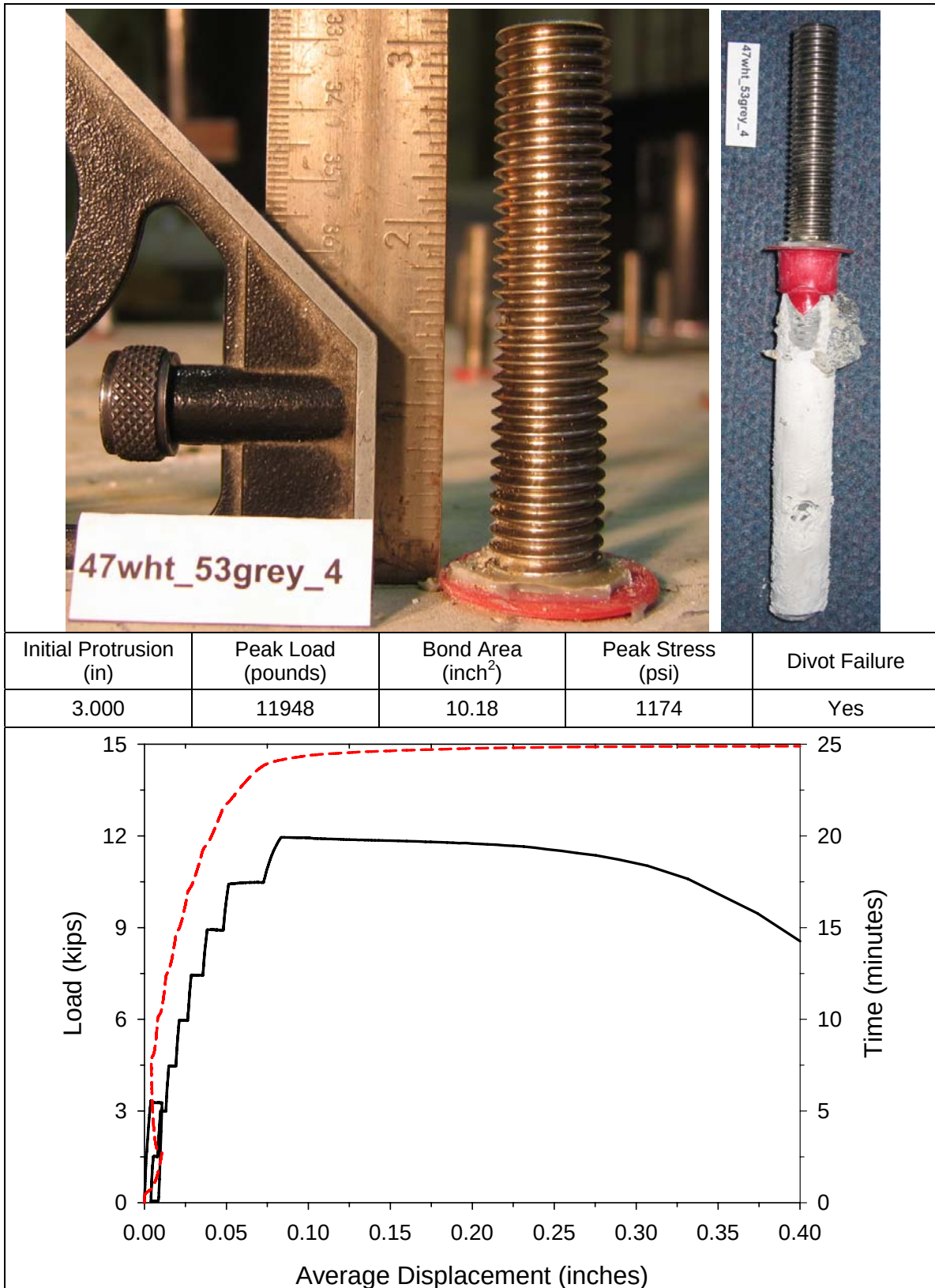


Table 92 53.3% wht / 46.6% gray 1

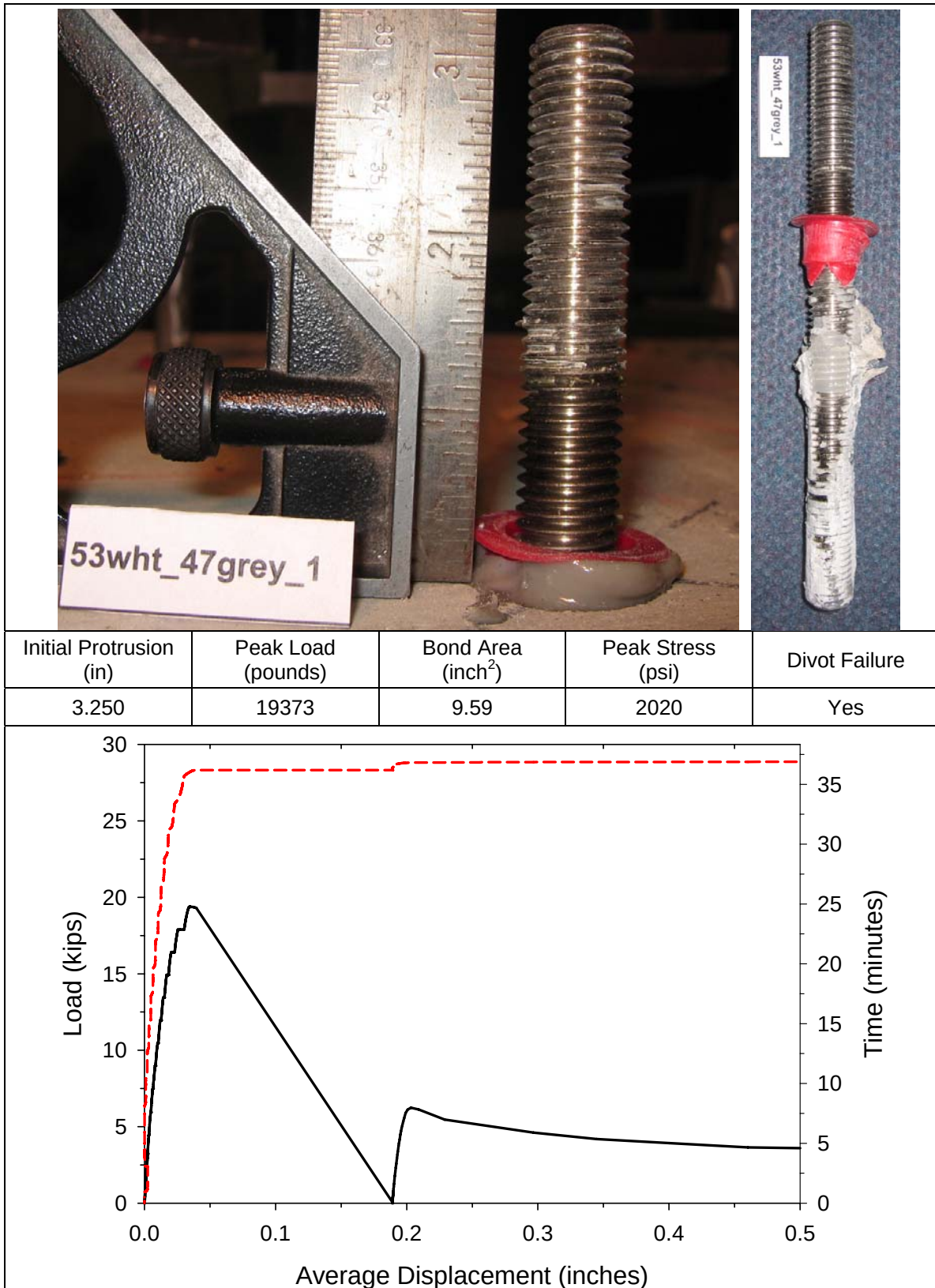


Table 93 53.3% wht / 46.6% gray 2

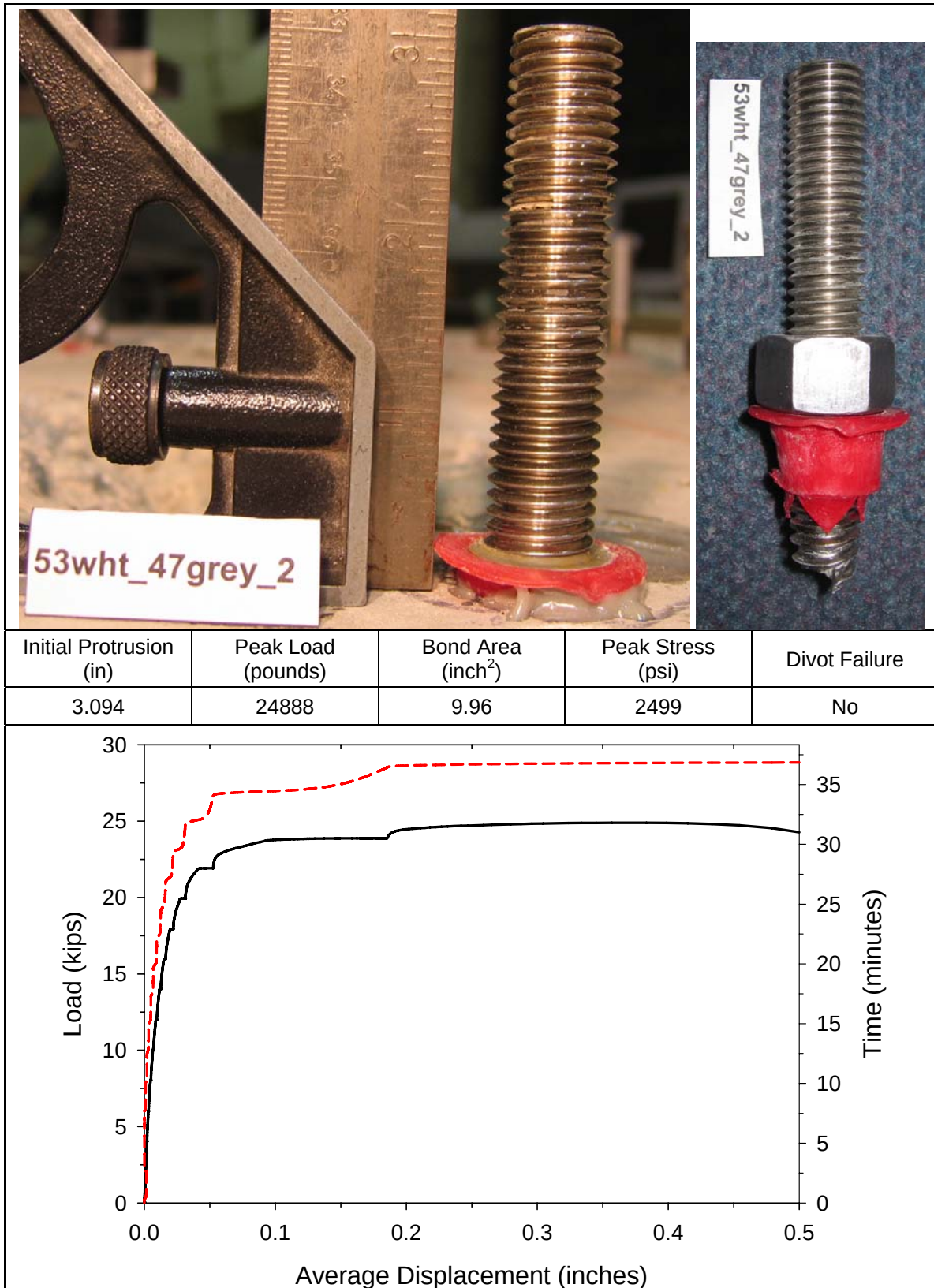


Table 94 53.3% wht / 46.6% gray 3

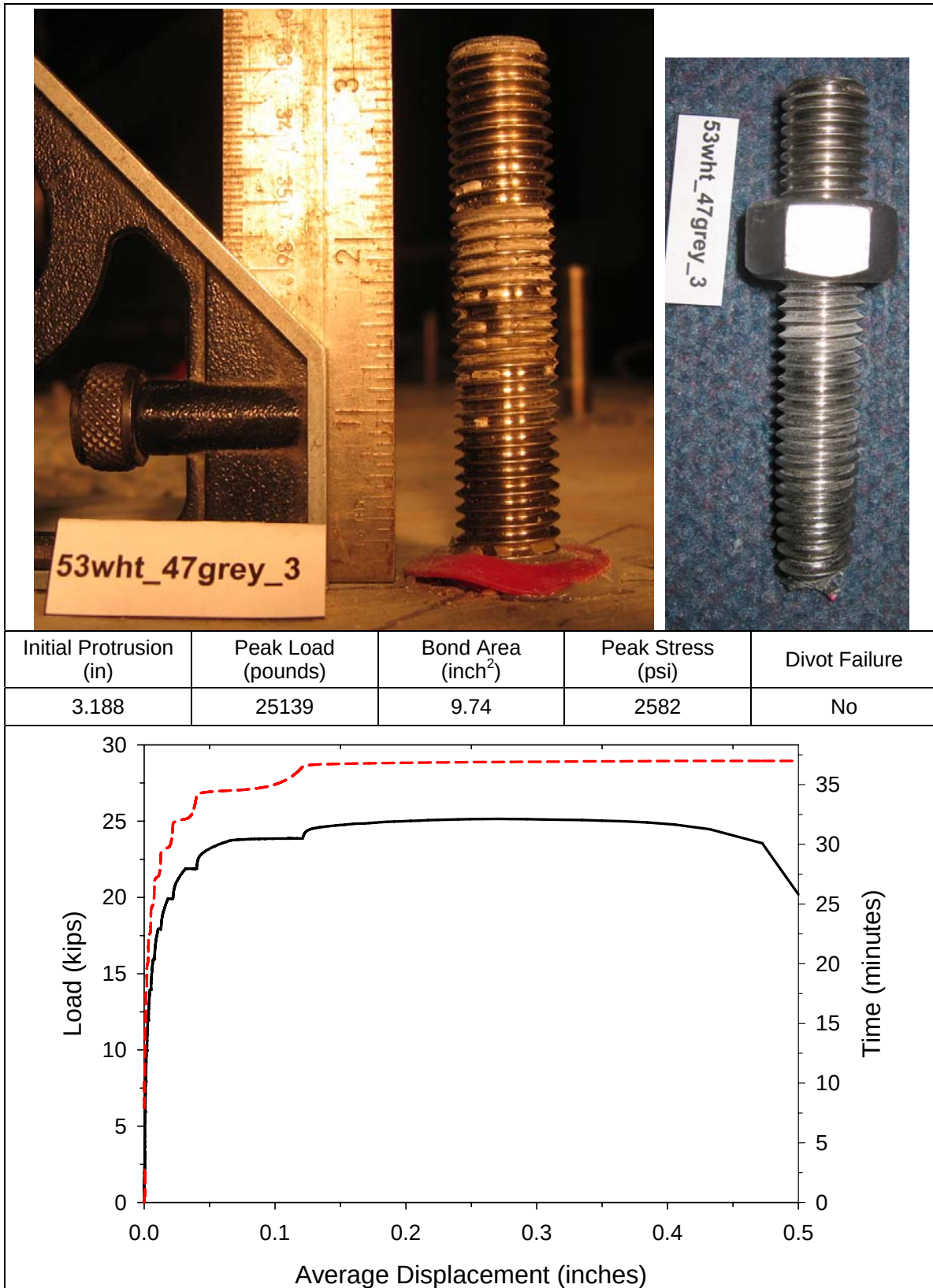


Table 95 53.3% wht / 46.6% gray 4

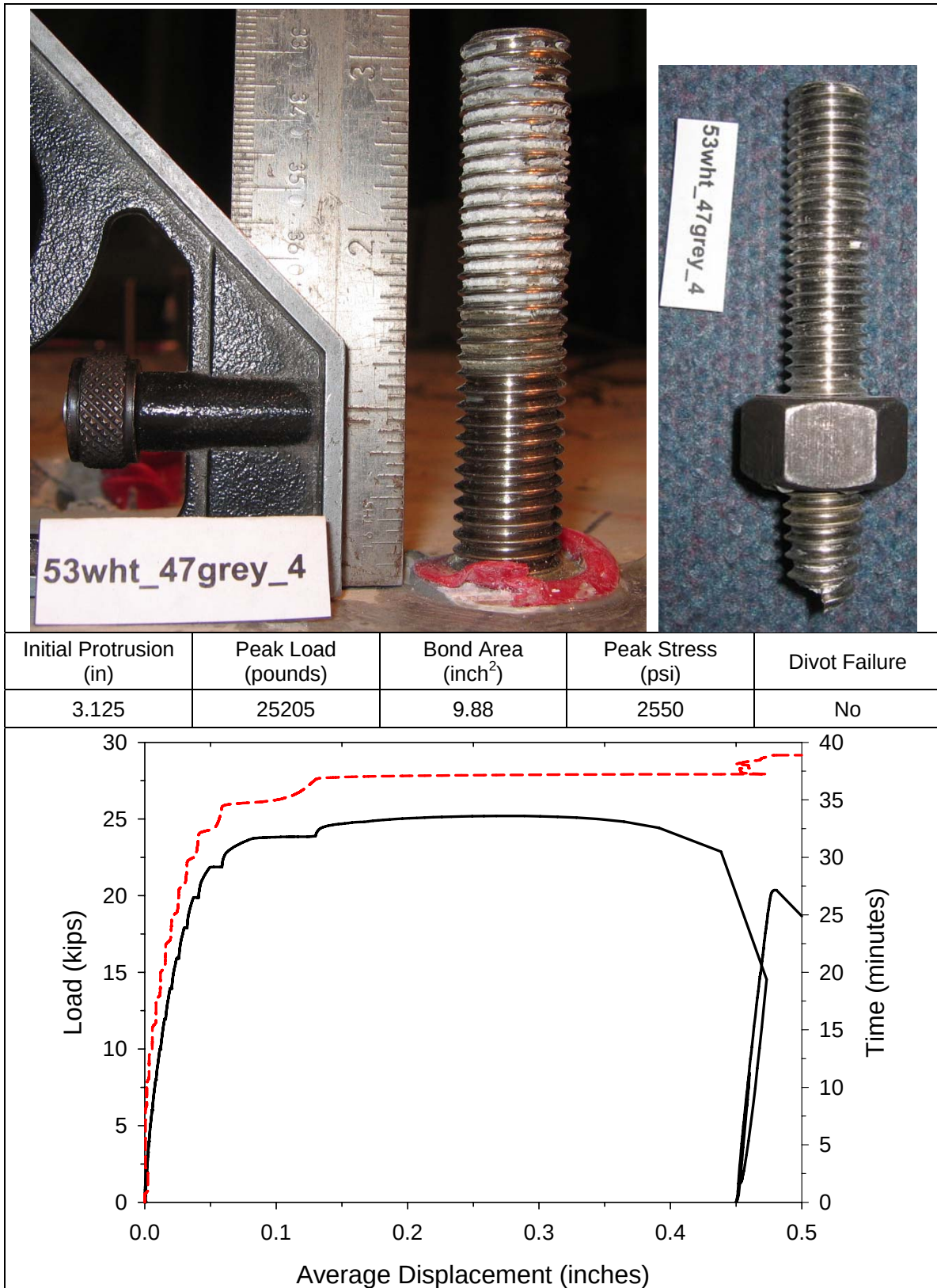


Table 96 40.0% wht / 60.0% gray 1

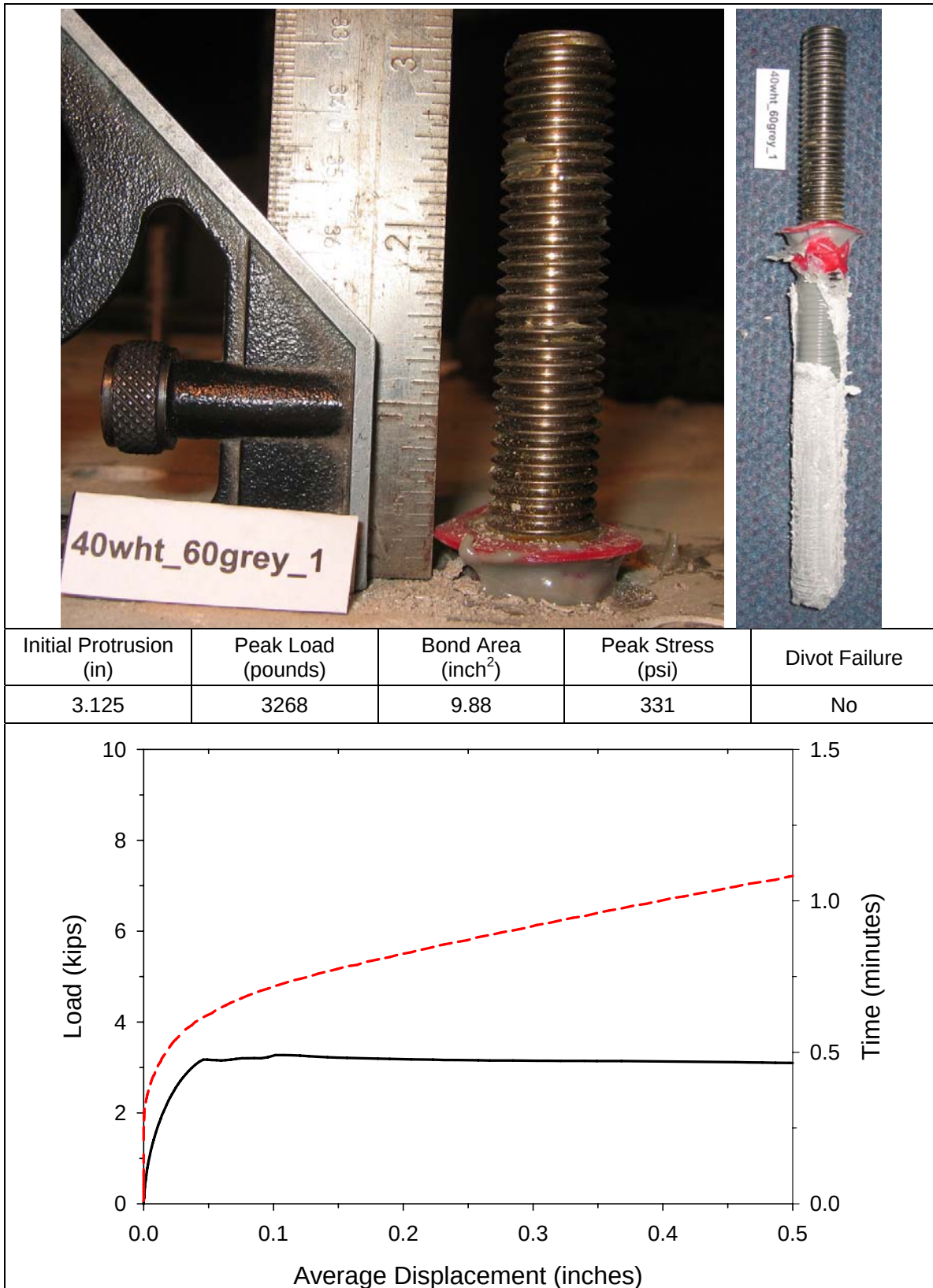


Table 97 40.0% wht / 60.0% gray 2

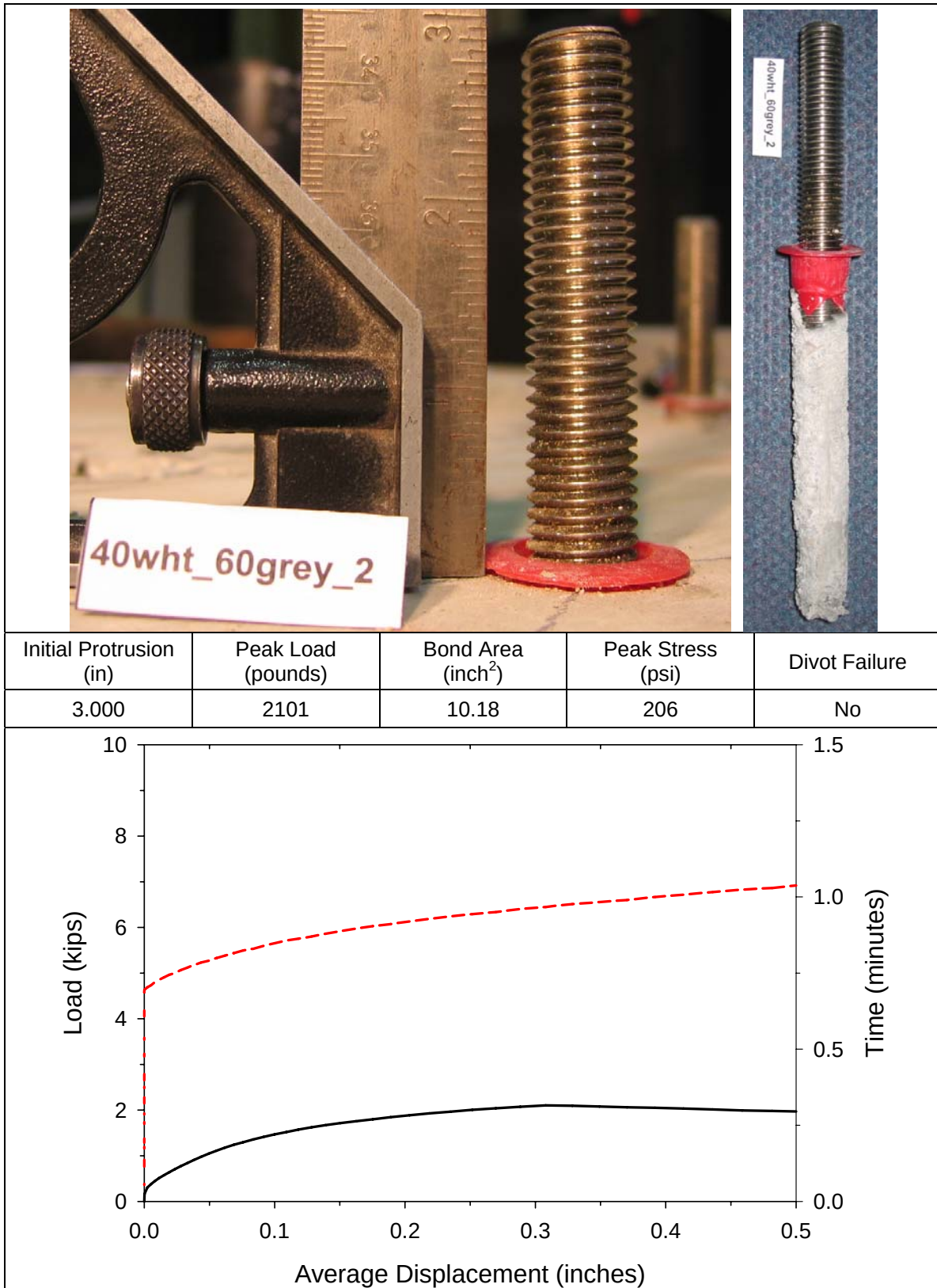


Table 98 40.0% wht / 60.0% gray 3

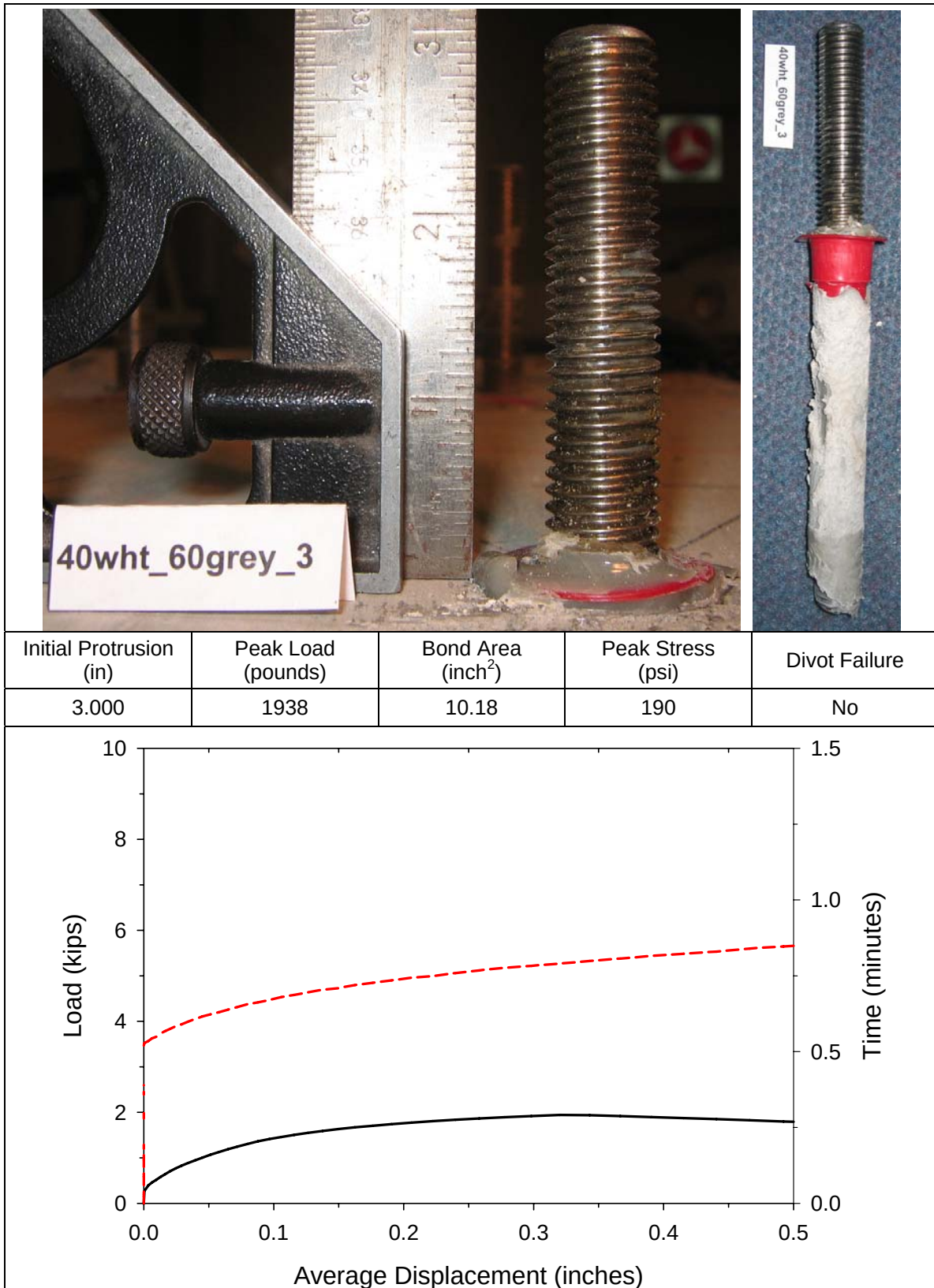


Table 99 60.0% wht / 40.0% gray 1

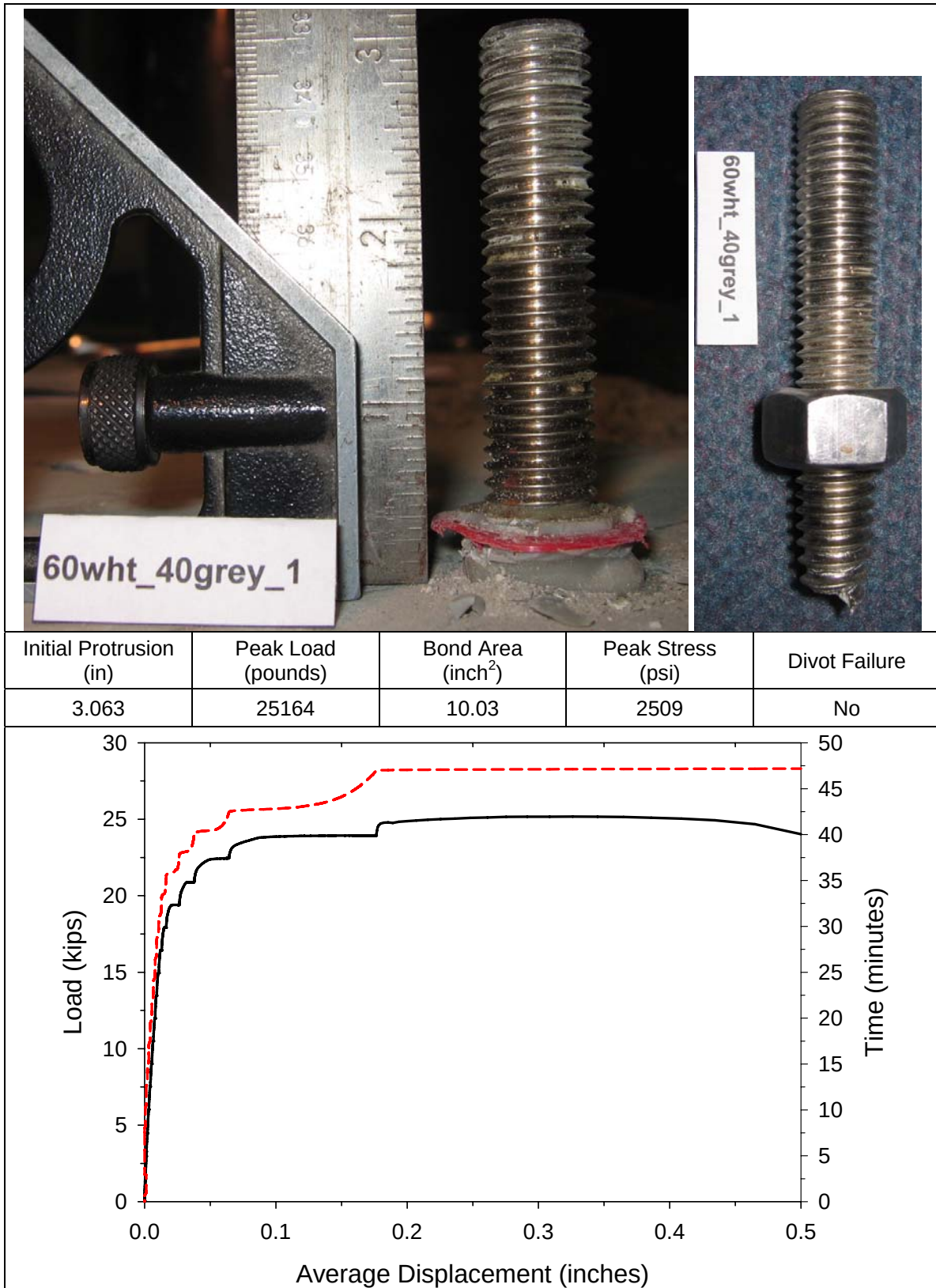


Table 100 60.0% wht / 40.0% gray 2

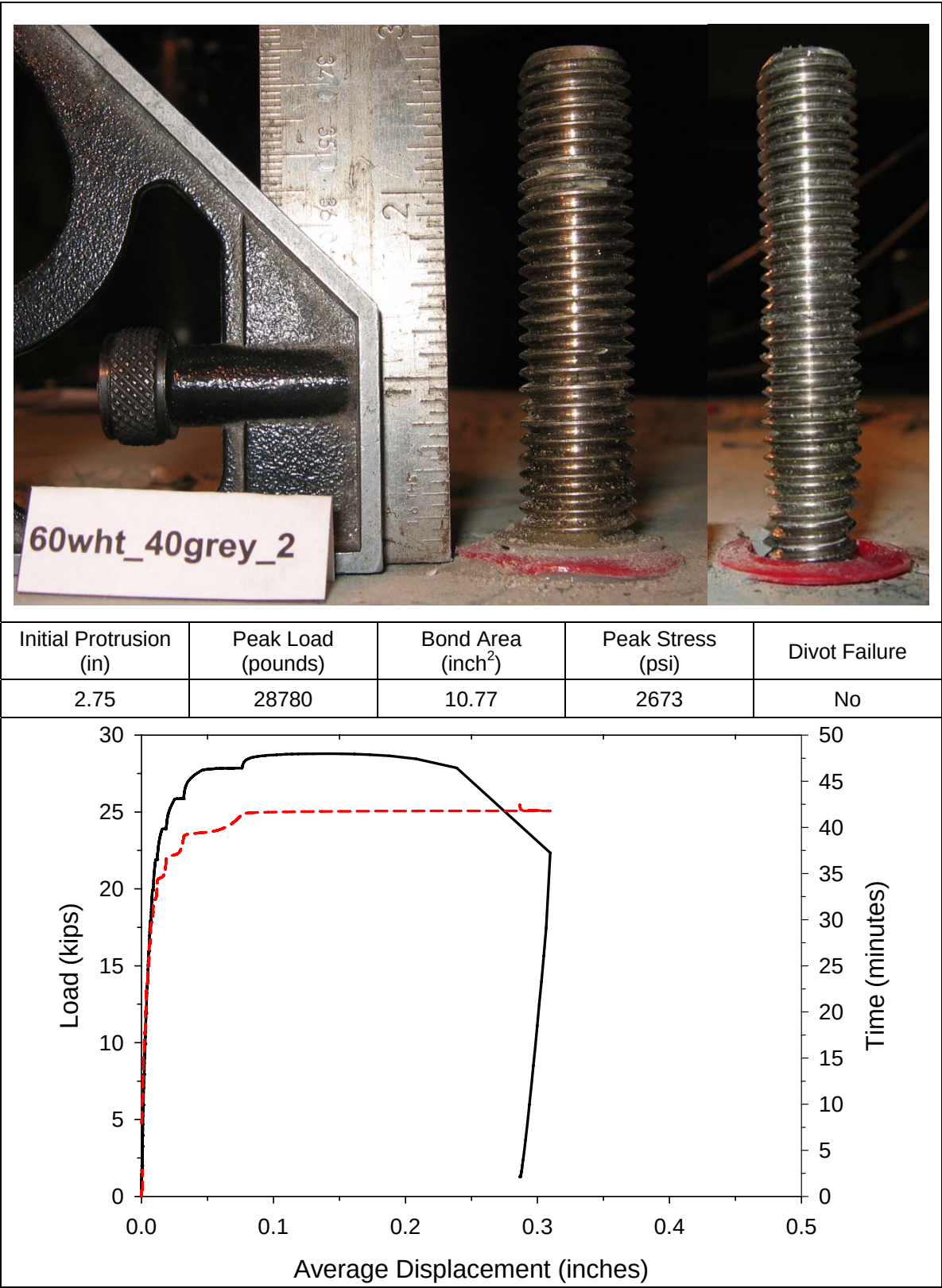


Table 101 60.0% wht / 40.0% gray 3

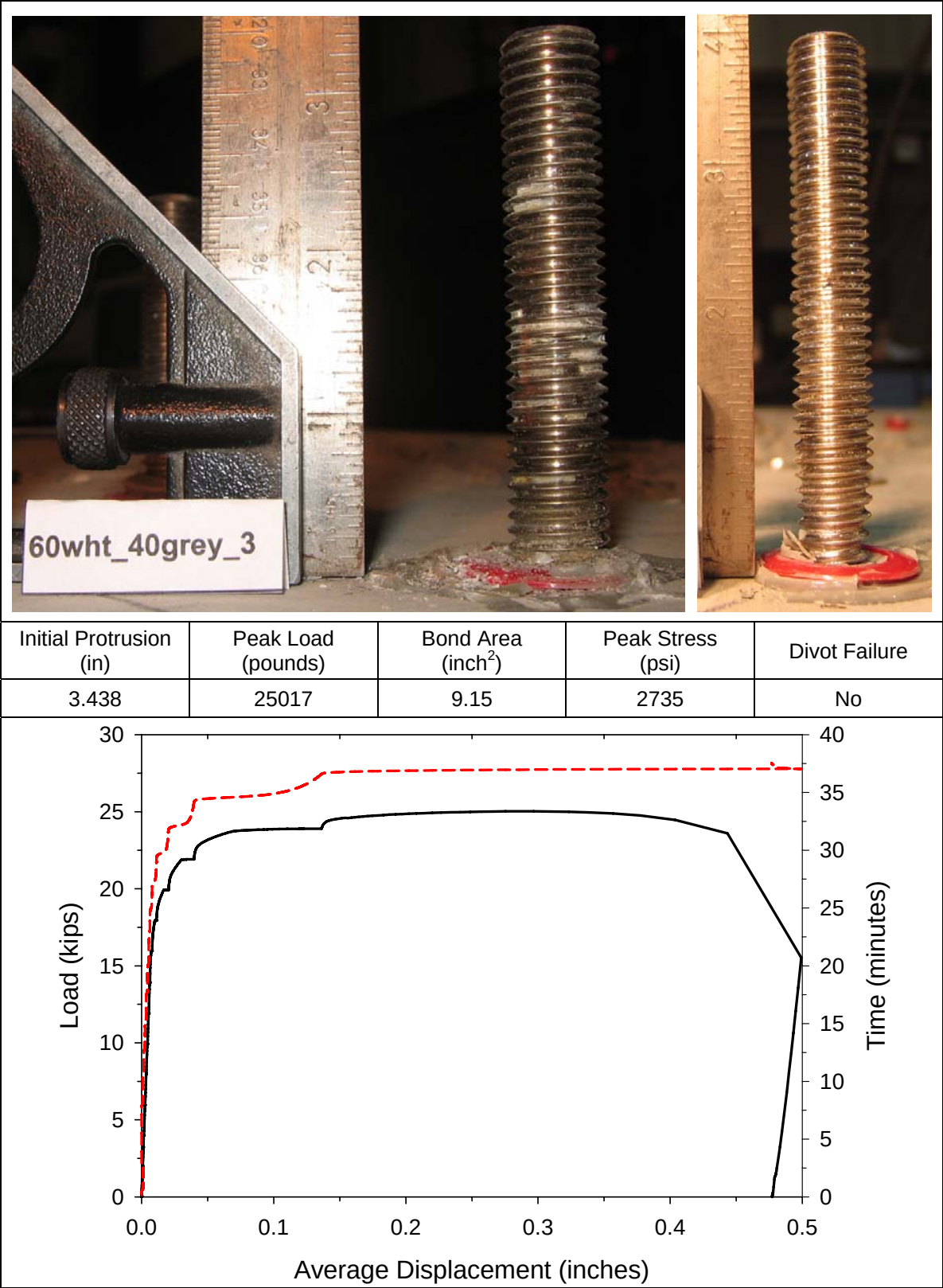


Table 102 Void Unturned Installation 1

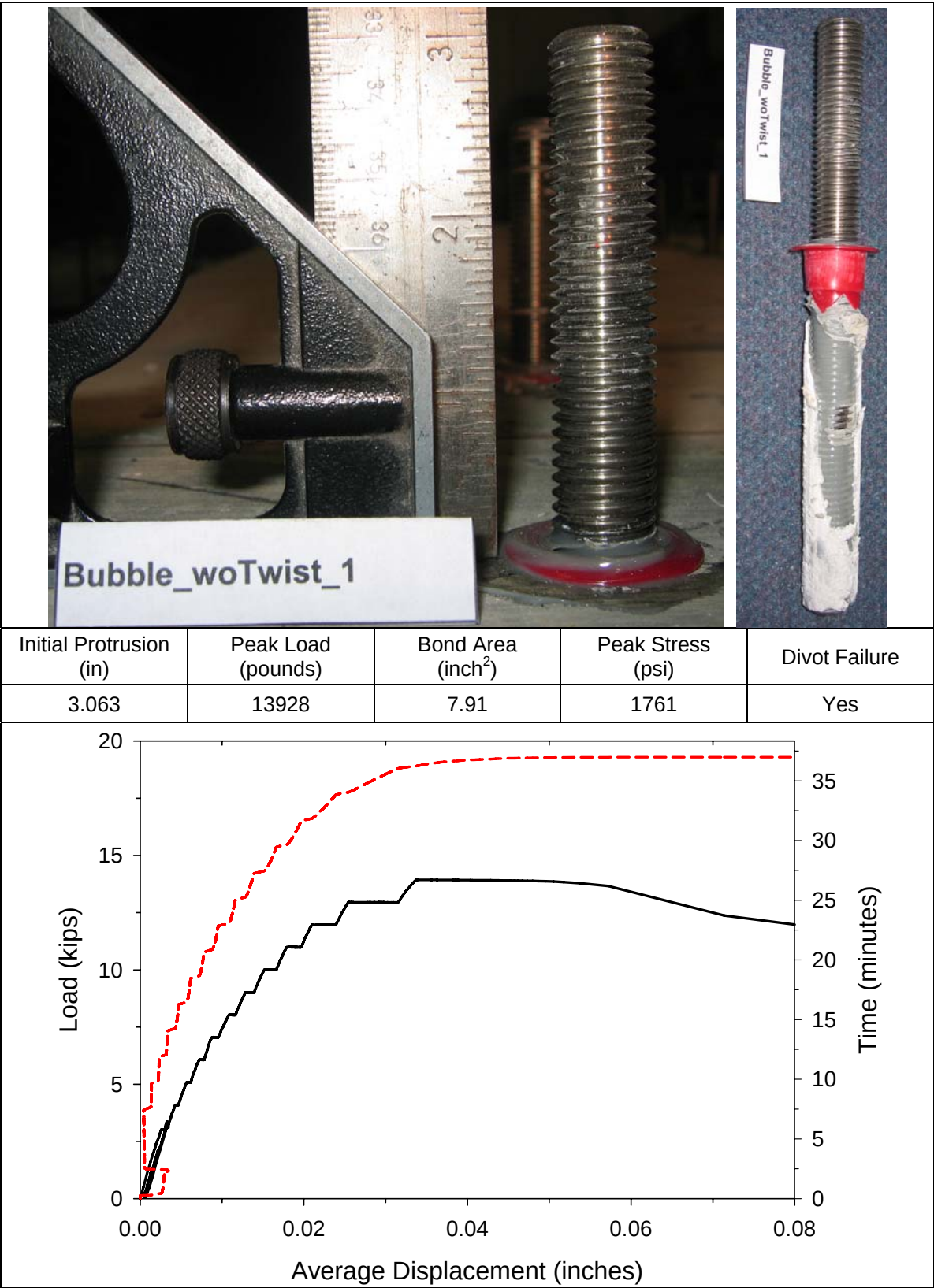


Table 103 Void Unturned Installation 2

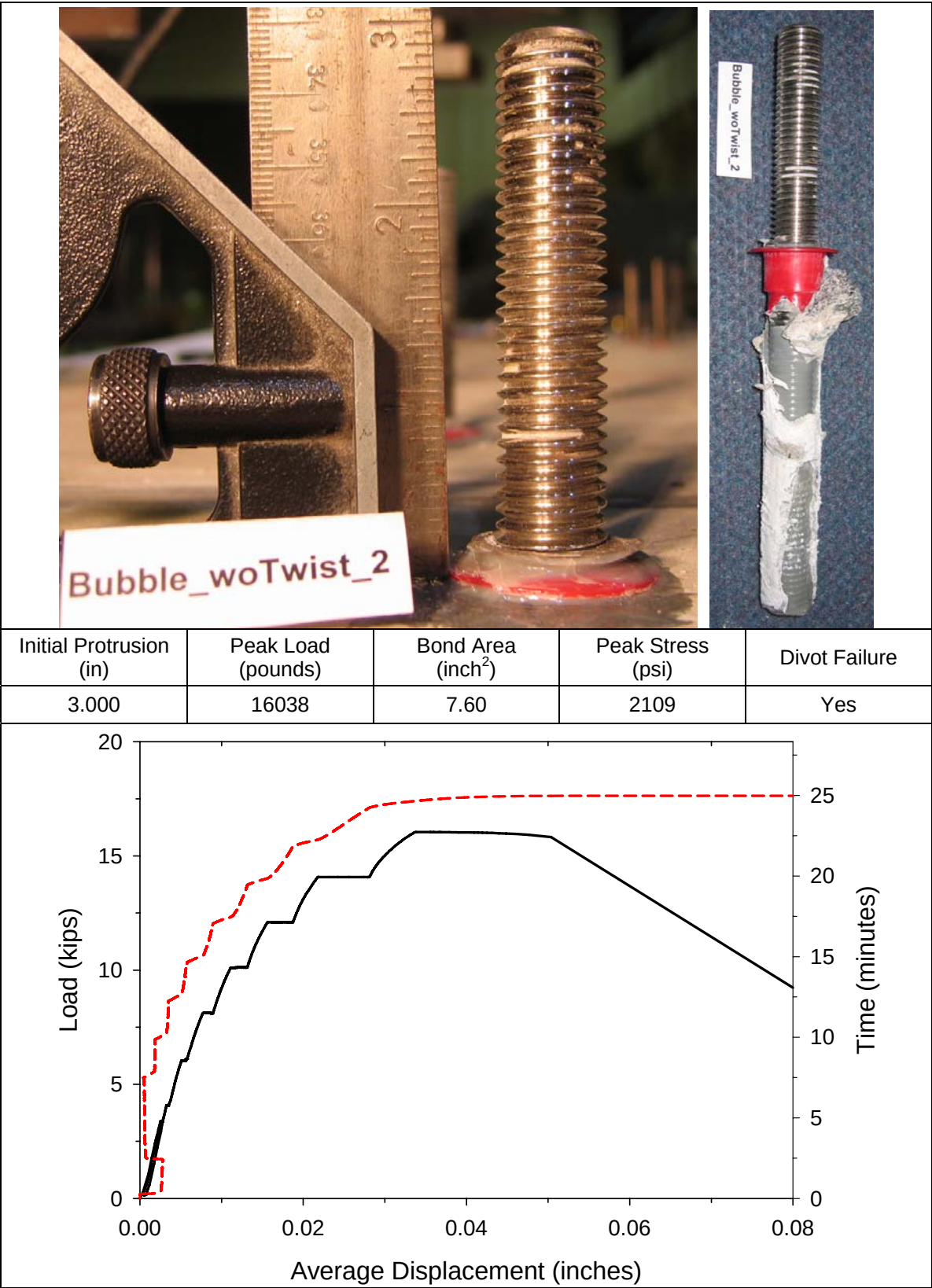


Table 104 Void Unturned Installation 3

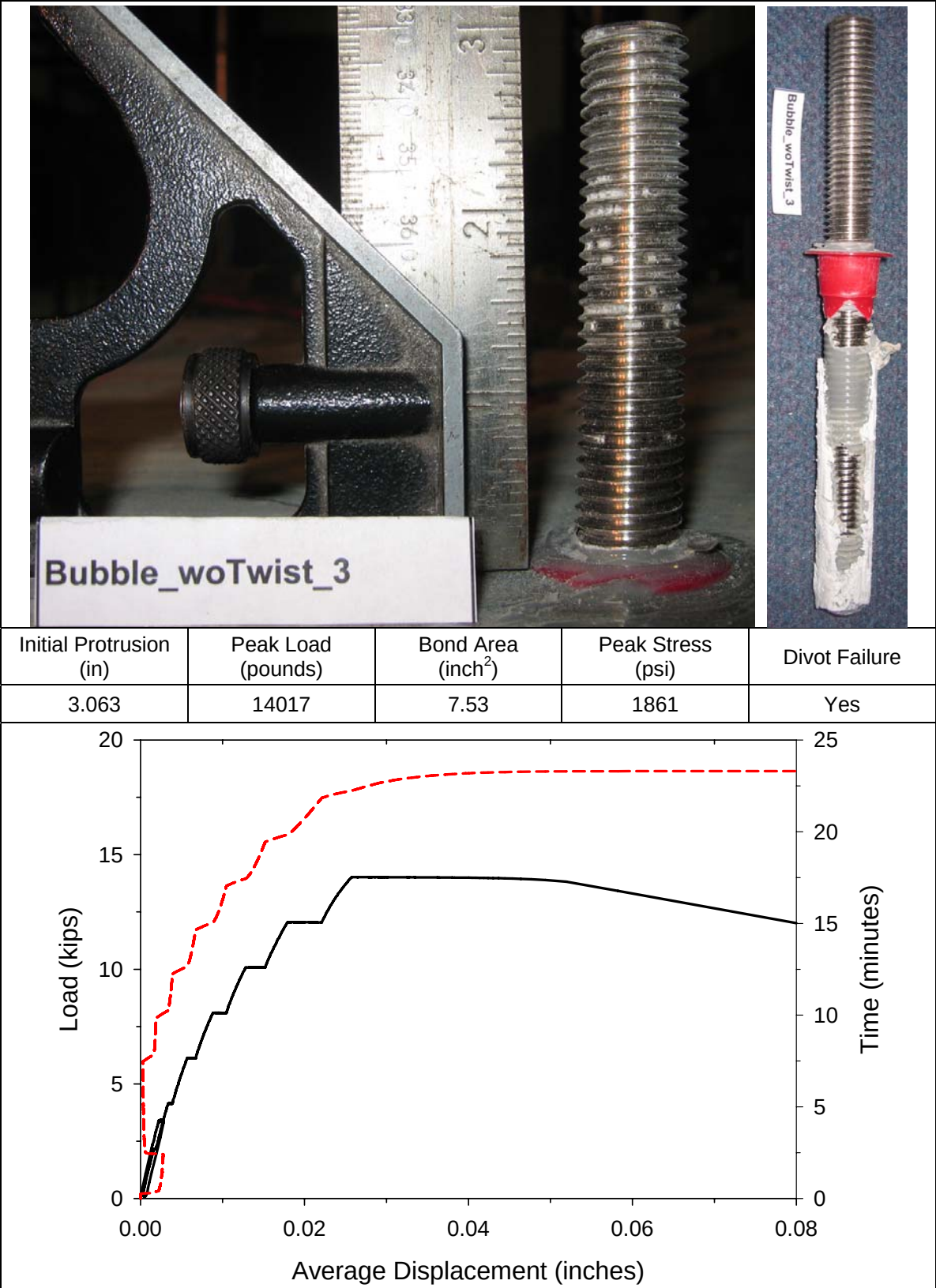


Table 105 Void Unturned Installation 4

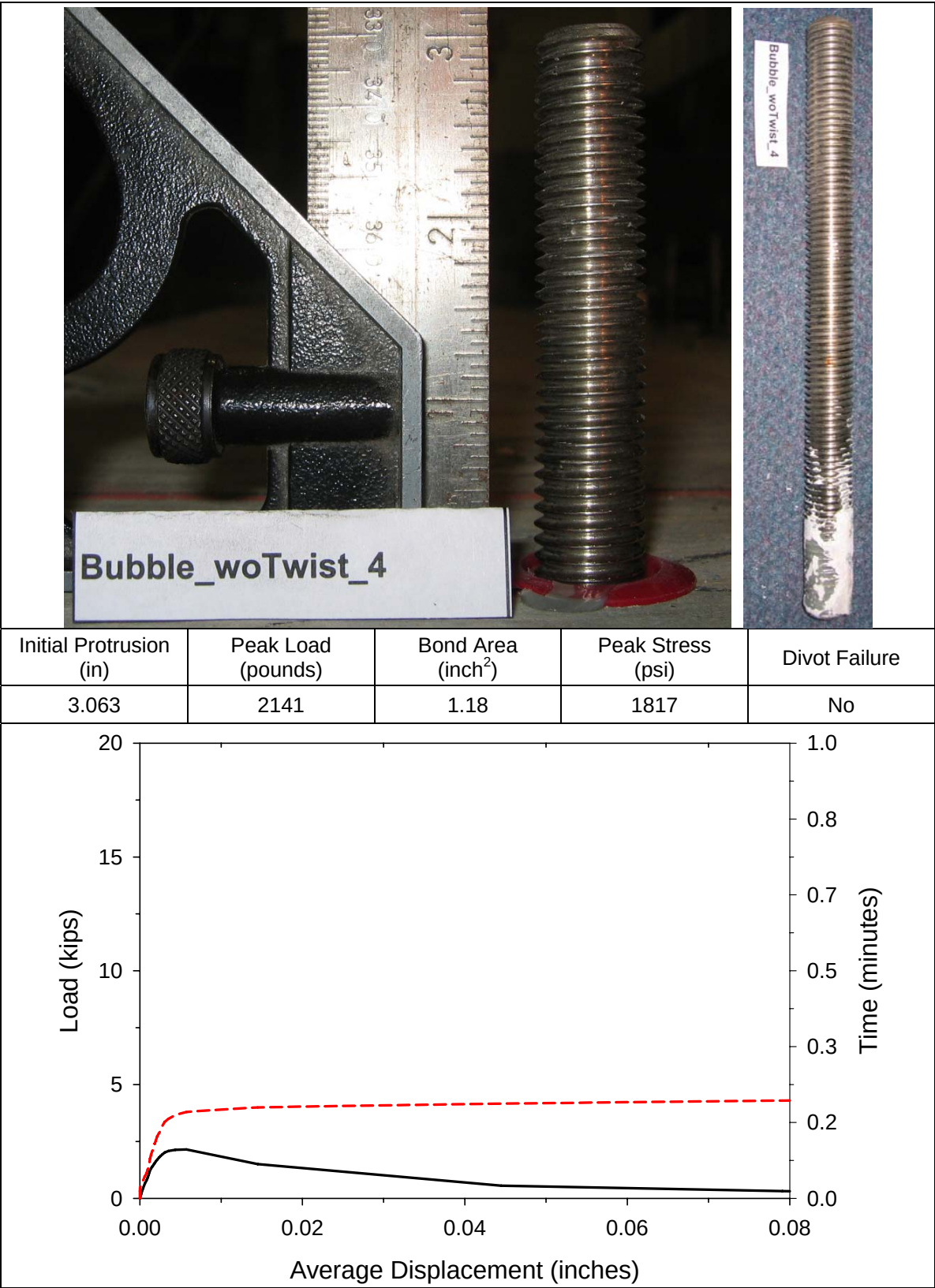


Table 106 Void Unturned Installation 5

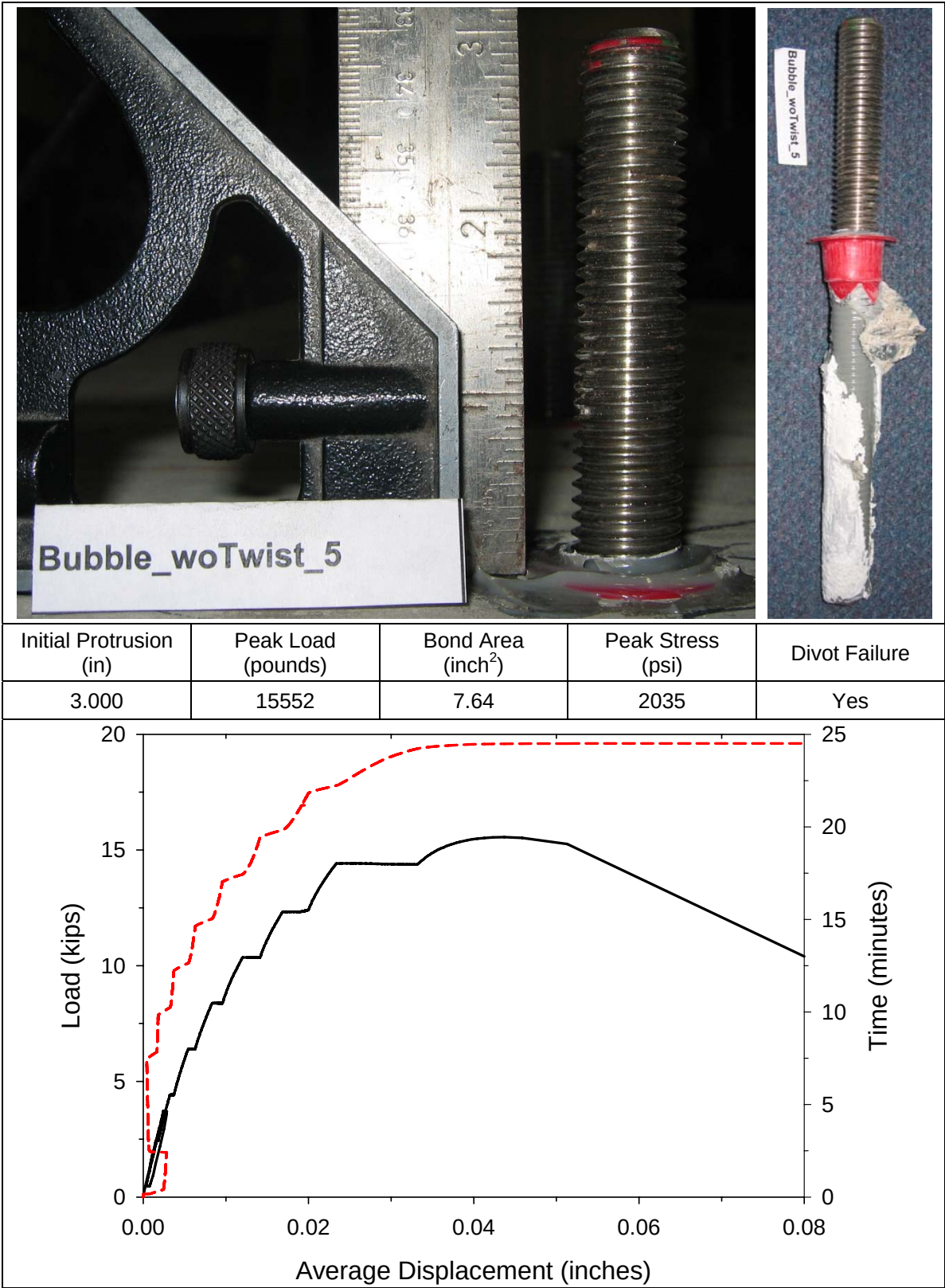


Table 107 Void Unturned Installation 6

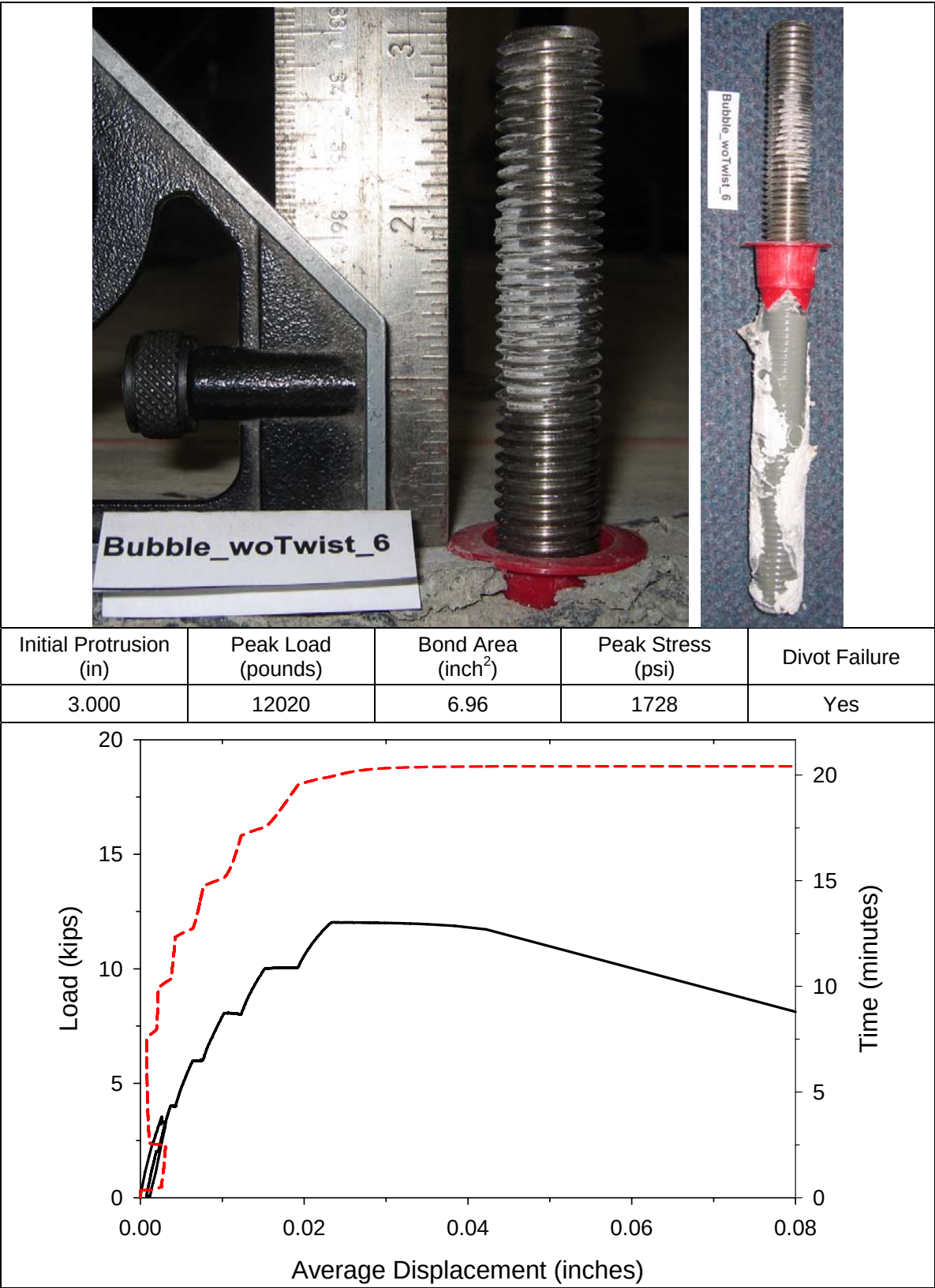


Table 108 Void Unturned Installation 7

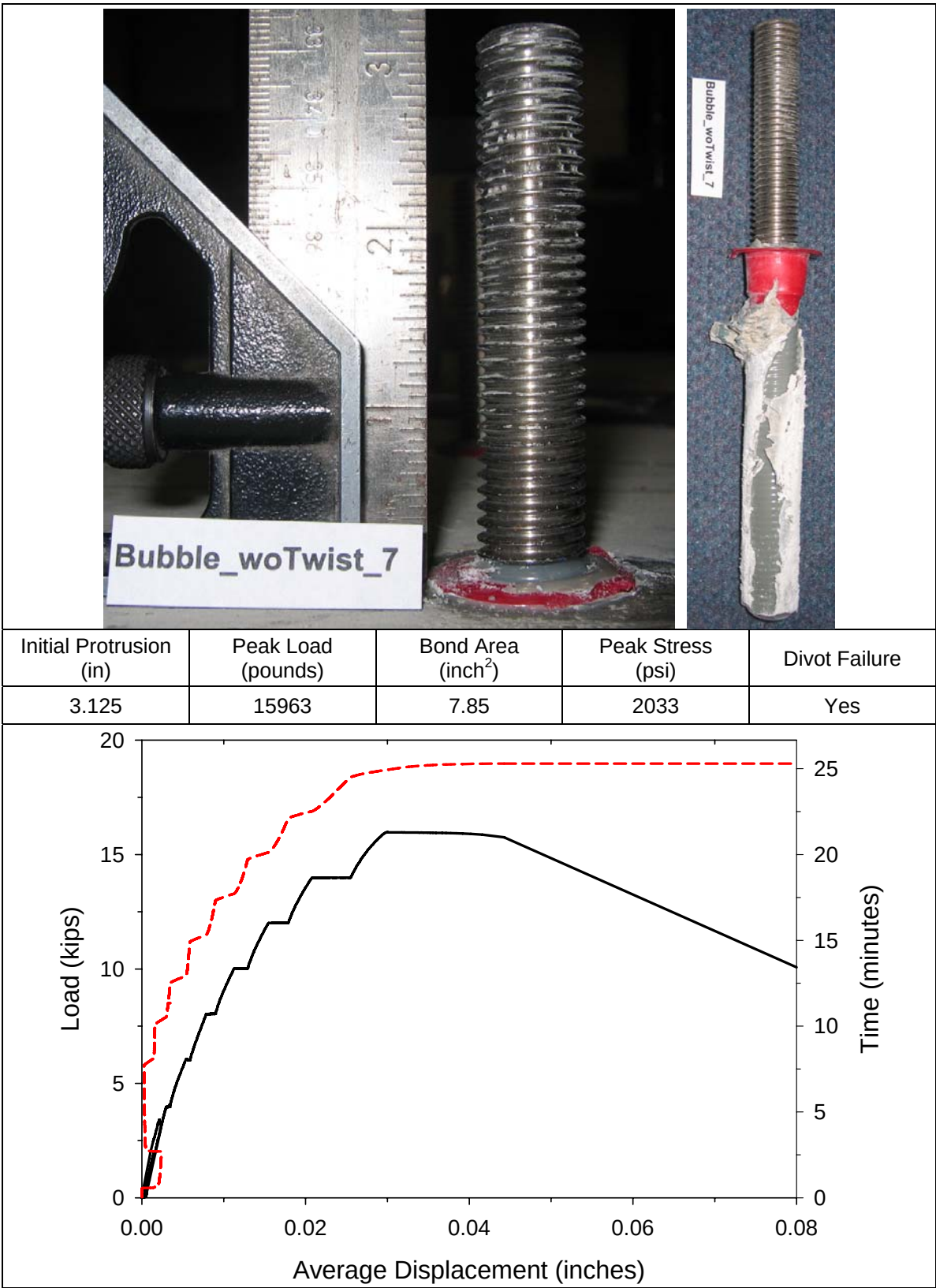


Table 109 Void Unturned Installation 8

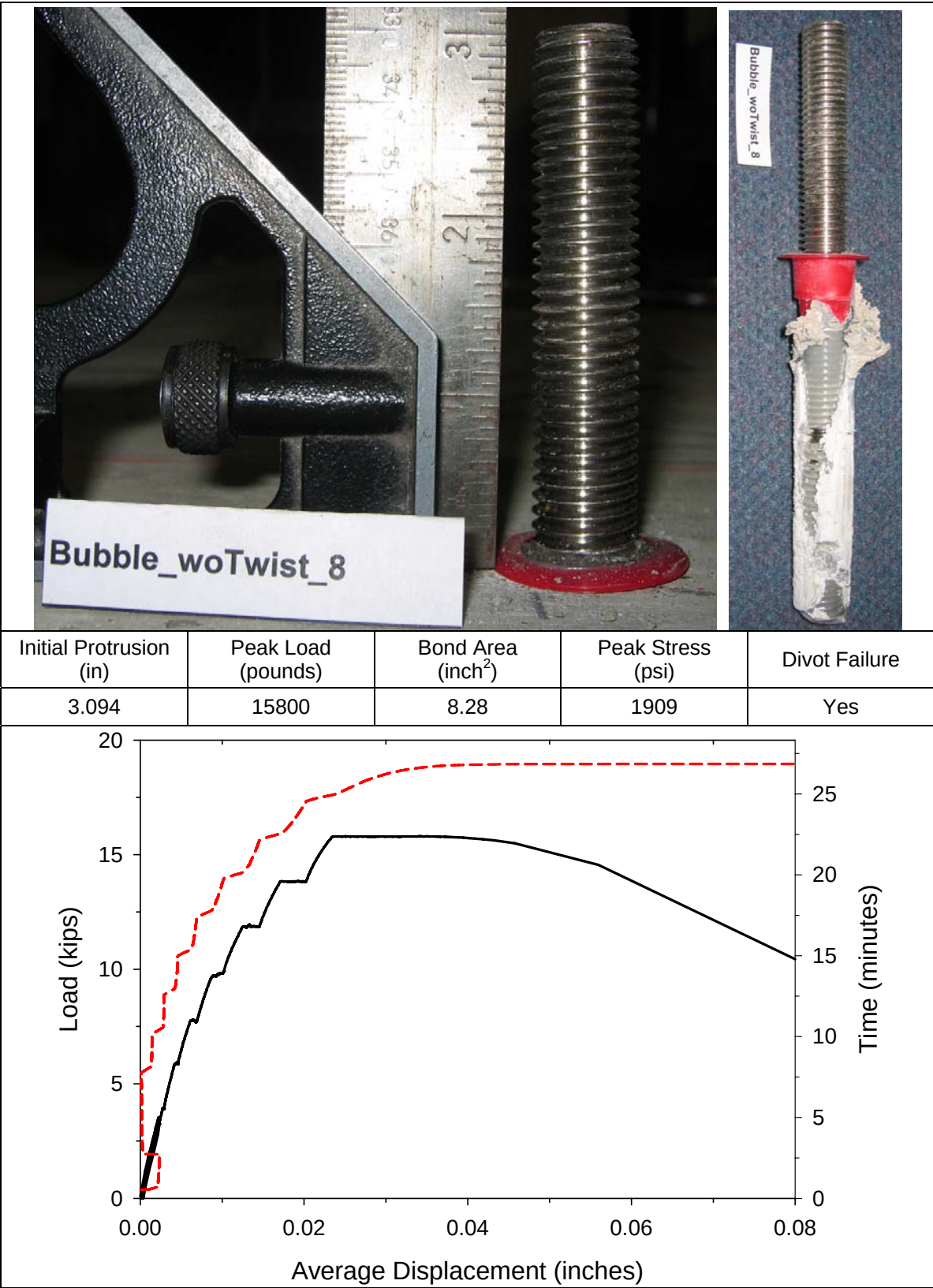


Table 110 Void Unturned Installation 9

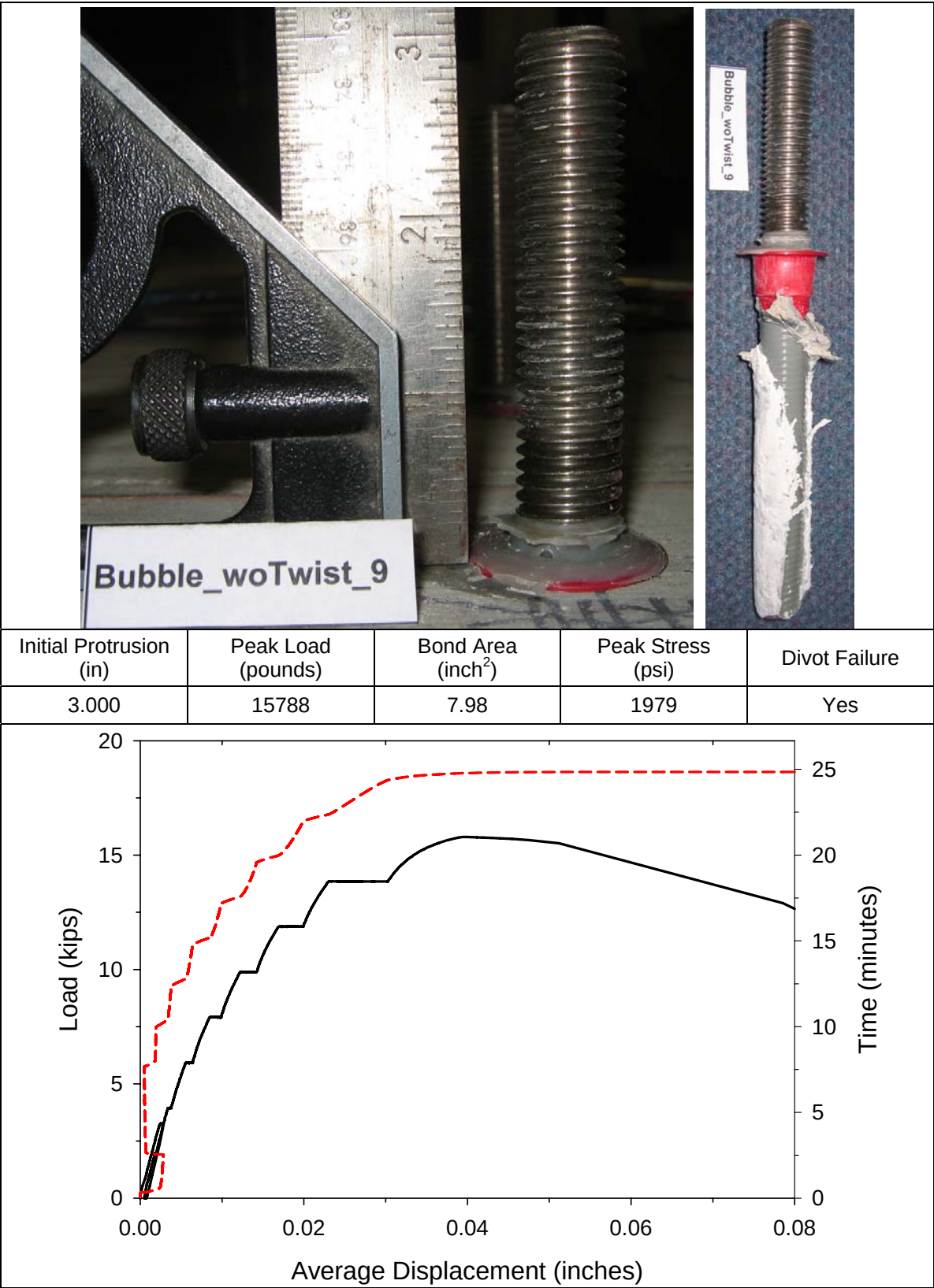


Table 111 Void Unturned Installation 10

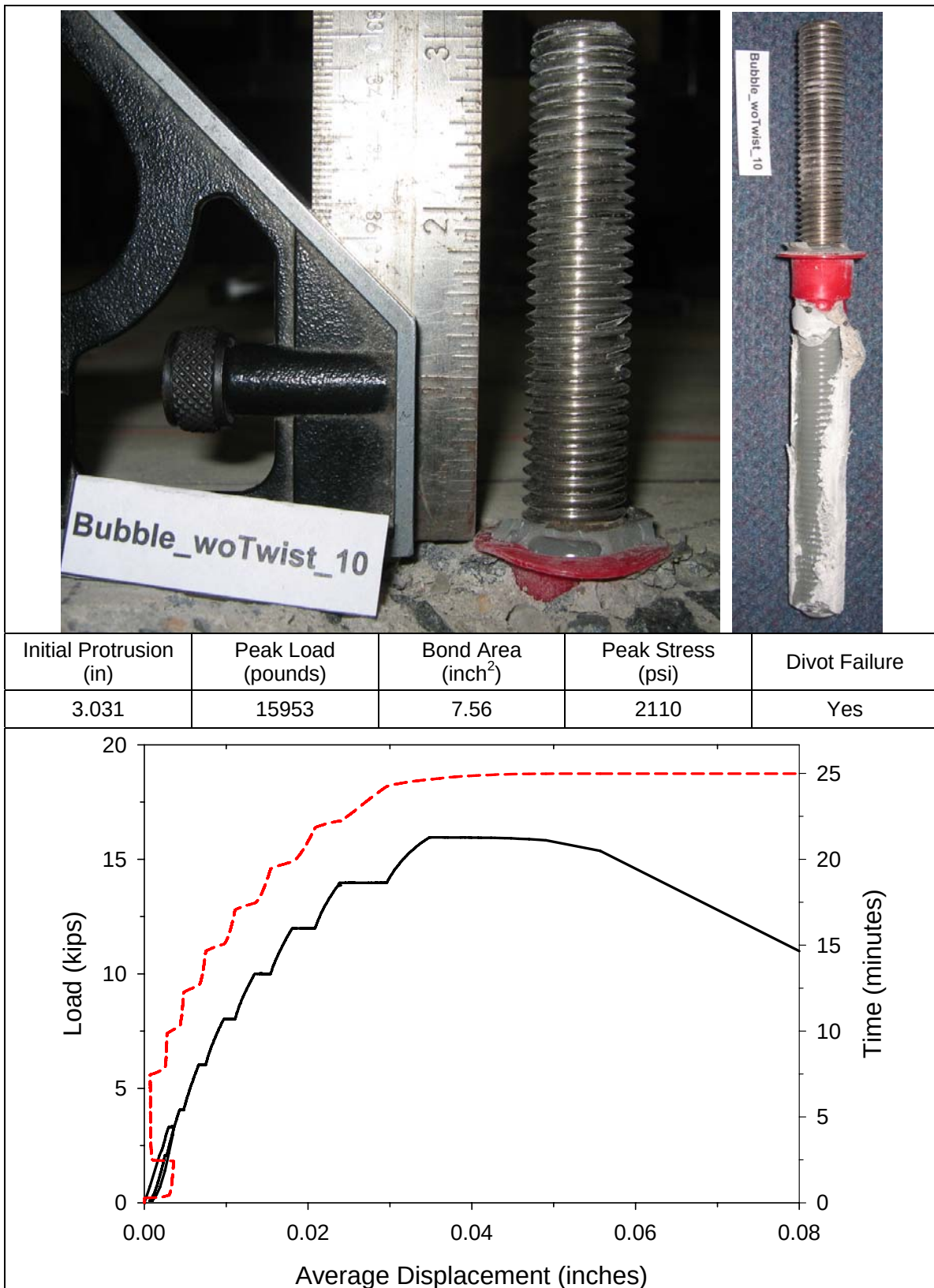


Table 112 Void with Twist Installation 1

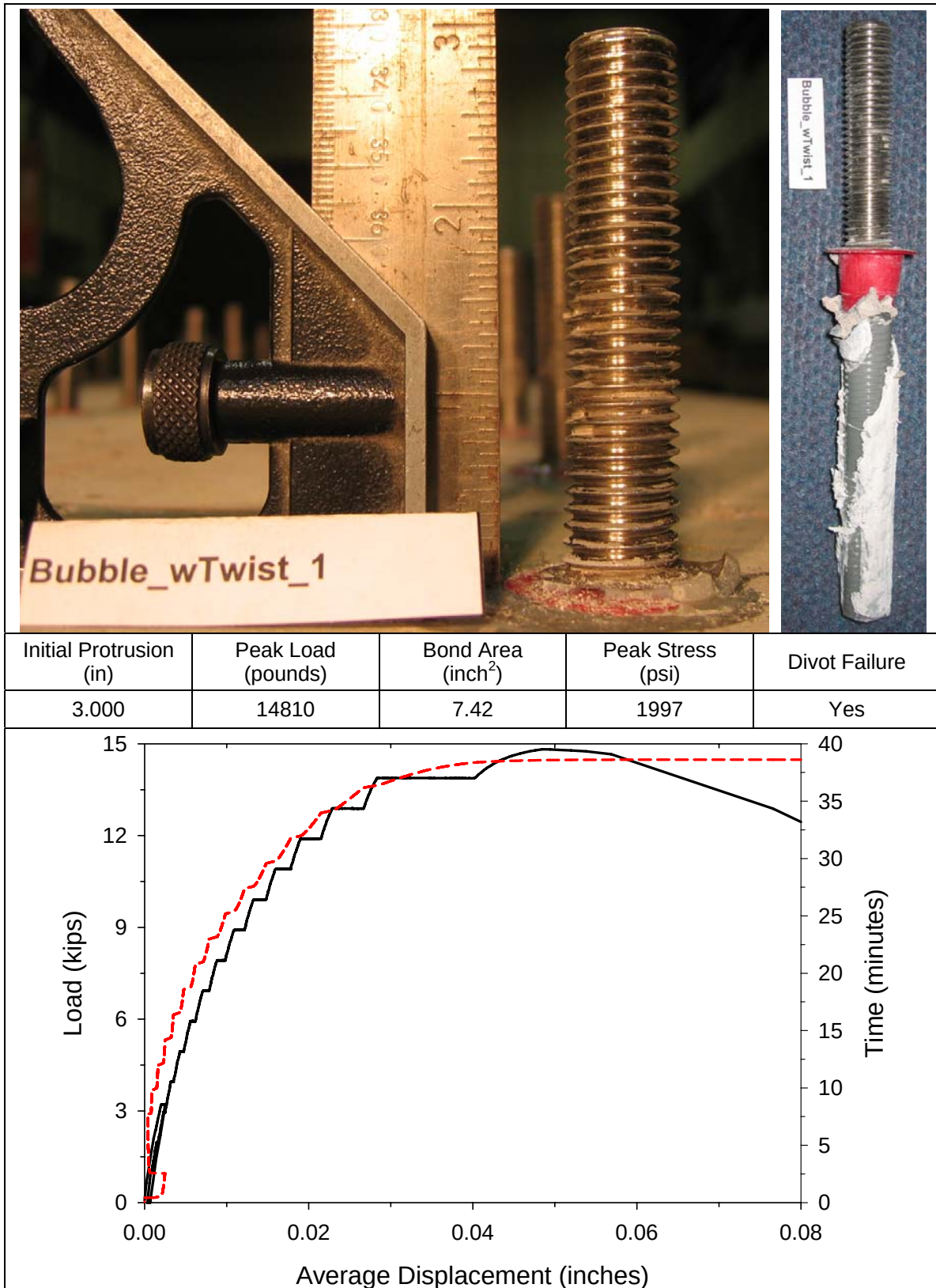


Table 113 Void with Twist Installation 2

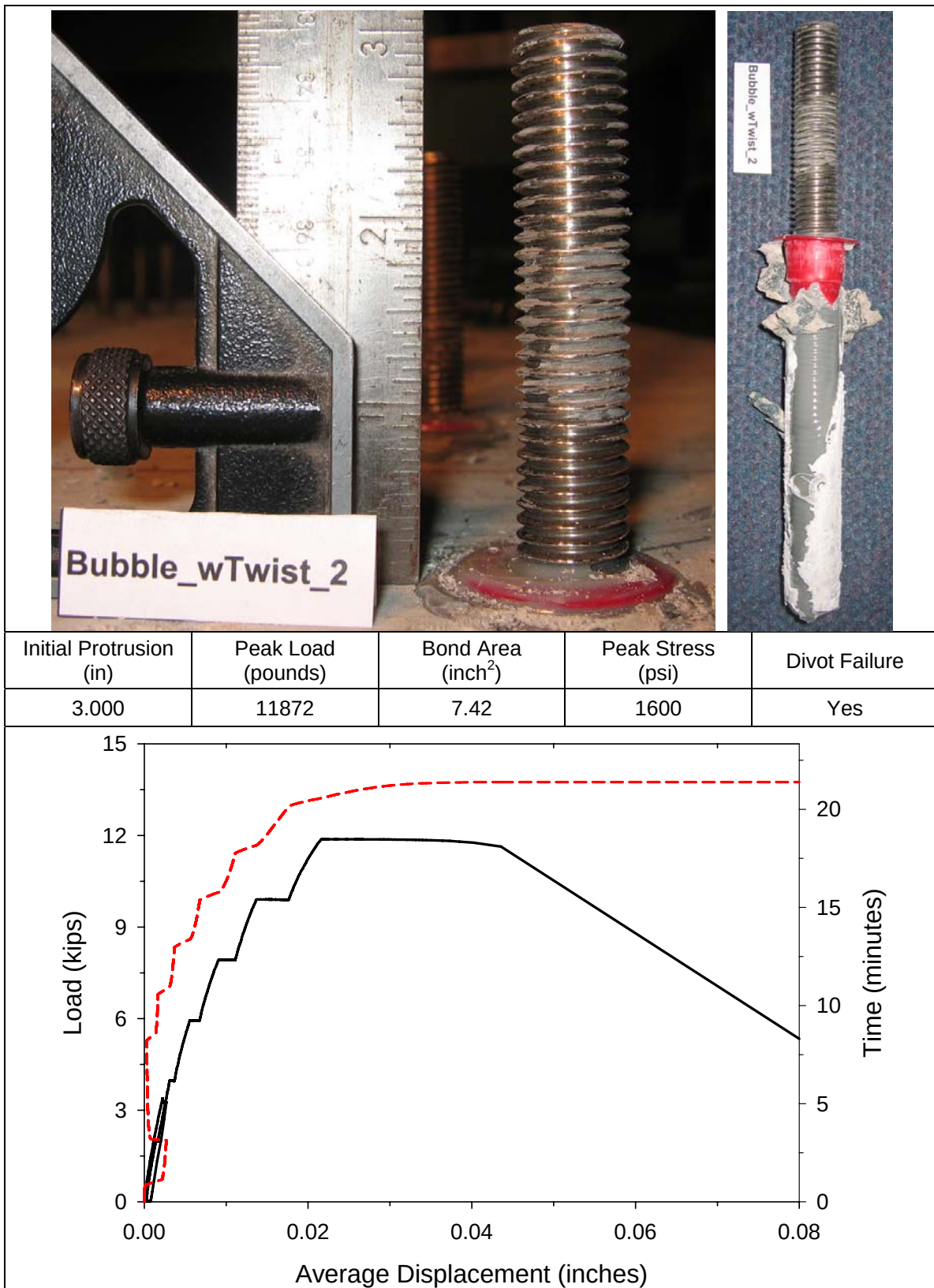


Table 114 Void with Twist Installation 3

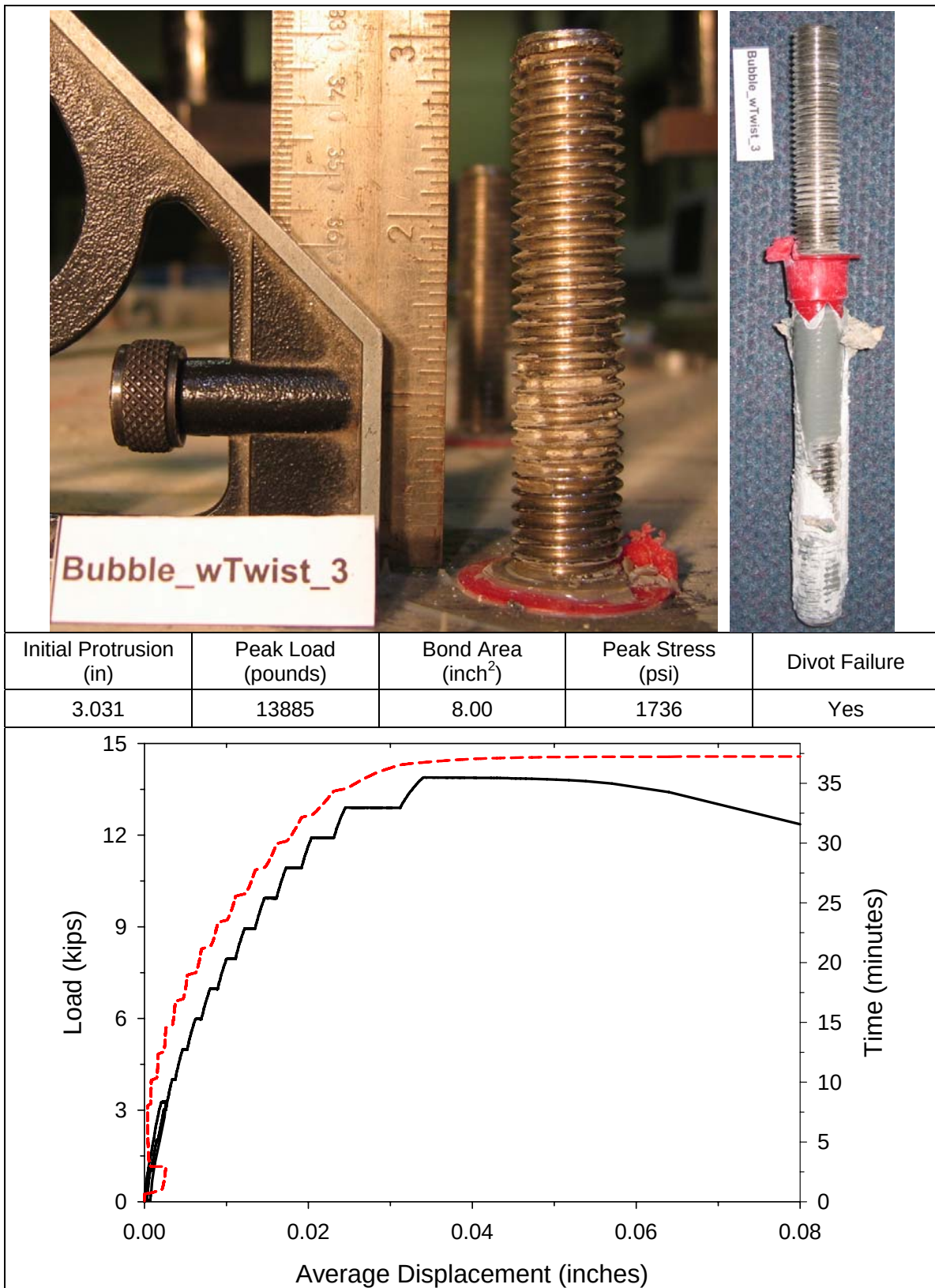


Table 115 Void with Twist Installation 4

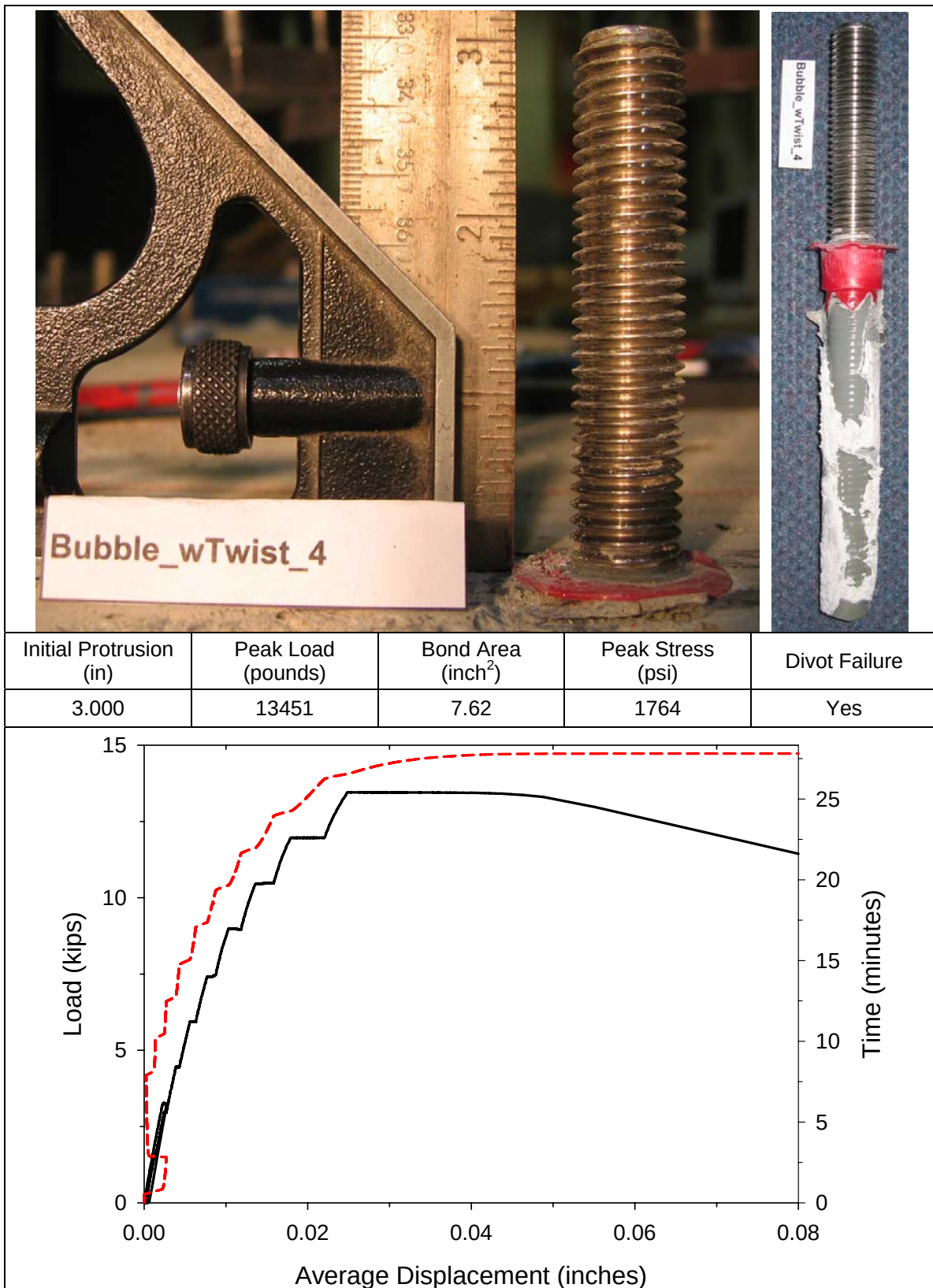


Table 116 Void with Twist Installation 5

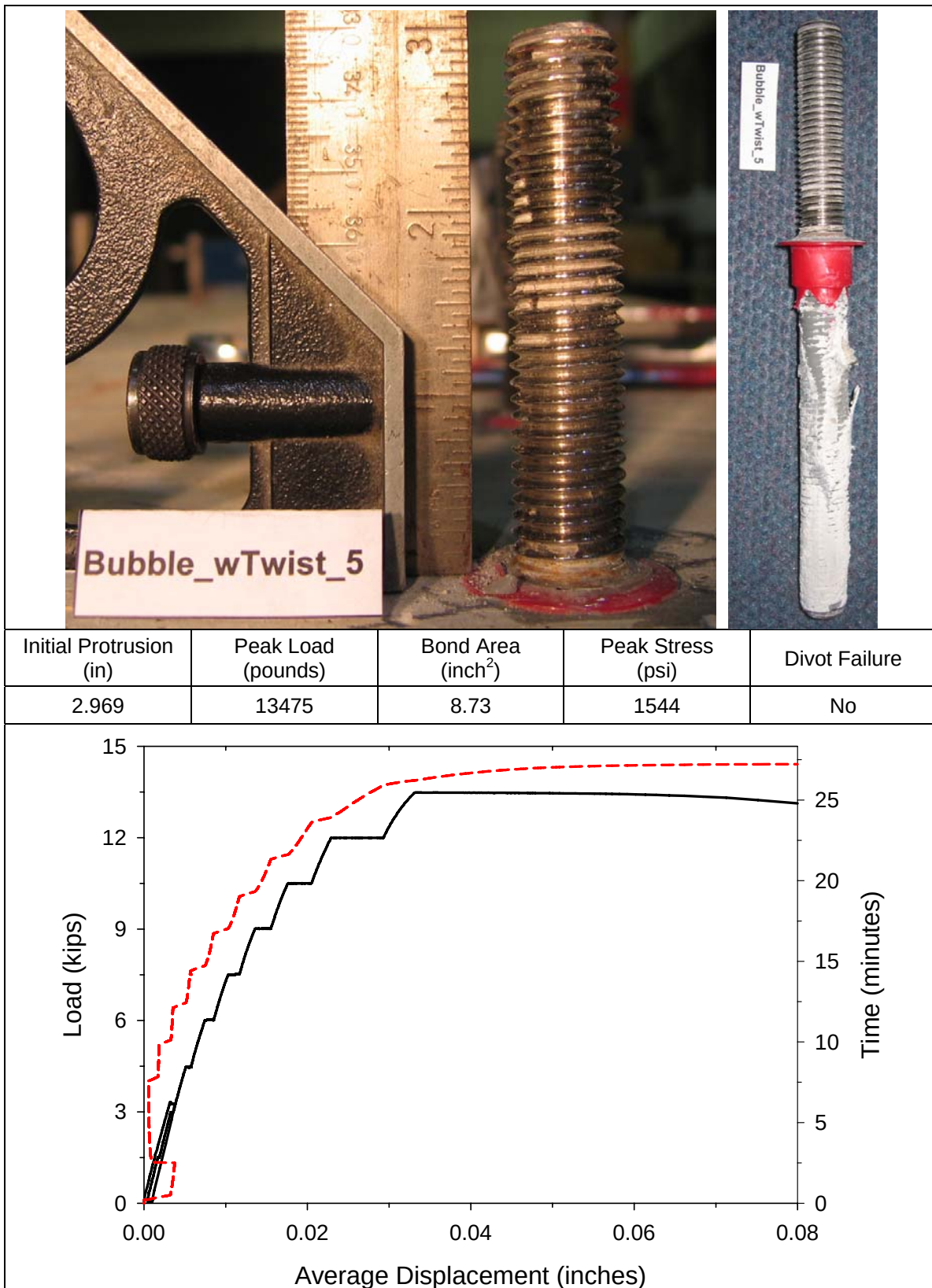


Table 117 Void with Twist Installation 6

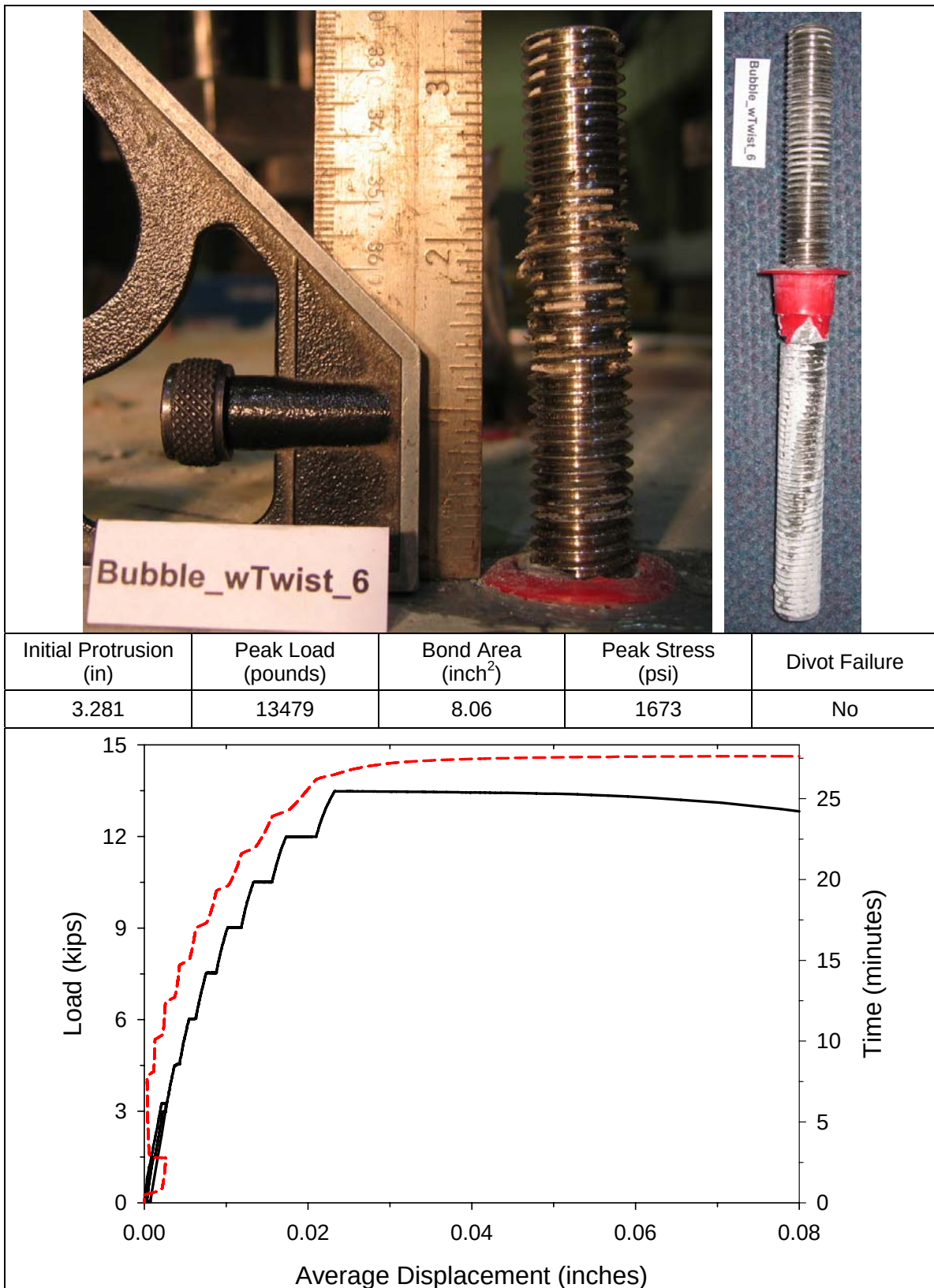


Table 118 Void with Twist Installation 7

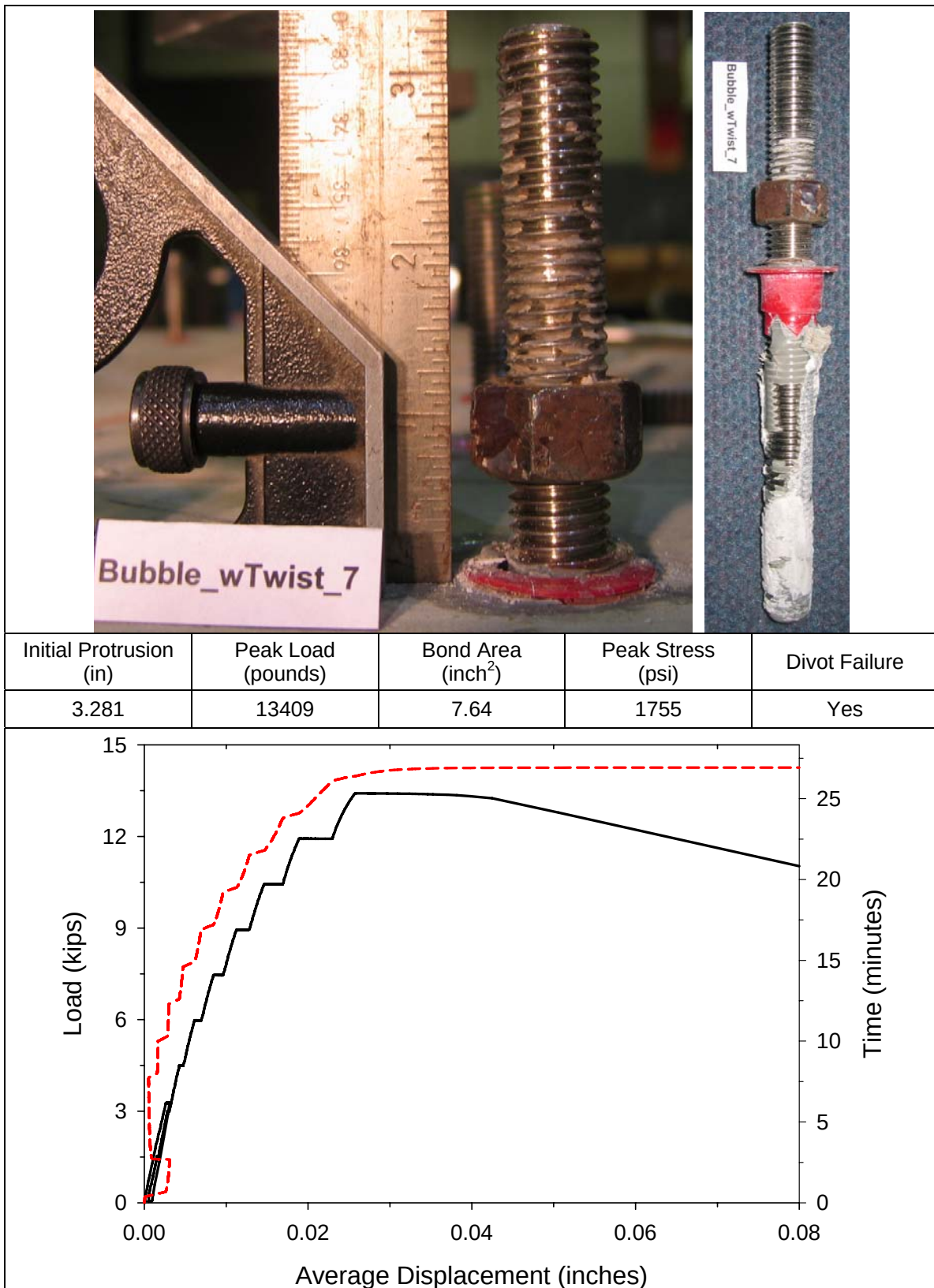


Table 119 Void with Twist Installation 8

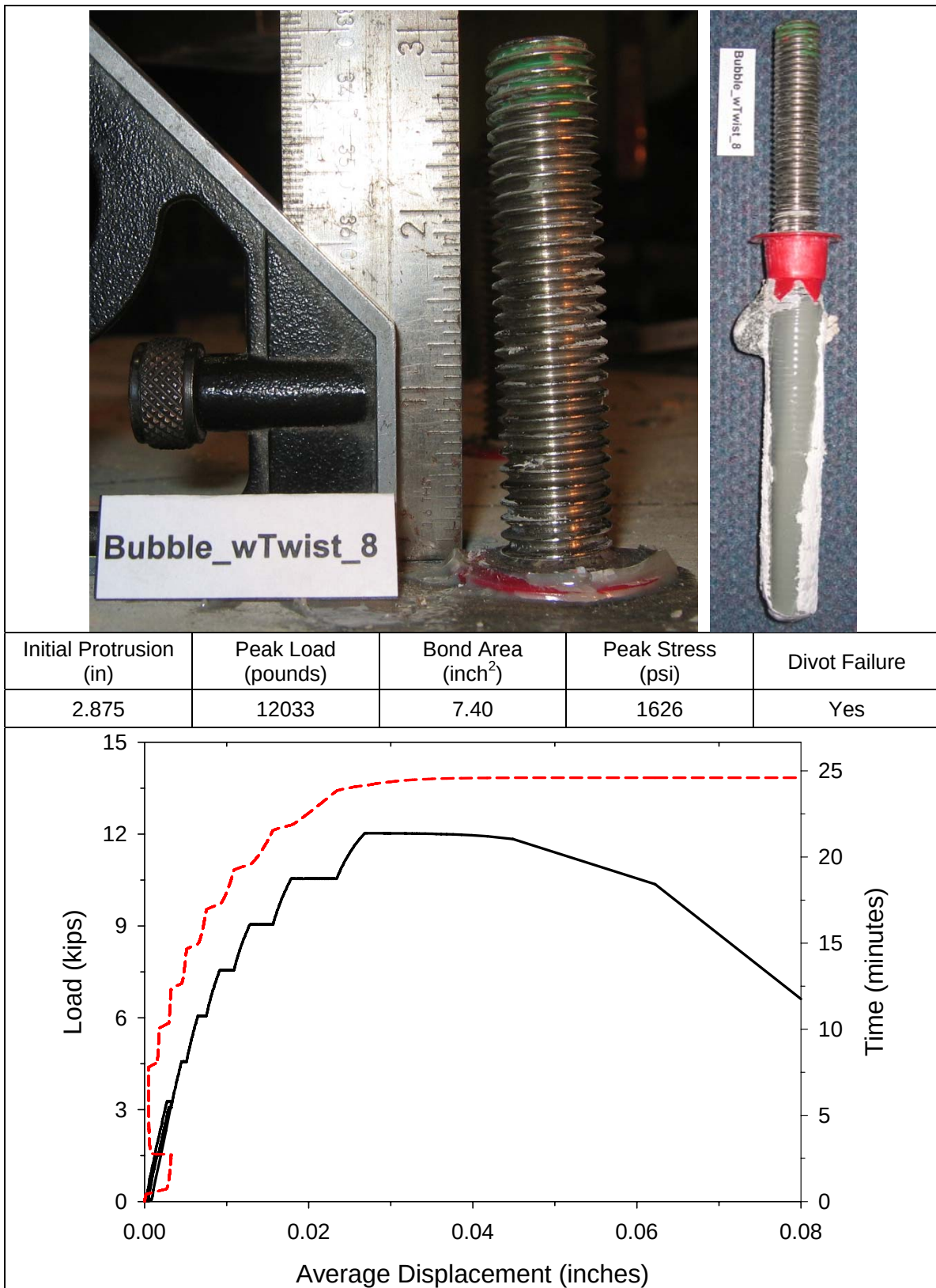


Table 120 Void with Twist Installation 9

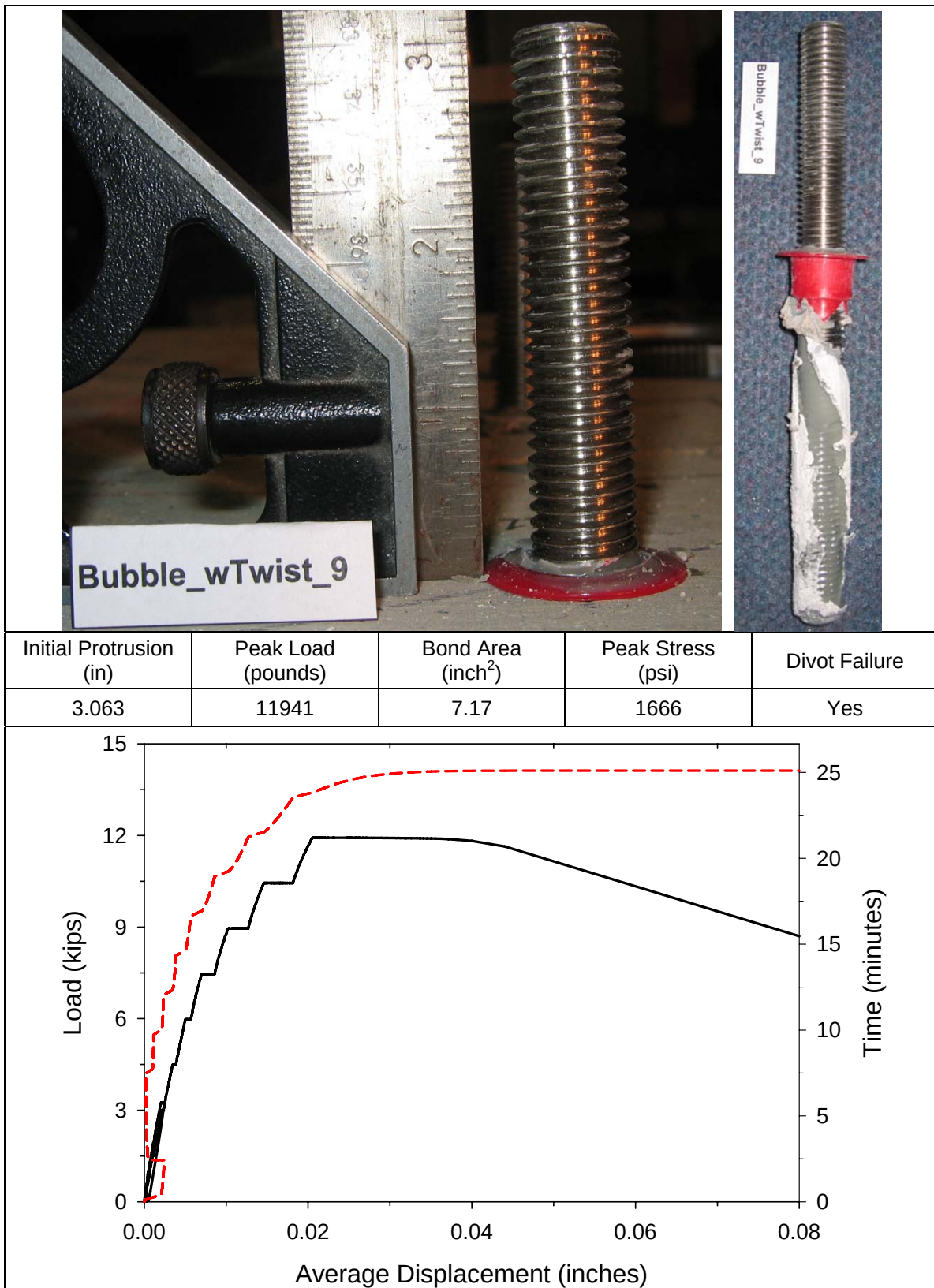


Table 121 Void with Twist Installation 10

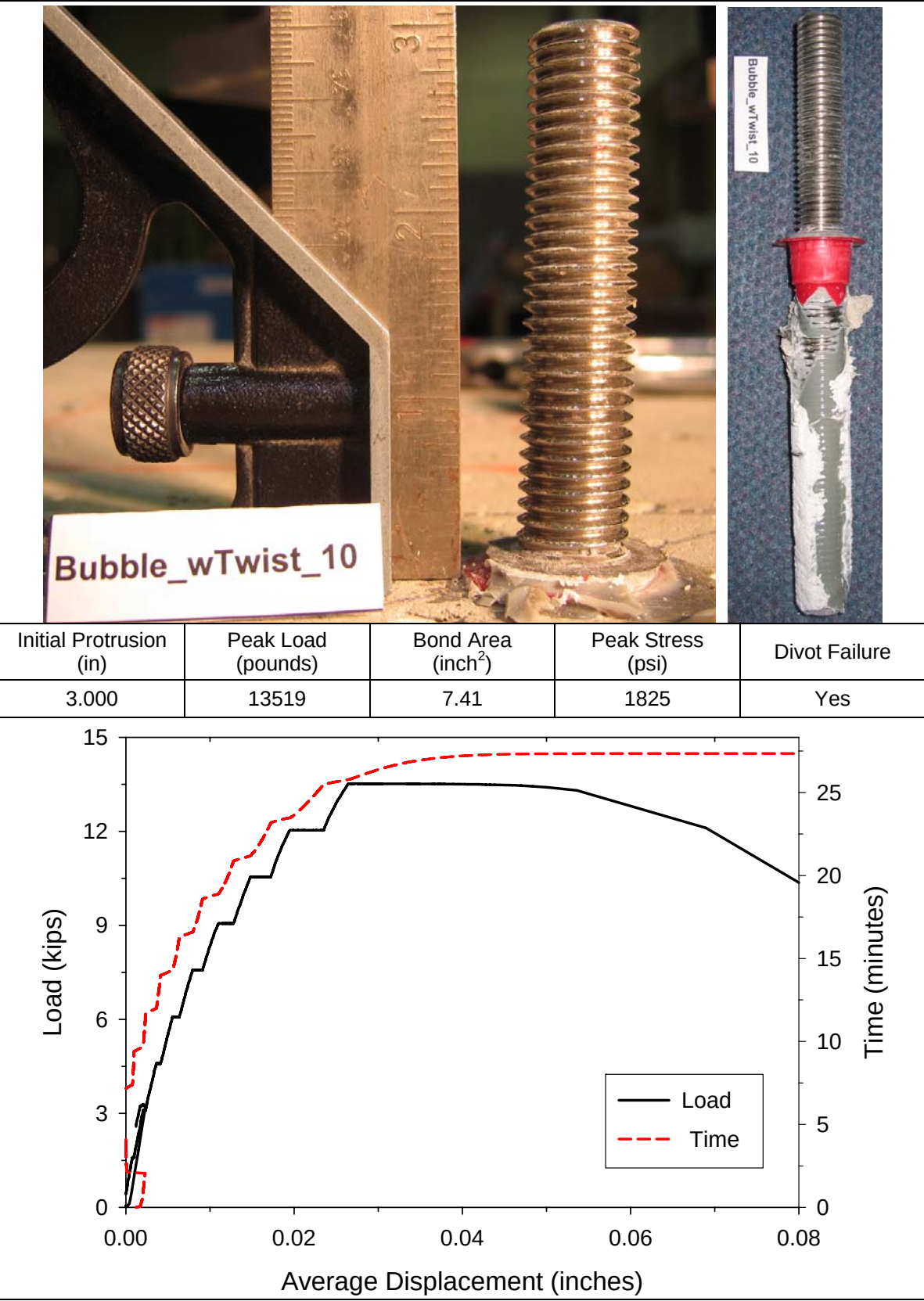


Table 122 Void with Twist and 6300lb Proof 1

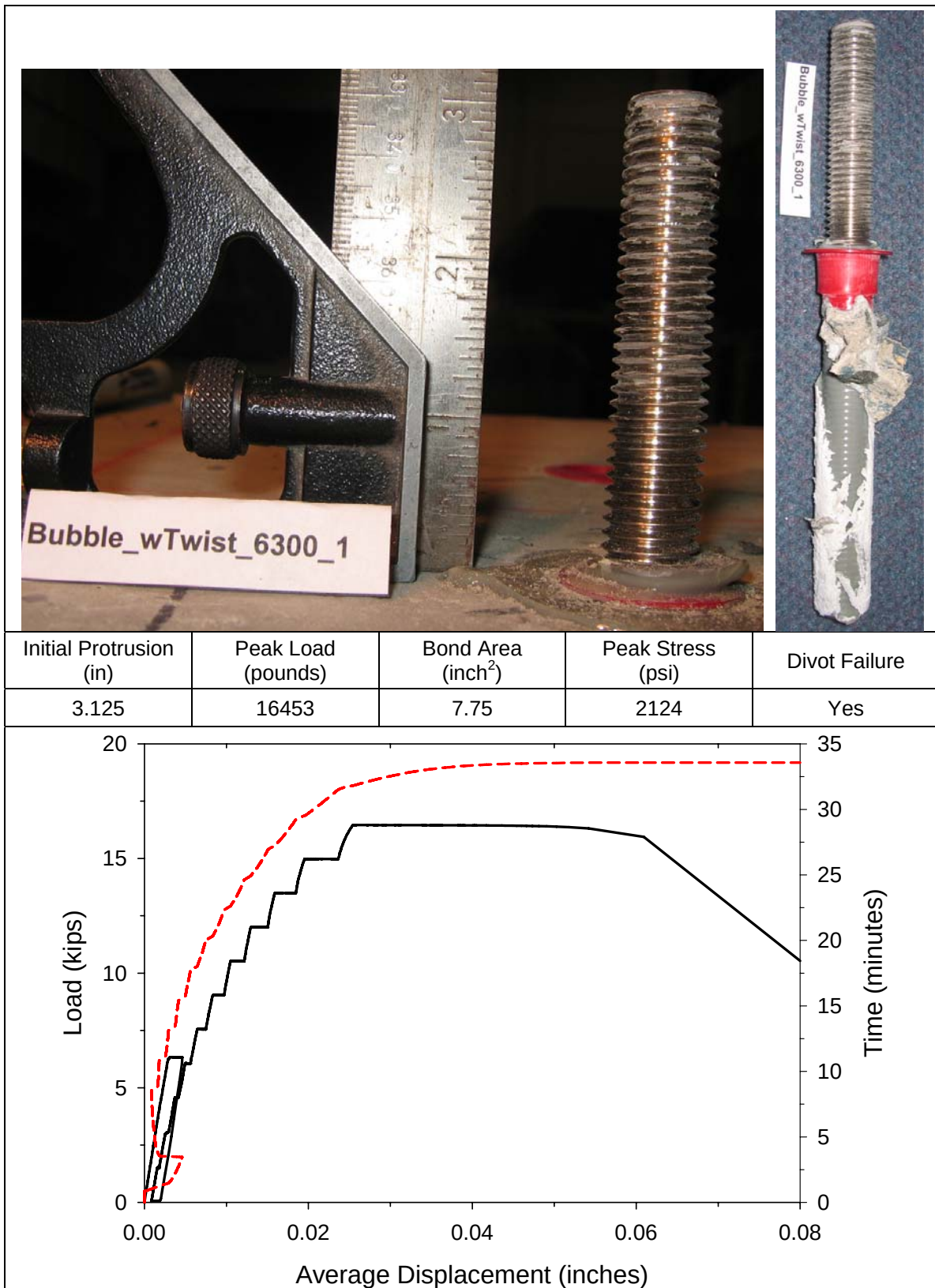


Table 123 Void with Twist and 6300lb Proof 2



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.938	10510	7.97	1319	No

This was accidentally overtorqued trying to remove the nut used during the anchor installation and freely spun within its hole. The anchor was thus considered damaged. Since the nut could not be removed, the LVDT bracket also not be used therefore displacements were not collected.

Table 124 Void with Twist and 6300lb Proof 3

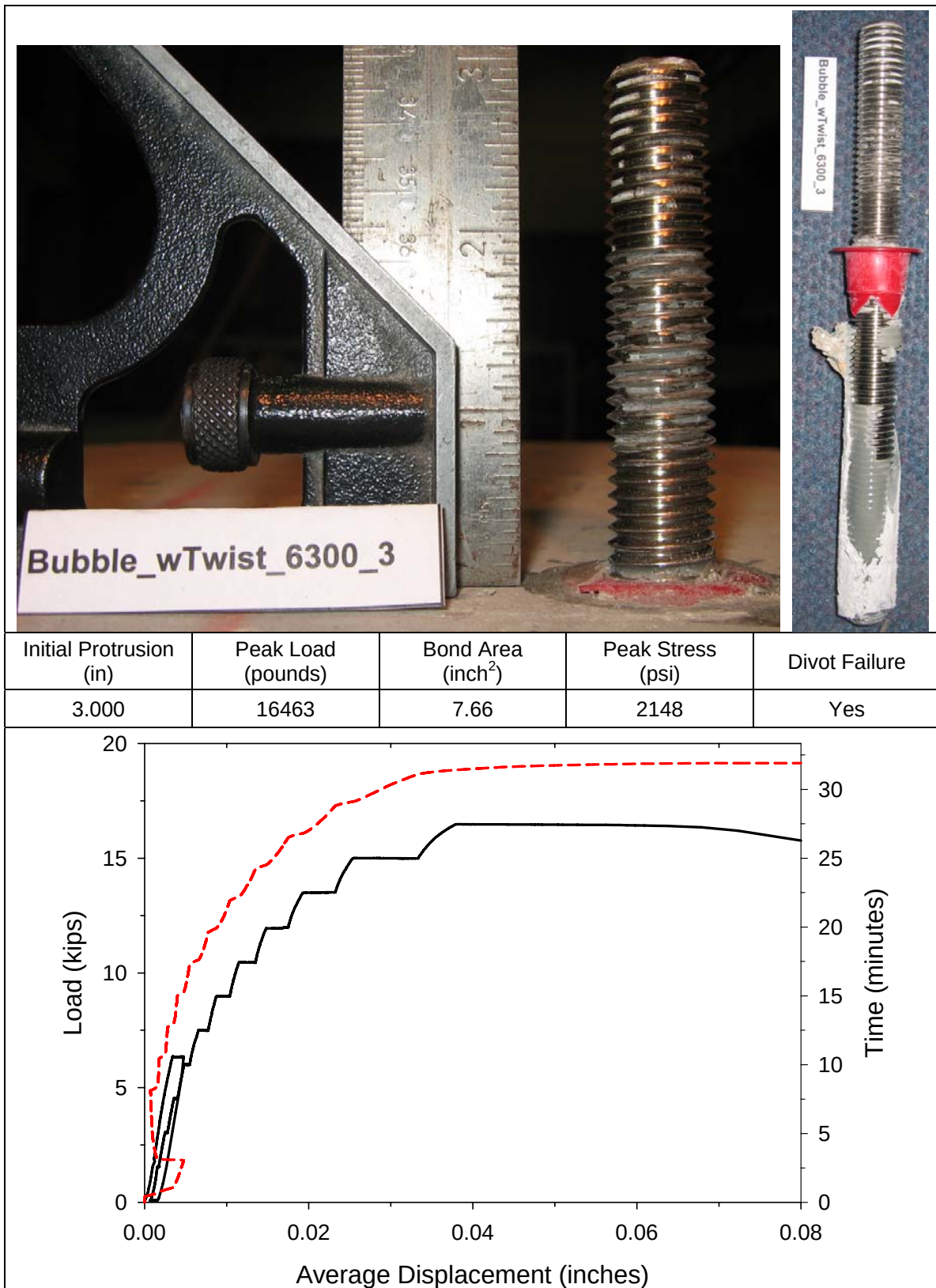


Table 125 Void with Twist and 6300lb Proof 4

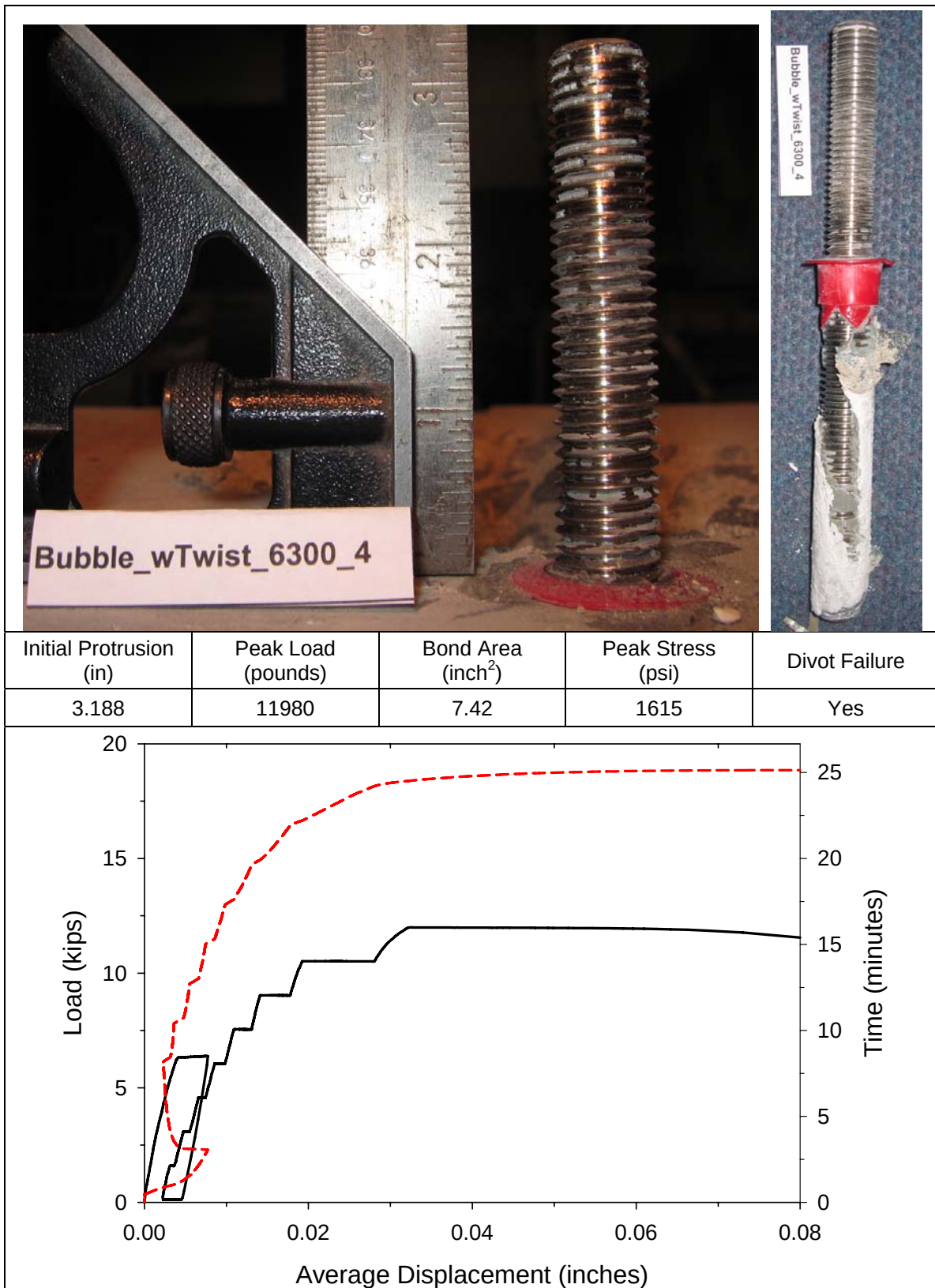


Table 126 Void with Twist and 6300lb Proof 5

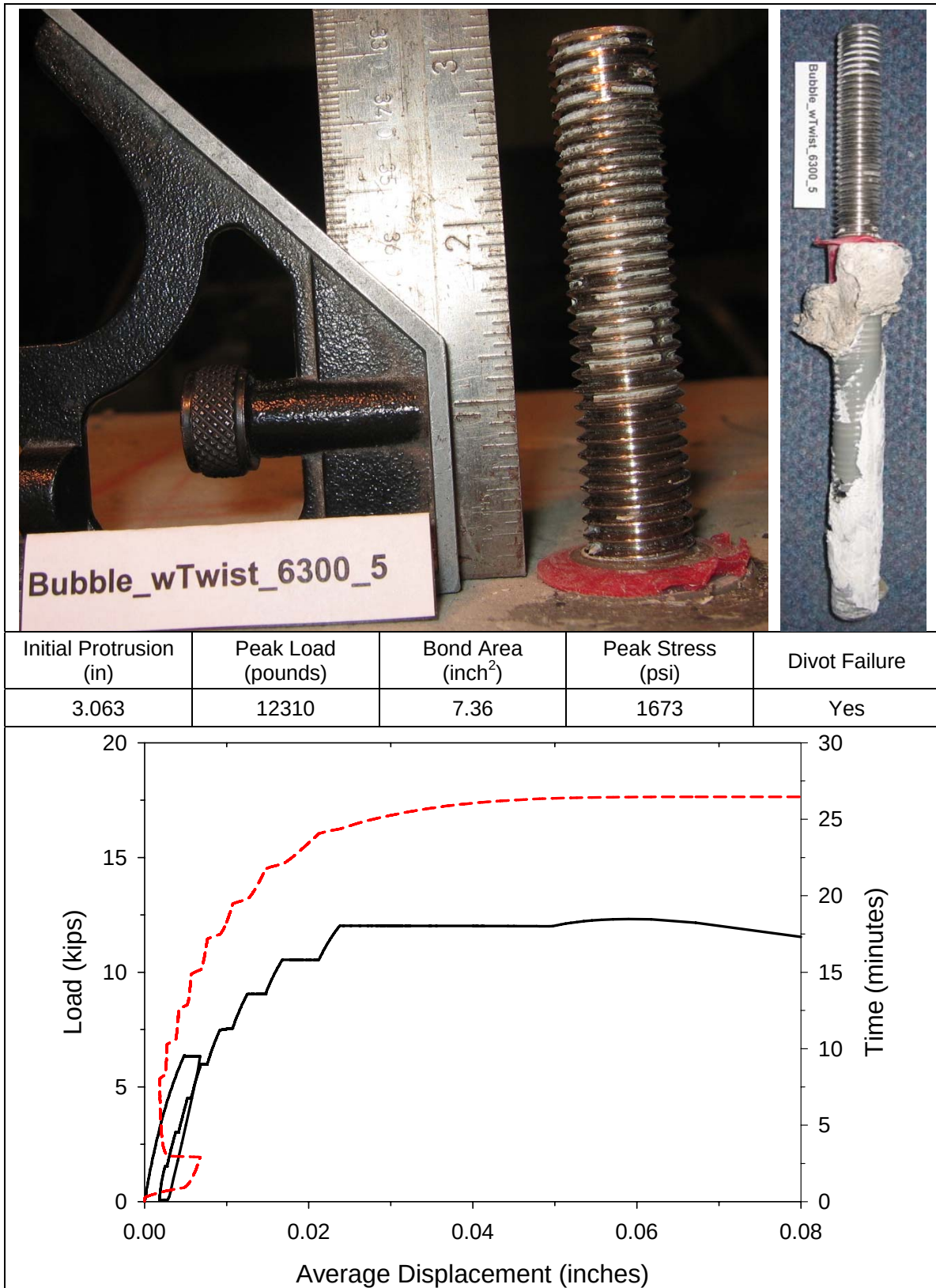


Table 127 Through #11 Rebar 1

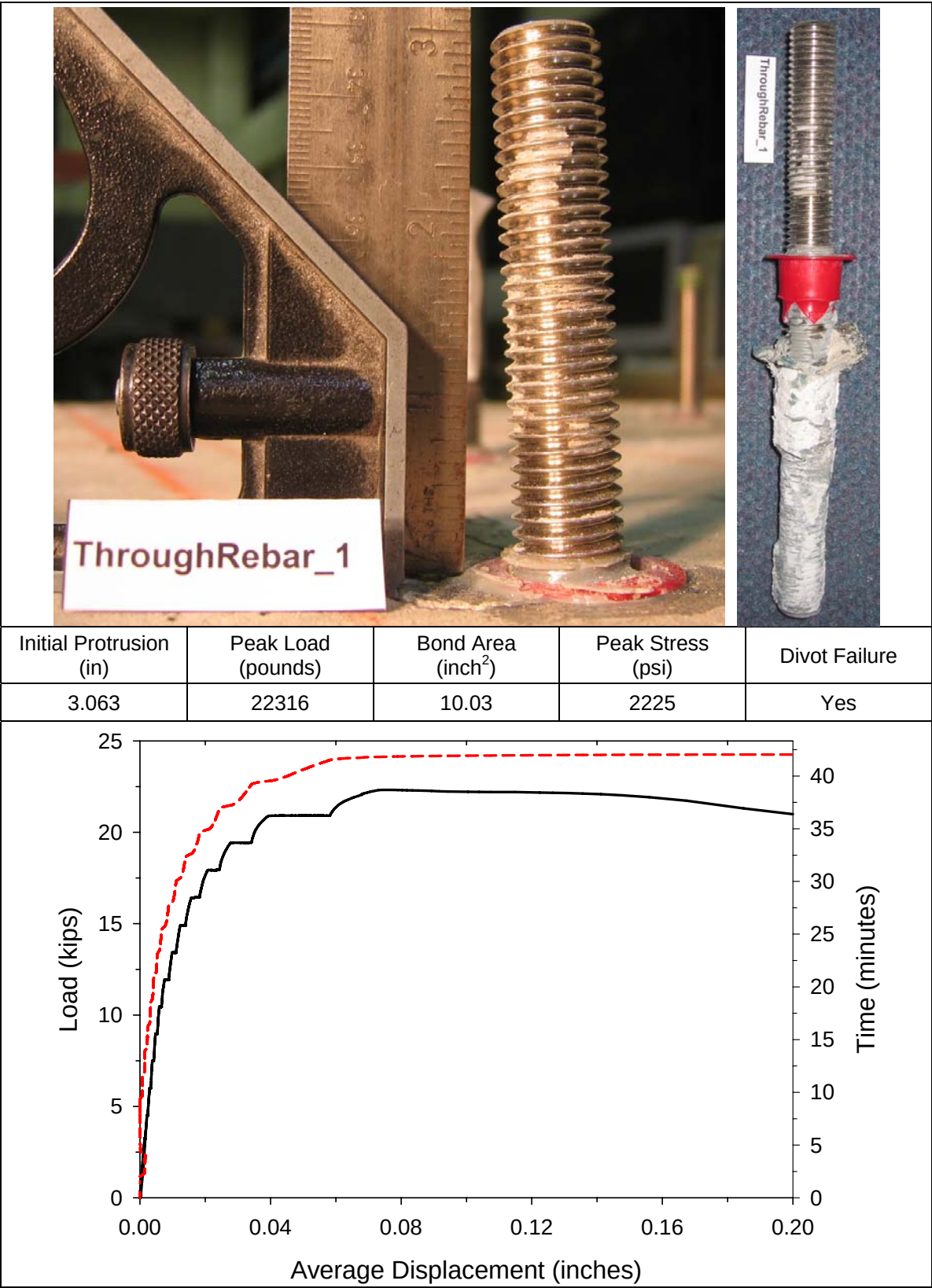


Table 128 Through #11 Rebar 2

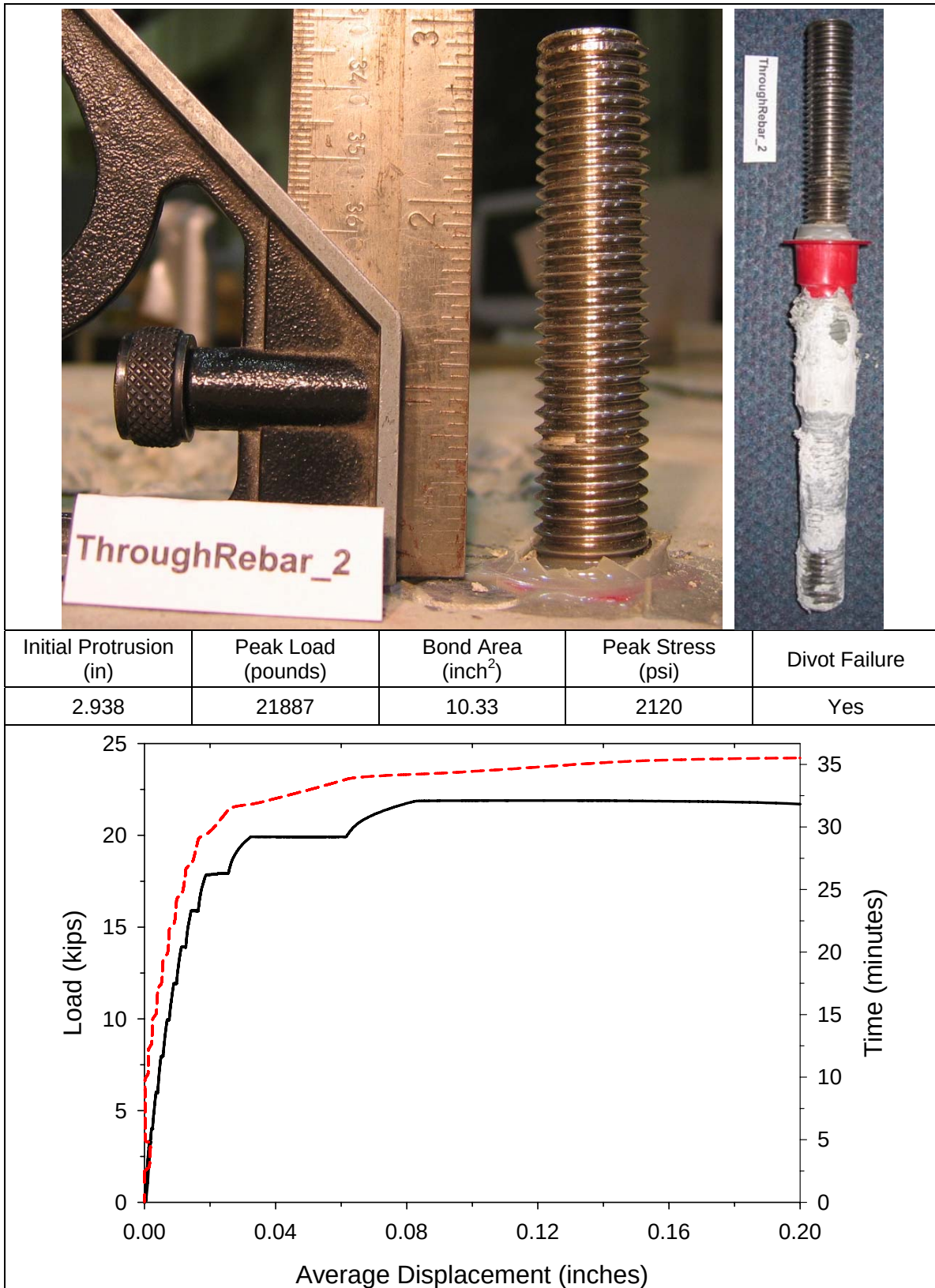


Table 129 Through #11 Rebar 3

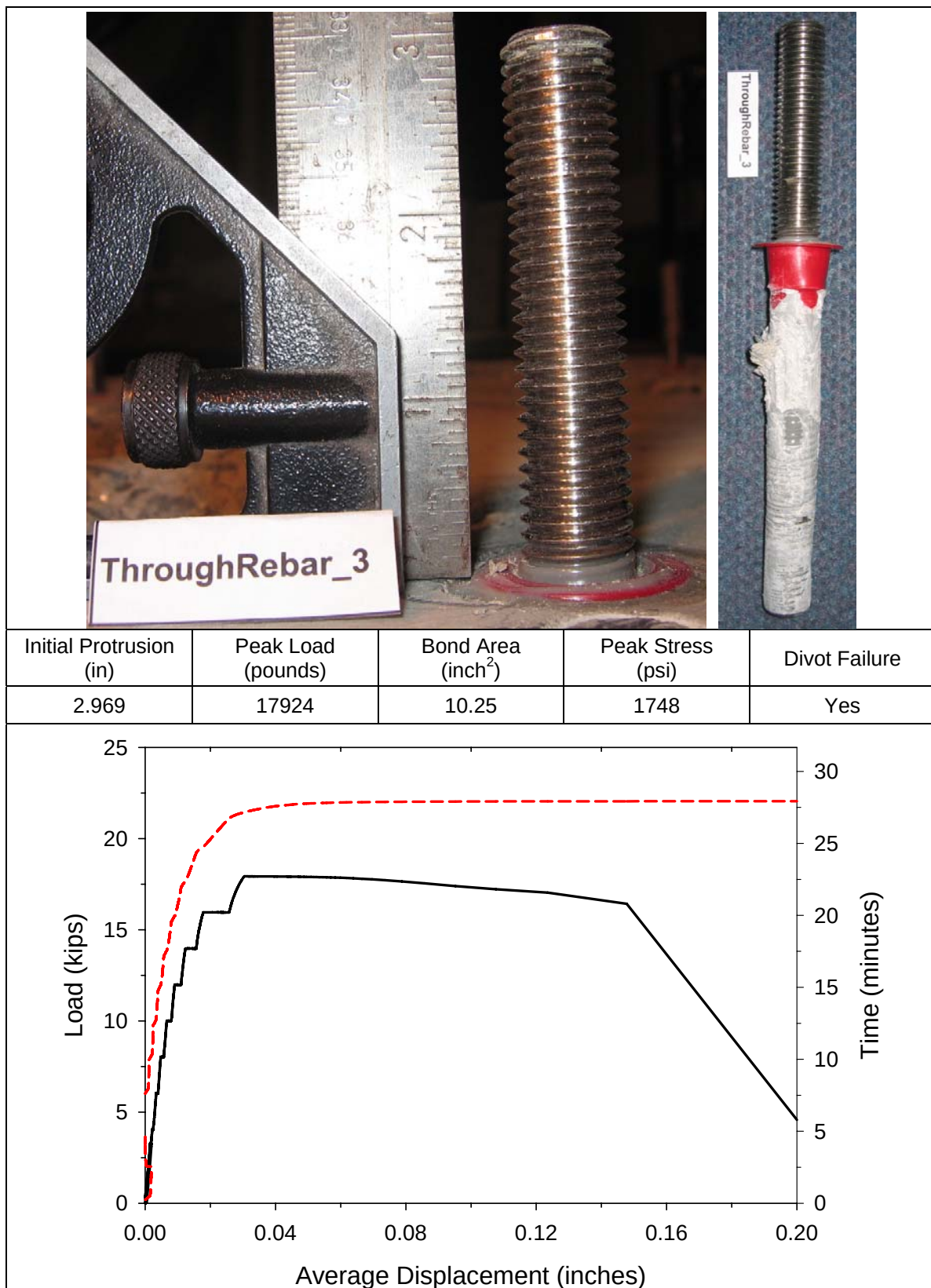


Table 130 Through #11 Rebar 4

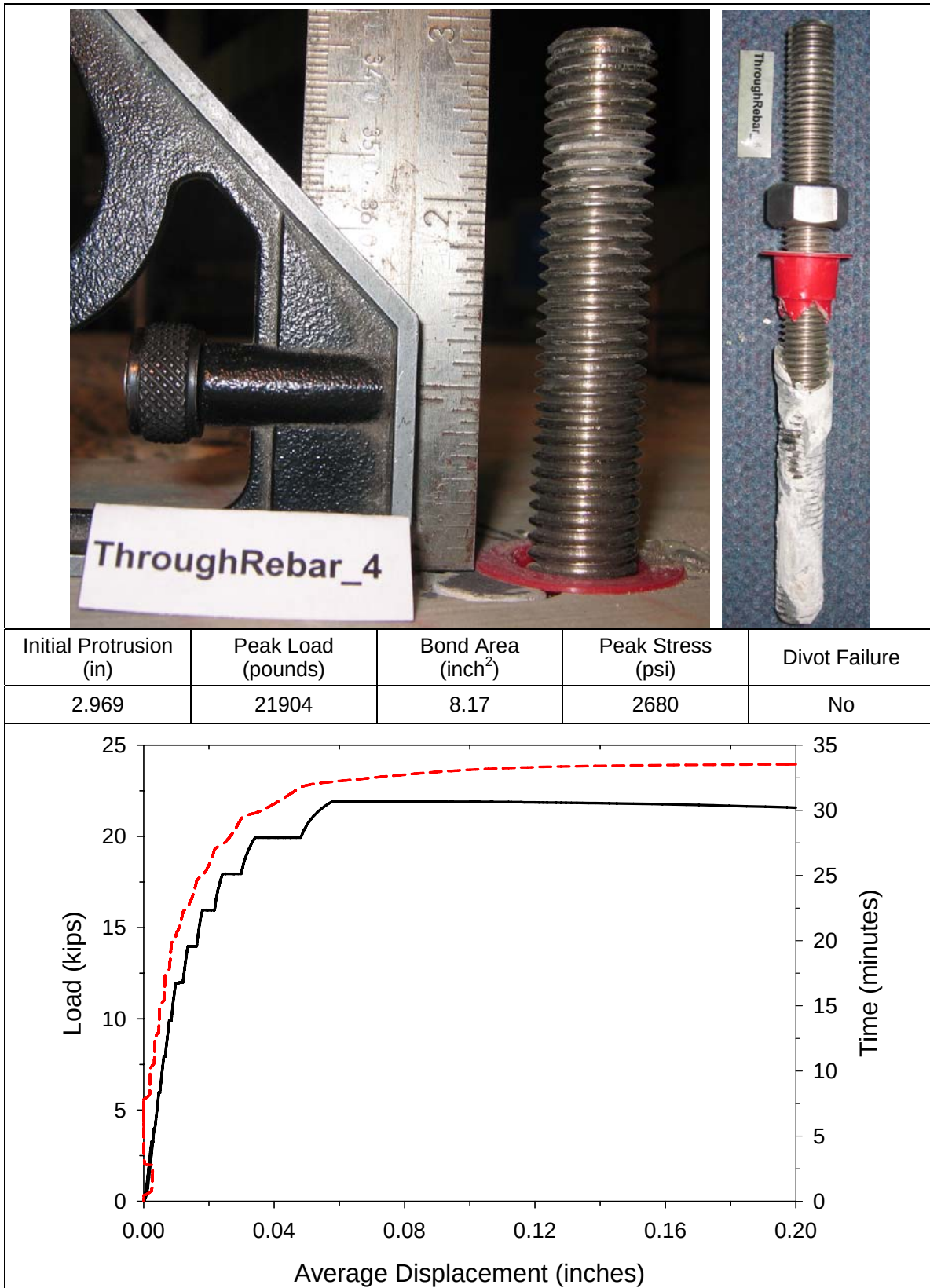


Table 131 Through #11 Rebar 5

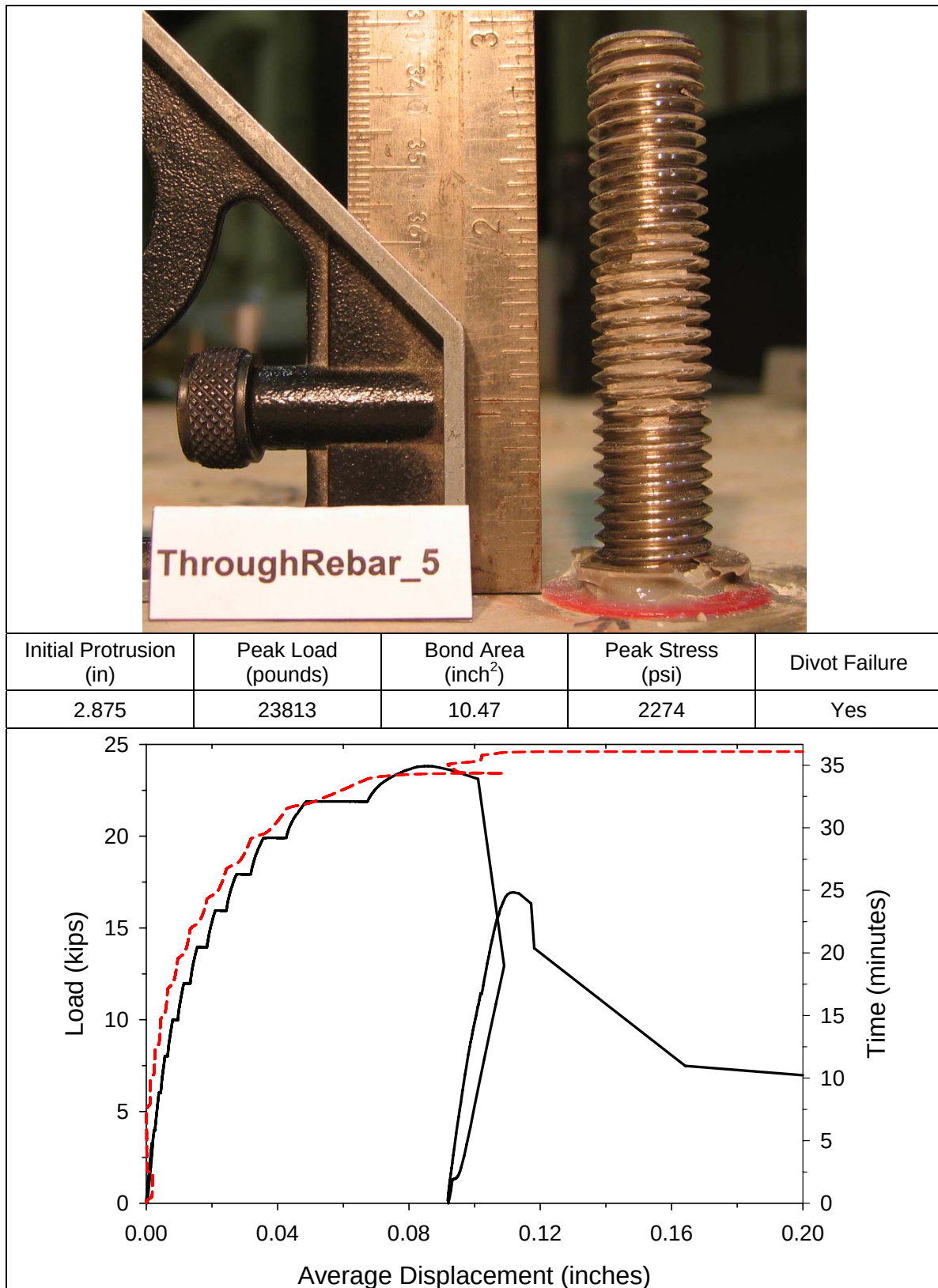


Table 132 Torque Pull 1

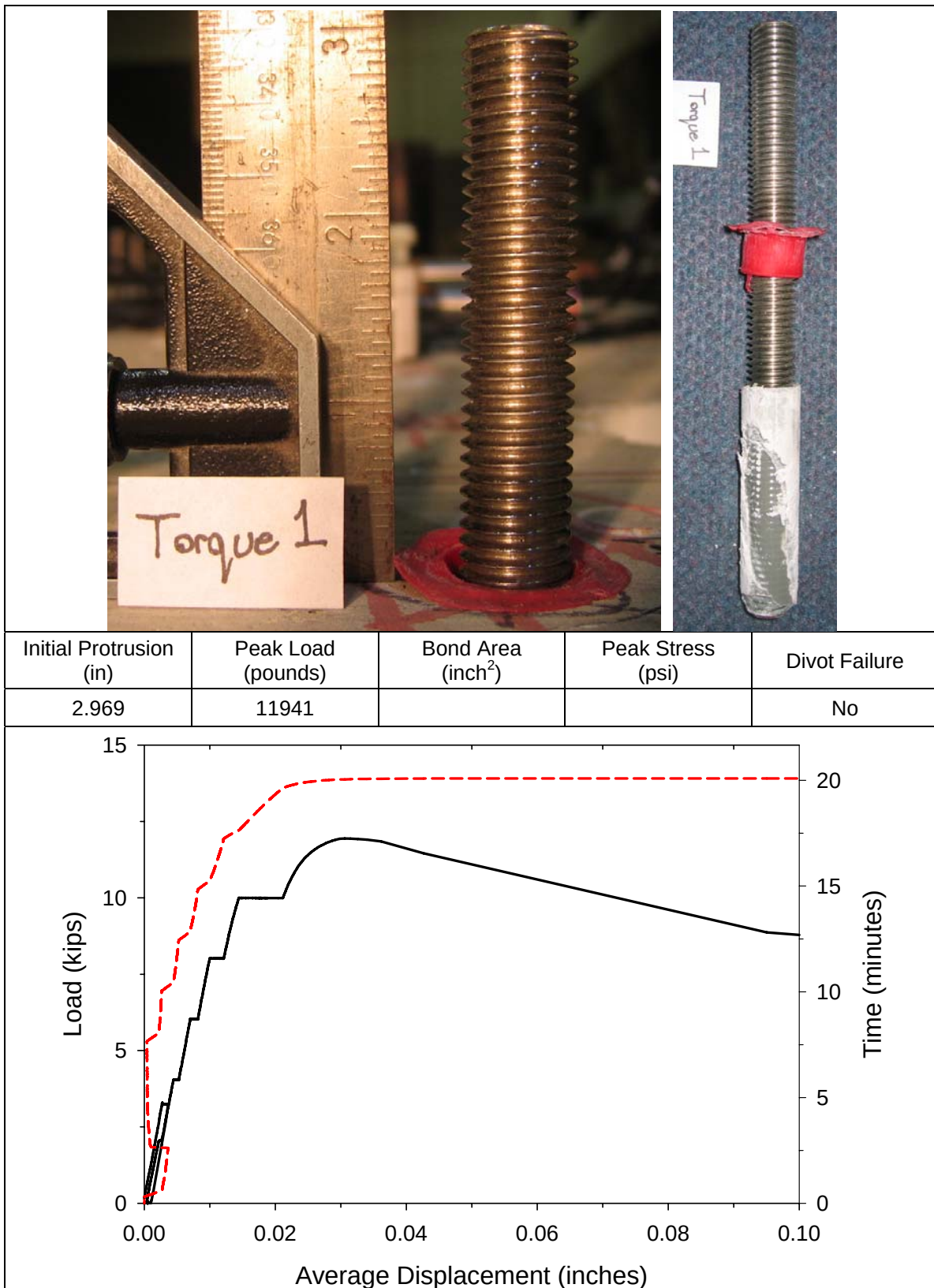


Table 133 Torque Pull 2

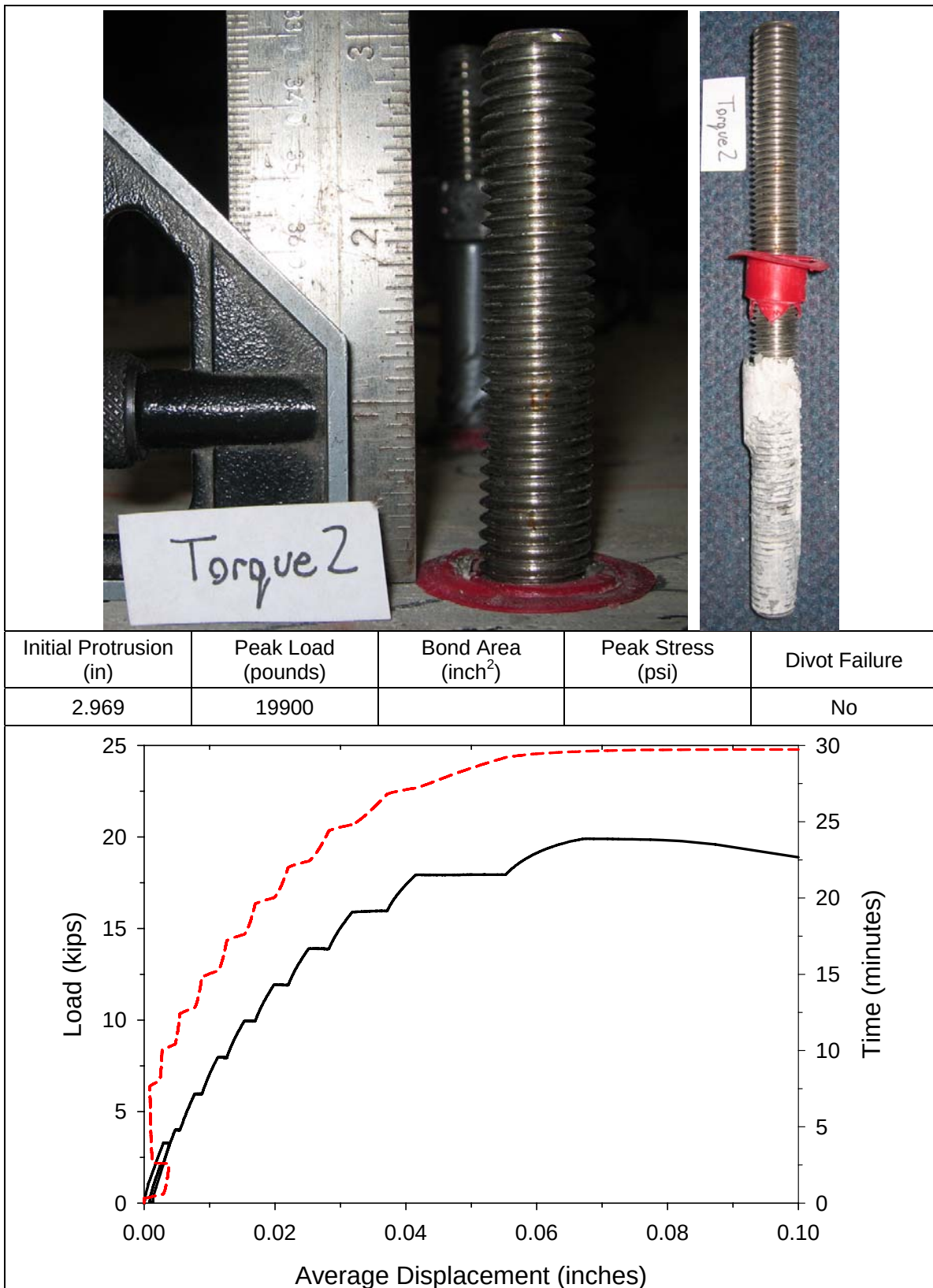


Table 134 Torque Pull 3

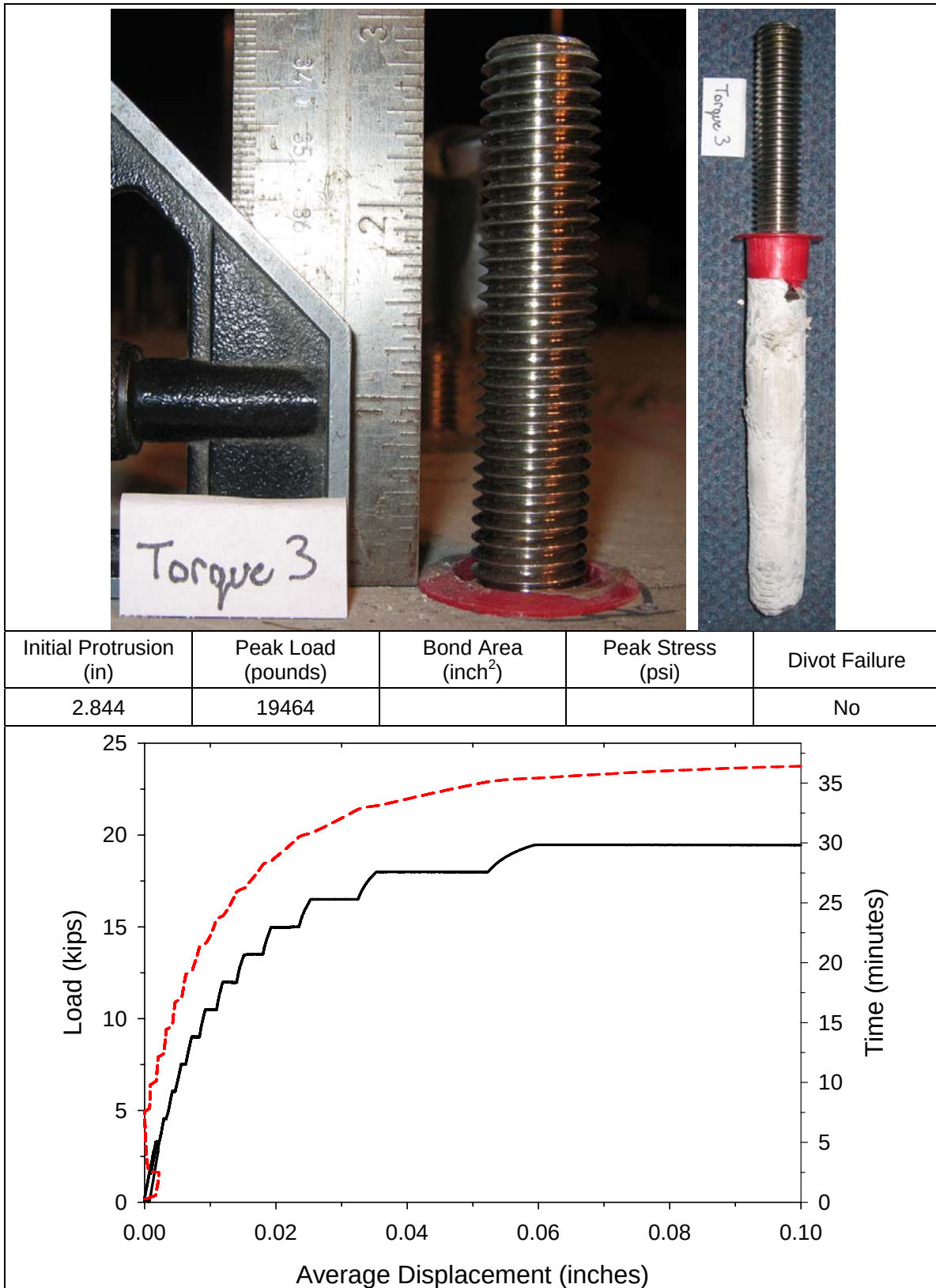
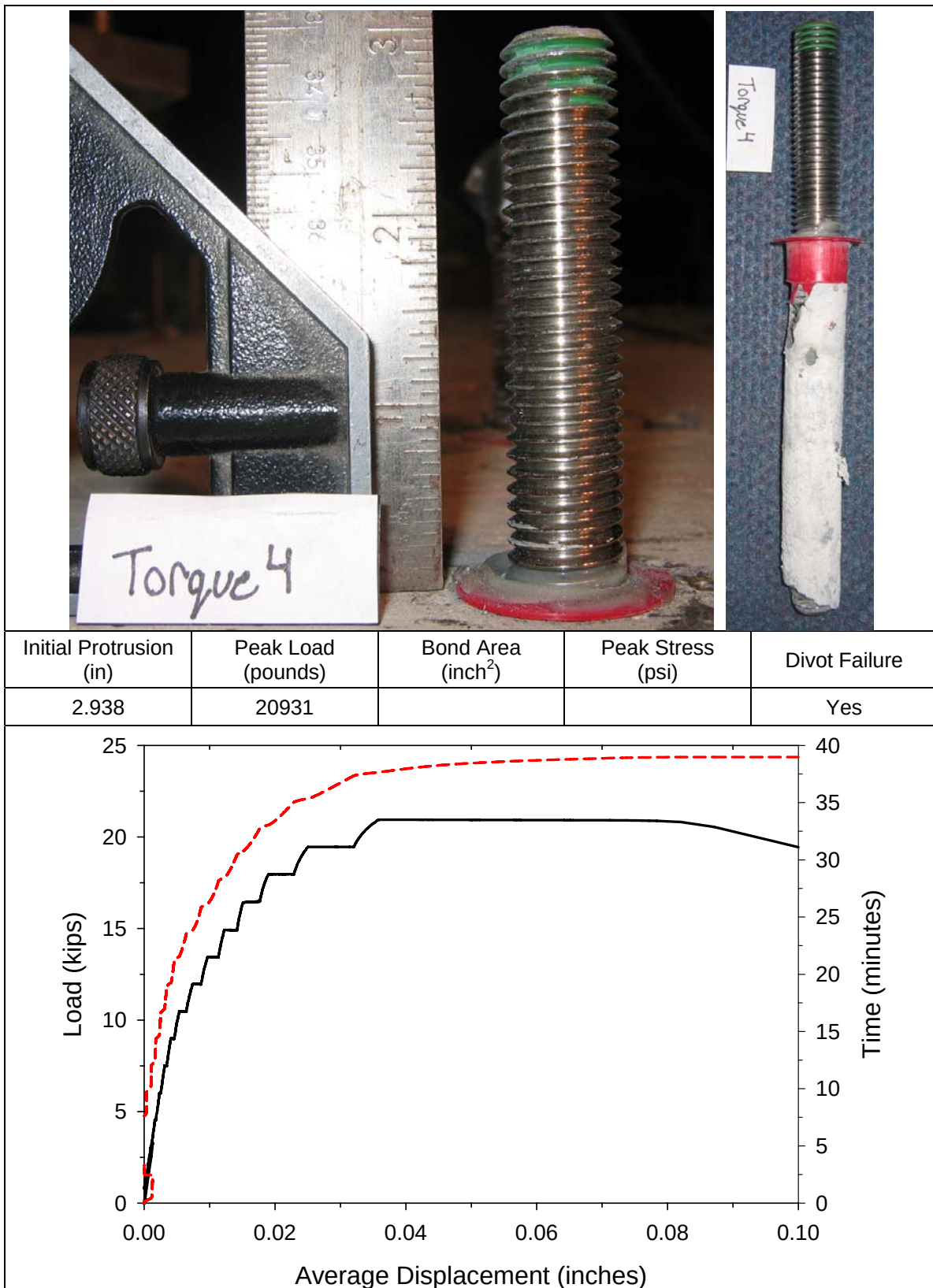


Table 135 Torque Pull 4



APPENDIX 2

In the time between testing and report writing, the names of the parameters either changed or were streamlined. In this appendix, pre- and post-pull pictures were taken of each anchor but the labels in the pictures do not match what is used with the test of this report. The following table serves as a cross-reference between the picture label and how it was referred to in the report text.

Installation Parameter	Picture Label in Appendix 2 Tables
SET22 Epoxy	SET22
Damp Dirty Hole	DampDirty
Torqued to 50 ft-lbs	50 Ft-Lbs
Torqued to 90 ft-lbs	90 Ft-Lbs
Torqued to 125 ft-lbs	125 Ft-Lbs
3 white : 2 gray	3wht_2gray
2 white : 3 gray	2wht_3gray
Baked Epoxy	BakedEpoxy
Incompletely Mixed Epoxy	PoorlyMixed
Worst Case Install	WorstCaseInstall

Table 136 SET22 Epoxy 1

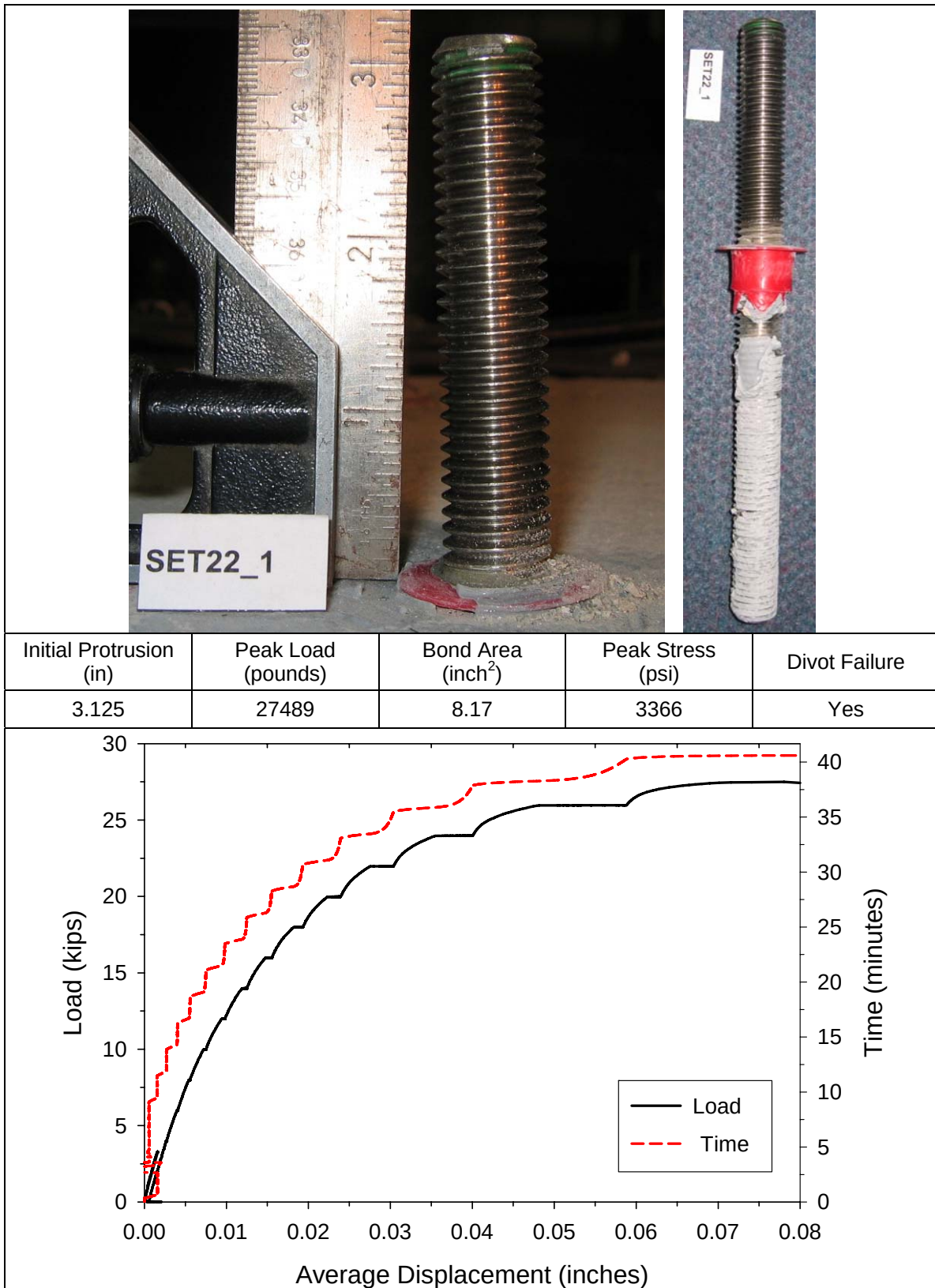


Table 137 SET22 Epoxy 2

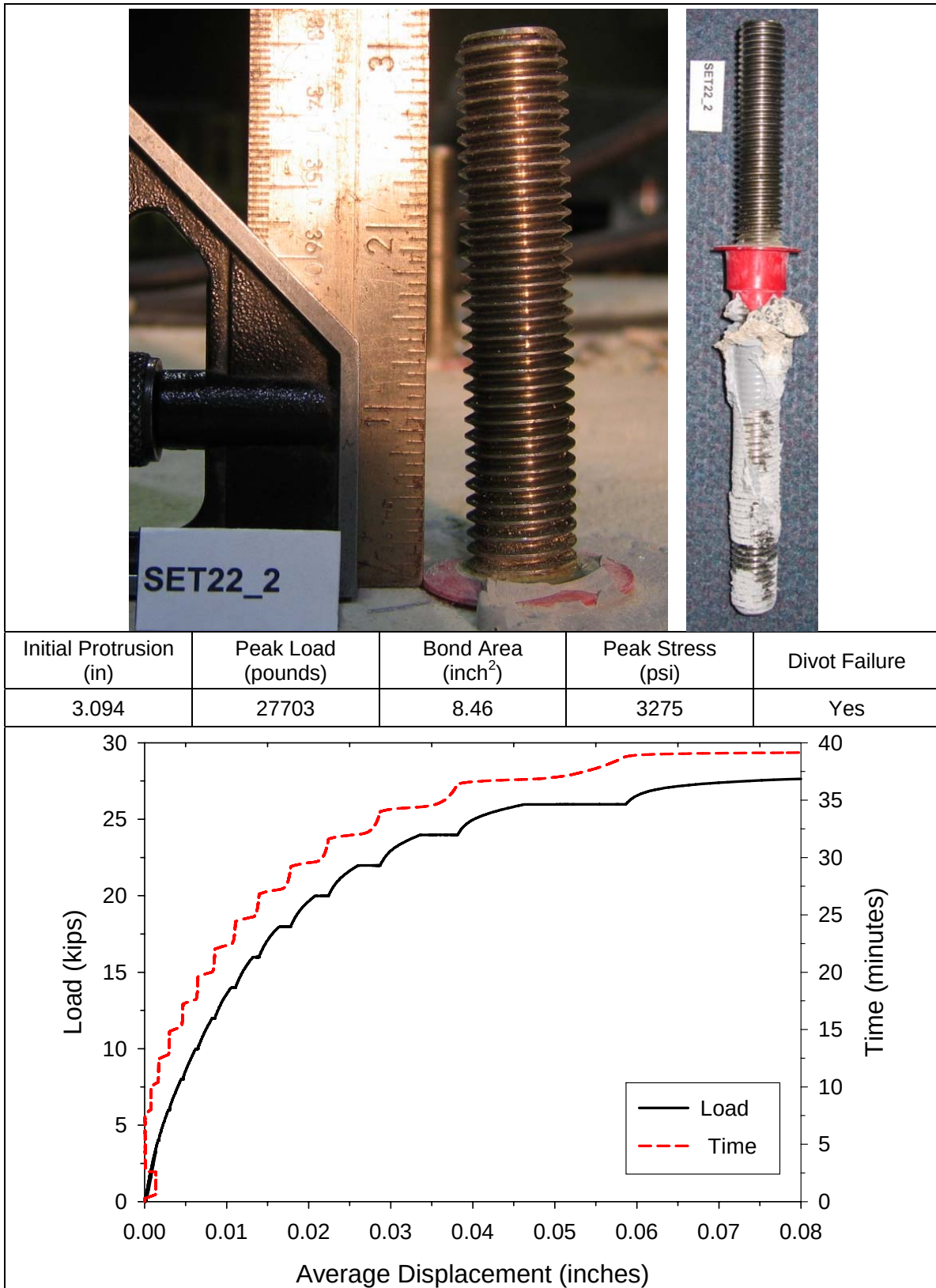


Table 138 SET22 Epoxy 3

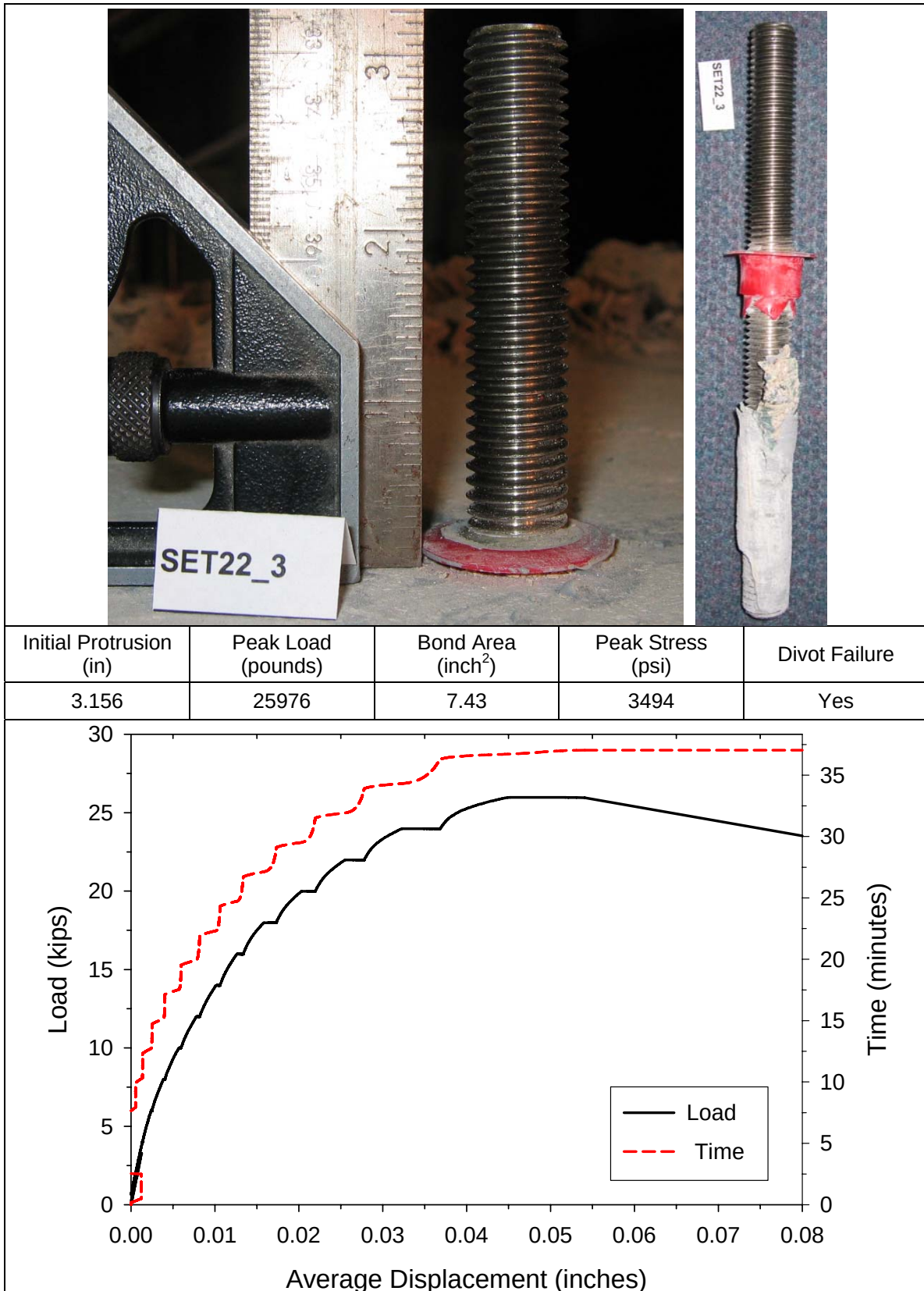


Table 139 SET22 Epoxy 4

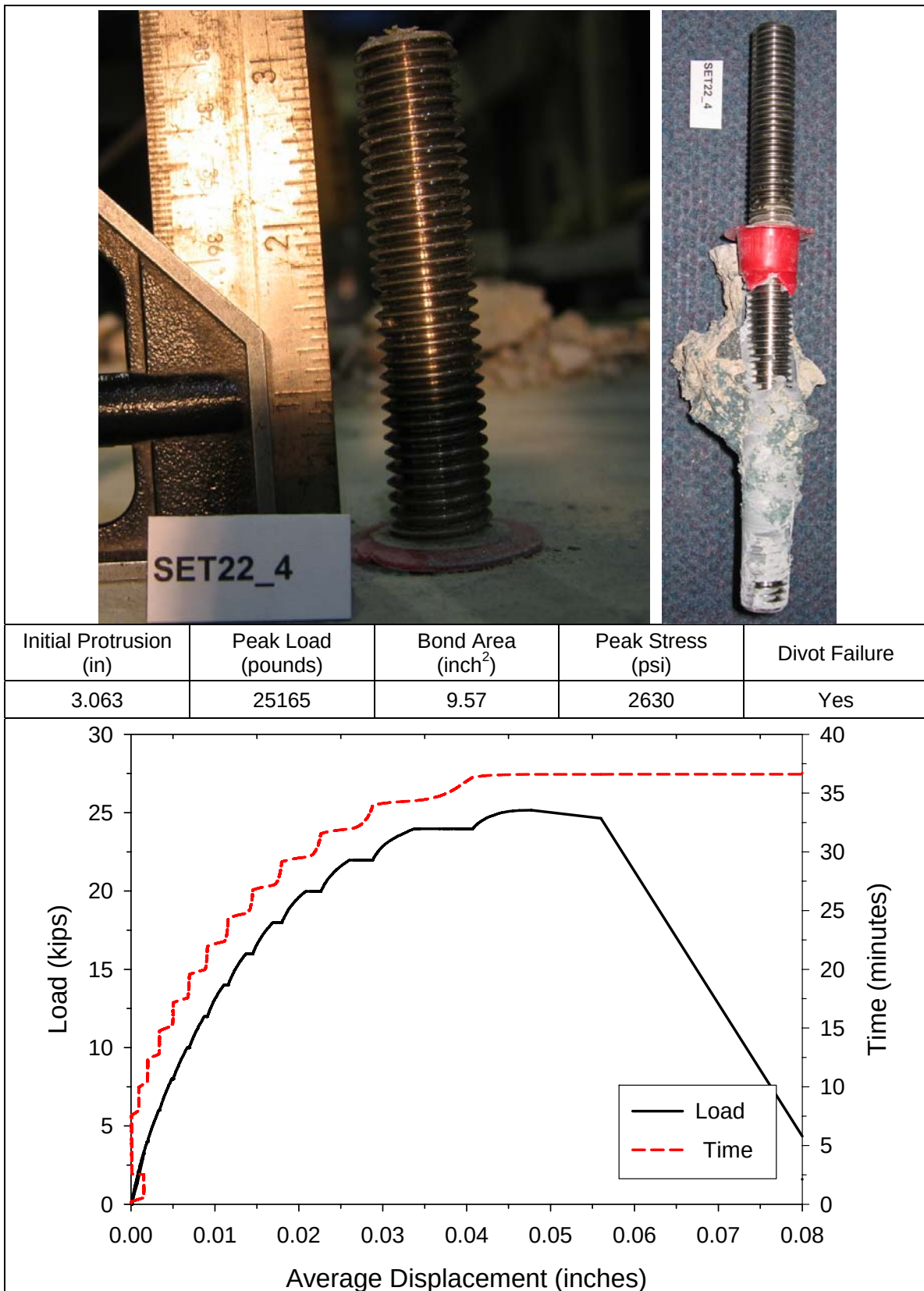


Table 140 SET22 Epoxy 5

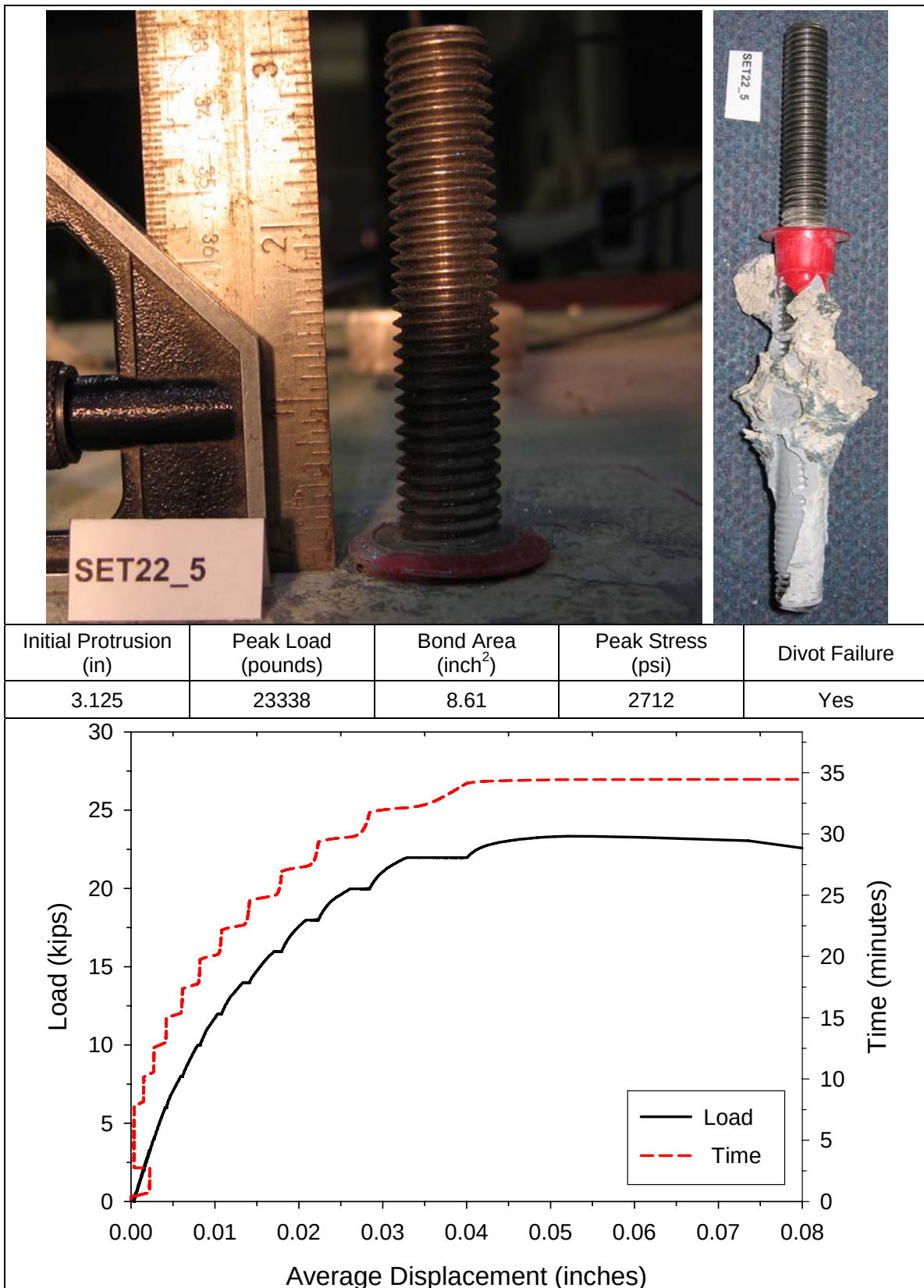


Table 141 Damp Dirty Hole 1

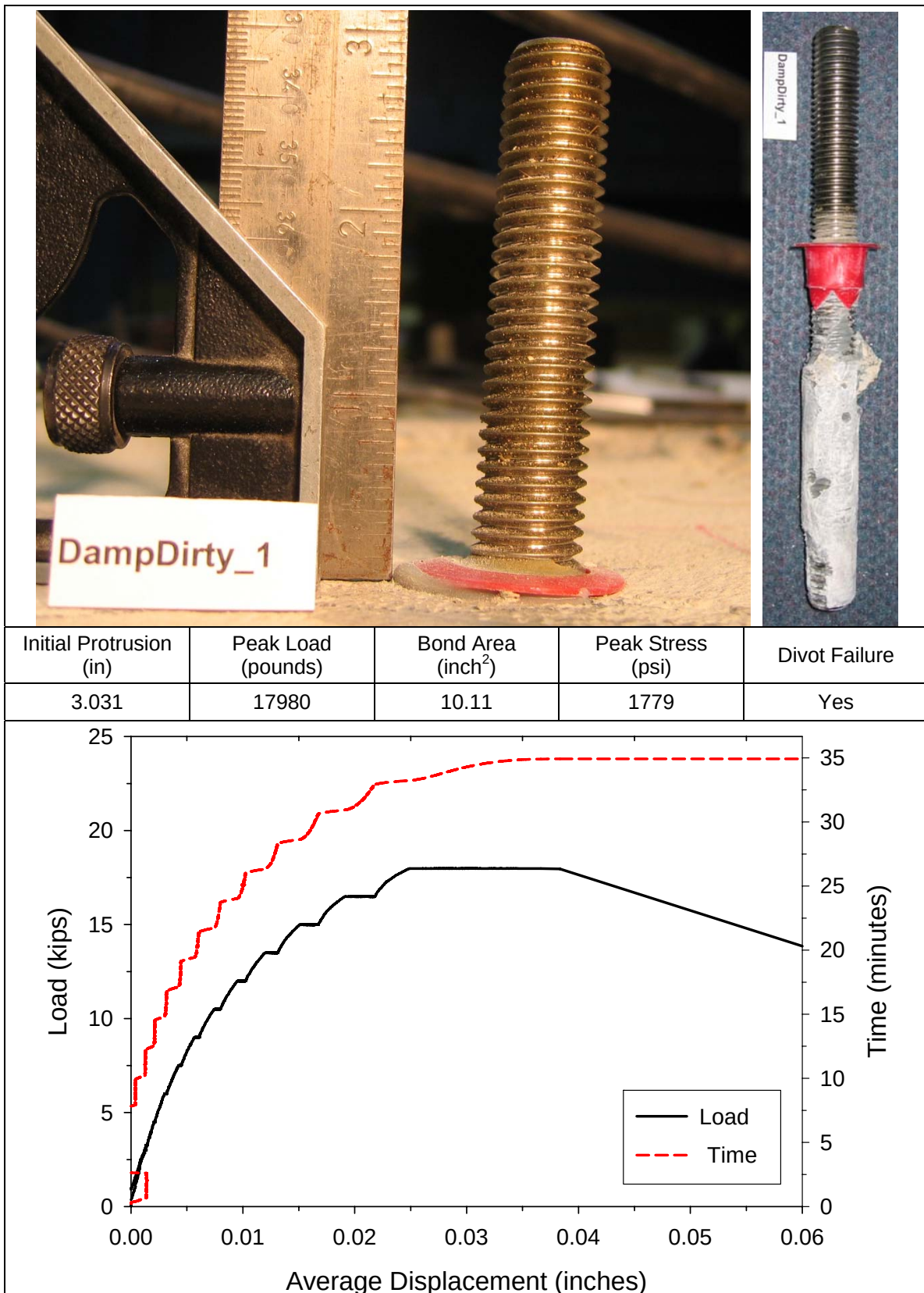


Table 142 Damp Dirty Hole 2

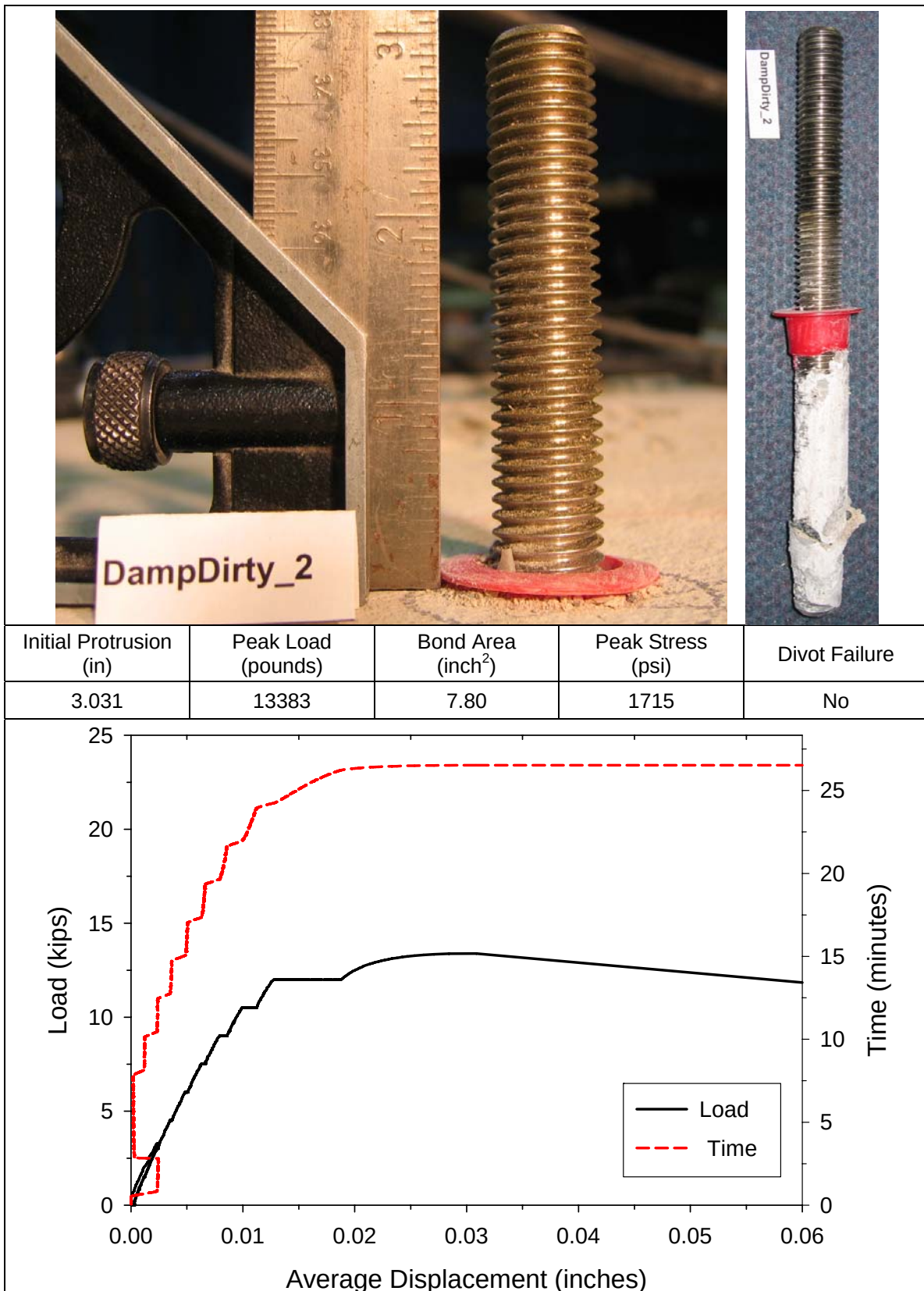


Table 143 Damp Dirty Hole 3

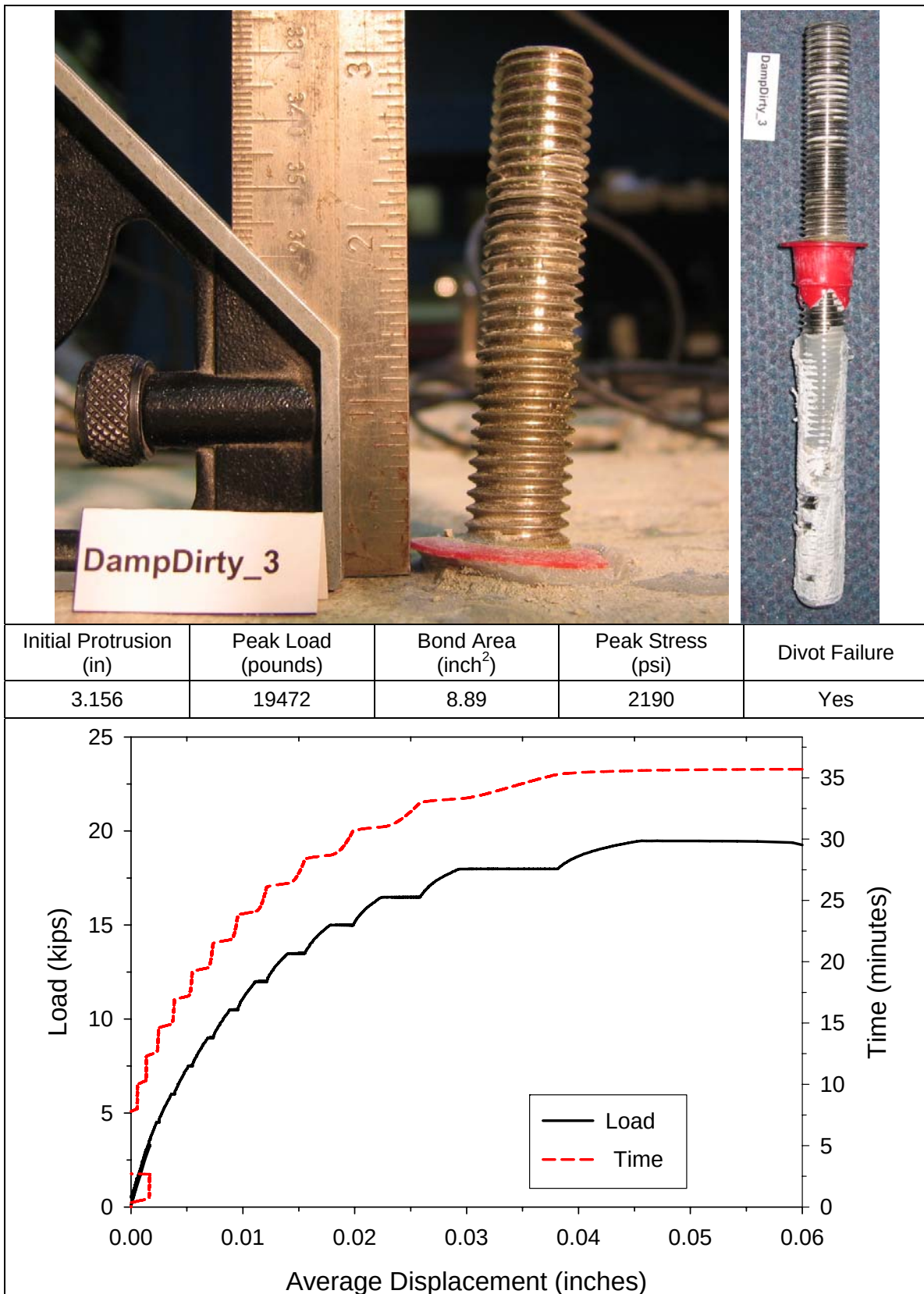


Table 144 Damp Dirty Hole 4

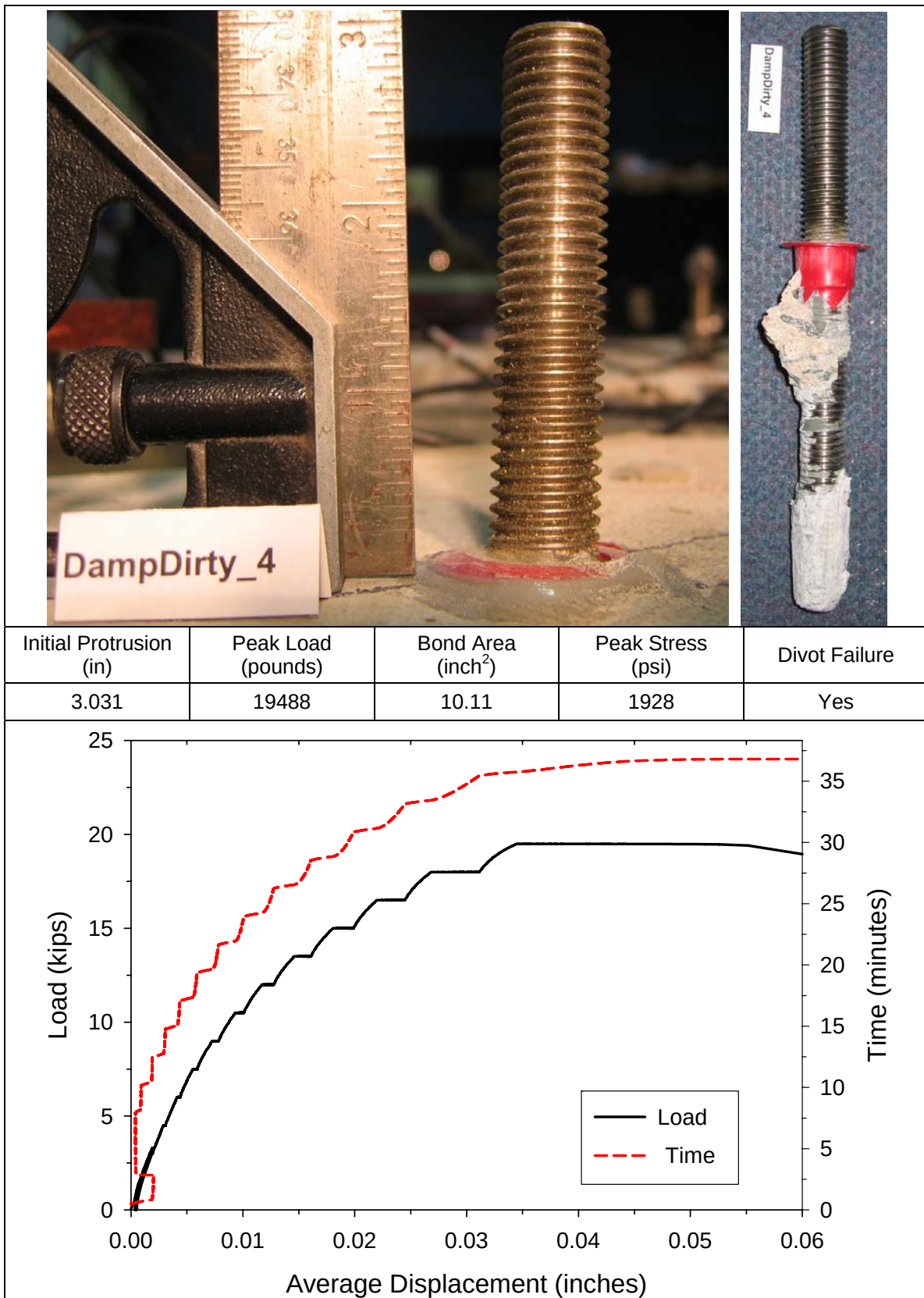


Table 145 Damp Dirty Hole 5

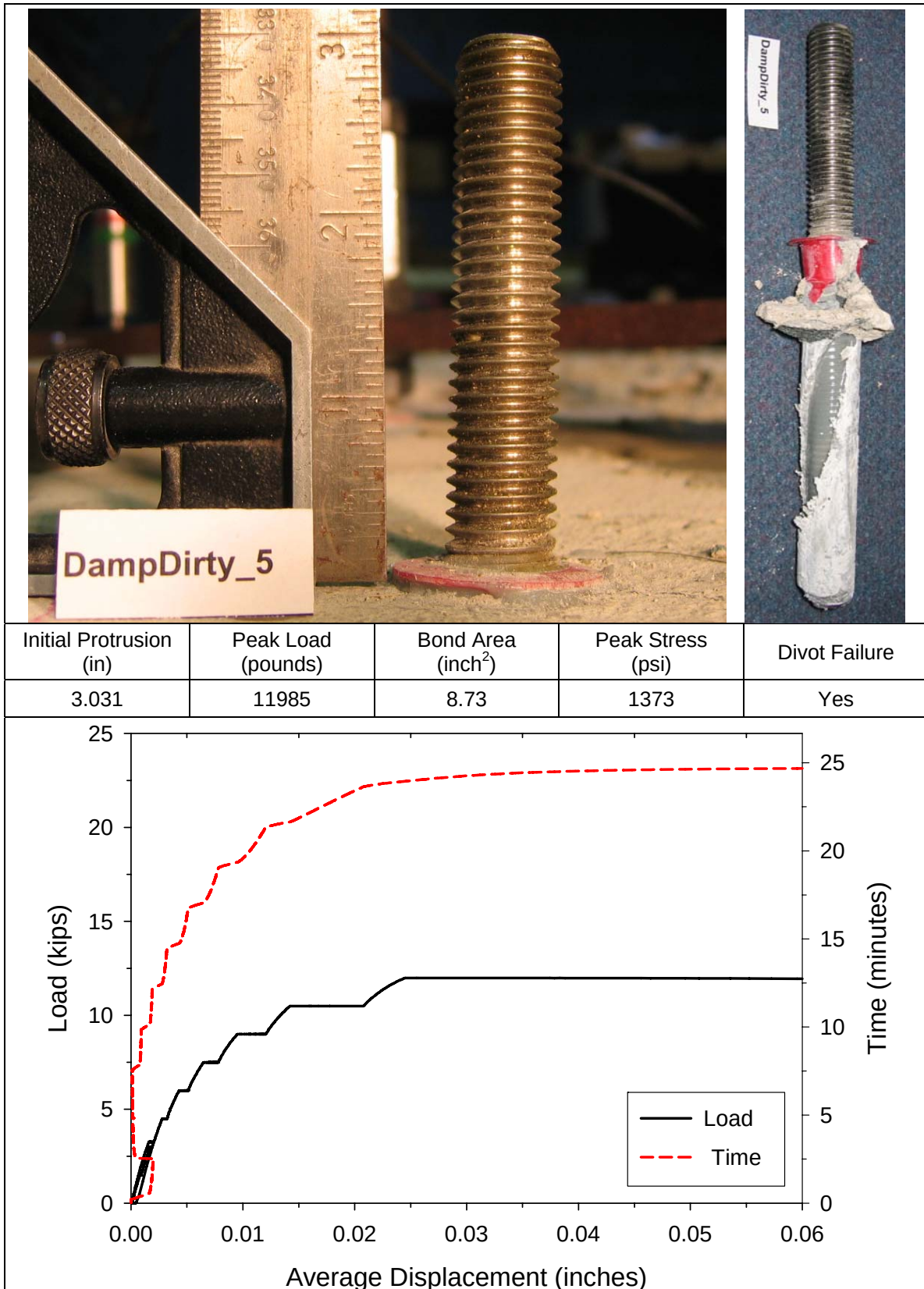


Table 146 Torqued to 50 Ft-Lbs 1

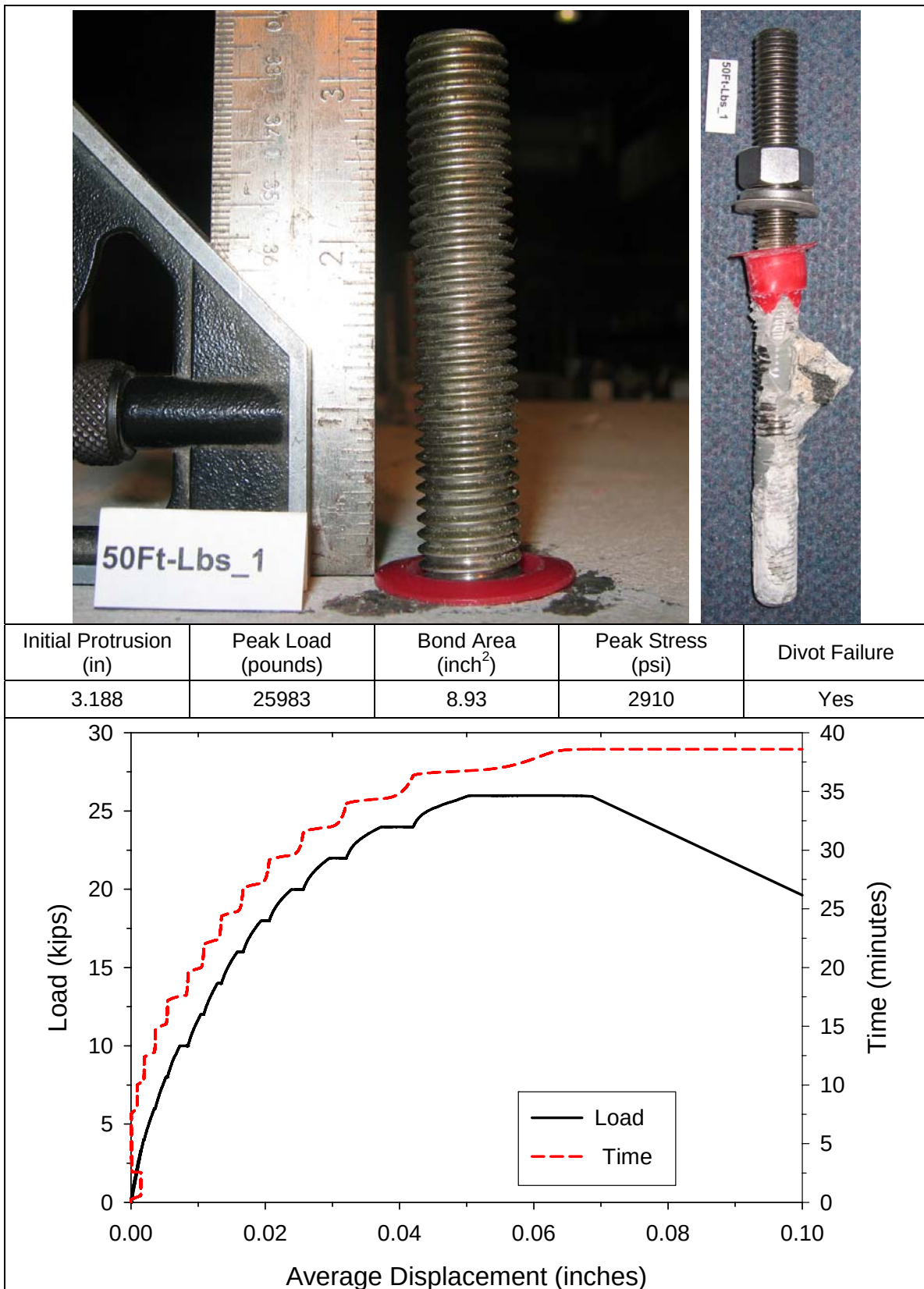


Table 147 Torqued to 50 Ft-Lbs 2

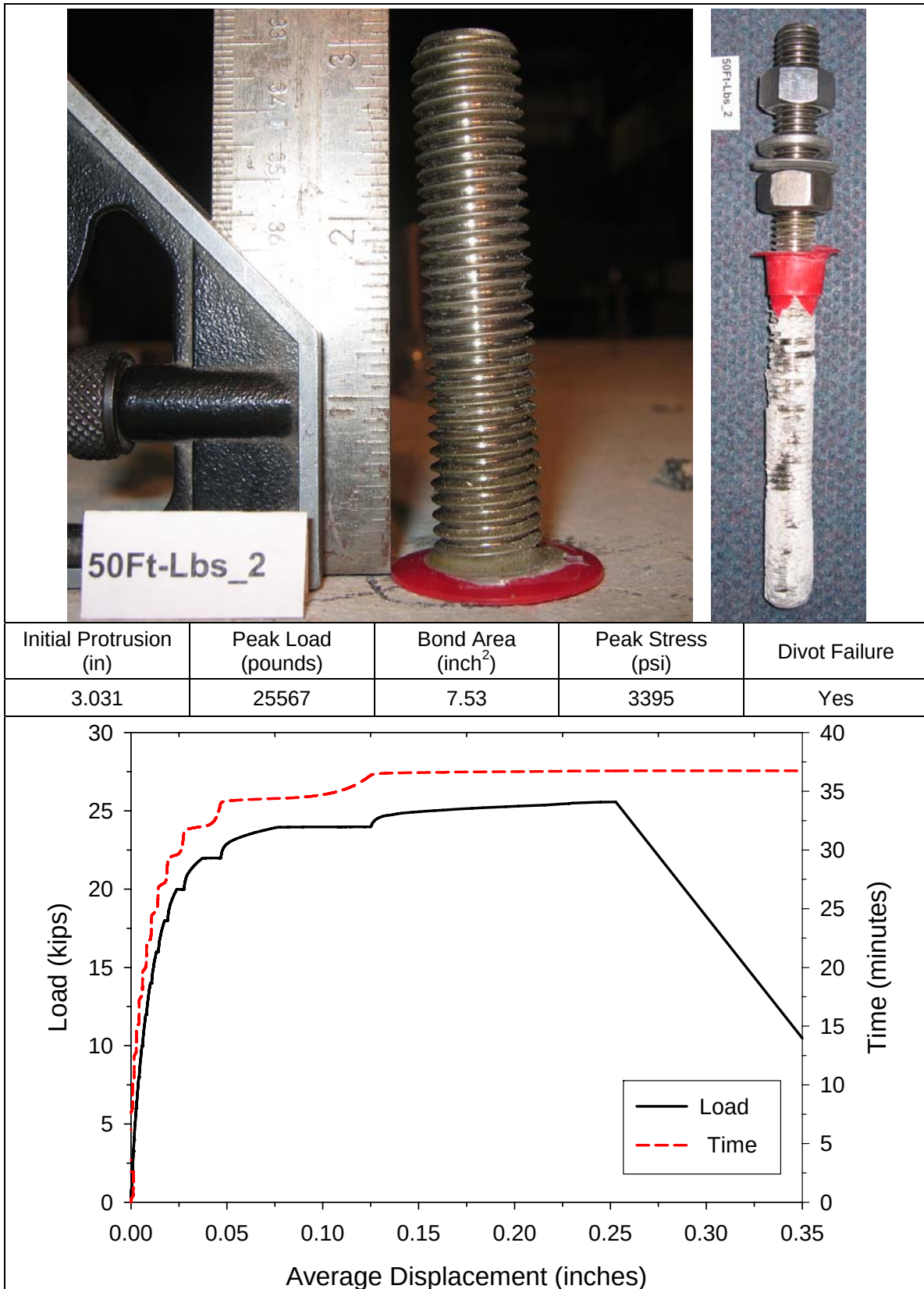


Table 148 Torqued to 50 Ft-Lbs 3

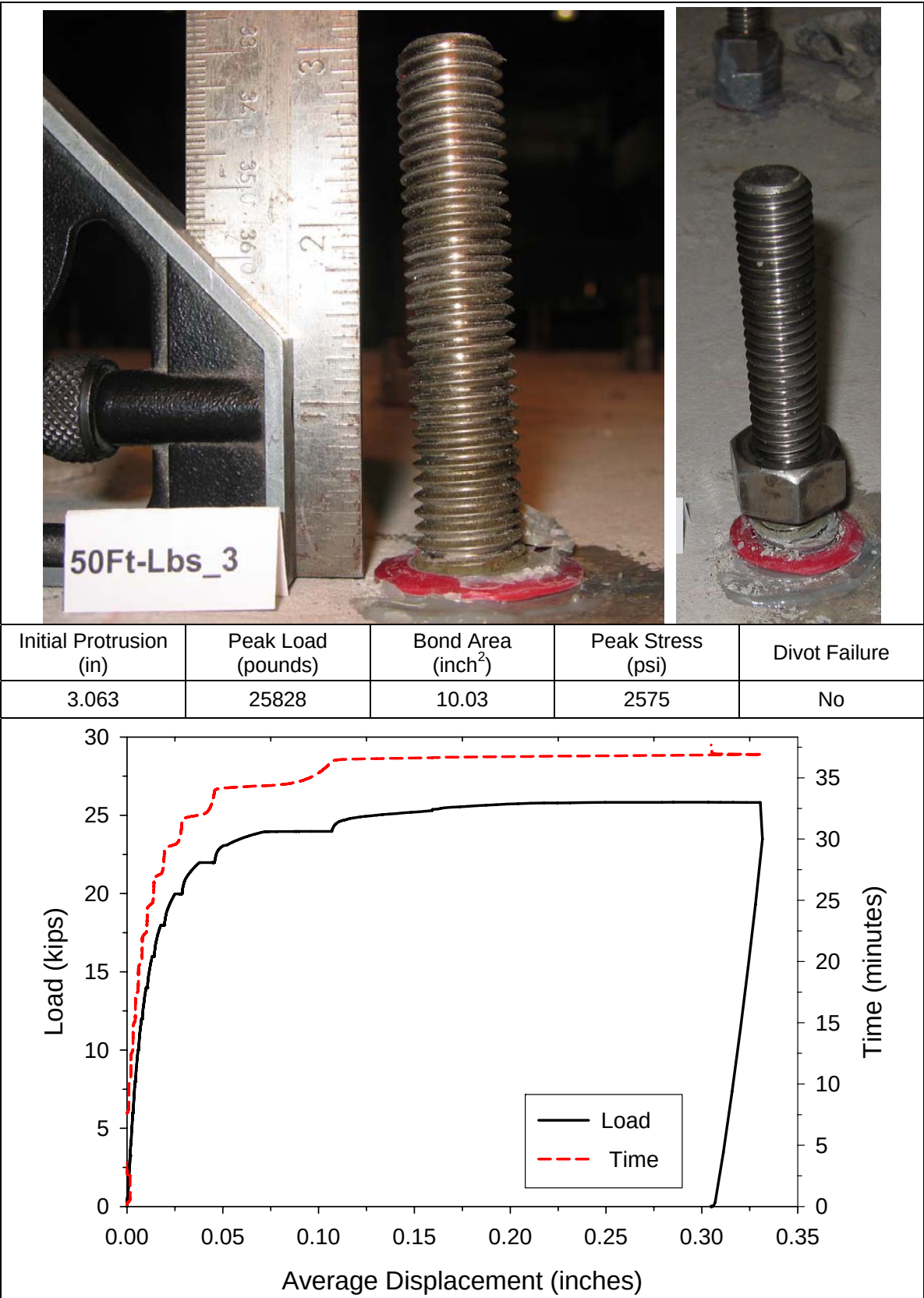


Table 149 Torqued to 50 Ft-Lbs 4

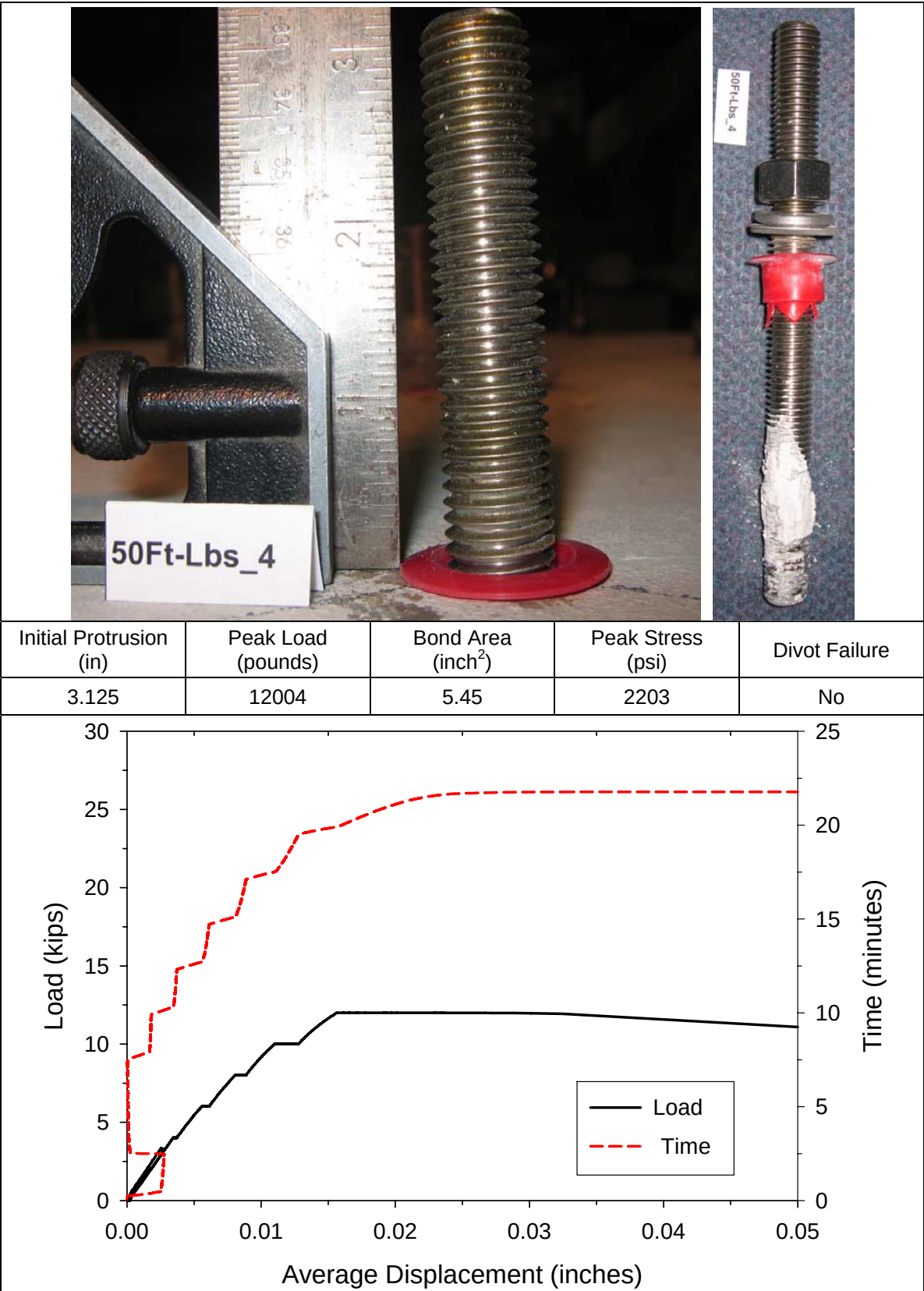


Table 150 Torqued to 50 Ft-Lbs 5

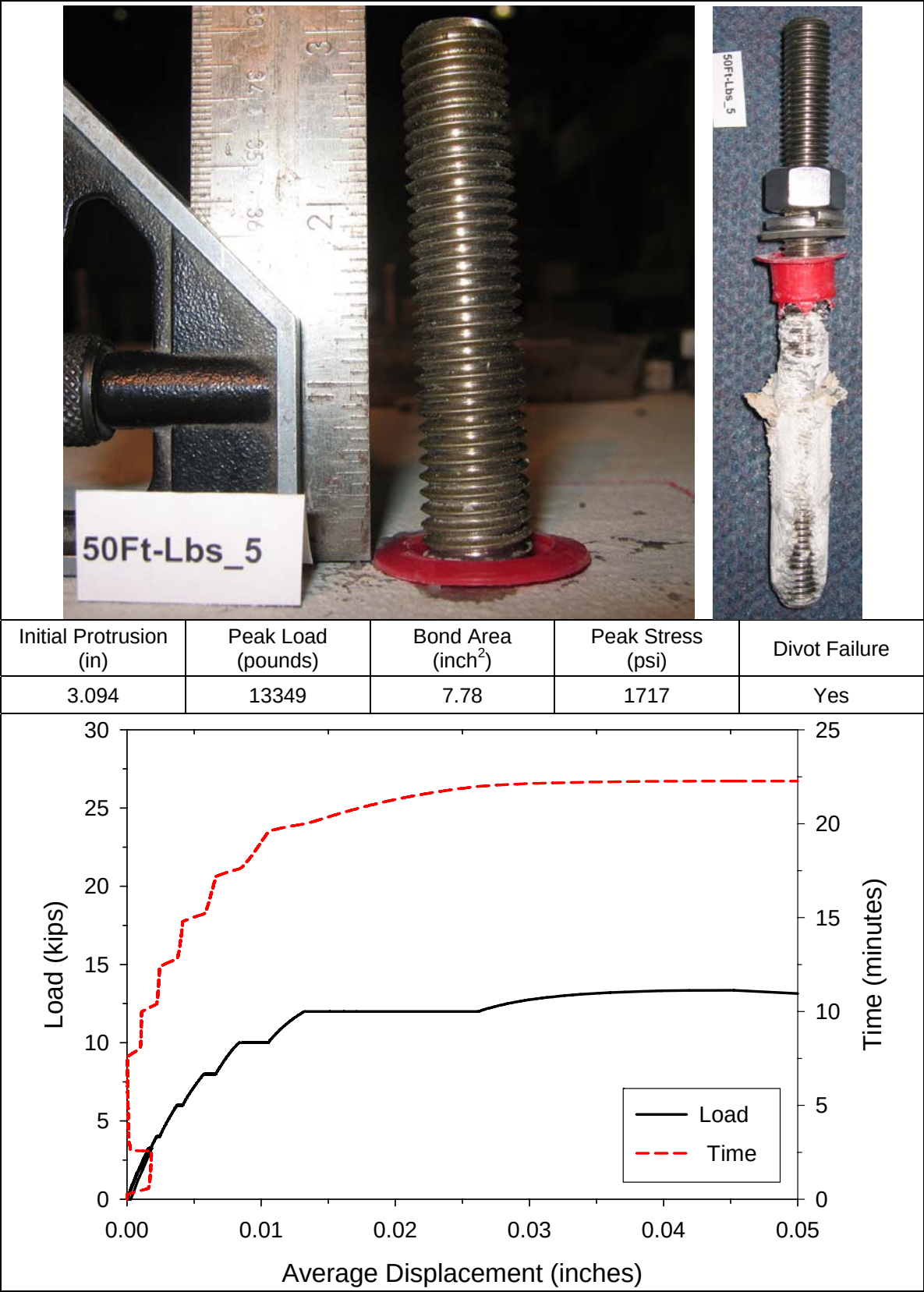


Table 151 Torqued to 90 Ft-Lbs 1

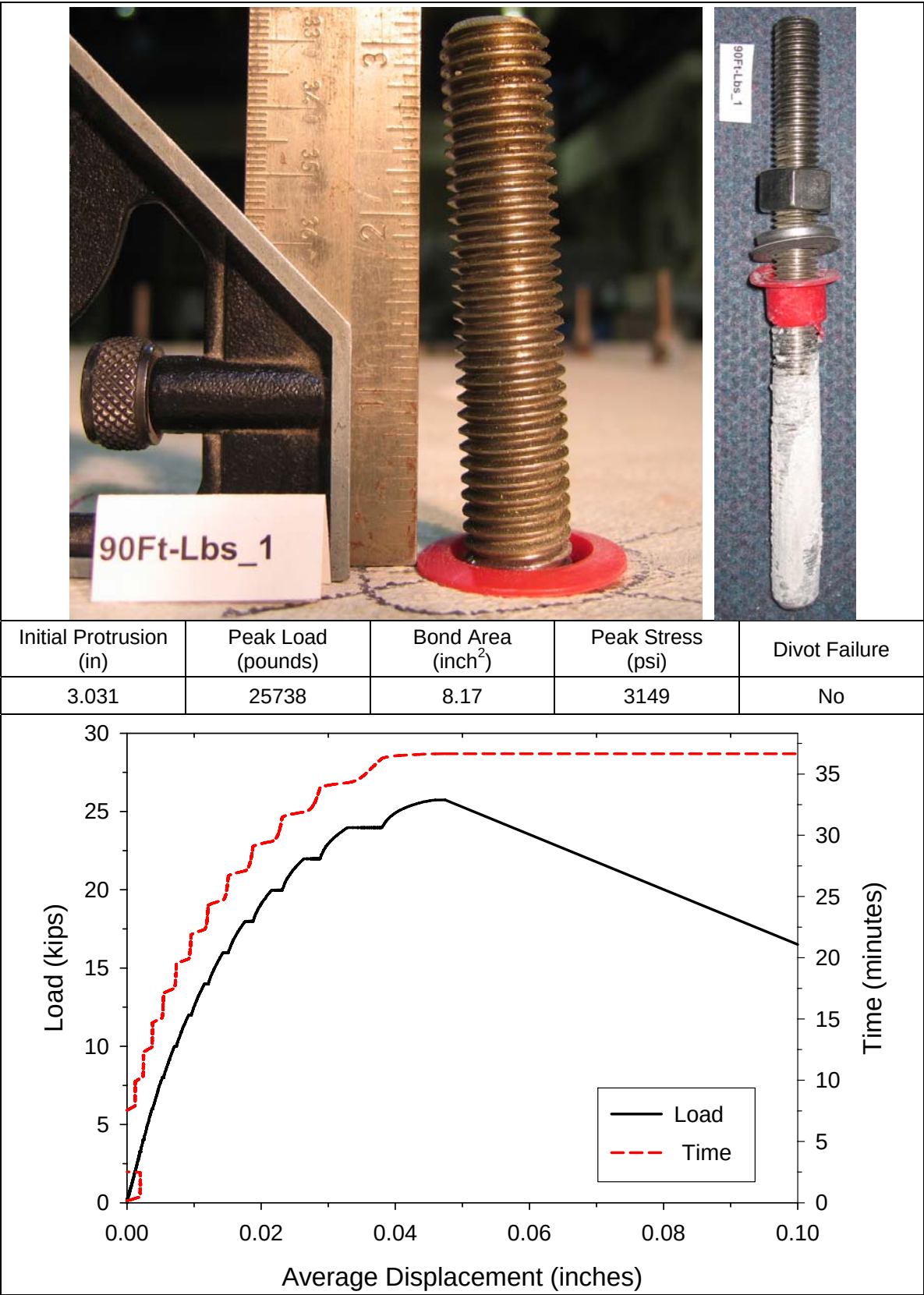


Table 152 Torqued to 90 Ft-Lbs 2

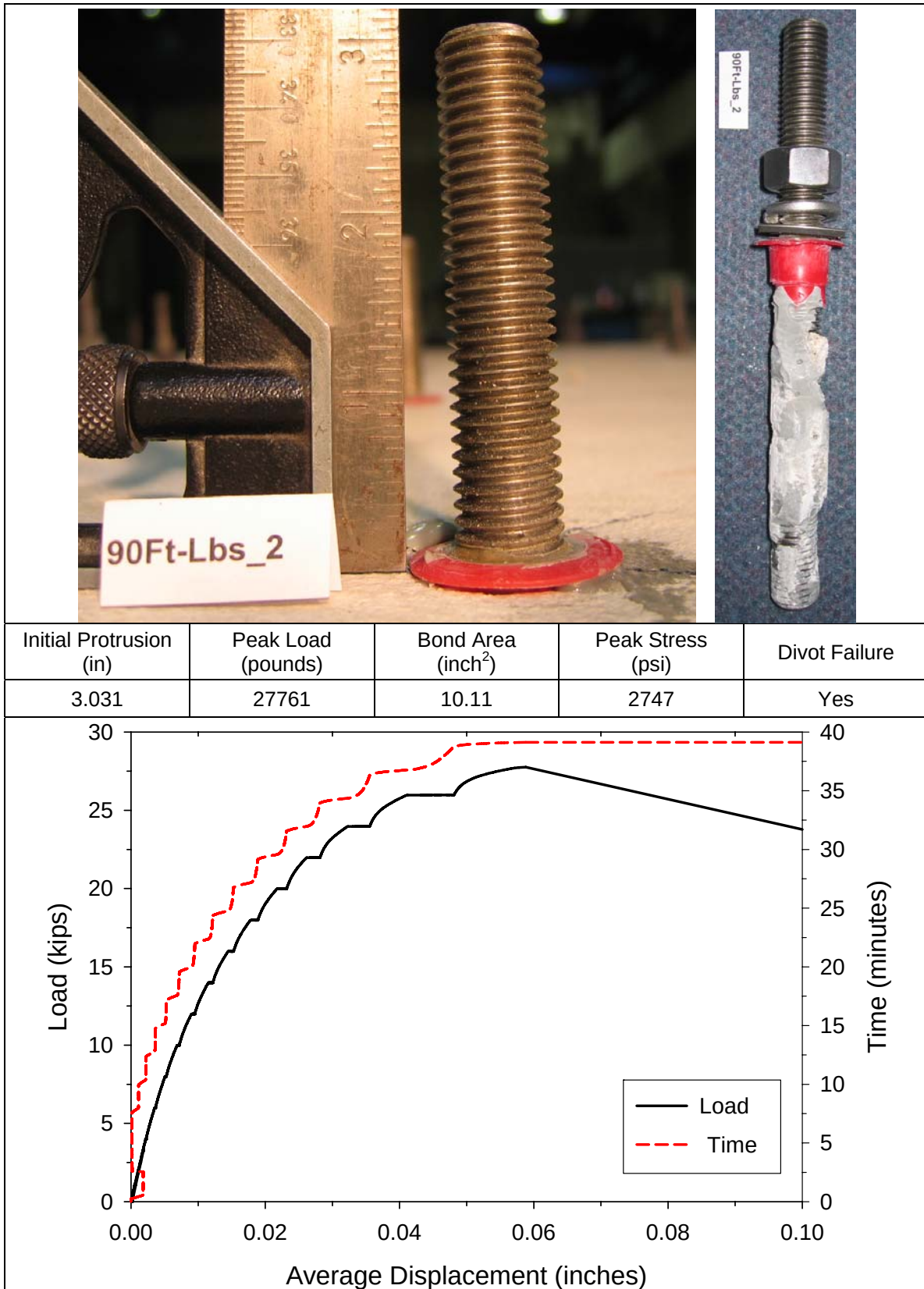


Table 153 Torqued to 90 Ft-Lbs 3

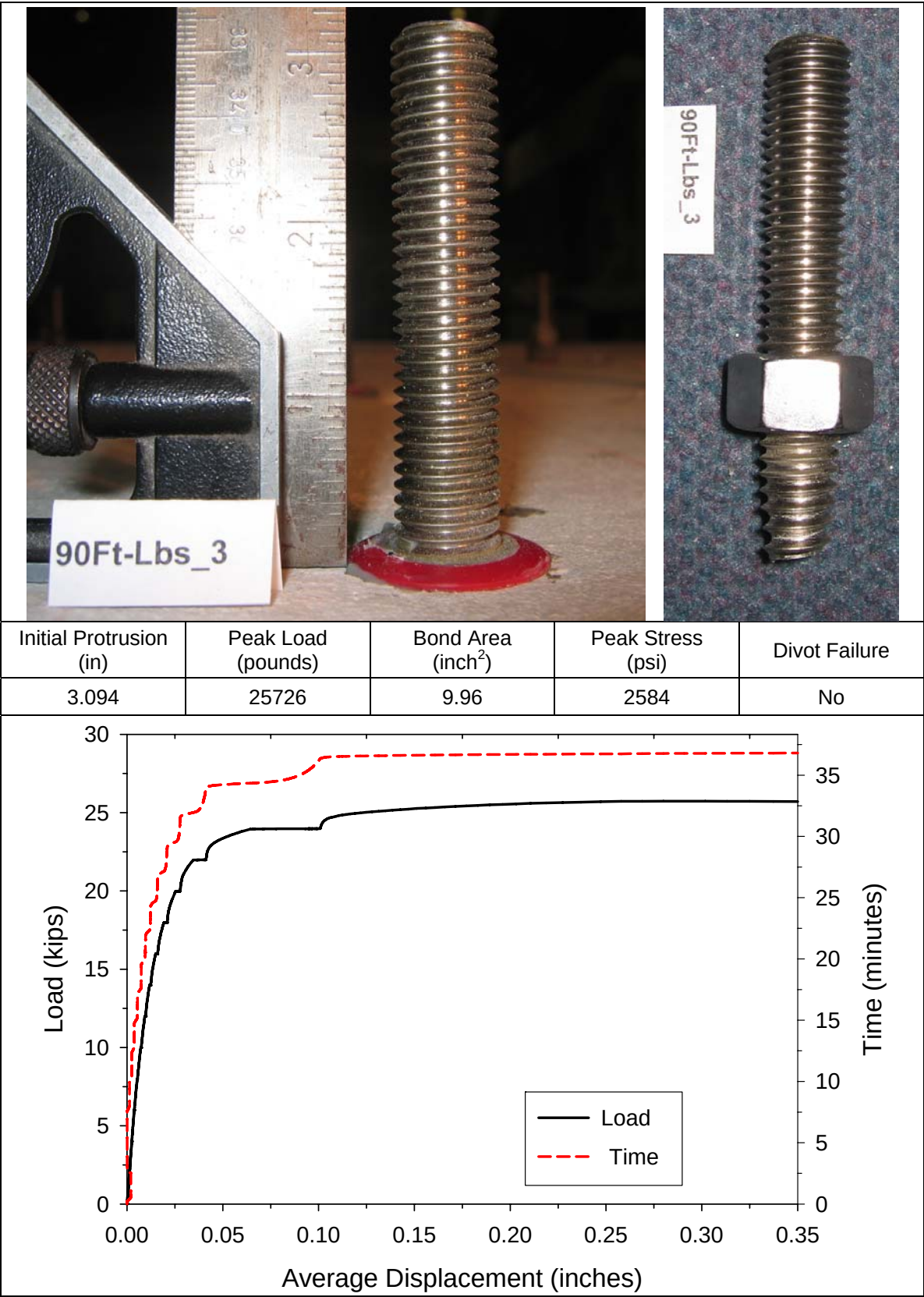


Table 154 Torqued to 90 Ft-Lbs 4

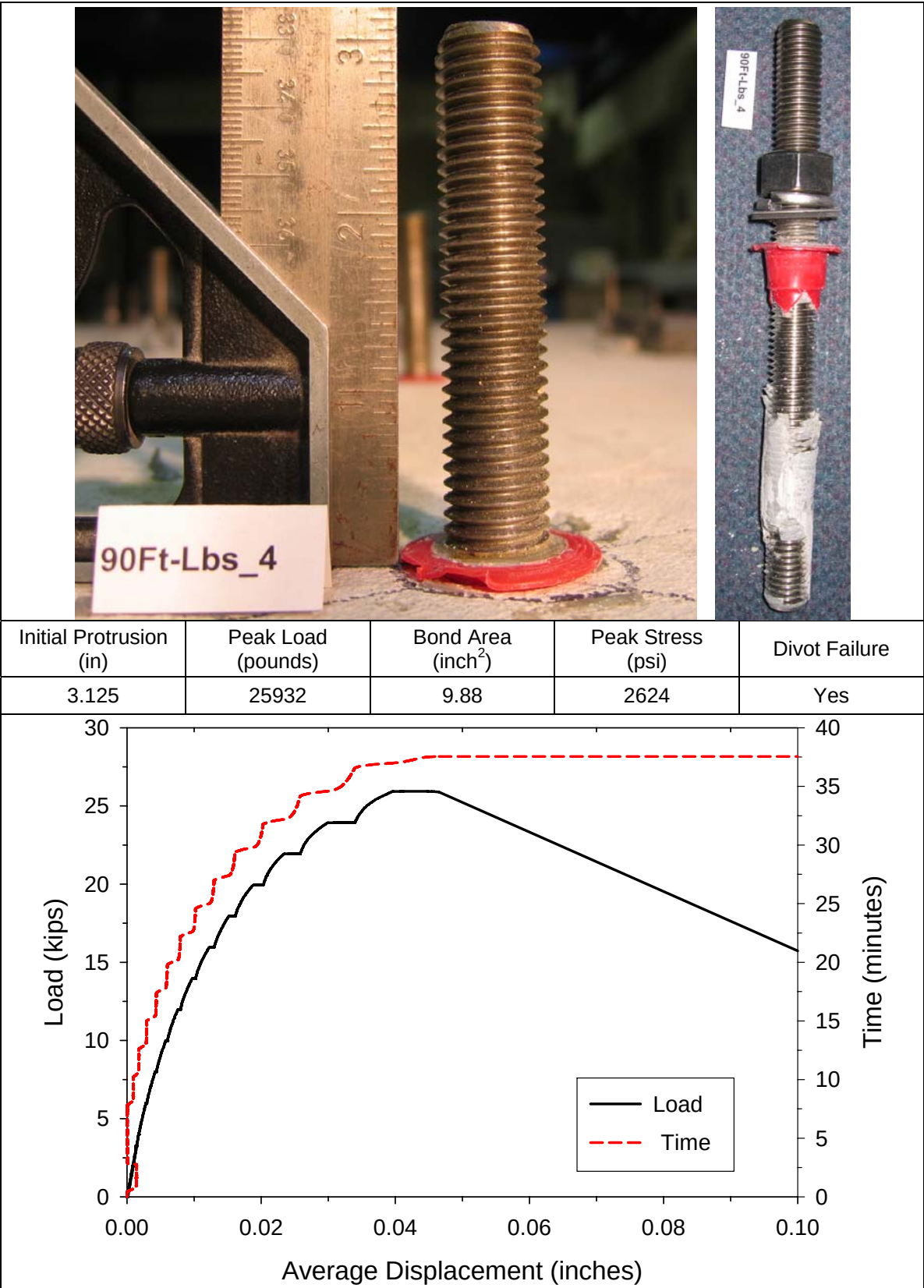


Table 155 Torqued to 90 Ft-Lbs 5

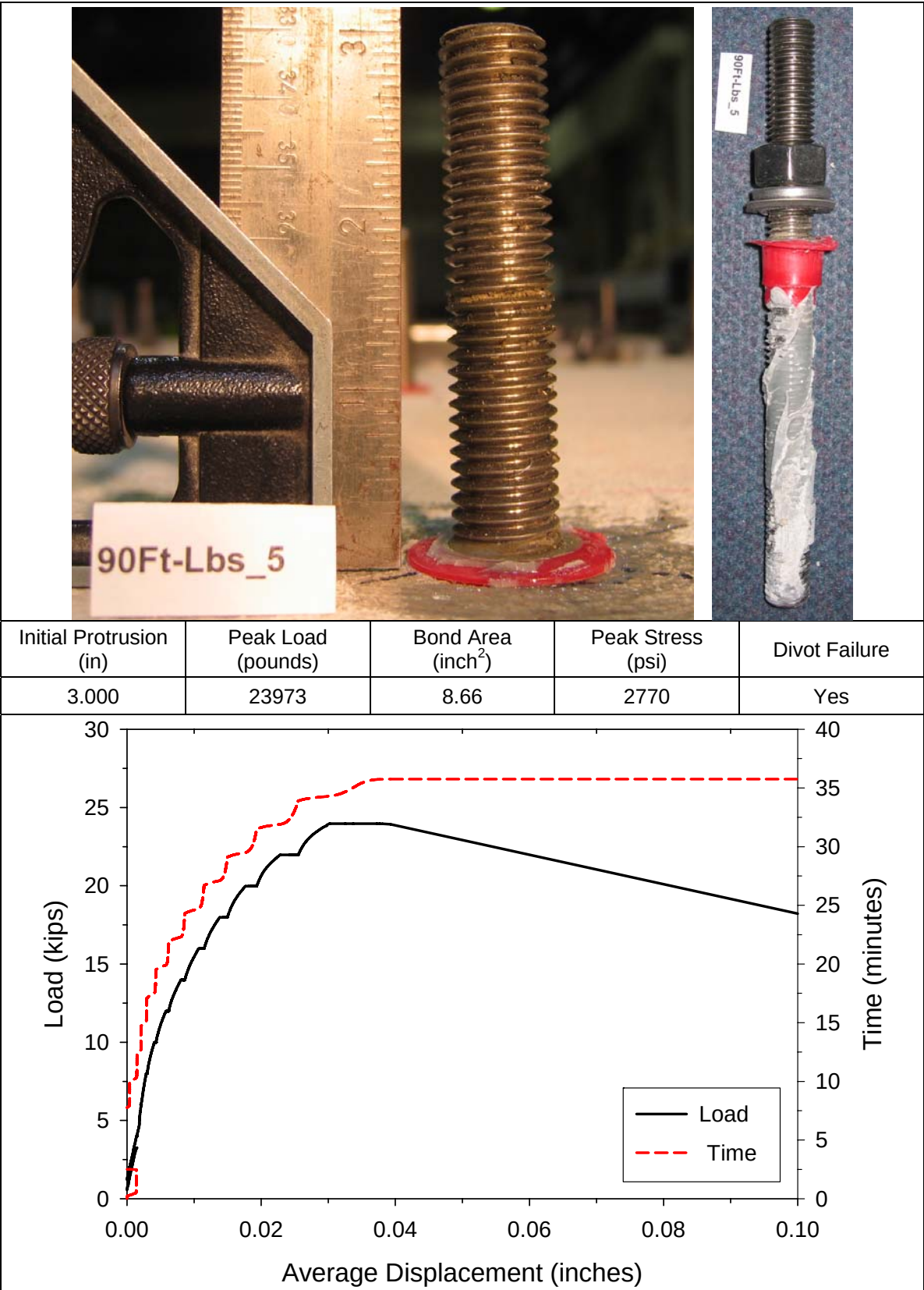


Table 156 Torqued to 125 Ft-Lbs 1

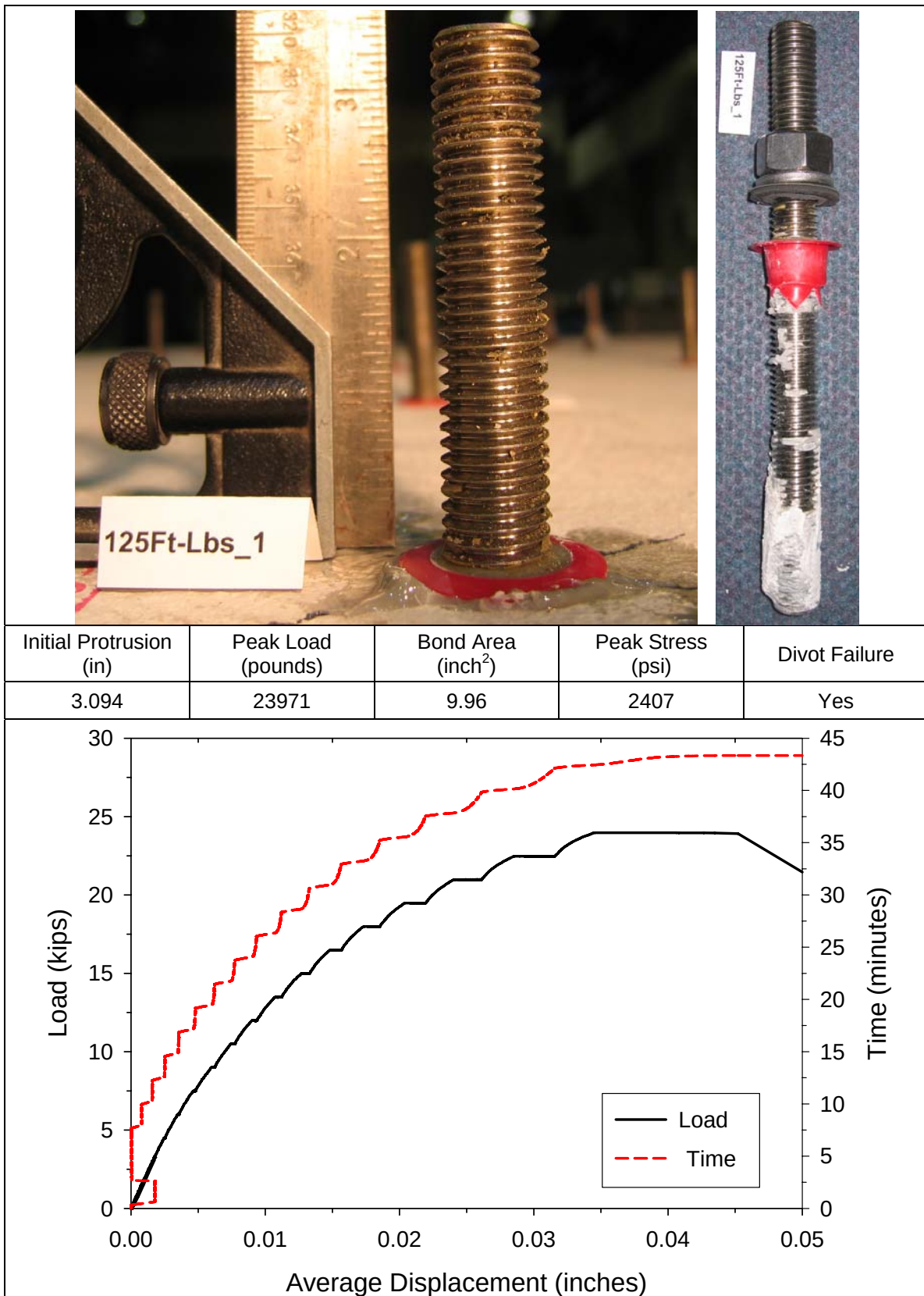


Table 157 Torqued to 125 Ft-Lbs 2

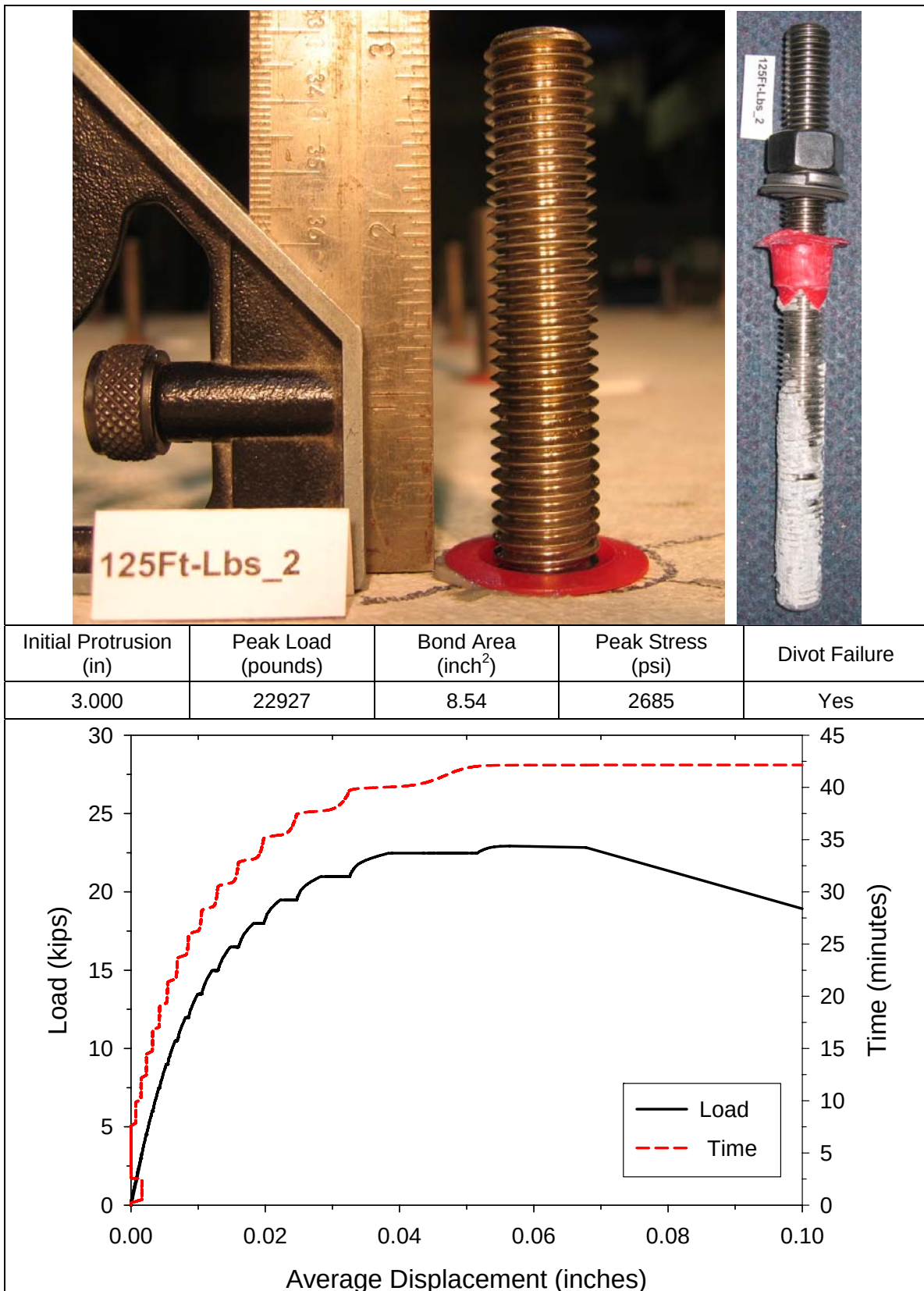
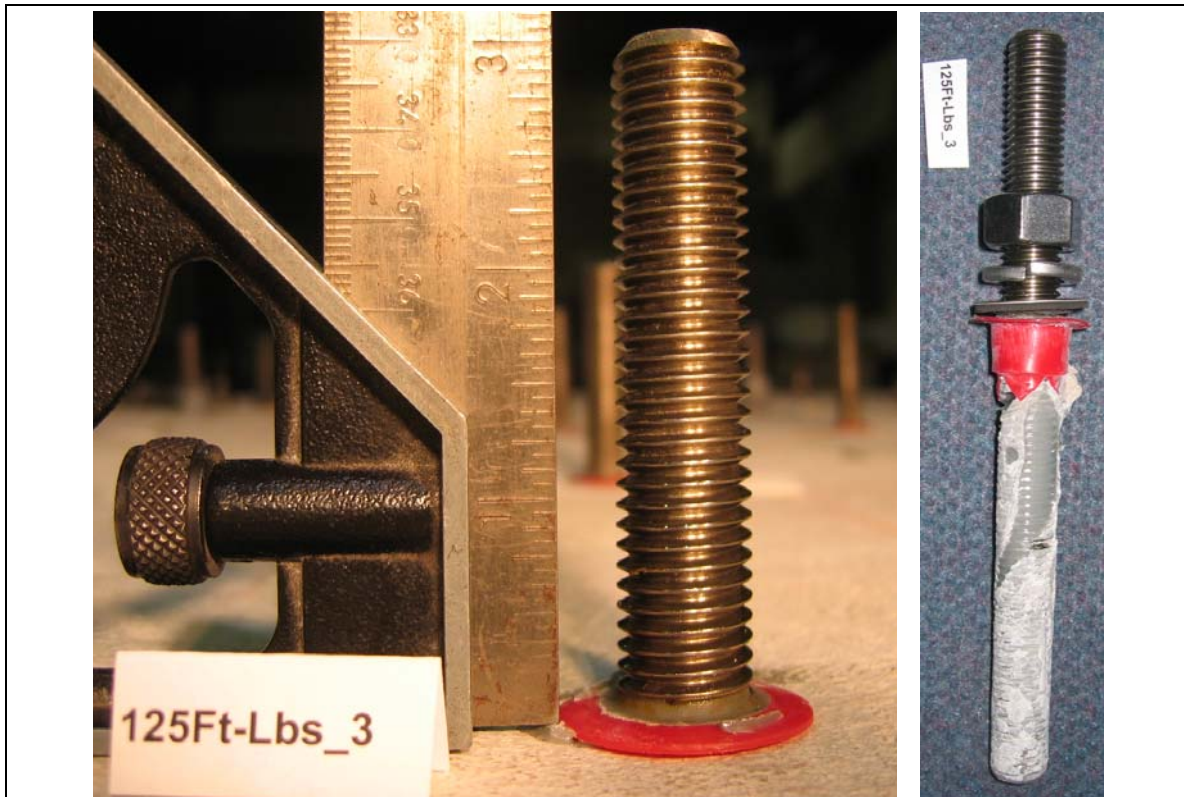


Table 158 Torqued to 125 Ft-Lbs 3



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.094	21988	7.72	2849	Yes

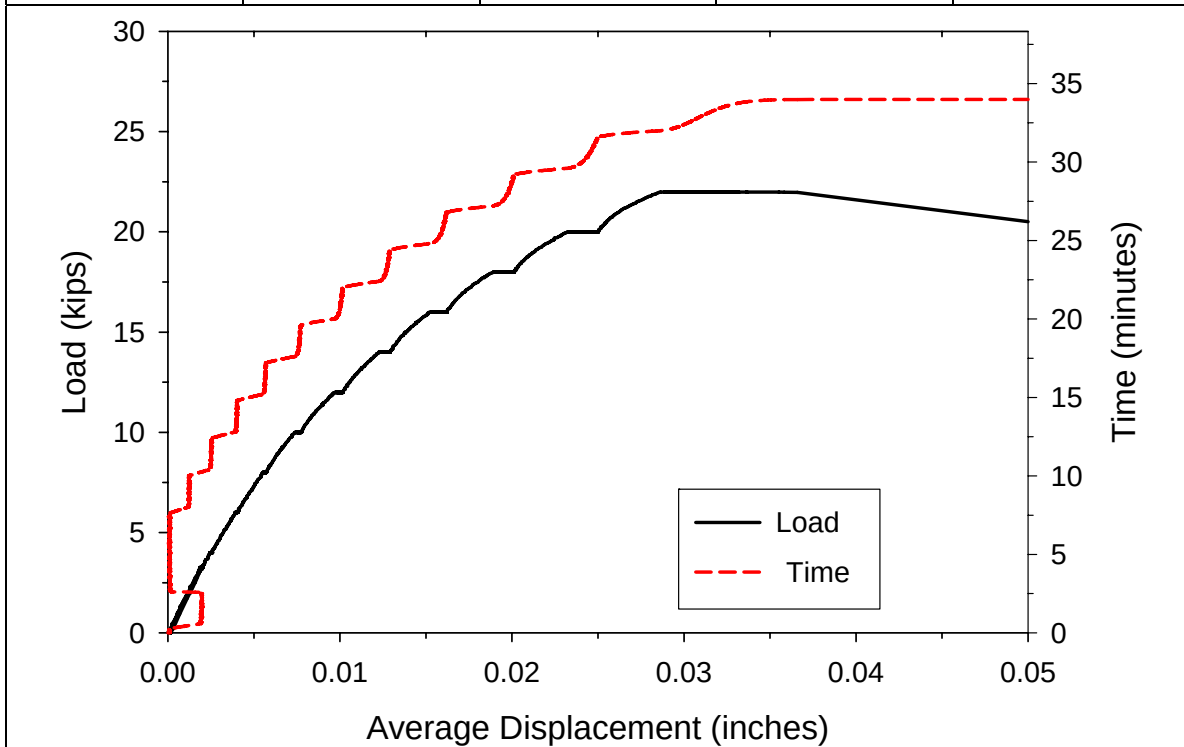


Table 159 Torqued to 125 Ft-Lbs 4

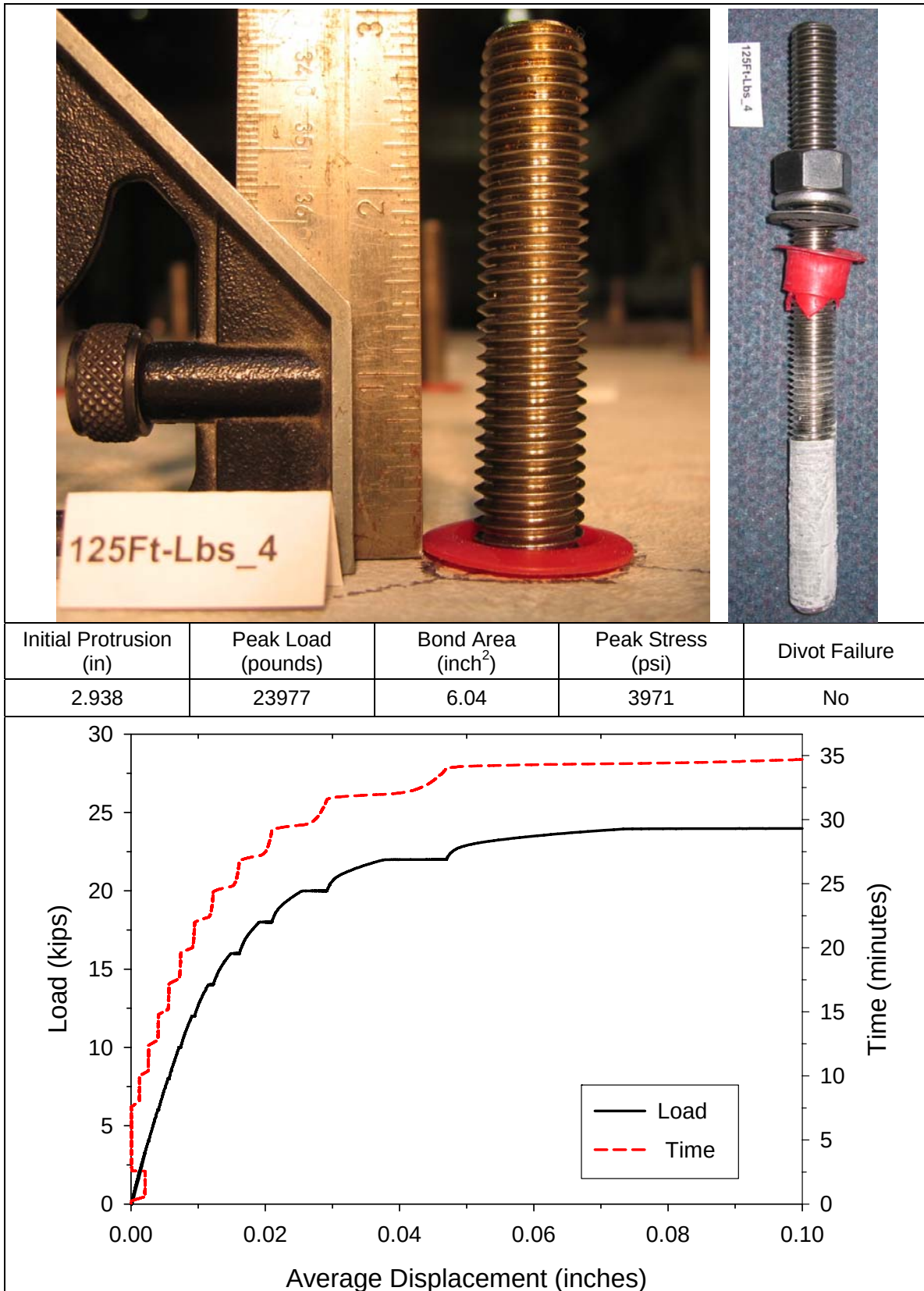


Table 160 Torqued to 125 Ft-Lbs 5

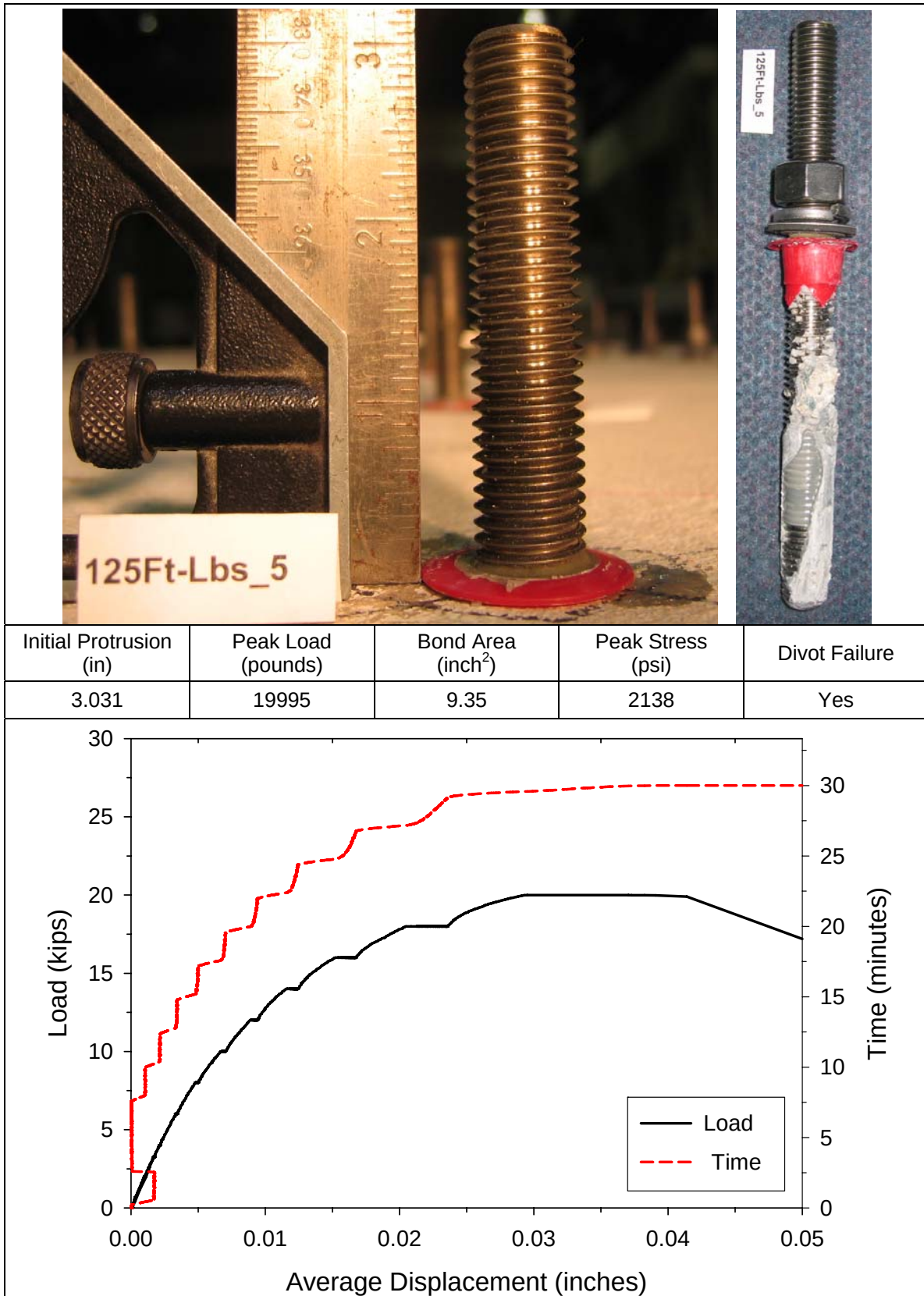


Table 161 3 white : 2 gray 1

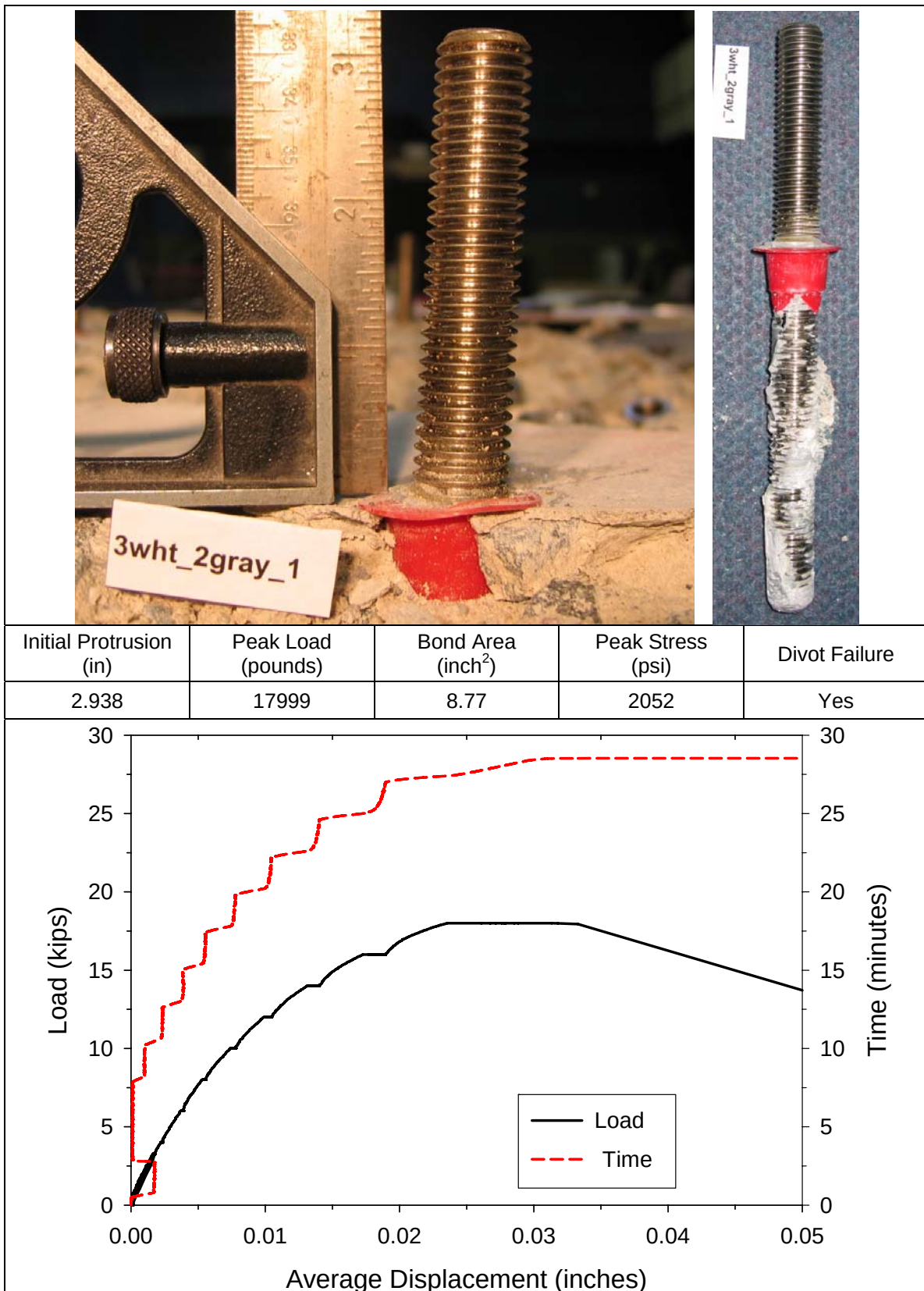


Table 162 3 white : 2 gray 2

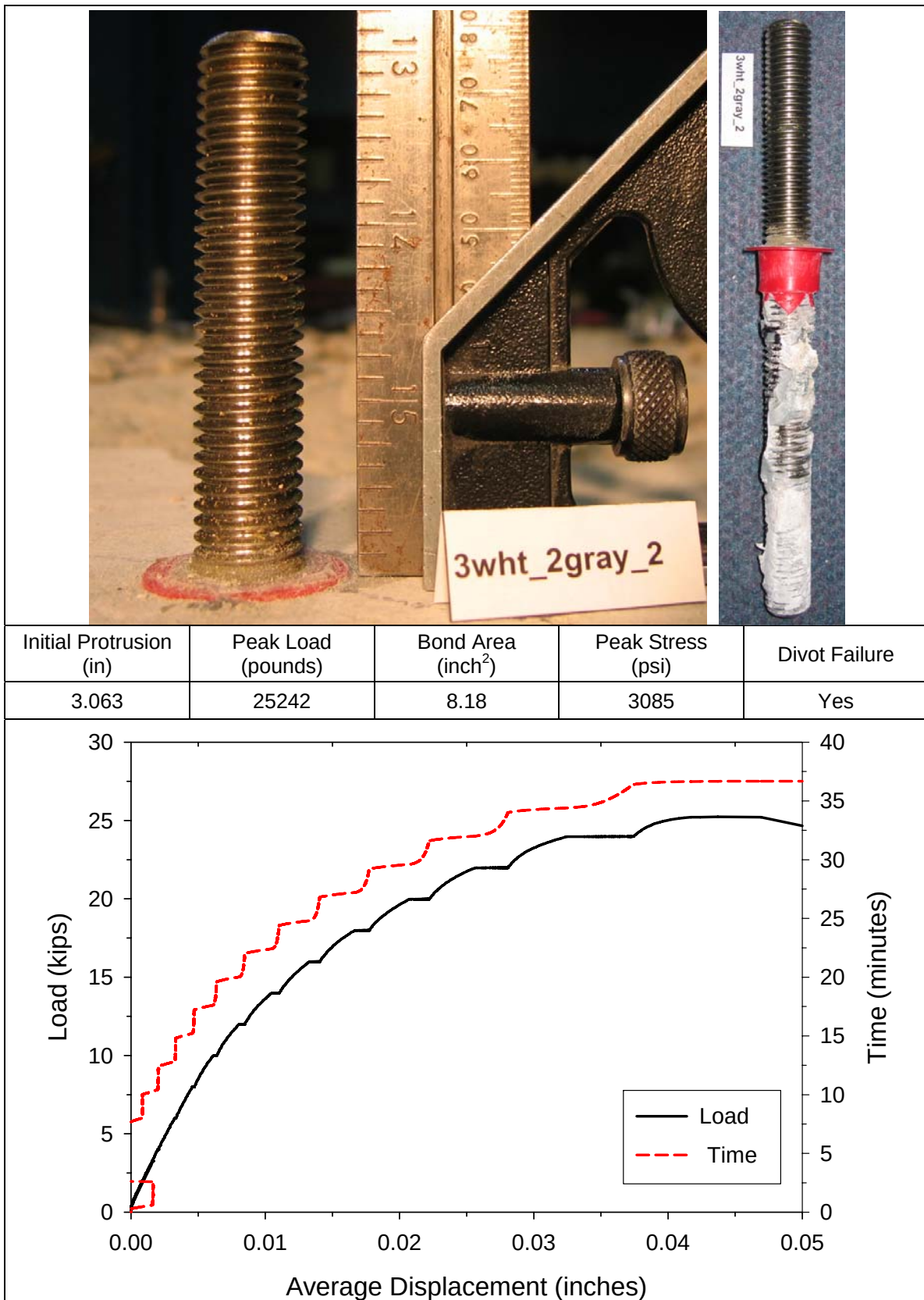


Table 163 3 white : 2 gray 3

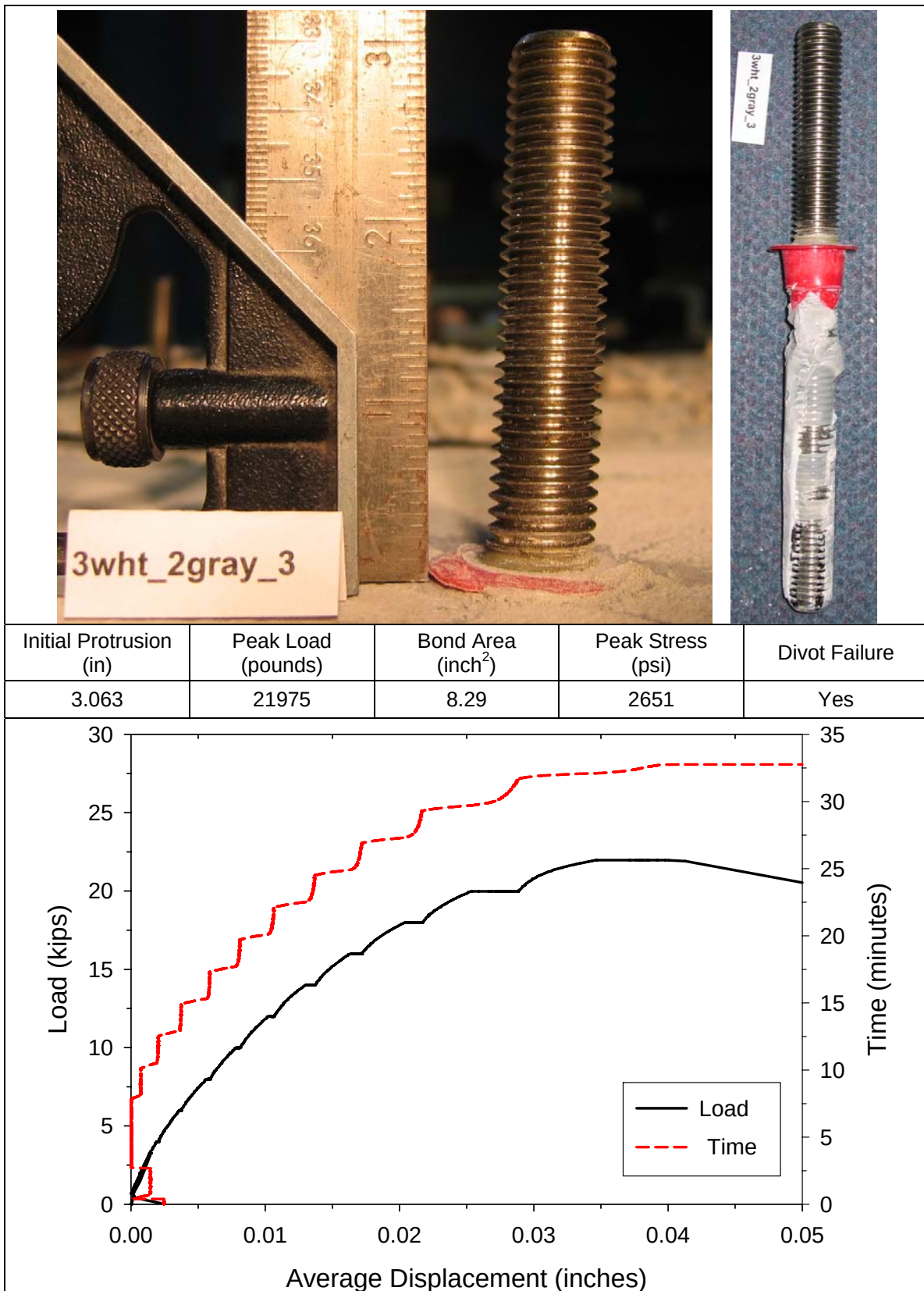


Table 164 3 white : 2 gray 4

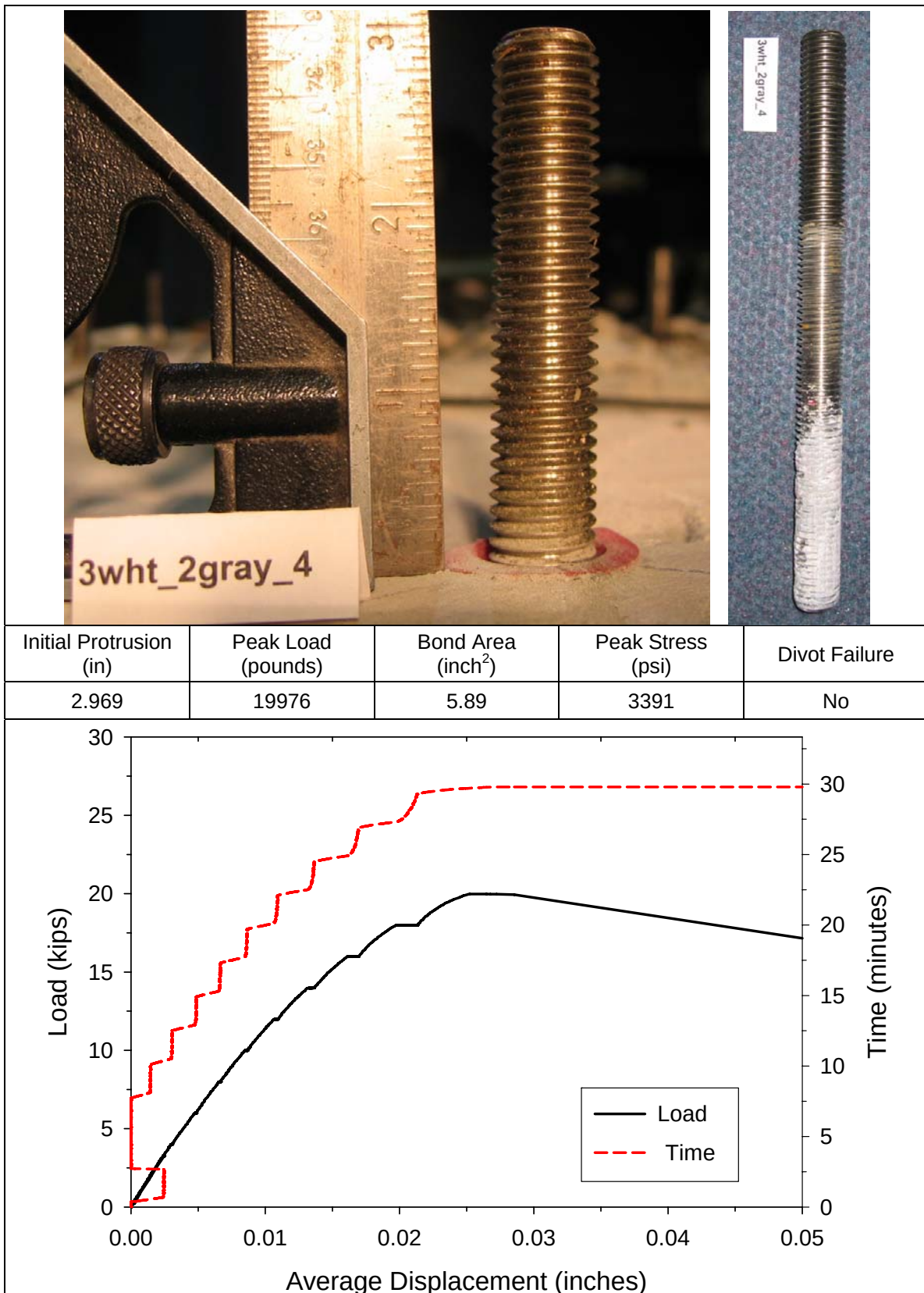


Table 165 3 white : 2 gray 5

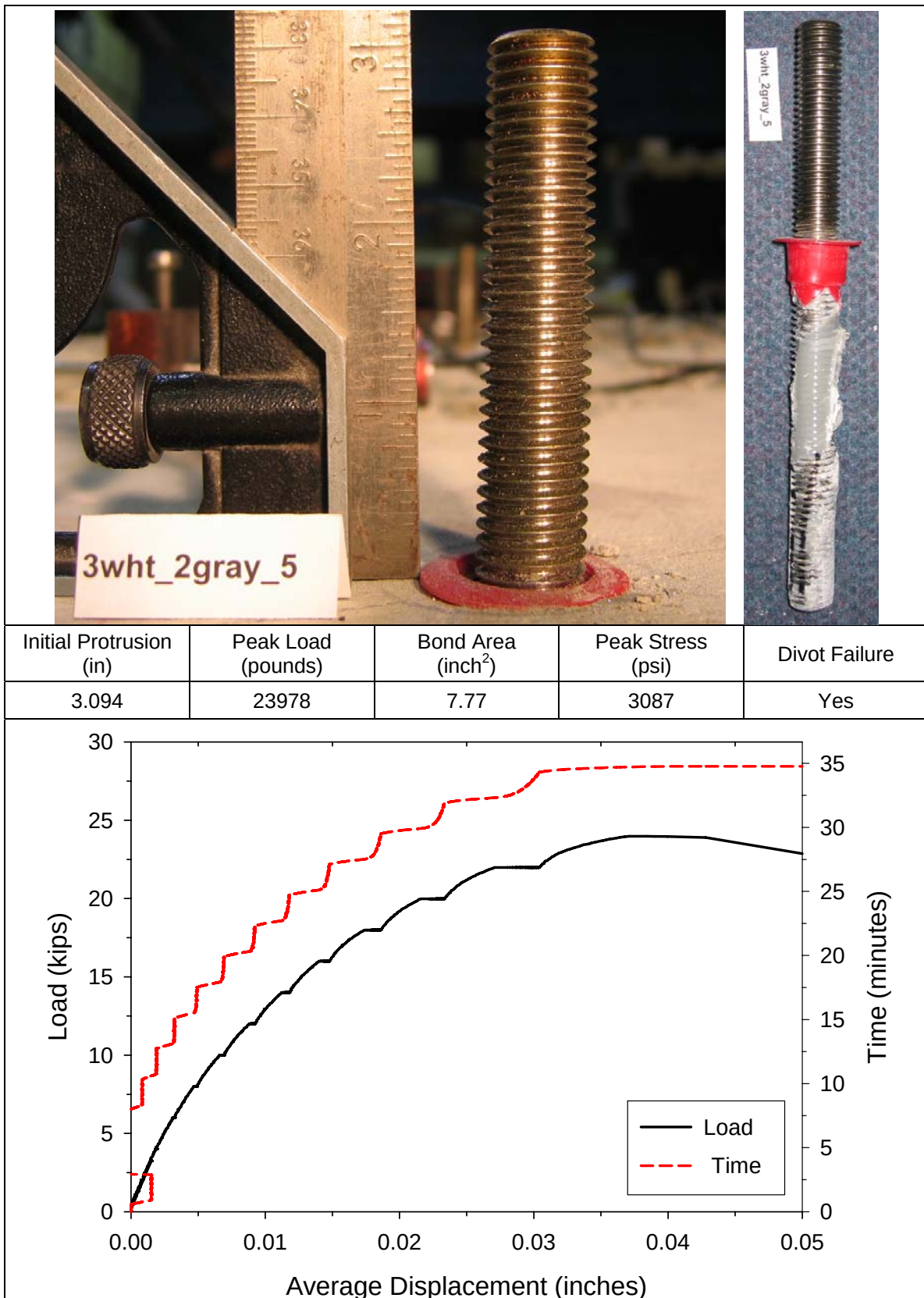


Table 166 2 white : 3 gray 1

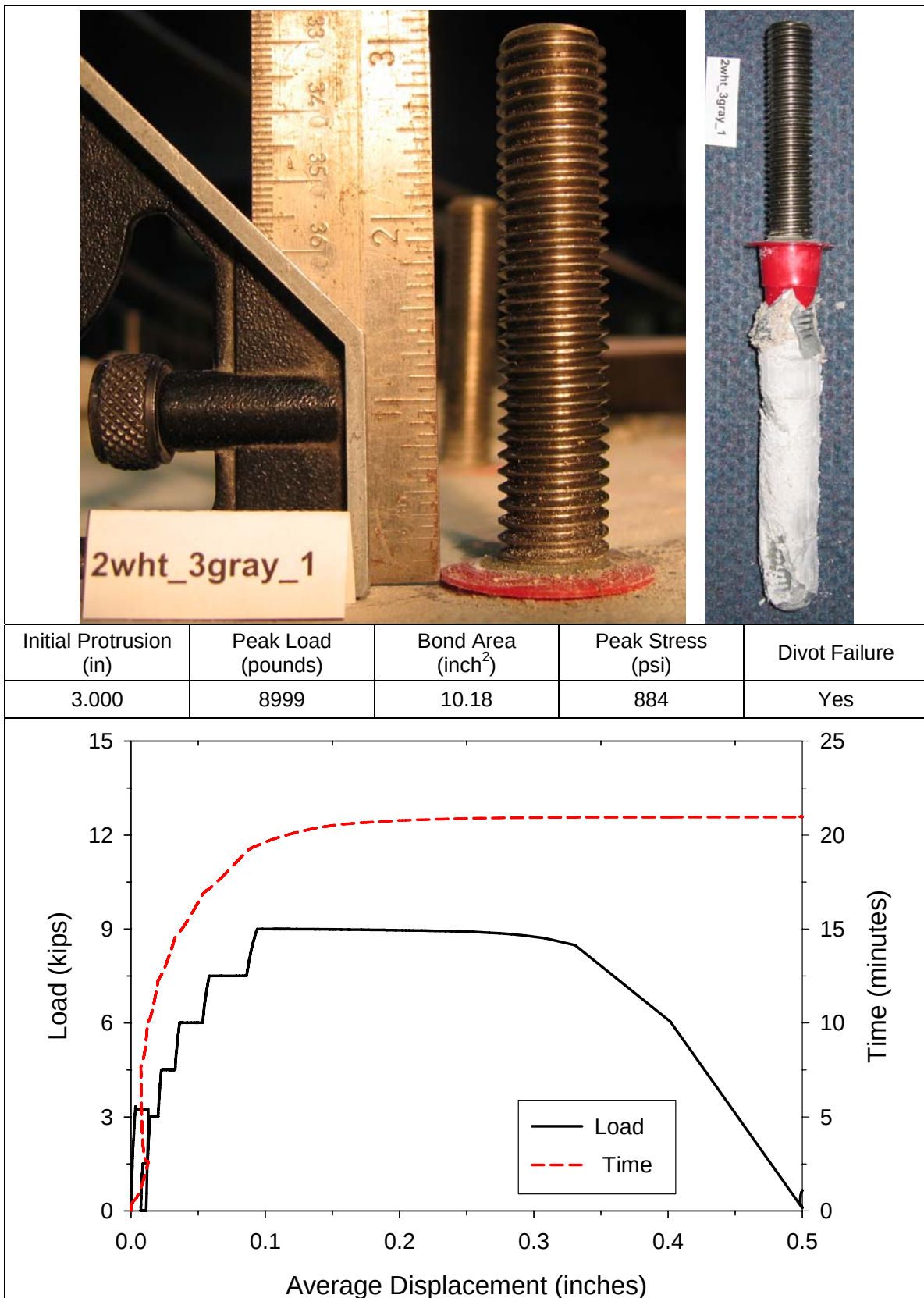


Table 167 2 white : 3 gray 2

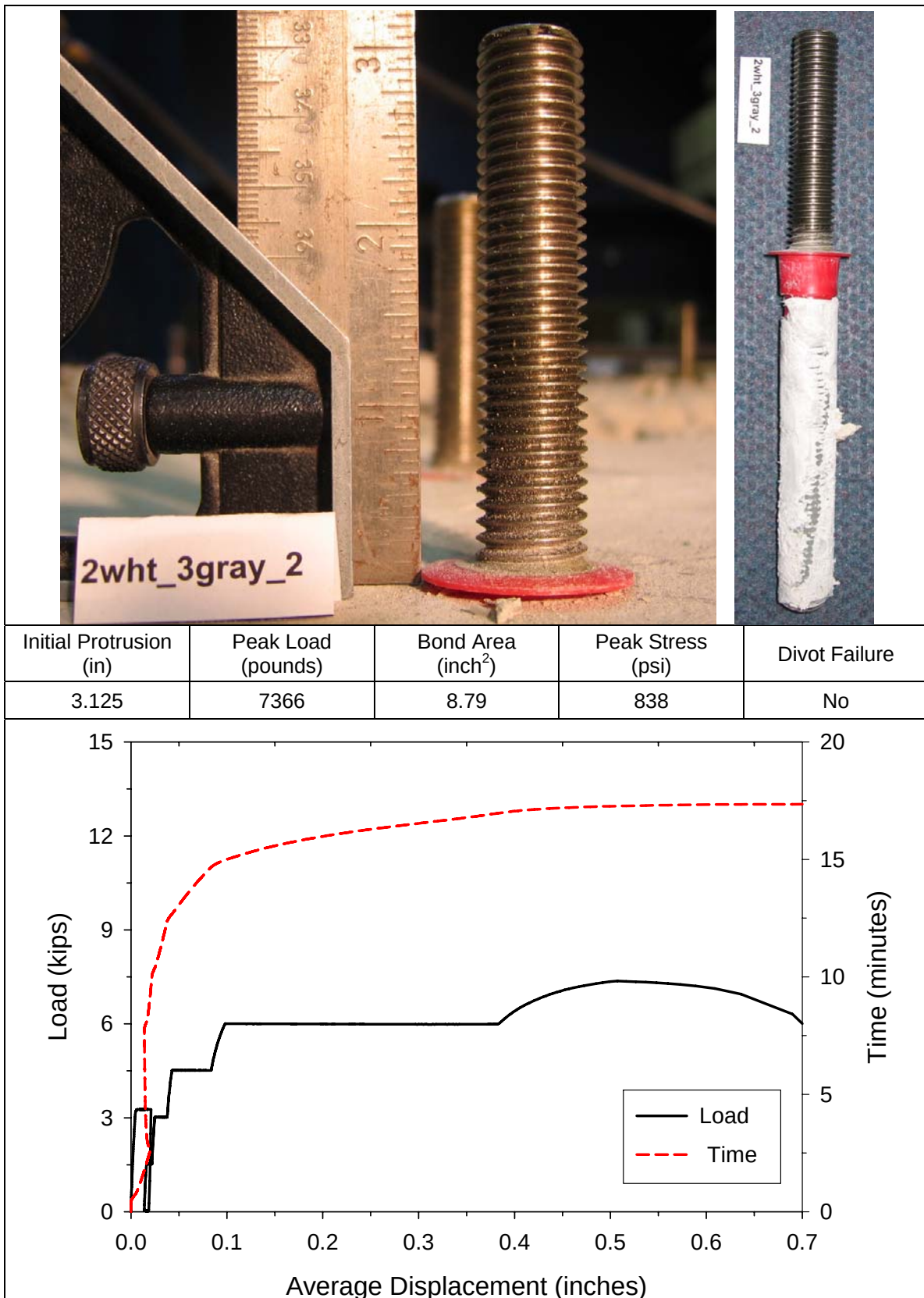


Table 168 2 white : 3 gray 3

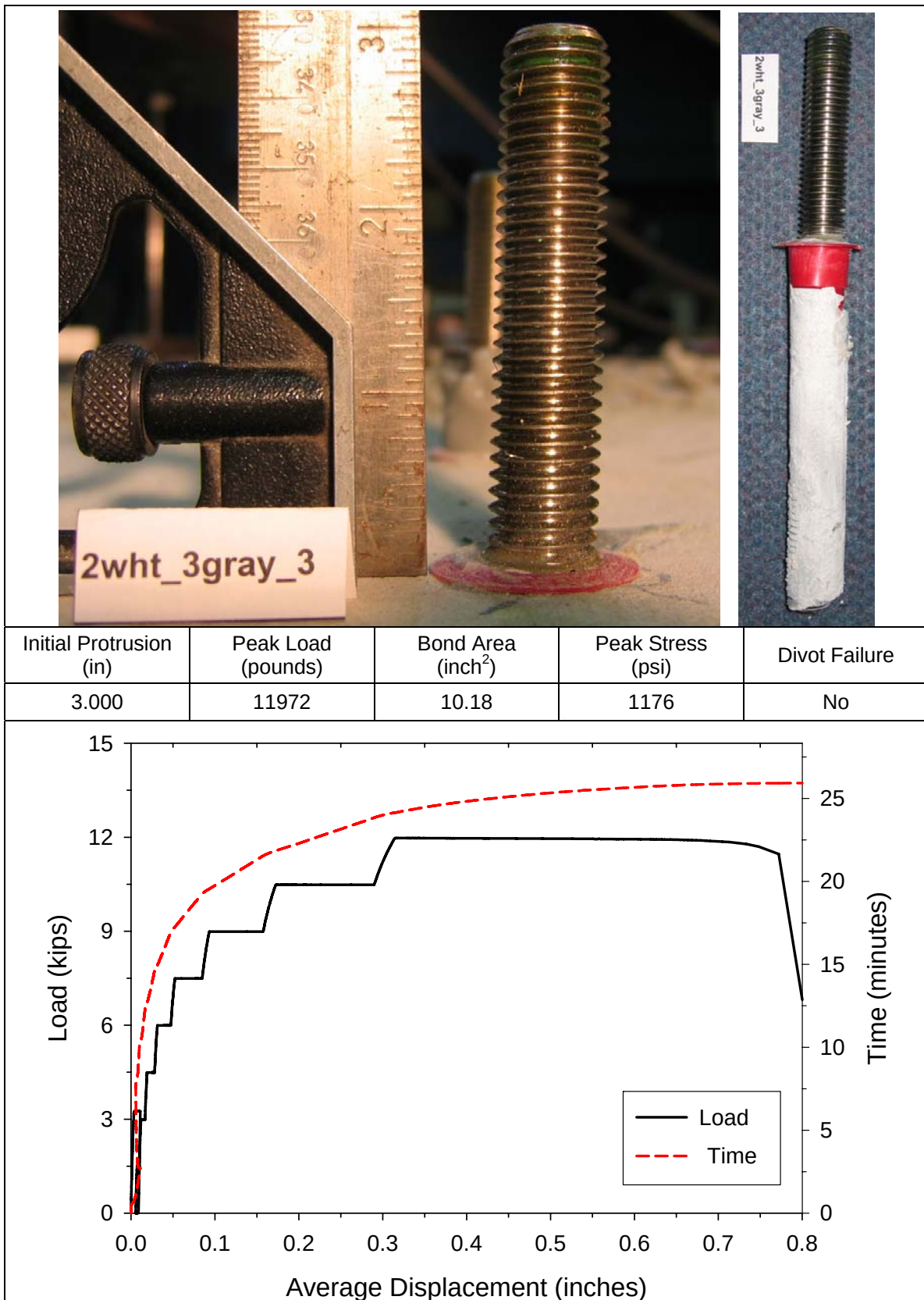


Table 169 2 white : 3 gray 4

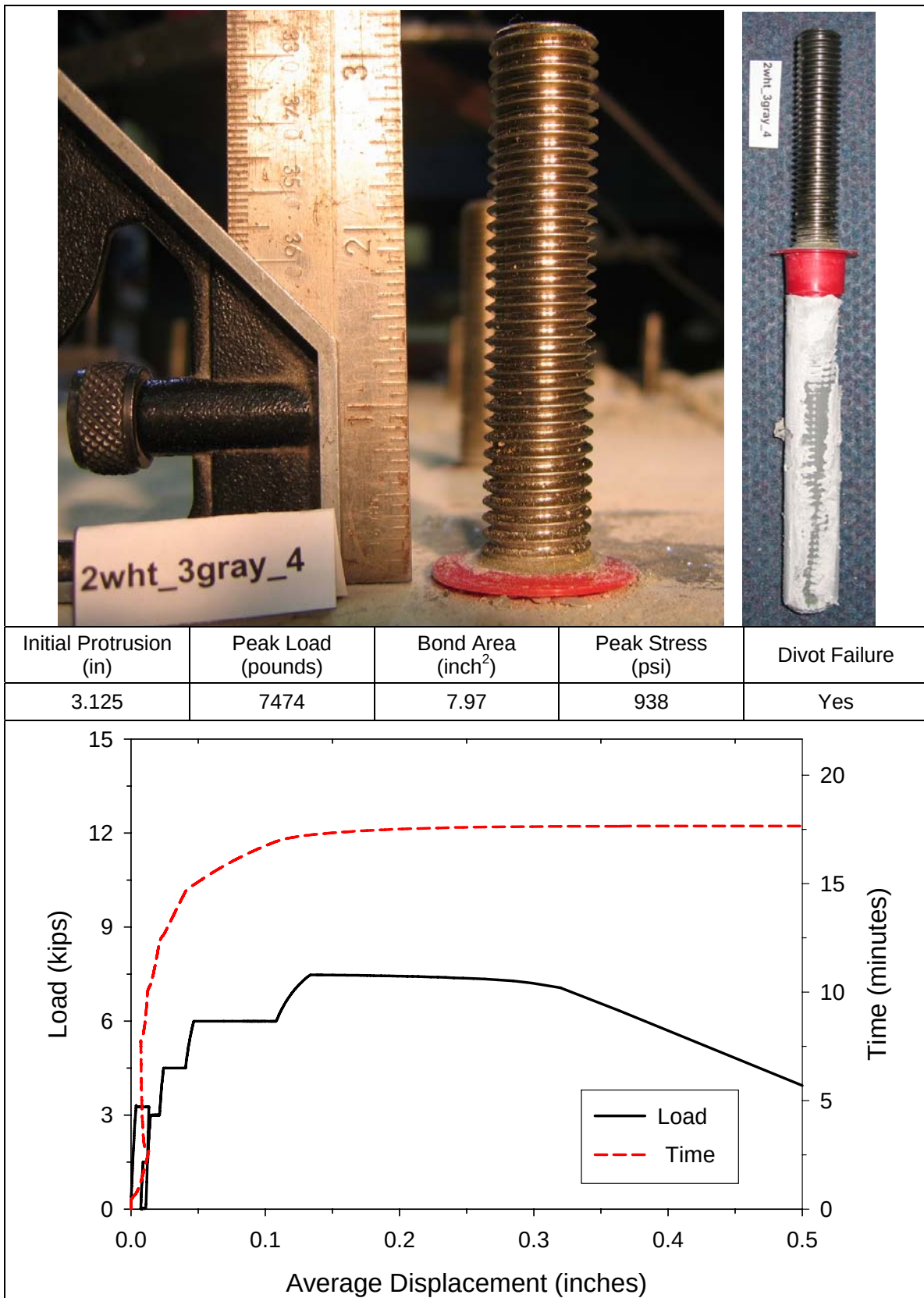


Table 170 2 white : 3 gray 5

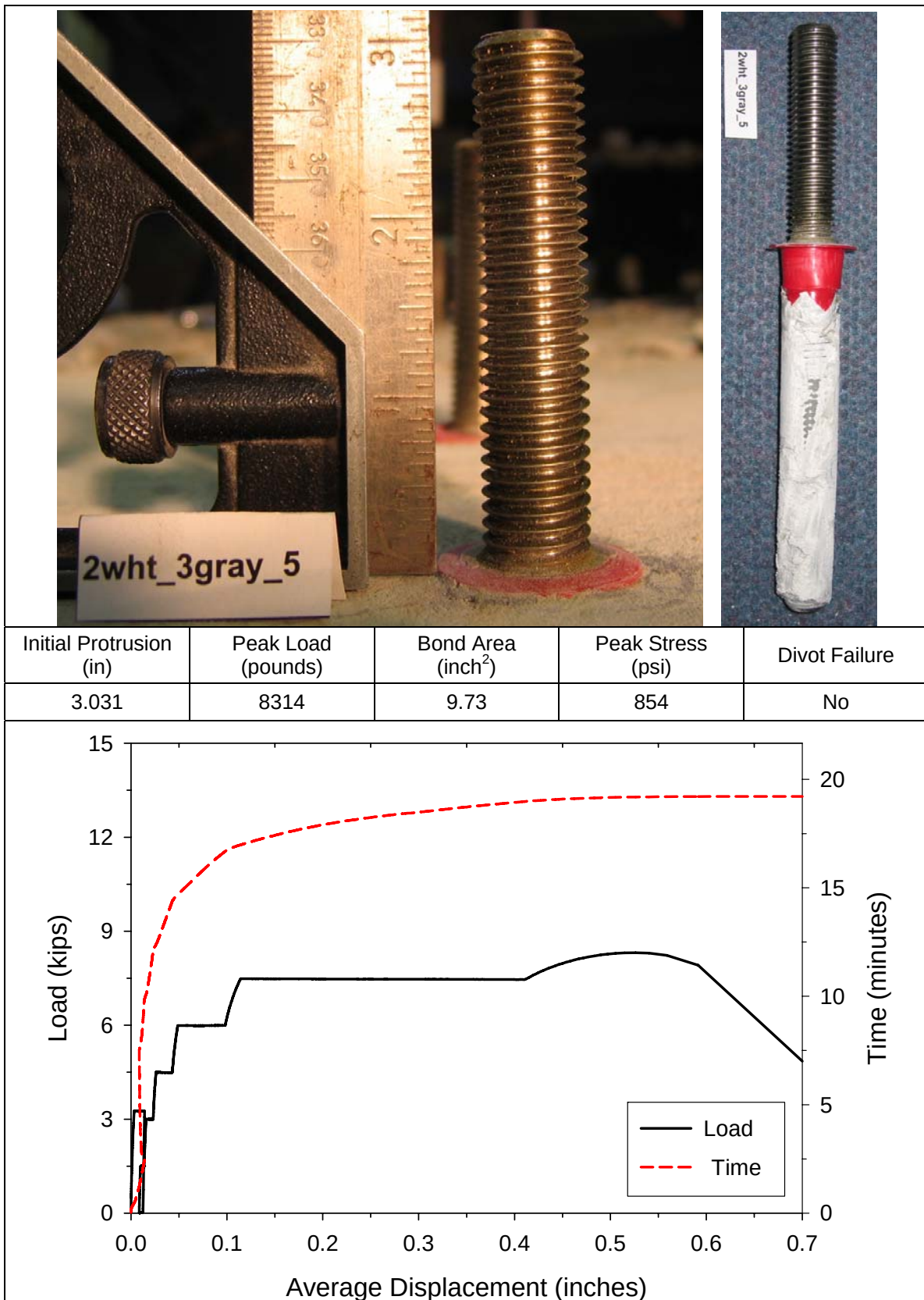


Table 171 Baked Epoxy 1

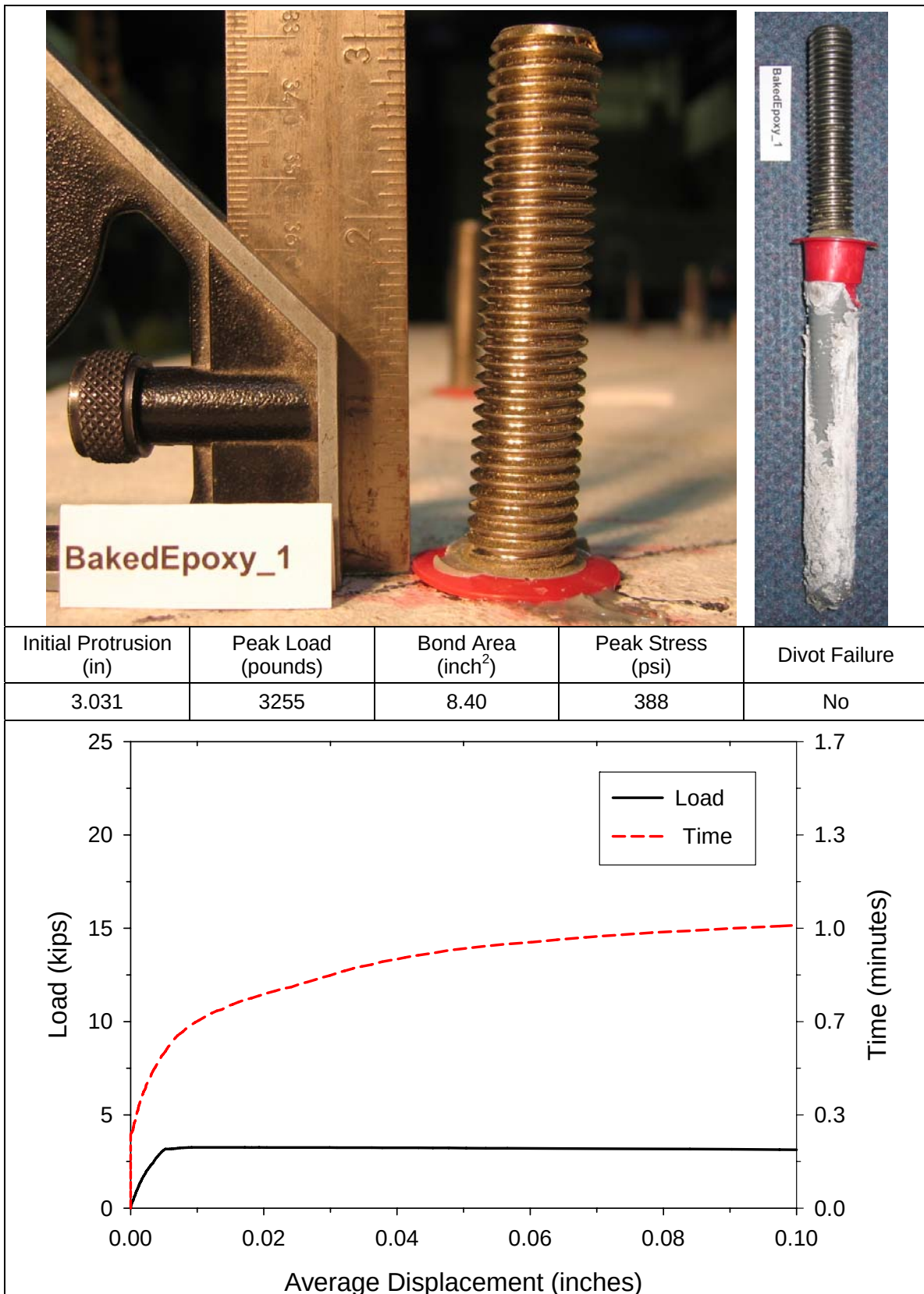


Table 172 Baked Epoxy 2

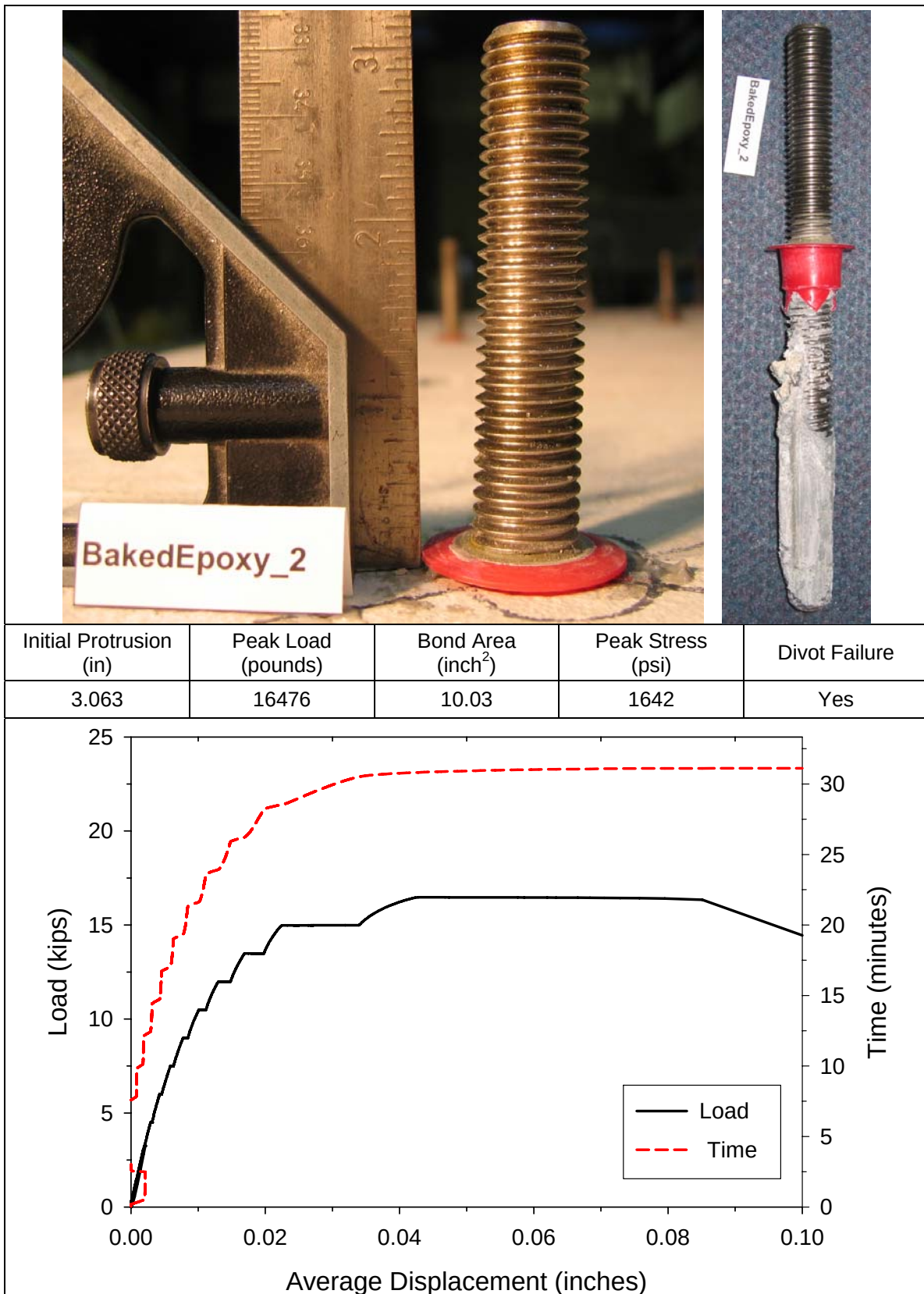


Table 173 Baked Epoxy 3

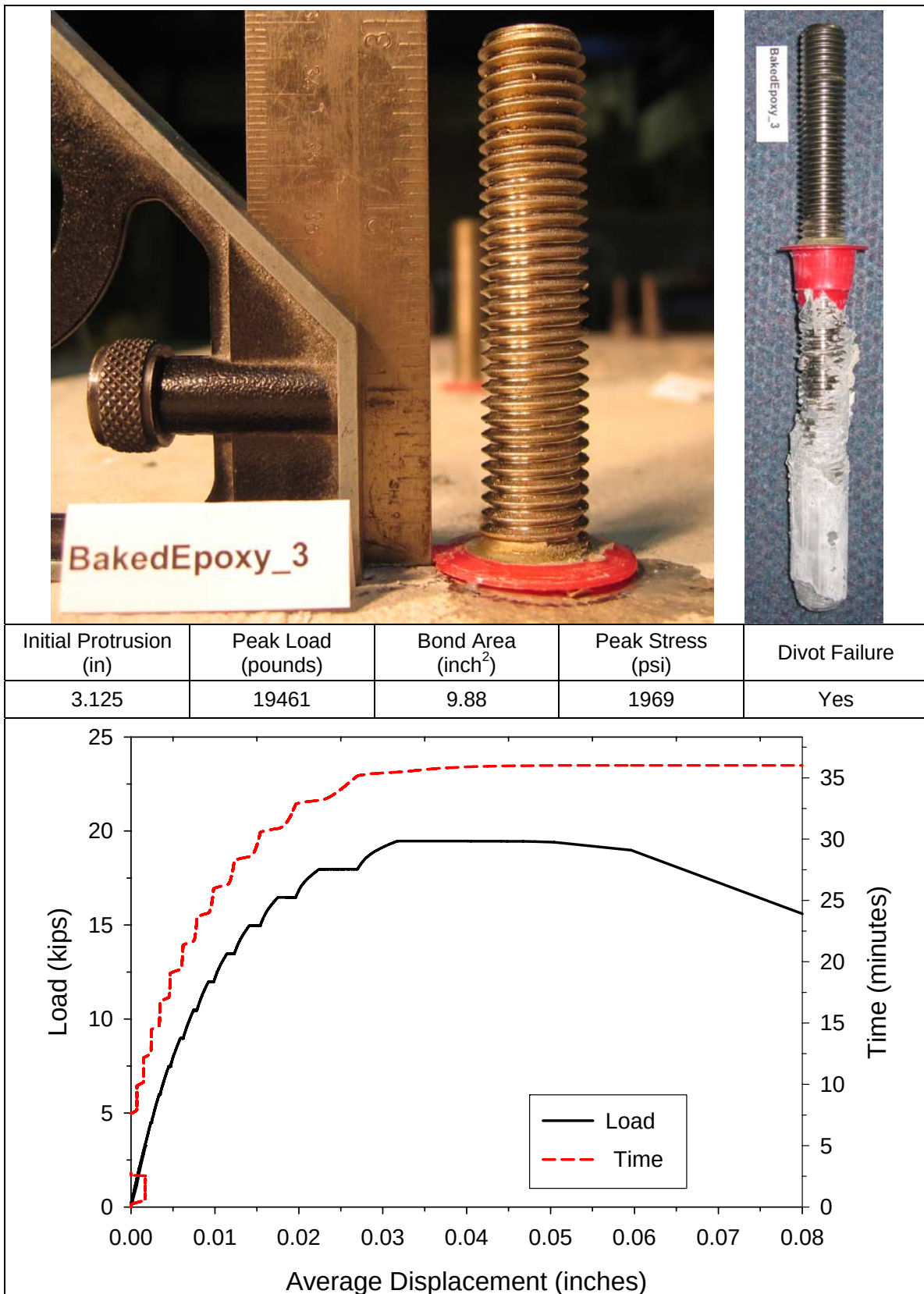


Table 174 Baked Epoxy 4

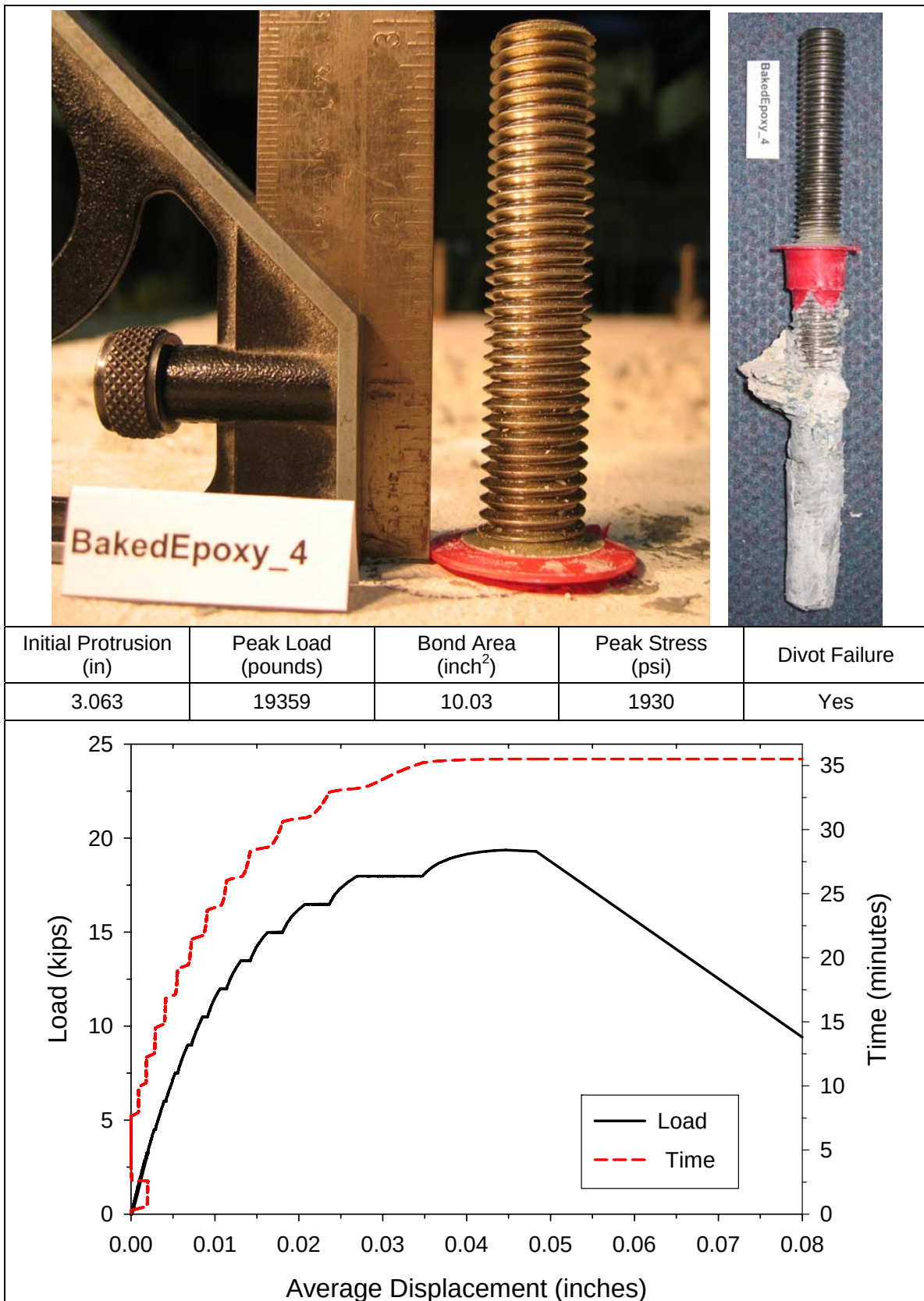


Table 175 Baked Epoxy 5

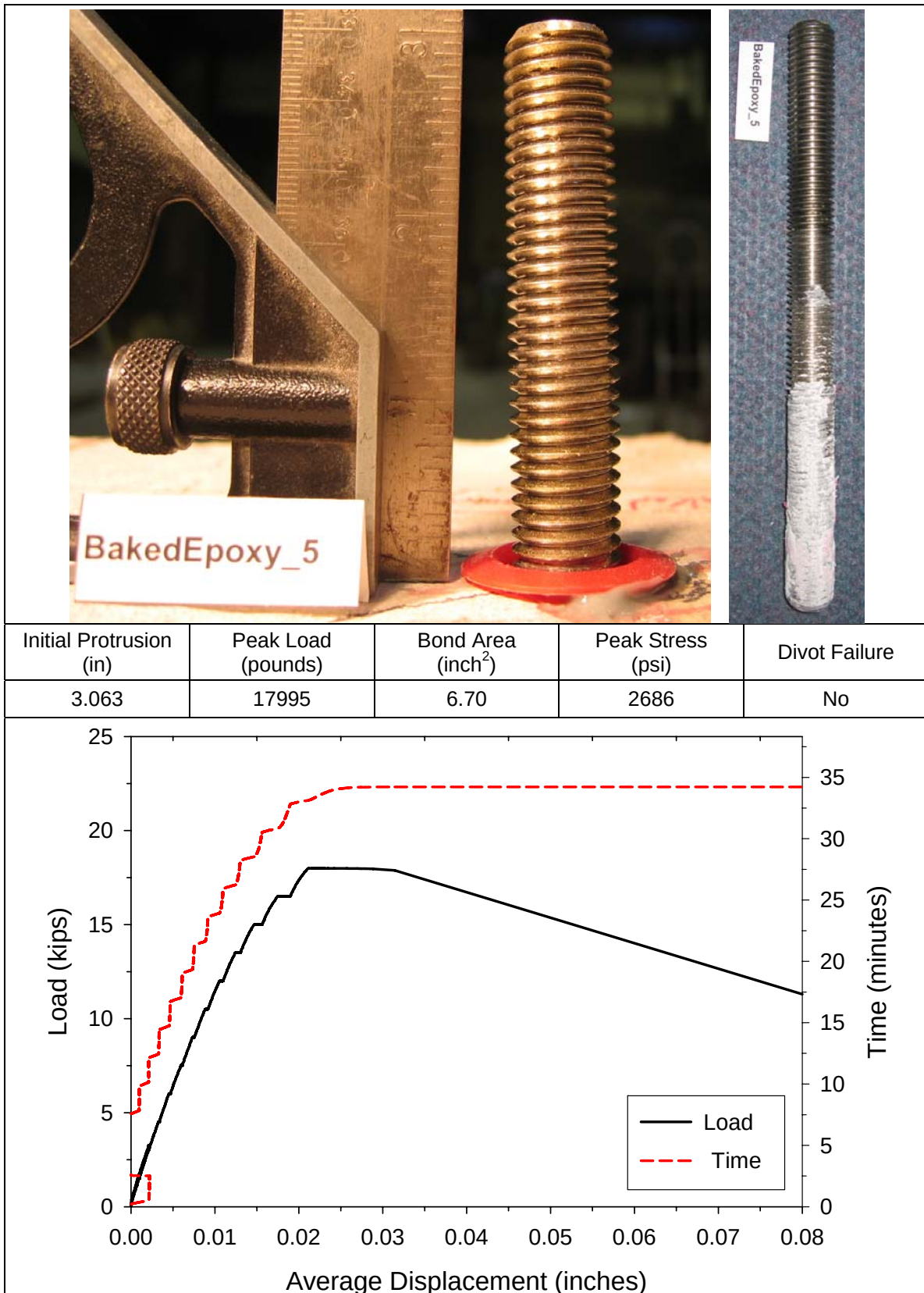


Table 176 Incompletely Mixed Epoxy 1

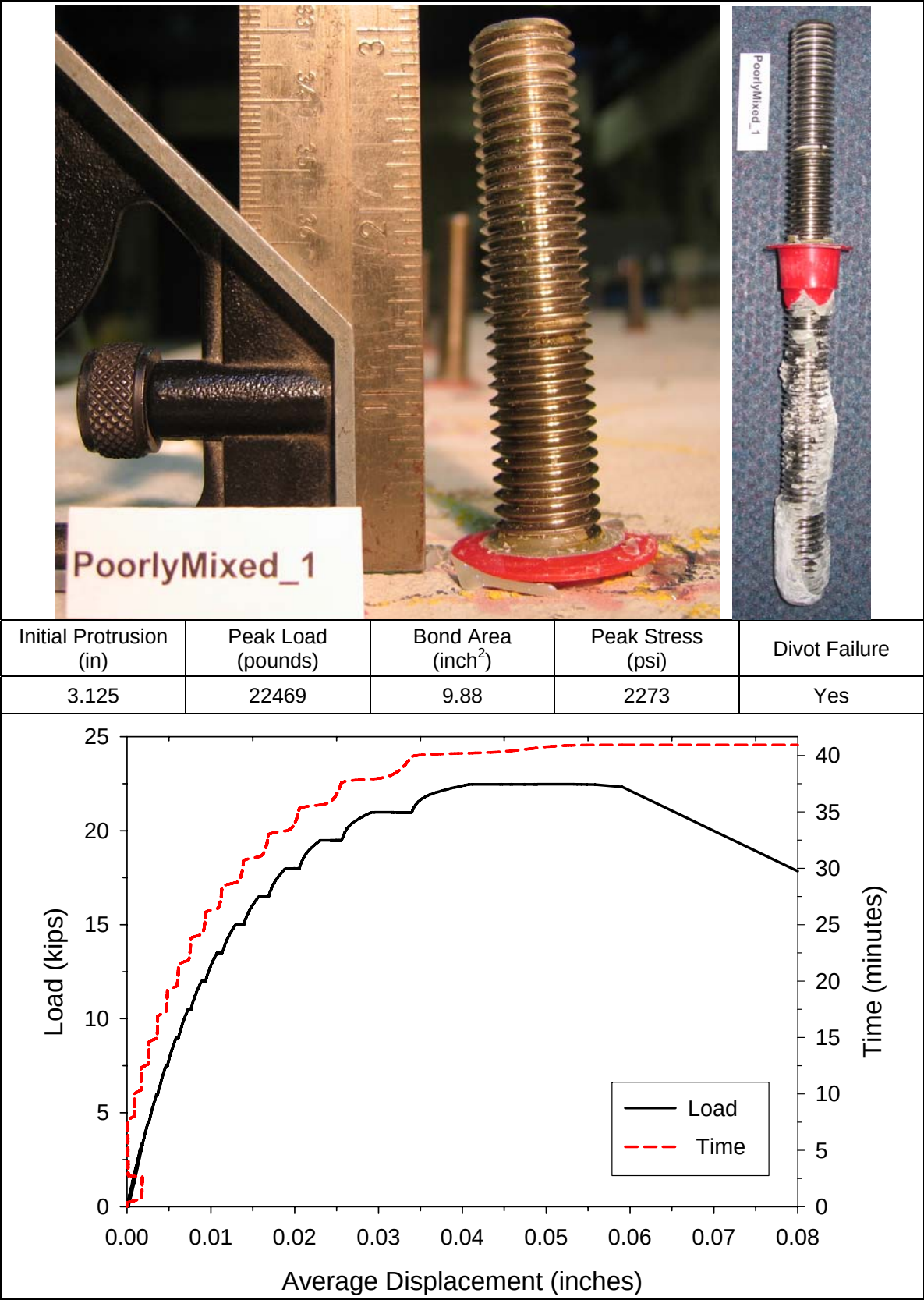


Table 177 Incompletely Mixed Epoxy 2

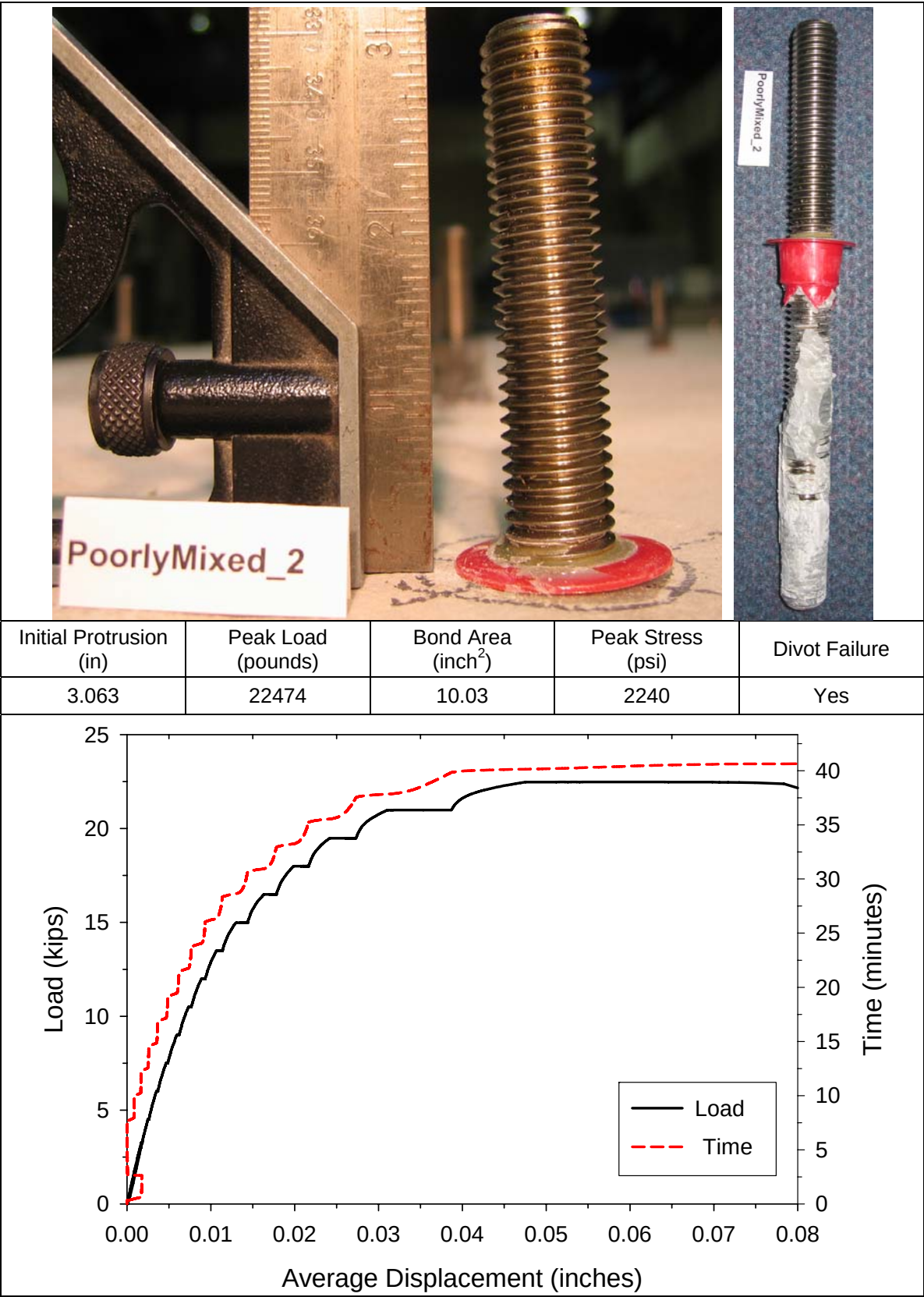


Table 178 Incompletely Mixed Epoxy 3

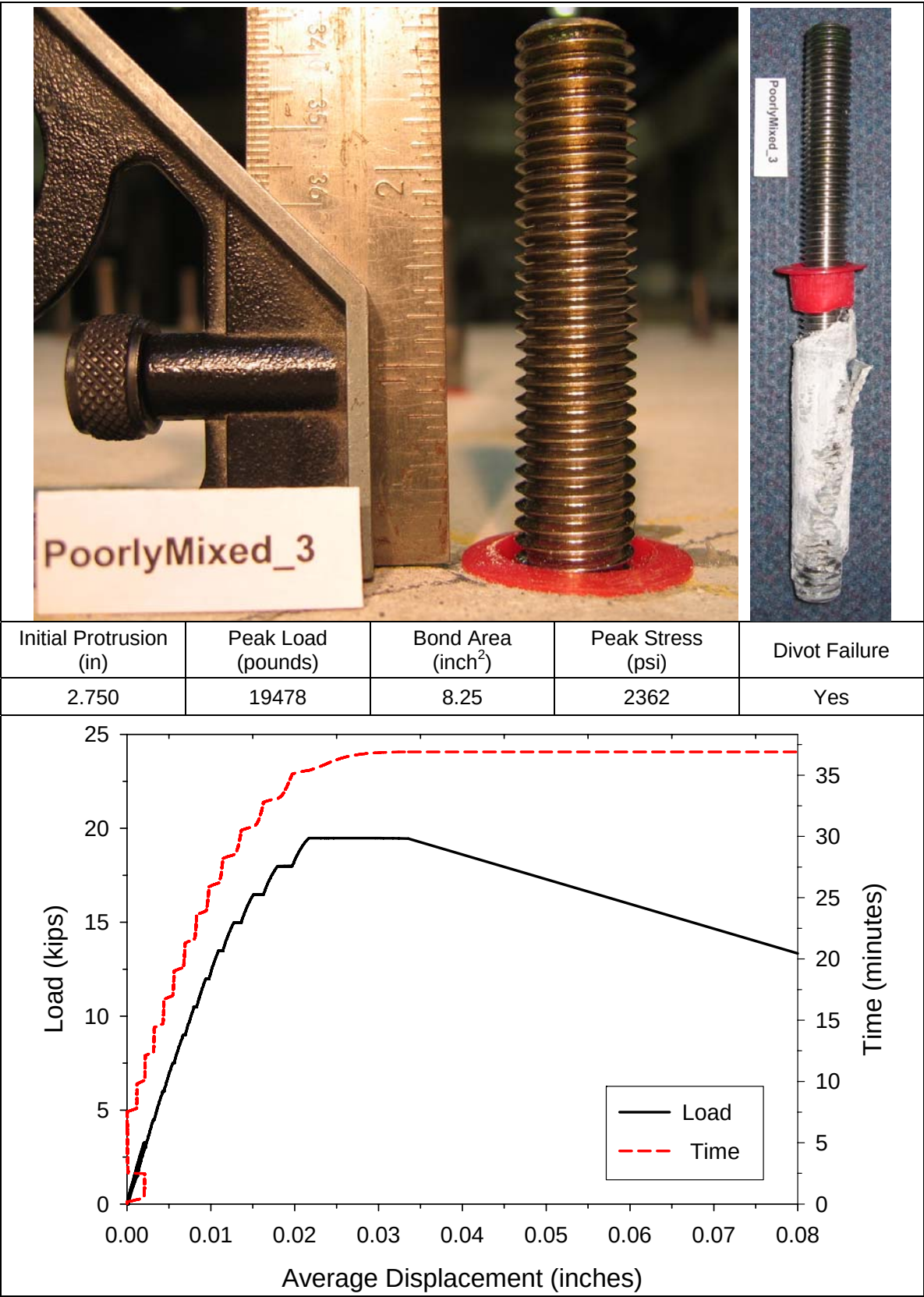


Table 179 Incompletely Mixed Epoxy 4

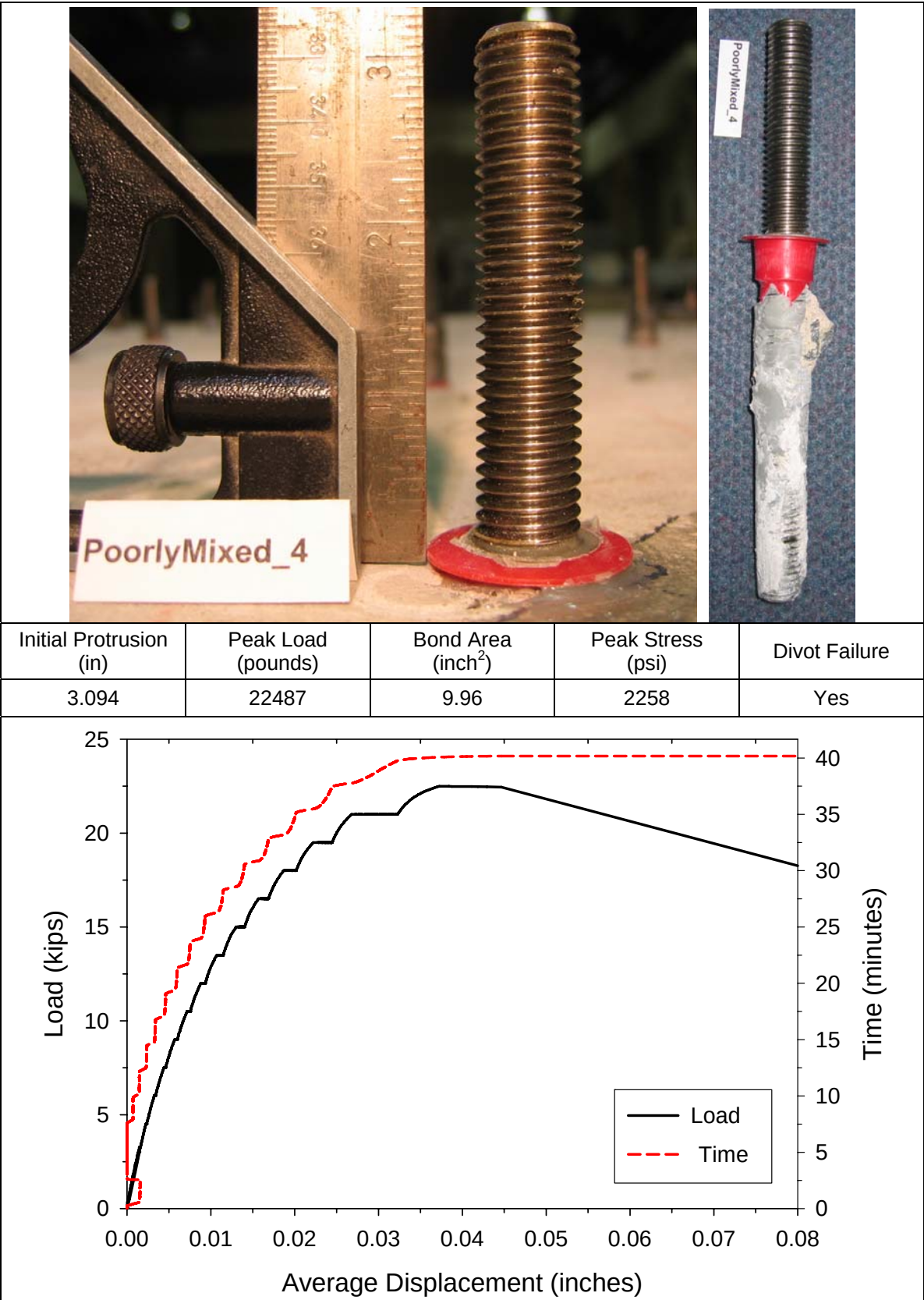


Table 180 Incompletely Mixed Epoxy 5

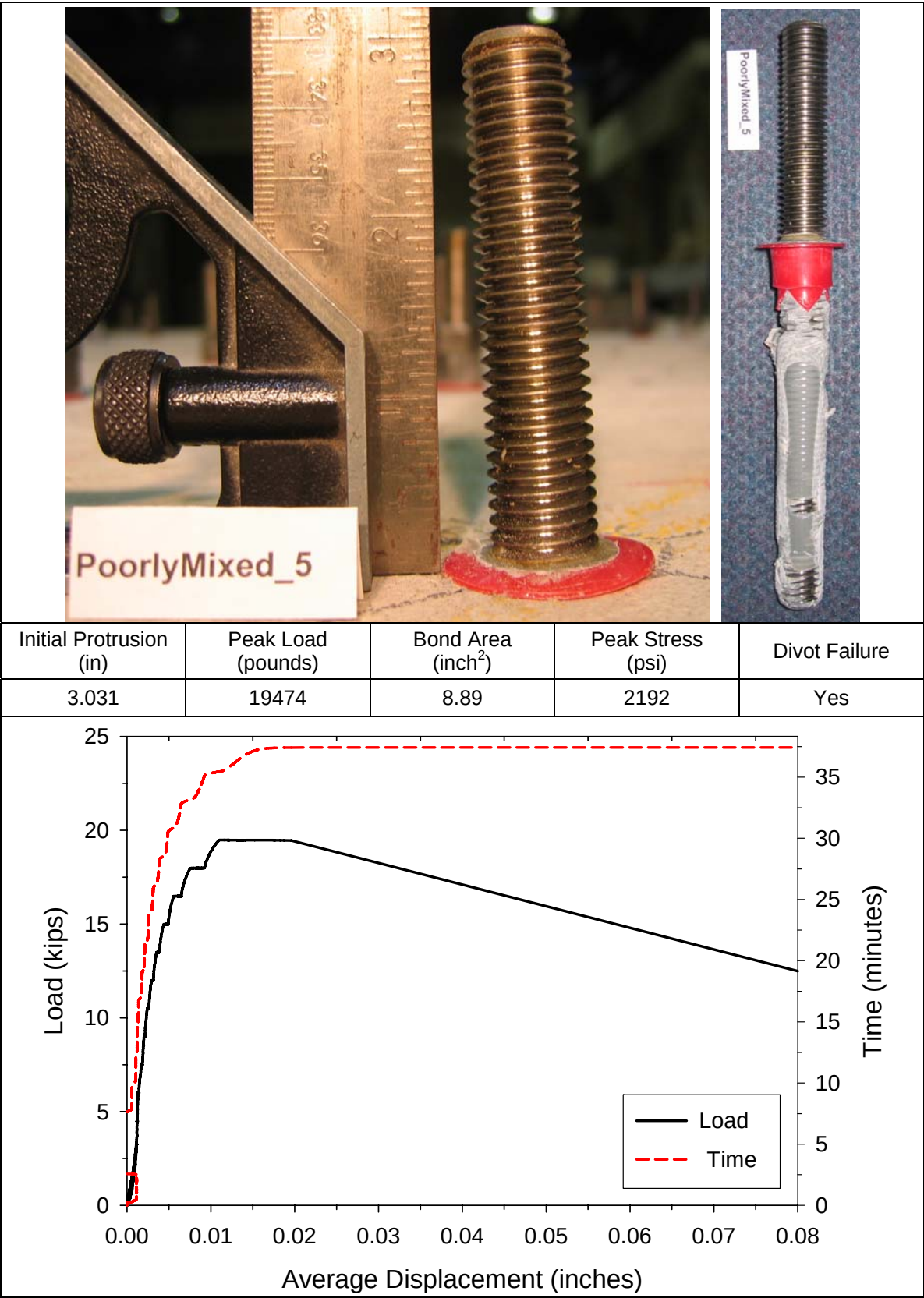


Table 181 Worst Case Install 1

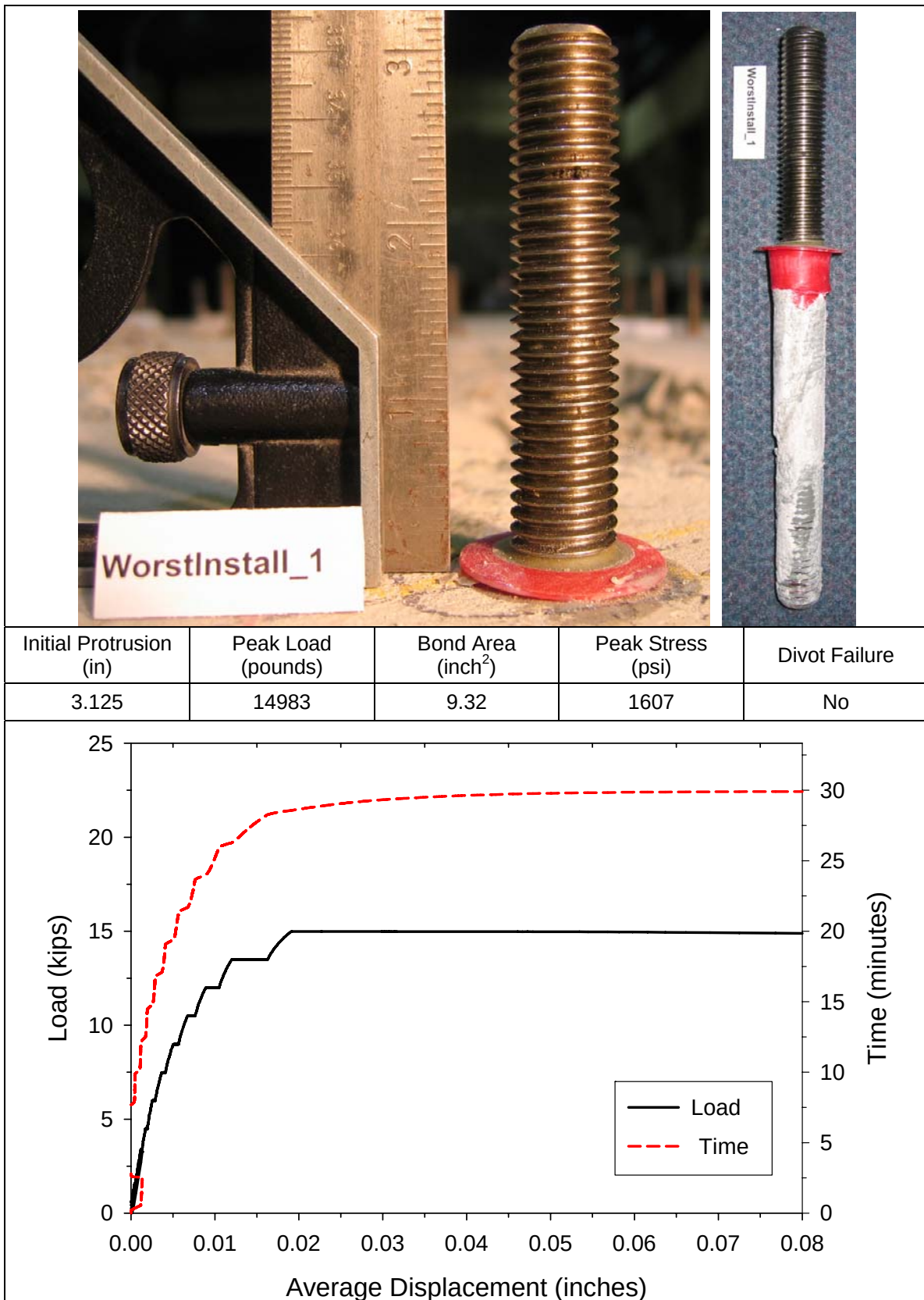


Table 182 Worst Case Install 2

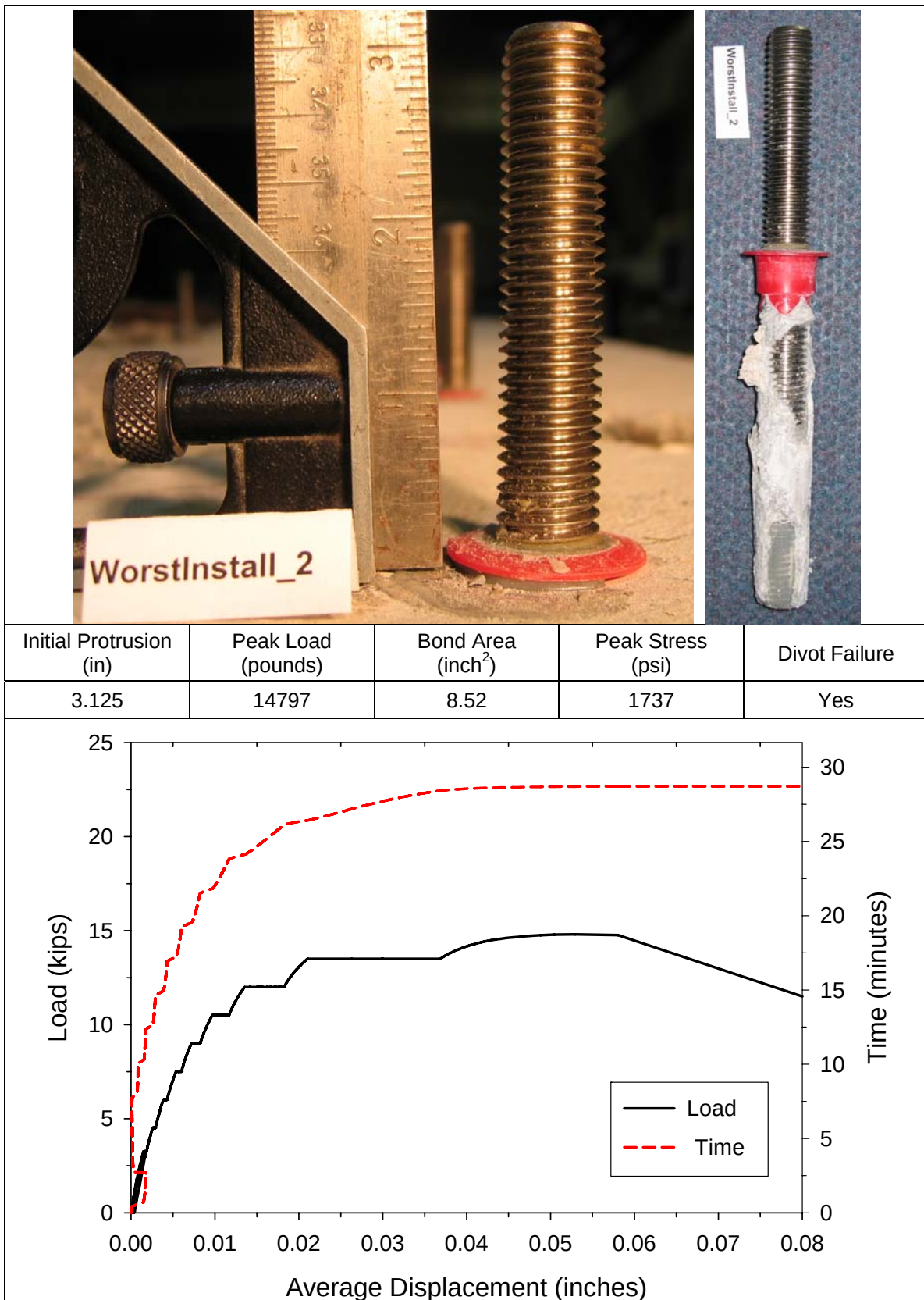
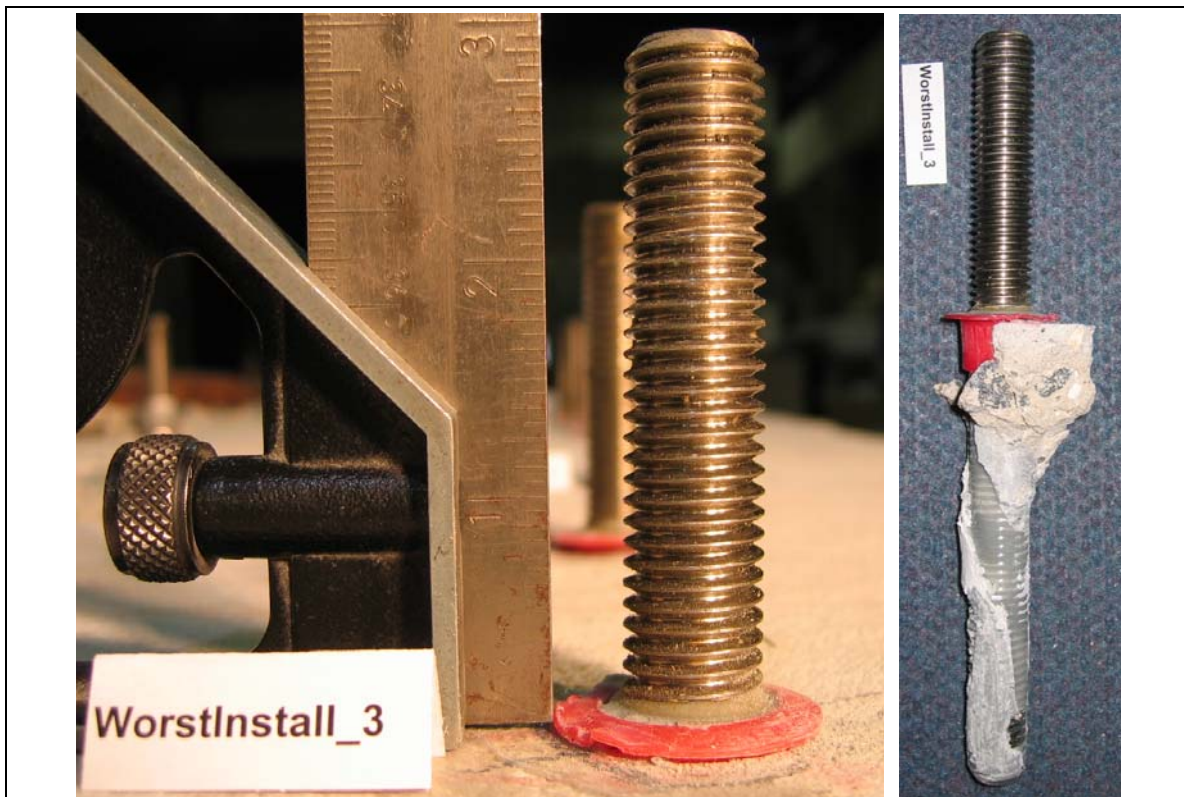


Table 183 Worst Case Install 3



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.000	13298	7.48	1779	Yes

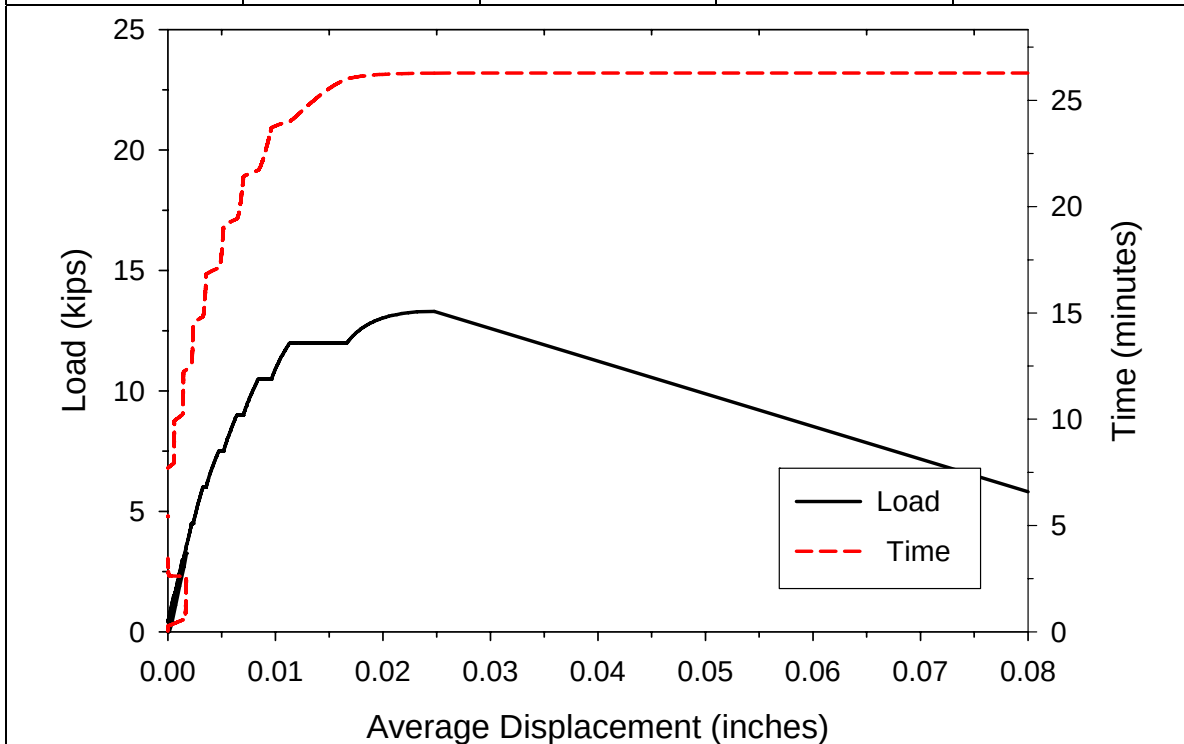


Table 184 Worst Case Install 4

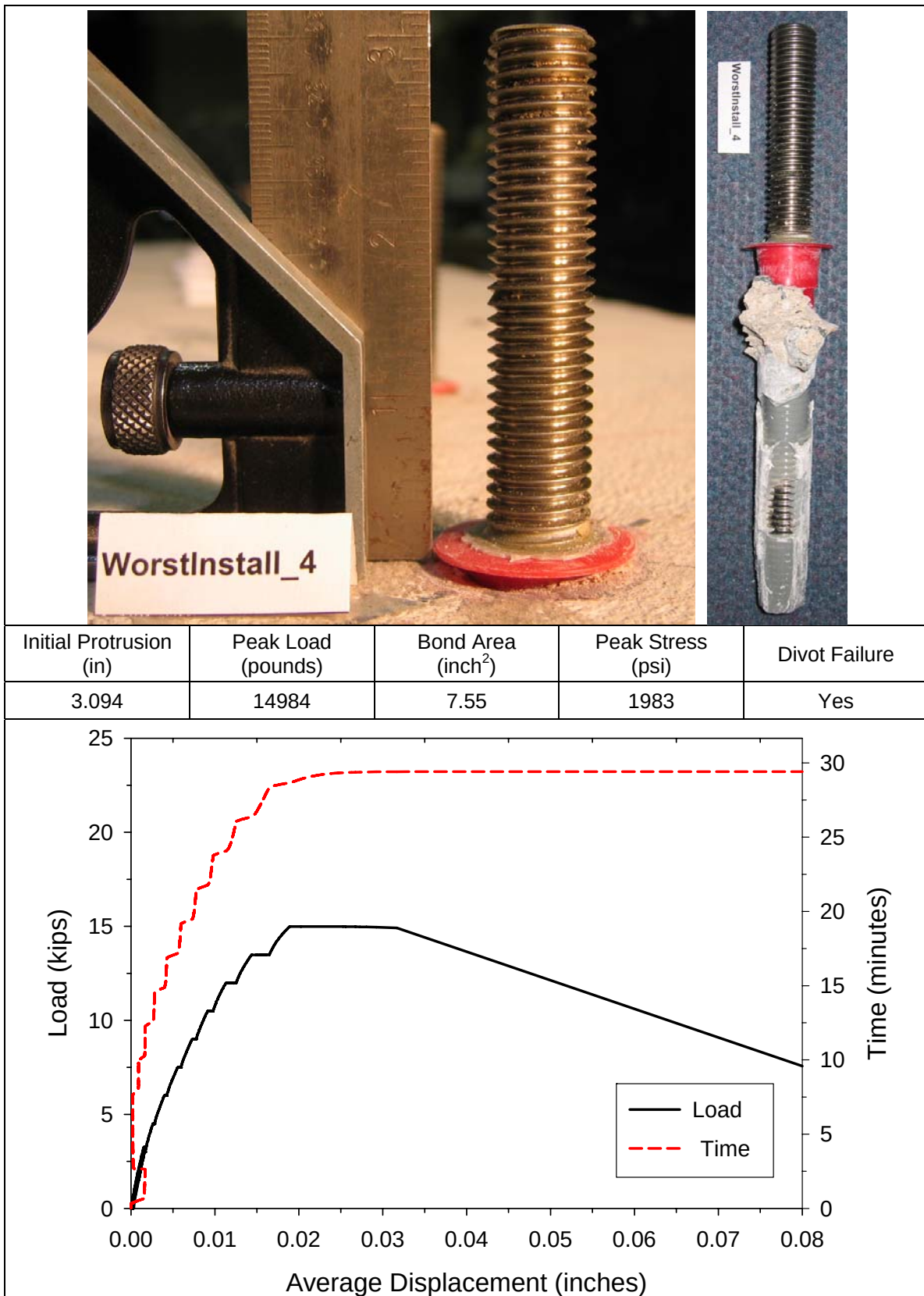


Table 185 Worst Case Install 5

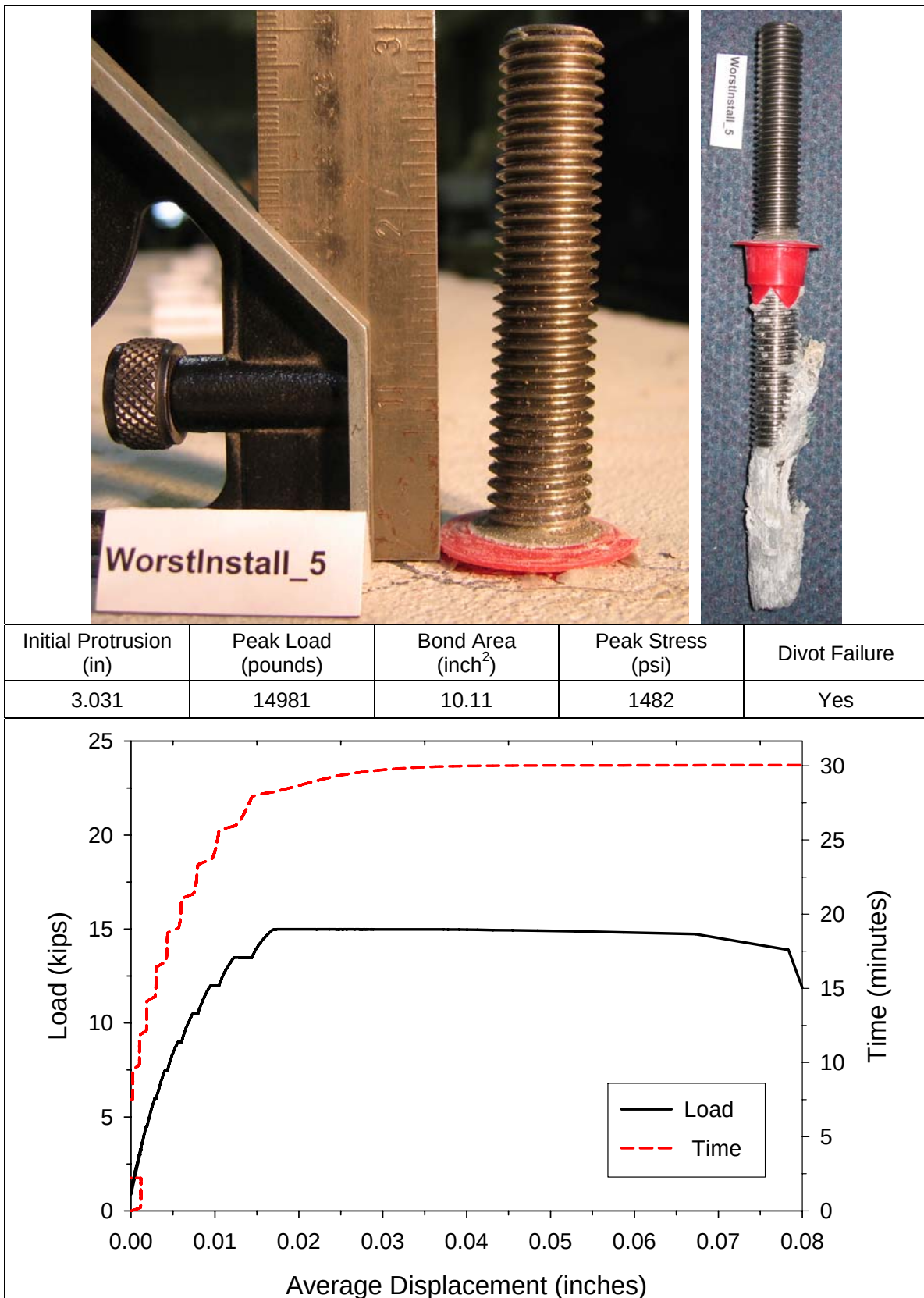


Table 186 Worst Case Install 5

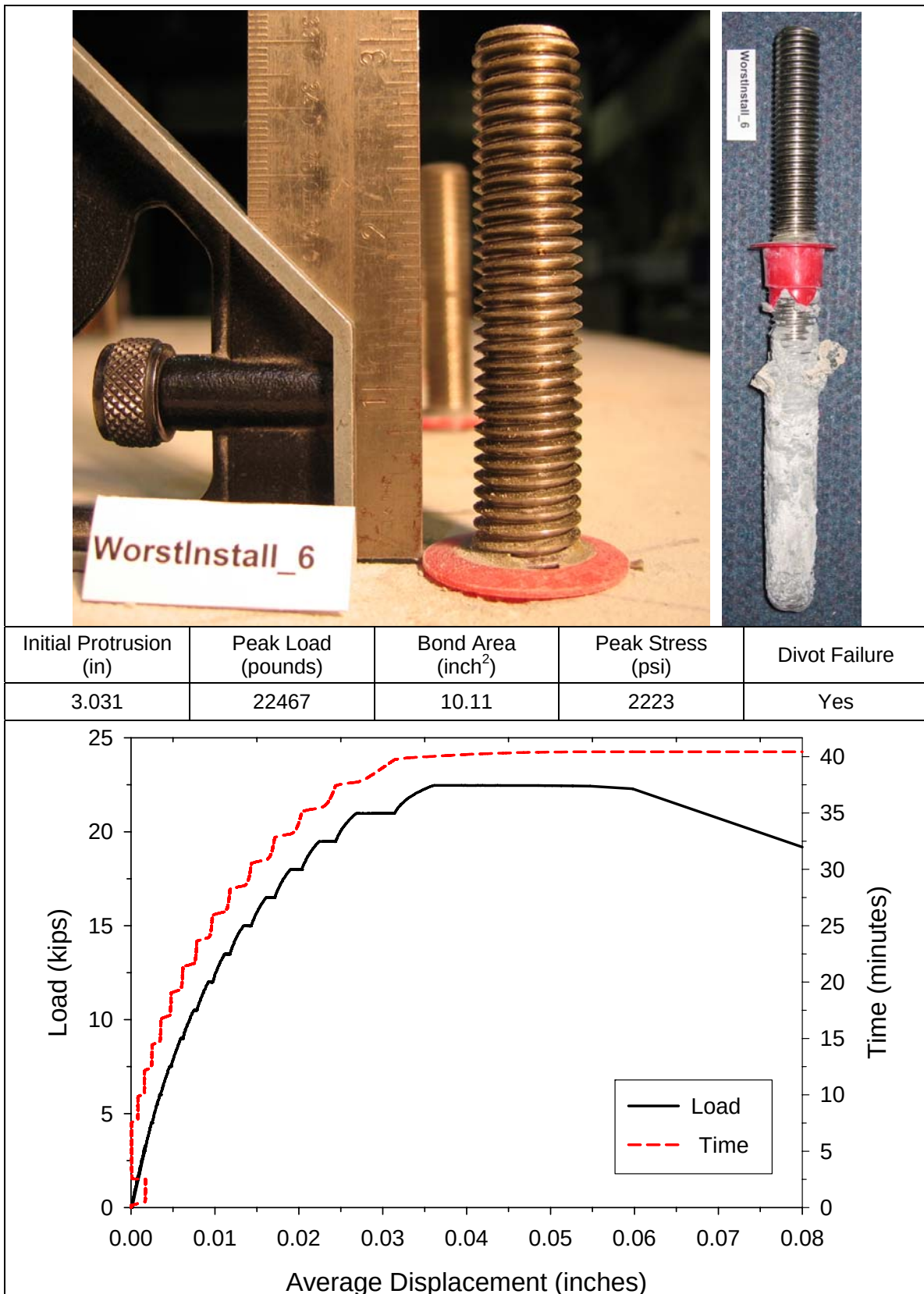


Table 187 Worst Case Install 7

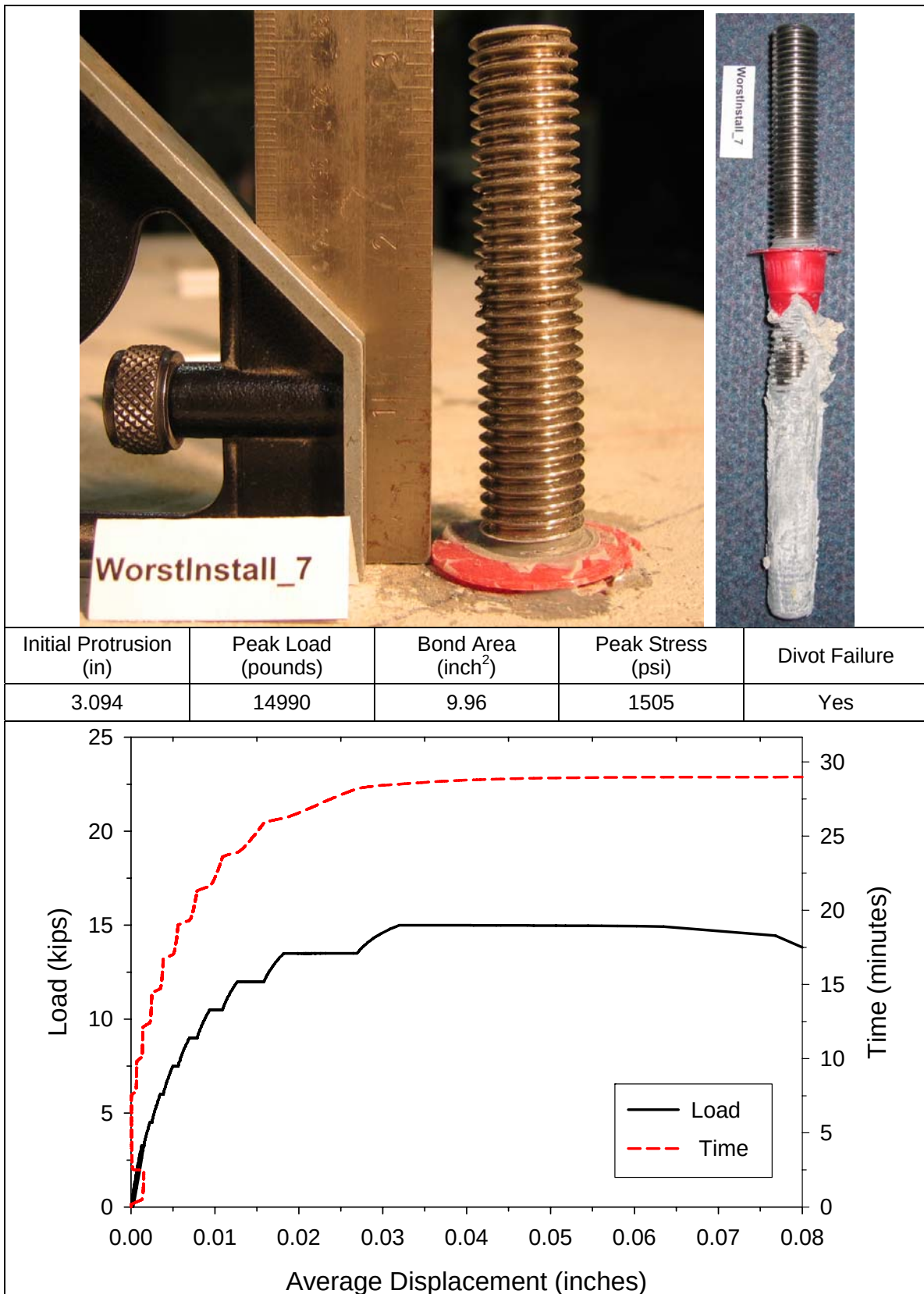


Table 188 Worst Case Install 8

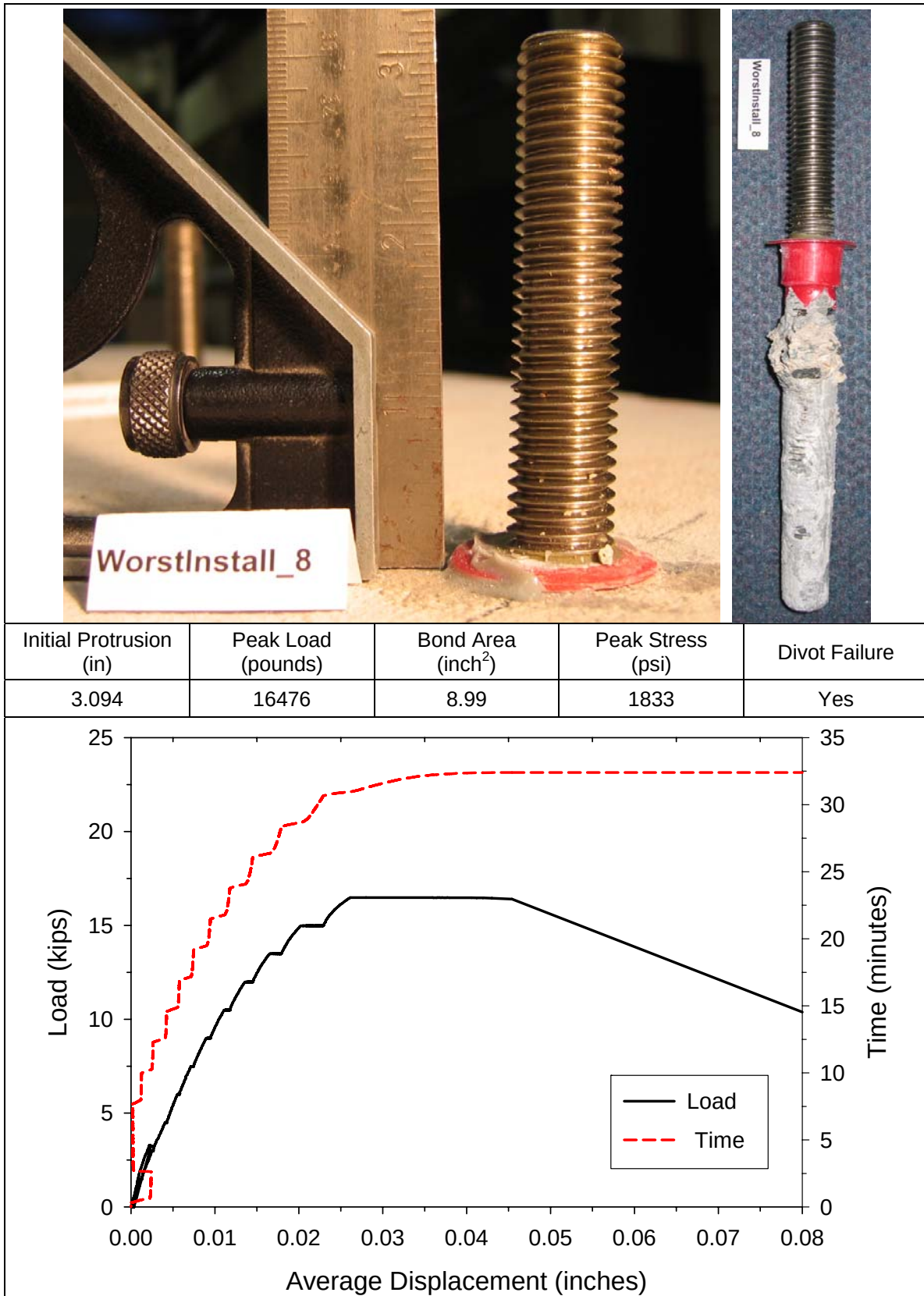


Table 189 Worst Case Install 9

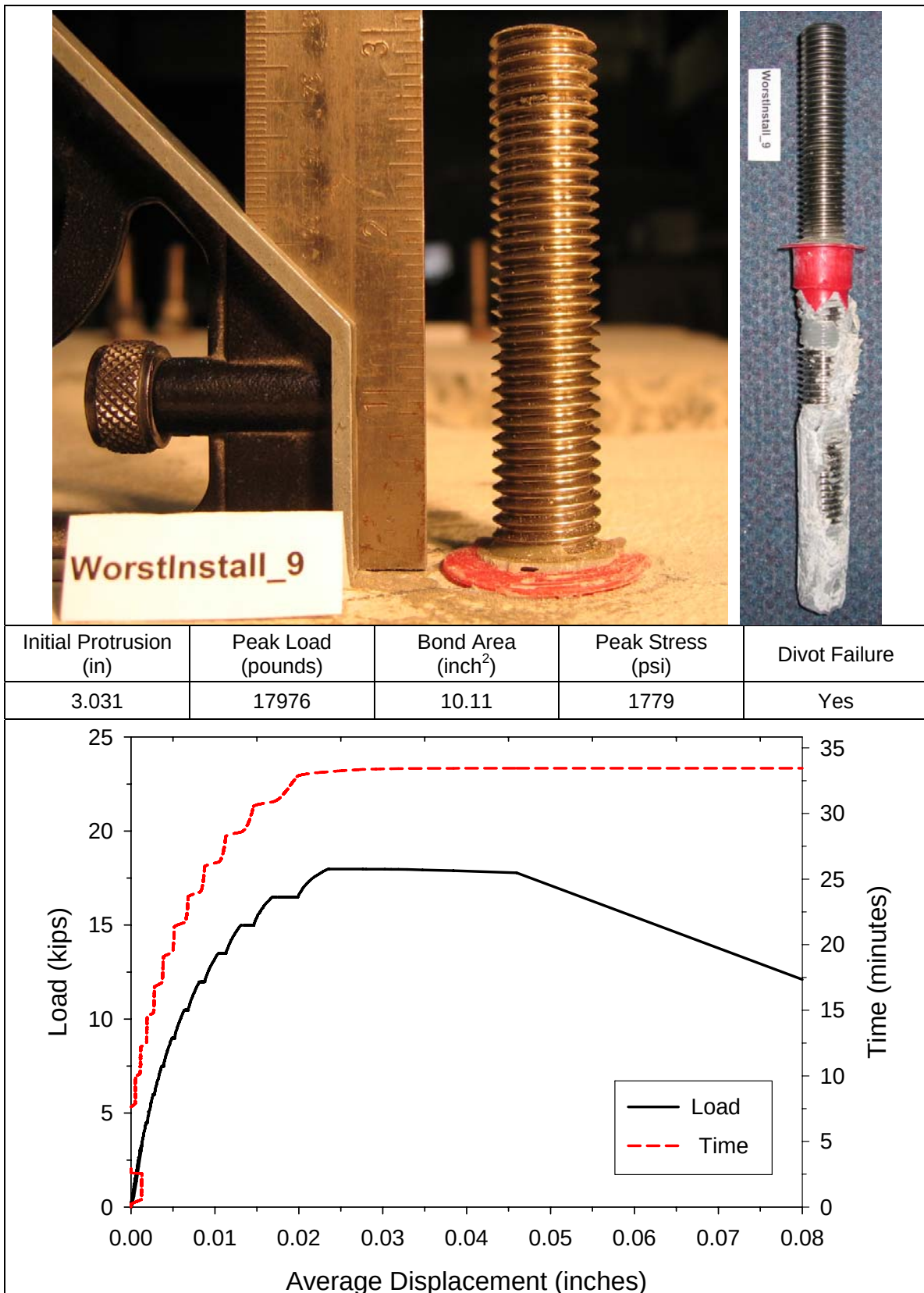


Table 190 Worst Case Install 10

